



PhD Dissertation

The Power of Input Perturbation for Data Privacy

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THE POWER OF INPUT PERTURBATION FOR DATA PRIVACY

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PhD Dissertation

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Dedicated to my family,
for their love and unwavering support,
on every path I chose.

Differential privacy protects individuals by releasing statistics that have been perturbed with carefully calibrated random noise. The thesis studies *input-perturbation* mechanisms, which first perturb the input, then run an arbitrary non-private algorithm on the perturbed input, and only then release its output. Canonical Additive Noise Mechanisms such as the Laplace or Gaussian mechanisms are special cases where the final step is the identity map. We ask when these approaches achieve optimal privacy-utility trade-offs, and develop matching error upper and lower bounds in three different settings. The main contributions of this thesis are:

- Private Graph Optimization.** We establish nearly tight upper and lower bounds in the edge-weight model of Sealfon [Sea16], where a graph topology $G = (V, E)$ with n vertices and m edges is public and only the edge weights $\mathbf{w} \in \mathbb{R}^m$ contain sensitive information. Under ℓ_∞ neighboring weights, we show that privately releasing the edges of a Minimum-Weight Spanning Tree reduces to adding Gumbel noise to each edge *once* before running any non-private MST algorithm $\mathcal{A}$ on top: the output distribution exactly matches a privatized Kruskal algorithm that iteratively calls the exponential mechanism. This runs in time $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, where $t_{\mathcal{A}}(m, n)$ is the running time of $\mathcal{A}$, improving on the previous $\mathcal{O}(m \cdot n)$. Through a connection to private top- k selection, we prove a matching lower bound, showing the $\tilde{\Theta}(n^{3/2})$ expected additive error is optimal up to log factors. Under ℓ_1 neighbors, we give novel reconstruction-based lower bounds: for Minimum-Weight Spanning Tree and Minimum-Weight (Perfect) Matching we close a remaining logarithmic gap, matching the Laplace mechanism and proving that input perturbation is optimal. For Minimum-Cost Hierarchical Clustering under Dasgupta's cost function we give the first approximate DP lower bound. Our bound is parameterized by the balanced min-cut. We show that the pure DP lower bound given in [Den+25] also holds for approximate DP as well as for sparse graphs with high connectivity, such as $\Theta(1)$ -degree expanders.
- Private Lossless Multiple Release.** We introduce the framework of Private Lossless Multiple Release, providing a general characterization of when the same statistic can be released under multiple privacy levels, in any order, and without *any* privacy loss beyond that of the least private release. We show that this is possible for every additive noise mechanism whose noise distribution satisfies a decomposition property we call *convolution preorder*, which is satisfied by, e.g., the Gaussian, Laplace, and Poisson distributions. Furthermore, we initiate the study of non-additive mechanisms like *Randomized Response* via a Markov-chain formulation and provide a counterexample where lossless release is provably impossible.
- Sparse Gaussian Histograms with Correlated Noise.** We improve the private release of sparse histograms by exploiting the sensitivity structure induced by the add/remove neighboring relationship. Combining insights about correlating noise from [Leb25] together with the exact analysis of the (Sparse) Gaussian mechanism [Wil+24; BW18], we obtain a utility improvement of up to a factor of two. We extend these techniques to aggregate and top- k queries. Furthermore, our experiments confirm these gains.

Differentiel privathed beskytter individer ved at offentliggøre statistikker, der er blevet forstyrret med nøje kalibreret tilfældig støj. Denne afhandling studerer *input-perturbation*-mekanismer, som først tilføjer støj til inputtet, derefter kører en vilkårlig ikke-privat algoritme på det perturberede input og først derefter offentliggør algoritmens output. Kanoniske *additive støjmekanismer*, såsom Laplace- eller Gauss-mekanismen, er specialtilfælde, hvor det sidste trin er identitetsafbildningen. Vi undersøger, hvornår disse tilgange opnår optimale afvejninger af privathed mod nytte i tre forskellige problemstillinger. Afhandlingens hovedbidrag er:

- **Privat grafoptimering.** Vi etablerer næsten tætte øvre og nedre grænser i modellen fra Sealfon [Sea16], hvor en graftopologi $G = (V, E)$ med n knuder og m kanter er offentlig, og kun kantvægtene $\mathbf{w} \in \mathbb{R}^m$ indeholder sensitiv information.

Under ℓ_∞ -naborelationer mellem vægte viser vi, at privat offentliggørelse af et minimum udspændende træ reduceres til at tilføje støj fra en Gumbel-fordeling til hver kant *én gang*, før en vilkårlig ikke-privat **MST**-algoritme $\mathcal{A}$ køres. Outputfordelingen stemmer præcist overens med en privatiseret Kruskal-algoritme, som iterativt kalder den eksponentielle mekanisme. Dette kører i tid $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, hvor $t_{\mathcal{A}}(m, n)$ er køretiden for $\mathcal{A}$, og forbedrer den tidligere $\mathcal{O}(m \cdot n)$. Vi etablerer en forbindelse til privat top- k -udvælgelse og beviser en matchende nedre grænse, som viser, at den forventede additive fejl $\tilde{\Theta}(n^{3/2})$ er næsten optimal. Under ℓ_1 -naboer giver vi nye rekonstruktionsbaserede nedre grænser: For Minimum-Weight Spanning Tree og Minimum-Weight (Perfect) Matching lukker vi et tilbageværende logaritmisk gab, matcher Laplace-mekanismen og beviser, at inputperturbation er optimal. For Minimum-Cost Hierarchical Clustering under Dasguptas omkostningsfunktion giver vi den første nedre grænse for approksimativ DP parameteriseret ved det balancerede min-cut. Vi viser, at den nedre grænse for ren DP givet i [Den+25] også gælder for approksimativ DP og sparse grafer med høj konnektivitet, såsom tilfældige expandere af grad $\Theta(1)$.

- **Privat tabsfri multipel offentliggørelse.** Vi introducerer frameworket *Private Lossless Multiple Release*, som giver en generel karakterisering af, hvornår den samme statistik kan offentliggøres under flere privathedsniveauer, i vilkårlig rækkefølge og uden *noget* privathedstab ud over tabet fra den mindst private offentliggørelse. Vi viser, at dette er muligt for enhver mekanisme med additiv støj, hvis støjfordeling opfylder en dekompositionsegenskab, som vi kalder *convolution preorder*, og som blandt andet opfyldes af Gaussian-, Laplace- og Poissonfordelingen. Desuden påbegynder vi studiet af ikke-additive mekanismer såsom *Randomized Response* via en Markovkædeformulering og giver et modeksempel, hvor tabsfri offentliggørelse beviseligt er umulig.

- **Sparse Gaussian-histogrammer med korreleret støj.** Vi forbedrer den private offentliggørelse af sparse histogrammer ved at udnytte sensitivitetsstrukturen induceret af add/remove-naborelationen. Ved at kombinere indsigter om korreleret støj fra [Leb25] med den eksakte analyse af den (Sparse) Gaussian-mekanisme [Wil+24; BW18], opnår vi en nytteforbedring på op til en faktor to. Vi udvider disse teknikker til aggregat- og top- k -forespørgsler. Desuden bekræfter vores eksperimenter disse forbedringer empirisk.

Differential Privacy (DP) schützt Individuen, indem Statistiken durch sorgfältig kalibriertes zufälliges Rauschen verändert werden, bevor sie veröffentlicht werden. Dabei lässt sich der entstehende Fehler gegen ein formales Maß an Privatsphäre abwägen. Intuitiv führt eine stärkere Veränderung zu mehr Privatsphäre, wohingegen wenig Veränderung mehr Rückschlüsse auf die Personen zulässt, die Teil des Datensatzes sind.

Diese Doktorarbeit untersucht Algorithmen, die zunächst nur die Eingabedaten verändern, anschließend einen beliebigen, nicht privaten Algorithmus auf der verrauschten Eingabe ausführen und erst danach die Ergebnisse veröffentlichen. *Additive Rauschmechanismen*, wie der Laplace- oder Gauß-Mechanismus, sind Spezialfälle, in denen der letzte Schritt die Identitätsabbildung ist. Wir untersuchen, wann diese Ansätze eine optimale Abwägung zwischen Privatsphäre und Fehler erreichen. Die Hauptbeiträge dieser Dissertation sind:

- **Netzwerkoptimierungsprobleme.** Wir etablieren nahezu optimale Fehlerschranken im Kantengewichtsmodell von Sealfon [Seal16]. Hierbei ist eine Graphtopologie $G = (V, E)$ mit n Knoten und m Kanten öffentlich und die Kantengewichte $\mathbf{w} \in \mathbb{R}^m$ enthalten sensible Informationen. Für ℓ_∞ -benachbarte Kantengewichte zeigen wir, dass die private Veröffentlichung der Kanten eines minimalen Spannbaumes (MST) darauf reduziert werden kann, *einmal* Zufallsvariablen einer Gumbelverteilung zu jeder Kante hinzuzufügen und anschließend einen beliebigen nicht privaten MST-Algorithmus darauf auszuführen. Der resultierende Algorithmus hat eine Laufzeit von $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, wobei $t_{\mathcal{A}}(m, n)$ die Laufzeit von $\mathcal{A}$ bezeichnet. Dies verbessert den früheren Algorithmus mit Laufzeit $\mathcal{O}(m \cdot n)$. Über eine Verbindung zu privater Top- k -Selektion beweisen wir eine untere Schranke und zeigen, dass der erwartete additive Fehler von $\tilde{\Theta}(n^{3/2})$ bis auf logarithmische Faktoren optimal ist. Für ℓ_1 -Nachbarn geben wir neue untere Schranken an, die auf Datenrekonstruktionen basieren. Für MST und das Problem eines perfekten Matchings mit minimalem Gewicht (MWPM) schließen wir eine verbleibende logarithmische Lücke in der oberen und unteren Schranke. Für hierarchisches Clustering unter Dasguptas Kostenfunktion geben wir die erste untere Schranke für (ϵ, δ) -DP an. Unsere Schranke ist durch den minimalen balancierten Schnitt parametrisiert. Wir zeigen, dass die untere Schranke für ϵ -DP aus [Den+25] auch für approximatives DP sowie für dünne Graphen mit hoher Konnektivität gilt, etwa für Expander mit konstantem Grad.
- **Private Verlustfreie Mehrfachveröffentlichung.** Wir führen das Framework der Private Lossless Multiple Release (Private Lossless Multiple Release) ein und geben eine allgemeine Charakterisierung dafür, wann dieselbe Statistik unter mehreren Privatsphärenniveaus, in beliebiger Reihenfolge und ohne *jedlichen* zusätzlichen Verlust veröffentlicht werden kann. Wir zeigen, dass dies für jeden additiven Rauschmechanismus möglich ist, dessen Rauschverteilung eine Zerlegungseigenschaft erfüllt. Wir bezeichnen diese Eigenschaft *convolution preorder* und zeigen, dass sie etwa von der Gauß-, Laplace- und der Poisson-Verteilung erfüllt wird. Darüber hinaus initiieren wir die Untersuchung nicht-additiver Mechanismen wie *Randomized Response* mittels Markovketten und geben ein Gegenbeispiel an, in dem PLMR nachweislich unmöglich ist.
- **Dünne Histogramme mit korreliertem Rauschen.** Wir verbessern die private Veröffentlichung dünner Histogramme, indem wir die Sensitivitätsstruktur ausnutzen, die durch eine Nachbarschaftsrelation induziert wird, die nur Hinzufügen und Entfernen zulässt. Indem wir Beobachtungen aus Lebeda [Leb25] mit der genauen Analyse des Gaußschen Mechanismus für dünnbesetzte Histogramme kombinieren [Wil+24; BW18], erzielen wir eine konstante Verbesserung des maximalen Fehlers um bis zu einen Faktor von zwei. Des Weiteren bestätigen Experimente diese theoretischen Ergebnisse empirisch.

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Part I

239

SYNOPSIS

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243
244 INTRODUCTION

245 Data often reveals more about individuals than it appears to at first.
246 In 2013, a Freedom of Information request forced the public release of all 173
247 million New York City taxi trips from that year. To protect privacy, each medal-
248 lion number was replaced by a seemingly random identifier [Pan14; Who14]. It
249 turned out that this *pseudonymization* was too weak: Trotter [Tro14] quickly recov-
250 ered the original identifiers. Combined with side information such as paparazzi
251 photographs of celebrities entering a cab, individual trips could be recovered —
252 including destinations, fares, and even tip amounts [Toc14]. Even the drivers were
253 exposed: their routes, working patterns, and income could be inferred. Similar
254 failures had already exposed Netflix viewers by linking movie preferences to public
255 IMDb accounts [NSo8], and AOL users through the partial release of search query
256 logs [Arro6].

257 These early *deanonymization attacks* required substantial manual effort: auxiliary
258 datasets had to be located, linked, and interpreted by hand. More recently, Lermen,
259 Paleka, Swanson, Aerni, Carlini, and Tramèr [Ler+26] demonstrated that large
260 language models can be very effective at these deanonymization tasks when
261 equipped with agentic abilities such as web search and reasoning capabilities. The
262 authors showed that given only an anonymized *Hacker News* profile, such an agent
263 can autonomously find the corresponding *LinkedIn* account in roughly 70% of the
264 cases. Their experiments further suggest that scaling to larger candidate pools
265 may be possible with only a minuscule loss in recall. Concurrent work reports
266 similar results in other domains [NSN24; Li26], suggesting that deanonymization
267 increasingly requires less manual effort.

268 **STATISTICAL RELEASES** Since removing identifiers is insufficient, a natural
269 response is to release an aggregated statistic, which is a summary over many
270 records. Examples include the average fare of taxi trips between two neighborhoods
271 or the count of trips leaving a specific area late at night. At first sight, such
272 summaries seem much safer than simply releasing the raw data. To see that this
273 is not the case, suppose, for instance, that we want to know whether a particular
274 person took a taxi between two neighborhoods. If the released count is zero, then it
275 rules out that possibility immediately. Thus, even an aggregated value can disclose
276 information about individual records.

277 A statistic is usually supposed to describe an overall population. For instance, if
278 we want to know which streets to avoid on a busy Monday morning, it is not a
279 single person, assuming no accidents, who causes the traffic jam, but rather that
280 many people use this street at the same time. Furthermore, its value should only
281 change marginally whether a particular record is present. We call this maximal
282 possible change the *sensitivity* of a statistical release. For example, a single trip
283 between two places only changes the total count by one. Among the thousands
284 of trips that may occur every day, the contribution of a single trip is negligible,
285 and in almost all cases, it would be sufficient to only get a rough estimate. This
286 motivates the idea of carefully altering the statistic. Intuitively, just from observing

the release, it should be provably difficult to infer anything about whether any single individual’s data has been used to compute it. This is precisely the intuition captured by the robust definition of *privacy* we use in this thesis.

One way to achieve this goal is to randomly perturb the release. Instead of publishing the exact value of a statistic, we publish a randomized version. This introduces uncertainty, usually called *noise*, which masks the effect that any single individual can have on the output. Intuitively, if the impact of a single record is small, we will get away with only a small amount of noise that covers its impact. This intuition was formalized as Differential Privacy (DP) by Dwork, McSherry, Nissim, and Smith [Dwo+06b]. DP requires that the released statistic of two datasets that differ in only one individual’s record should look very similar. We defer the precise definitions to Chapter 2. Releases under DP have some desirable properties. For example, *post-processing* ensures that any function applied to a release (independent of the dataset) preserves privacy and *composition* theorems allow us to quantify how much privacy is lost if multiple mechanisms are run on the same dataset.

ADDITIVE NOISE MECHANISMS Among the many mechanisms that satisfy DP, Additive Noise Mechanisms (ANMs) play a particularly important role. Given a statistical query $f : \mathcal{X} \rightarrow \mathbb{R}^d$, an ANM releases $\mathcal{A}(x) = f(x) + Z$ where Z is a random data-independent noise vector. Canonical examples are the *Laplace* and *Gaussian* mechanisms where Z is drawn from the corresponding distribution. The noise is calibrated to the sensitivity of f , i.e., the largest possible change of f under neighboring inputs. This makes ANMs a very general way to provide formal privacy guarantees. As long as the sensitivity of f can be bounded, we can directly construct a private mechanism.

In this direct form, however, additive noise uses only a worst-case sensitivity bound and ignores much of the structure of the task. Many private release problems, such as *Private Selection*, do not ask for a numerical vector at all, but for example for an object that maximizes some utility score. In such cases, adding noise to every score and releasing the entire perturbed vector leaks more information than required. Even when the desired output is numerical, the query might impose additional structure we can exploit. For example, the *Sparse Vector Technique* allows releasing private answers to only those queries that exceed some defined threshold without “paying” privacy cost for those queries whose answers are not released. This is strictly better than the naive ANM-based approach that simply adds noisy answers to all the queries.

THE POWER OF INPUT PERTURBATION In this thesis we study more general *input perturbation* mechanisms: add data-independent noise and then apply a private operation on top of it. The ANMs from above are special cases where this last step is the identity map. Otherwise, the noisy internal quantities are not directly released, but only an output that has been computed on top of them. This final step is required to obtain the privacy guarantees we are looking for. The *Exponential Mechanism*, which solves the *Private Selection* problem, can be implemented by first adding appropriate noise to each score and then outputting the element with the maximum perturbed score. Likewise, the *Sparse Vector Technique* can be implemented by adding noise once to the threshold and once to each query, releasing those answers that exceed the threshold.

These examples show that perturbing an internal representation before post-processing can be substantially more powerful than releasing a noisy statistic directly. This motivates the driving question of this thesis.

When is *Input Perturbation* optimal for a differentially private release?

We investigate this question in three fundamentally different private release settings. We defer an overview of the exact results to Chapter 3.

NETWORK DATA. We first study the private release of approximate solutions to fundamental graph optimization problems. We assume a public graph topology $G = (V, E)$ with n vertices and m edges together with edge weights $\mathbf{w} \in \mathbb{R}^m$ that carry sensitive information. Two graphs over the same topology are neighboring if their corresponding weight vectors differ by at most some constant in ℓ_1 or ℓ_∞ distance. This model, introduced by Sealfon [Sea16], arises naturally when edge weights represent aggregated statistics, for example, traffic intensities collected from the taxi drivers in New York.

We first study the problem of releasing an approximate Minimum-Weight Spanning Tree (**MST**) under privacy constraints. A spanning tree $T \subseteq E$ is an acyclic subset of the edges of size $n - 1$ making the graph connected. We want to release a spanning tree T that minimizes the sum of the weights $\sum_{t \in T} w(t)$ and consider the additive error $|w(T) - w(T^*)|$ relative to the optimal solution T^* .

Our results show that input-perturbation based mechanisms are surprisingly powerful for this problem. Under ℓ_1 neighbors, simply privatizing the weights with independent Laplace noise gives an additive error of $\mathcal{O}(\frac{m}{n} \log(m/n))$ (for fixed privacy parameters). We start by showing a reconstruction-based lower bound proving that this error is asymptotically optimal. Similar to existing (worst-case) lower bounds [Sea16], we encode a random dataset into a fixed dense graph. Releasing the **MST** with too high accuracy would then immediately contradict privacy.

Under ℓ_∞ , previous works [Pin+18; MMS21] already showed that private variants of standard textbook algorithms achieve asymptotically better utility than the naive approach of simply adding noise to each edge. Intuitively, these better bounds can be achieved by exploiting a tighter composition theorem and injecting the noise during each round of the algorithm directly. However, these algorithms require fresh noise in each round and are therefore computationally expensive. In a first preprint [PR24] we used a simulation argument to improve the running time of a private version of Prim–Jarník’s **MST** algorithm down from $\mathcal{O}(m \cdot n)$ to $\mathcal{O}(m + n^{3/2} \log(n))$. In follow-up work, we realized that there is an even simpler algorithm [Pag+24]. We prove that an optimal algorithm can be obtained by adding noise from a Gumbel distribution to each weight exactly *once* and then only releasing the output of any **MST** algorithm. We show that this output distribution exactly matches that of a private version of Kruskal’s algorithm. Note that similar to the *Exponential Mechanism*, releasing the perturbed weights directly does not give the desired privacy guarantees. It is only through the algorithm that the actual privacy guarantees are established. We complement this upper bound with a corresponding lower bound showing that the resulting asymptotic error is optimal up to a logarithmic factor.

Furthermore, we show that several of these bounds also hold for the problem of privately releasing a Minimum-Weight (Perfect) Matching.

380 **MULTIPLE PRIVACY LEVELS.** Our second setting studies the problem of re-
 381 leasing the same statistic under different privacy levels simultaneously. A naive
 382 approach is to generate both releases independently, but a standard composition
 383 result implies that the overall privacy budget accumulates across those releases. We
 384 show how to bypass this for a large family of **ANMs**, so that the overall guarantee is
 385 solely determined by the least private release. Intuitively, we add a small amount of
 386 additional noise on top of previous releases, which does not cause any additional
 387 privacy costs as it only post-processes some previous output. We show that this
 388 way of correlating the noise works for all noise distributions that satisfy a very
 389 general decomposition property, which we call *convolution preorder*. This works,
 390 among others, for *Laplace*, *Gaussian* and *Poisson* noise. Our algorithm works even
 391 in a streaming setting, where the required privacy levels arrive one at a time and
 392 in arbitrary order.

393 **SPARSE GAUSSIAN HISTOGRAMS** Finally, we study the *Gaussian mechanism* un-
 394 der the add-remove neighboring relationship in a sparse setting. We combine some
 395 recent results [Leb25] with the exact analysis of the Gaussian mechanism [Wil+24]
 396 in the sparse setting. For example, consider releasing how often each term has
 397 been searched for in a search engine. Since only a small fraction of all possible
 398 terms ever appears, the resulting histogram is *sparse*, and we only want to release
 399 its few non-zero counts rather than a value for every possible term. We show how
 400 a small amount of correlated noise can help to improve the utility by a constant
 401 factor of up to two. Again, the algorithm is based on a simple perturbation of the
 402 input using additive noise.

403

404 **Thesis Organization.** We introduce Differential Privacy (**DP**) in more detail in
 405 Chapter 2 and then give a technical summary of the exact papers included in this
 406 thesis in Chapter 3. Many standard probability distributions are frequently used in
 407 all chapters. We defer their definitions to Part iii. The rest of the thesis includes
 408 the corresponding papers chapter-wise with small modifications to the published
 409 versions to reduce redundancy.

410 But if the difference between the heap [of sand grains] and the heap-minus-one is miniscule,
 411 how can you ever transform a heap into a non-heap?
 412 — “Sorites Paradox”, R. F. Kuang, *Katabasis* [Kua25]

2

413 DIFFERENTIAL PRIVACY

415 Introduced in seminal work by Dwork, McSherry, Nissim, and Smith [Dwo+06b],
 416 Differential Privacy (DP) gives a framework for providing privacy guarantees
 417 by limiting how much a data release may reveal about any individual. DP has
 418 been deployed in real-world settings such as the release of *Wikipedia usage* statis-
 419 tics [Ade+23], *Israeli birth data* [HC25], *product analytics in Brave* [Sha+25], and
 420 probably most notably, the *2020 US Census* [Abo+22]. A community-maintained
 421 list of deployments has recently been established by Howarth [How25].

422 2.1 THREE DIFFERENT ANGLES ON DIFFERENTIAL PRIVACY

423 We will approach a formal definition of DP from three equivalent perspectives. But
 424 first, we need some definitions. Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be a randomized algorithm that
 425 releases a statistic computed on a dataset. For simplicity, we assume that $\mathcal{Y}$ is a
 426 discrete output space, but all statements can be extended to a continuous domain.
 427 We write $\mathbf{x} \sim \mathbf{x}'$ to indicate that two datasets are *neighboring*; the treatment of
 428 possible neighboring relationships is deferred to Section 2.2. For now, it suffices to
 429 think of $\mathbf{x}'$ as $\mathbf{x}$ but with a single record added or removed.

430 **HYPOTHESIS TESTING** One way to view Differential Privacy is from the per-
 431 spective of an adversary who, given only a released statistic, must decide whether
 432 a particular record was part of the underlying dataset. If she can answer correctly
 433 with high confidence, the release leaks information about that record. If she can
 434 do no better than random guessing, the release reveals nothing about it. This is
 435 precisely the binary hypothesis test setting [KOV15; WZ10; Ste22; DRS19]: the
 436 adversary must decide whether the observed output came from a dataset where
 437 one record was present, or from one in which it wasn’t.

438 Suppose we release $Y \leftarrow \mathcal{A}(\mathbf{x})$ and want to hide whether some record was
 439 present in $\mathbf{x}$. Formally, an adversary who only sees the output Y faces a binary
 440 hypothesis test where the null hypothesis $\mathbf{H}_0$ states that Y was drawn from $\mathcal{A}(\mathbf{x})$.
 441 We test it against the alternative $\mathbf{H}_1$ that Y was drawn from $\mathcal{A}(\mathbf{x}')$ for a neighboring
 442 dataset $\mathbf{x}'$ in which the record is absent, i.e.,

$$\mathbf{H}_0 : Y \leftarrow \mathcal{A}(\mathbf{x}) \quad \mathbf{H}_1 : Y \leftarrow \mathcal{A}(\mathbf{x}').$$

443 A test can be specified by a *rejection region*: we reject $\mathbf{H}_0$ if the observed outcome
 444 lies in that region and accept it otherwise. For a rejection region S , the *type-I* error
 445 $\alpha(S)$ is the probability of falsely rejecting $\mathbf{H}_0$, and the *type-II* error $\beta(S)$ that of
 446 failing to reject it when $\mathbf{H}_1$ holds. The adversary’s goal is to find an S for which
 447 both $\alpha(S)$ and $\beta(S)$ are as small as possible. Differential Privacy (DP) is now the
 448 requirement that no test can reliably distinguish $\mathbf{H}_0$ from $\mathbf{H}_1$, that is, for every
 449 rejection region $S \subseteq \mathcal{Y}$ we require that at least one of $\alpha(S)$ or $\beta(S)$ is large.

450 **PRIVACY LOSS DISTRIBUTIONS** A classic result by Neyman and Pearson [NP33]
 451 from 1933 states that (in the binary hypothesis test setting) for any target type-
 452 I error level, the test that minimizes the type-II error is the one that thresholds
 453 $\log(\mathbf{P}[\mathcal{A}(\mathbf{x}) = y] / \mathbf{P}[\mathcal{A}(\mathbf{x}') = y])$, also known as the log-likelihood ratio. This identity
 454 is closely related to the concept of *Privacy Loss*, which leads us to the next way to
 455 characterize privacy. This perspective is often used to give tighter guarantees for
 456 the composition of multiple private mechanisms.

Definition 2.1 (Privacy Loss). *For a pair of distributions $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$ we define the privacy loss as*

$$\mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}(\mathbf{y}) := \begin{cases} -\infty & \text{if only } \mathbf{P}[\mathcal{A}(\mathbf{x}) = y] = 0, \\ \log\left(\frac{\mathbf{P}[\mathcal{A}(\mathbf{x}) = y]}{\mathbf{P}[\mathcal{A}(\mathbf{x}') = y]}\right) & \text{if both are positive,} \\ +\infty & \text{if only } \mathbf{P}[\mathcal{A}(\mathbf{x}') = y] = 0. \end{cases}$$

The loss is undefined if both $\mathbf{P}[\mathcal{A}(\mathbf{x}) = y] = \mathbf{P}[\mathcal{A}(\mathbf{x}') = y] = 0$.

457 The privacy loss measures how much an observation favors $\mathcal{A}(\mathbf{x})$ over $\mathcal{A}(\mathbf{x}')$: it is
 458 positive when y is more likely under $\mathcal{A}(\mathbf{x})$, and negative when it favors $\mathcal{A}(\mathbf{x}')$. A
 459 loss close to zero indicates that distinguishing $\mathbf{x}$ and $\mathbf{x}'$ is hard, while higher values
 460 tell us the opposite. Note that the case of infinite privacy loss occurs when one of
 461 the probabilities is zero. In that case, an attacker can immediately tell where the
 462 data must have come from. As we will see in Chapter 7, these kinds of events can
 463 happen, for instance, in algorithms that preserve sparsity by a thresholding step.
 464 The *Privacy Loss Random Variable* can now be defined as the loss we get when we
 465 sample from $\mathcal{A}(\mathbf{x})$:

Definition 2.2 (Privacy Loss Random Variable). *For a pair of distributions $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$, the privacy loss random variable $Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}$ is obtained by drawing $Y \sim \mathcal{A}(\mathbf{x})$ and setting $Z := \mathcal{L}_{\mathcal{A}(\mathbf{x})\|\mathcal{A}(\mathbf{x}')}(\mathbf{Y})$.*

466 Intuitively, privacy requires that, when we release statistics computed on any
 467 pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$, the *Privacy Loss Random Variable* is not too large.
 468 We would expect strong guarantees if we can bound the loss by some constant
 469 $\varepsilon > 0$, but even a probabilistic bound on the loss should give us “something”. The
 470 privacy loss distribution is closely related to the following textbook definition of
 471 **DP**.
 find a survey of different notions in [Des20]

472 **THE CLASSIC DEFINITION OF DIFFERENTIAL PRIVACY.** We now introduce the
 473 standard textbook definition of (ε, δ) -**DP**. The parameter ε bounds the likelihood
 474 ratio multiplicatively and $\delta > 0$ accounts for a small additive error (in the sense of
 475 total variation distance).

Definition 2.3 $((\varepsilon, \delta)$ -indistinguishability). *We say that two distributions $\mathcal{A}(\mathbf{x})$*

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and $\mathcal{A}(\mathbf{x}')$ over $\mathcal{Y}$ are (ϵ, δ) -indistinguishable (denoted $\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}')$) if for all $S \subseteq \mathcal{Y}$,

$$\mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x})} [Y \in S] \leq e^\epsilon \cdot \mathbf{P}_{Y \leftarrow \mathcal{A}(\mathbf{x}')} [Y \in S] + \delta.$$

DP is satisfied if the output distributions for every pair of neighboring datasets are (ϵ, δ) -indistinguishable.

Definition 2.4 ([Dwo+o6b]) (ϵ, δ) -differential privacy). Given $\epsilon \in \mathbb{R}$ and $\delta \in [0, 1]$, a randomized mechanism $\mathcal{A} : \mathcal{U}^* \rightarrow \mathcal{Y}$ satisfies (ϵ, δ) -DP if for every pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$, $\mathcal{A}(\mathbf{x})$ is (ϵ, δ) -indistinguishable from $\mathcal{A}(\mathbf{x}')$, i.e.,

$$\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}').$$

We refer to the case $\delta = 0$ as *pure DP*. It is easy to see that pure DP immediately recovers the requirement that the privacy loss $\mathcal{L}_{\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')}$ is bounded by ϵ in absolute value for every pair of neighboring datasets. We refer to the other case where $\delta > 0$ as *approximate DP*. It turns out that interpreting δ directly is non-trivial but thinking in terms of the privacy loss gives some intuition. One can show that if the probability that the privacy loss exceeds some fixed ϵ can be bounded by δ , then the mechanism satisfies (ϵ, δ) -DP, but the reverse direction is not true. We will give the exact characterization in Lemma 2.1.

Meiser [Mei18] gives more intuition about interpreting δ .

We usually think of $\delta = n^{-\Omega(1)}$ where n is the size of the dataset. Otherwise, there are algorithms satisfying strong privacy guarantees that we would not consider private. Consider, for example, the following “Name and Shame” algorithm that releases each record with probability δ independently. It can be shown that this algorithm satisfies $(0, \delta)$ -DP [Sm12]. So, unless δ is much smaller than $1/n$, we have a significant chance of exactly releasing the private records, something we would definitely not consider private.

All of these three perspectives — the hypothesis test, the privacy loss, and the standard (ϵ, δ) -formulation — are closely related. In fact, the following lemma states that all of these characterizations of DP are equivalent.

Lemma 2.1 ([KOV15; WZ10]). Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be any randomized algorithm. The following statements are equivalent for any neighboring relationship $\sim$ and fixed privacy parameters ϵ, δ .

1. (classic definition) $\mathcal{A}(\mathbf{x}) \approx_{\epsilon, \delta} \mathcal{A}(\mathbf{x}')$ for each pair of neighboring datasets $\mathbf{x} \sim \mathbf{x}'$.
2. (privacy loss)

$$\delta \geq \mathbf{P}_{Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')}} [Z > \epsilon] - e^\epsilon \cdot \mathbf{P}_{Z \leftarrow \mathcal{L}_{\mathcal{A}(\mathbf{x}') \parallel \mathcal{A}(\mathbf{x})}} [-Z > \epsilon].$$

3. (binary hypothesis test) For every rejection region $S \subseteq \mathcal{Y}$, we have

$$\alpha(S) + e^\epsilon \beta(S) \geq 1 - \delta \quad \text{and} \quad e^\epsilon \alpha(S) + \beta(S) \geq 1 - \delta,$$

$$\text{where } \alpha(S) := \mathbf{P}_{Y \sim \mathcal{A}(\mathbf{x})} [Y \in S] \text{ and } \beta(S) := \mathbf{P}_{Y \sim \mathcal{A}(\mathbf{x}')} [Y \notin S].$$

2.2 CLASSIC MECHANISMS

Arguable the most widely used mechanisms are the *Gaussian* and the *Laplace* mechanisms where the amount of noise is calibrated to the *sensitivity* of the released statistic.

SENSITIVITY Suppose we want to release some statistical query $f : \mathcal{X} \rightarrow \mathbb{R}^d$ computed from some private dataset $\mathbf{x}$. The *sensitivity* Δ_p of f captures how much $f(\mathbf{x})$ can change if instead, the dataset were some neighboring $\mathbf{x}'$.

Definition 2.5 (ℓ_p Sensitivity). The ℓ_p sensitivity of a function $f : \mathcal{X} \rightarrow \mathbb{R}^d$ is

$$\Delta_p := \sup_{\mathbf{x} \sim \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_p,$$

where the supremum is taken over all neighboring datasets $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$.

The concrete adjacency notion can have a big influence on the *sensitivity*. Consider, for example, the setting where two vectors $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^d$ are neighboring if they are close in ℓ_1 distance, i.e., $\|\mathbf{x} - \mathbf{x}'\|_1 \leq 1$ implies $\Delta_1 = \Delta_2 = 1$. On the other hand, if we define the neighboring relationship as $\|\mathbf{x} - \mathbf{x}'\|_\infty \leq 1$ where each coordinate can change by one, the sensitivity blows up to $\Delta_2 = \sqrt{d}$ and even $\Delta_1 \leq d$. This difference will become important in Chapter 4 and 5 where we show that optimal algorithms can be obtained by Additive Noise Mechanism.

Many algorithms in this thesis are based on *additive* noise mechanisms, which add noise from some distribution to the computed statistic $f(\mathbf{x})$ before releasing it. The sensitivity directly influences how the noise must be “calibrated” such that the output distribution satisfies privacy guarantees. This is one of the key contributions shown by Dwork, McSherry, Nissim, and Smith [Dwo+06b].

Definition 2.6 (Additive Noise Mechanism (ANM)). For some positive integer d , a randomized mechanism $\mathcal{A} : \mathcal{X} \rightarrow \mathbb{R}^d$ is an additive noise mechanism if

$$\mathcal{A}(\mathbf{x}) = f(\mathbf{x}) + Z, \quad Z \sim \mathcal{D},$$

where $f : \mathcal{X} \rightarrow \mathbb{R}^d$ is some statistical query and $\mathcal{D}$ an arbitrary (data-independent) probability distribution over $\mathbb{R}^d$.

We will mostly assume that $\mathcal{D}$ independently adds noise to each $f(\mathbf{x})_i$ but sometimes, like in Chapter 7, the noise might be correlated across different coordinates. The two classic instantiations of (independent) ANMs are the *Laplace* and *Gaussian* mechanisms.

Definition 2.7 ([Dwo+06b] Laplace mechanism). Let Δ_1 be the ℓ_1 sensitivity of f . The Laplace mechanism is the ANM where $Z = (Z_1, \dots, Z_d)$ with each $Z_i \sim \text{Lap}(\Delta_1/\epsilon)$ independently. It satisfies ϵ -DP.

more on correlated
noise in Chapter 7

Definition 2.8 ([DR14] Gaussian mechanism). Let Δ_2 be the ℓ_2 sensitivity of f . The Gaussian mechanism is the ANM where $Z = (Z_1, \dots, Z_d)$ with each $Z_i \sim \mathcal{N}(0, \sigma^2)$ independently.

The scaling of the noise can be exactly characterized. [BW18] showed that the Gaussian mechanism satisfies (ϵ, δ) -DP, if and only if

$$\Phi\left(\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) - e^\epsilon \Phi\left(-\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) \leq \delta, \quad (1)$$

where Φ denotes the CDF of the normal distribution. One can derive Definition 2.8 by observing that the *Privacy Loss Random Variable* of the Gaussian mechanism is also a Gaussian and then use the second characterization in Lemma 2.1. Although not giving the tightest guarantees, most of the times, one can think of $\sigma \geq \Delta_2 \sqrt{2 \ln(1.25/\delta)} \cdot \frac{1}{\epsilon}$, which is sufficient to achieve (ϵ, δ) -DP.

NEIGHBORING RELATIONSHIP Until now, we only considered datasets where we wanted to protect the membership of a record in the dataset and neighboring (or adjacent) datasets were defined to be exactly those that differ by the presence or absence of a single record. This setting is known as the *add/remove* neighboring relationship. Differential Privacy itself is oblivious to this notion. This freedom is meaningful because the choice of adjacency gives us the flexibility to exactly decide what to protect.

Another popular notion is that of *replacement* where the dataset size stays the same but a single record itself is changed. In this case, we no longer hide whether an individual contributed to the statistic at all, but rather what their record exactly contained. Note that a single replacement can be simulated by combining an add followed by a remove operation. A large part of this thesis is dedicated to studying differentially private releases when the input is a graph. Graphs admit several natural neighboring relations, like protecting the existence of edges (*edge-DP*), nodes (*node-DP*) or associated attributes such as edge weights (*edge-weight-DP*).

find an introduction
to Differential
Privacy in
Section 2.4

VARIANTS There are several other variants of DP that behave better in certain settings like composition or subsampling. We refer to the survey by Desfontaines and Pejó [DP20] for a comprehensive overview and how these notions relate to each other. For example, *f-DP* formalizes the binary hypothesis testing intuition previously described directly by describing privacy in terms of the best possible trade-off function between type-I and type-II errors [DRS19]. *Gaussian-DP* (μ -GDP) is a natural instantiation that requires that distinguishing the mechanism's output is at least as hard as distinguishing the two Gaussians $\mathcal{N}(0, 1)$ and $\mathcal{N}(\mu, 1)$. ρ -zero concentrated DP on the other hand, is closely related to the privacy loss distribution and provides a simple way to exactly characterize privacy guarantees under composition.

Definition 2.9 ([BS16], ρ -zero concentrated DP (ρ -zCDP)). Let $\rho > 0$. An algo-

rithm $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ satisfies ρ -zCDP, if for all $\alpha > 1$ and all pairs of neighboring inputs $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, it holds that

$$D_\alpha(\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')) \leq \rho\alpha,$$

where $D_\alpha(\mathcal{A}(\mathbf{x}) \parallel \mathcal{A}(\mathbf{x}')) = (1 - \alpha)^{-1} \cdot \ln \left(\mathbb{E}_{x \leftarrow \mathcal{A}(\mathbf{x})} \left[\left(\mathcal{A}(\mathbf{x}) / \mathcal{A}(\mathbf{x}') \right)^{\alpha-1} \right] \right)$ denotes the α -Rényi divergence between the two output distributions of $\mathcal{A}(\mathbf{x})$ and $\mathcal{A}(\mathbf{x}')$.

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The advantage is that if two mechanisms are ρ_1 and ρ_2 DP respectively, then the composition of those algorithms is exactly $(\rho_1 + \rho_2)$ -DP. This characterization often gives better bounds than the advanced composition theorems applied to two approximate DP mechanisms. Therefore, we need a way to convert between ρ -zCDP and (ϵ, δ) -DP.

Fact 2.2 ([BS16] Conversion ρ -zCDP $\Leftrightarrow$ approximate DP).

1. If $\mathcal{M}$ satisfies ρ -zCDP, then $\mathcal{M}$ is $(\rho + 2\sqrt{\rho \log(1/\delta)}, \delta)$ -DP for all $\delta > 0$.
2. Conversely, if $\mathcal{M}$ satisfies $(\epsilon, 0)$ -DP, then $\mathcal{M}$ satisfies ρ -zCDP for $\rho \leq \frac{1}{2}\epsilon^2$.

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2.3 PROPERTIES OF DIFFERENTIAL PRIVACY

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As already mentioned, DP satisfies many desirable properties. First, it prohibits *blatant non-privacy* and formally rules out any kind of linear reconstruction attack. It is immune to *post-processing*, meaning that applying any data-independent function to a private output cannot weaken the privacy guarantees. Furthermore, it provides nice composition properties and can easily extend to providing guarantees for groups.

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COMPOSITION What happens with the overall privacy guarantee of a mechanism that is built from several private subroutines that all have access to the same dataset? This question is answered by another key property *composition*. Composition theorems then provide a way to bound the overall privacy cost directly from the guarantees of the individual building blocks. This offers the advantage of allowing modularization: complex mechanisms can be assembled from known private primitives, whose privacy guarantees are already known.

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Proposition 2.3 (Basic Composition). Let $\mathcal{A}_1, \dots, \mathcal{A}_k : \mathcal{X} \rightarrow \mathcal{Y}$ be randomized algorithms, each of them (ϵ_i, δ_i) -DP. Then the composed mechanism $\mathcal{A}(x) : \mathcal{X} \rightarrow \mathcal{Y}^k$ with $\mathcal{A} := (\mathcal{A}_1(\mathbf{x}), \dots, \mathcal{A}_k(\mathbf{x}))$ is (ϵ', δ') -DP where

$$\epsilon' := \sum_{i \in [k]} \epsilon_i \quad \text{and} \quad \delta' := \sum_{i \in [k]} \delta_i. \quad (2)$$

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In the case of pure-DP, this bound is very clean and even optimal [Ste22], but it turns out that for approximate DP, we can do significantly better.

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Lemma 2.4 (Advanced Composition). Let $\mathcal{A}_1, \dots, \mathcal{A}_k : \mathcal{U}' \rightarrow \mathcal{Y}$ be randomized

algorithms, each of them $(\varepsilon_i, \delta_i)$ -DP. Then $\mathcal{A}(\mathbf{x}) := (\mathcal{A}_1(\mathbf{x}), \dots, \mathcal{A}_k(\mathbf{x}))$ is (ε', δ') -DP for all $\delta > 0$ where

$$\varepsilon' := \frac{1}{2} \cdot \sum_{j=1}^k \varepsilon_j^2 + \sqrt{2 \log(1/\delta) \sum_{j=1}^k \varepsilon_j^2} \quad \text{and} \quad \delta' = \sum_{j=1}^k \delta_j.$$

As mentioned, ρ -zCDP provides a particularly clean way of composition.

Fact 2.5 ([BS16] Composition). *If $\mathcal{A}_1$ and $\mathcal{A}_2$ satisfy ρ_1 -zCDP and ρ_2 -zCDP, respectively, then $\mathcal{A} = (\mathcal{M}_1, \mathcal{M}_2)$ satisfies $(\rho_1 + \rho_2)$ -zCDP.*

This concept will play a major role in Section 6.3 where we want to create multiple releases with different privacy levels simultaneously and avoid the loss imposed by composition.

POST-PROCESSING Naturally, privacy must be preserved, regardless of what (data-independent) function we apply to the output. This really is a basic requirement: an adversary without access to the dataset should not be able to weaken the privacy guarantee just by *post-processing* the output.

Proposition 2.6 ([DR14] Post-Processing). *Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be a randomized algorithm that is (ε, δ) -differentially private. Let $f : \mathcal{Y} \rightarrow \mathcal{Y}'$ be any (potentially randomized) data-independent function. Then the composition $f \circ \mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}'$ is (ε, δ) -differentially private.*

Many of our graph results are obtained by releasing a privatized version of the weight vector. Together with the public topology, we can compute combinatorial objects like an approximate MST by post-processing these perturbed weights.

GROUP PRIVACY Occasionally, we do not only want to protect the privacy of a single individual, but rather a whole group of individuals. One can also view this requirement as the similarity between the output for two inputs that are at most " k -hops away", where $k \in \mathbb{N}^+$. Of course, an increasing group size degrades the overall privacy guarantees, but the good news is that DP can already handle this privacy loss.

Fact 2.7 ([Vad17] Group Privacy). *Let $\mathcal{A} : \mathcal{X} \rightarrow \mathcal{Y}$ be any (ε, δ) -DP mechanism with $\varepsilon < e$. Given $k \in \mathbb{N}^+$ and $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$, if there exist $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k-1)}$ such that $\mathbf{x}^{(i-1)} \sim \mathbf{x}^{(i)}$ for each $i \in [k]$ (where we define $\mathbf{x}^{(0)} = \mathbf{x}$ and $\mathbf{x}^{(k)} = \mathbf{x}'$), then for each subset $S \subseteq \mathcal{Y}$, we have:*

$$\mathbf{P}[\mathcal{A}(\mathbf{x}) \in S] \leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \frac{e^{k\varepsilon} - 1}{e^\varepsilon - 1} \cdot \delta. \quad (3)$$

Proof. By simply applying the definition of (ε, δ) -DP k times, we have for all $S \subseteq \mathcal{Y}$

$$\begin{aligned} \mathbf{P}[\mathcal{A}(\mathbf{x}) \in S] &\leq e^\varepsilon \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}^{(1)}) \in S] + \delta \\ &\leq \dots \leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \delta \sum_{i=1}^k (e^\varepsilon)^i \\ &\leq e^{k\varepsilon} \cdot \mathbf{P}[\mathcal{A}(\mathbf{x}') \in S] + \frac{e^{k\varepsilon} - 1}{e^\varepsilon - 1} \cdot \delta. \end{aligned}$$

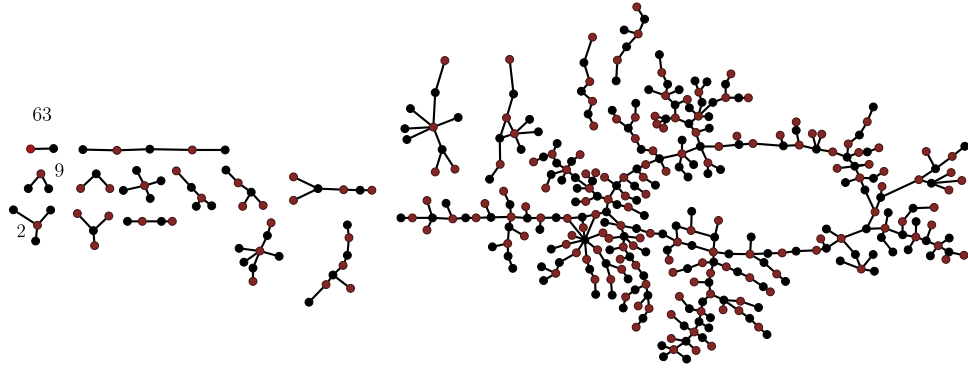


Figure 2.1: *Sexual relationships at a US high school [BMSo4] from the Add Health longitudinal study [Nat]. (Replicated with explicit permission of University of Chicago Press). The color indicates the reported gender.*

604 The last line follows from a geometric series. □

605 In Chapter 5, we use the simpler $\delta \sum_{i=1}^k (e^\epsilon) \leq \epsilon k e^{\epsilon k \delta} \delta$.

606 2.4 DIFFERENTIAL PRIVACY ON GRAPHS

607 Graphs are rich theoretical objects [Die12] that can be used to model complex rela-
 608 tionships. They have found applications across diverse domains such as social net-
 609 works [Gra73], fraud detection [Web+18], or protein structure prediction [Jum+21].
 610 Sensitive data also arises in these contexts, often encoded in the graph structure.

611 Consider, for example, the graph depicted in Figure 2.1. The vertices represent
 612 students at a US high school extracted from interview data. The edges indicate
 613 romantic or sexual partnerships that the students reported. This data is extremely
 614 sensitive and reidentifying even a single edge can reveal a student’s sexual orienta-
 615 tion or, via the chain of contacts, place them at inferred risk of sexually transmitted
 616 infection. Even years later, reidentification can have severe consequences. Similar
 617 to the reidentification attacks on tabular data as mentioned in the introduction, it
 618 has been repeatedly shown that such attacks are possible [Hay+08; NS09; JMo9].
 619 Protecting against such attacks motivates the use of Differential Privacy (DP) for
 620 graph-structured data.

621 **NEIGHBORING RELATIONSHIPS** As before, privacy guarantees can be given
 622 respecting some underlying neighboring relationship that dictates what exactly we
 623 want to protect. In the setting above, it is natural to protect the relationships itself,
 624 while we are content with revealing that someone participated in the study. Intro-
 625 duced in [NRS07], this setting has become known as *edge-DP* where neighboring
 626 graphs differ by the addition or removal of a single edge. Kasiviswanathan, Nissim,
 627 Raskhodnikova, and Smith [Kas+13] and Blocki, Blum, Datta, and Sheffet [Blo+13]
 628 independently initiated the study of graphs under the stronger notion of *node-DP*.
 629 Here, neighboring graphs are obtained by adding or removing a node, including all
 630 of its connections. In the above context, this would also protect, whether someone
 631 has taken part in the study or not.

632 This means that she has full access to the private dataset and it suffices to give
 633 privacy guarantees for the final release. A more stringent notion is that of local
 634 DP that can be traced back to [War65]. The idea is that privacy should already be
 635 satisfied, as soon as data leaves some individual. In other words, this requires that

the interviewee already lies about some of her answers. In the literature, mostly *local edge-DP* and, most recently, *local node-DP* [Ras+26] are considered. We refer to the recent surveys due to [DNK26; Mue+22; Li+23; BMM24] for more background.

EDGE-WEIGHT DIFFERENTIAL PRIVACY In this thesis, we study a slightly different model due to Sealfon [Sea16] that aims to protect information that is encoded in the weights of the edges. This models settings where the network topology is publicly known, while the data carried by its connections is sensitive. Consider again the example of the New York road network and releasing the count of taxi trips between two locations. The road network is known to everyone and definitely not considered private while the counts itself encode private information. In many settings, the edge-weights are computed by aggregating the data of an underlying private dataset. For example, a Chow-Liu tree [CL68] can be used to privately generate synthetic data [MMS21]. Here, a graph encodes the pairwise mutual information between all (aggregated) features of some underlying private dataset. This setting is known as the notion of *Edge-Weight Differential Privacy* as defined next.

Definition 2.10 ([Sea16] *Edge-Weight DP*). For some constant $\Delta > 0$ and two weight vectors $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^E$ are neighboring, denoted $\mathbf{w} \sim \mathbf{w}'$, if and only if $\|\mathbf{w} - \mathbf{w}'\|_1 \leq \Delta$. Two graphs $G = (V, E, \mathbf{w})$ and $G' = (V', E', \mathbf{w}')$ are neighboring if and only if $V = V'$, $E = E'$, and $\mathbf{w} \sim \mathbf{w}'$.

Analogously, two weight vectors are ℓ_∞ -neighboring if $\max_{e \in E} |w_e - w'_e| \leq \Delta$. Throughout the thesis, we usually consider $\Delta = 1$, but there are settings where this is a bit more intricate. For example, in the Chow-Liu setting above, $\Delta = \mathcal{O}(\log(s)/s)$ is the maximal amount the value of mutual information between two binary features can change if the underlying dataset had s users. In all our error bounds, Δ is simply an additional multiplicative factor. We sometimes overload notation and simply note $G \sim G'$ to be two neighboring graphs, implicitly meaning that the topology is the same and their weights are neighboring. Observe that the *edge-DP* setting above can be simulated by using a complete graph, where the existence of an edge is simply encoded by binary weight. Graphs would then be neighboring if their weight vectors $\mathbf{w}$ and $\mathbf{w}'$ are close in Hamming distance.

GRAPH OPTIMIZATION PROBLEMS There is a plethora of graph analysis tasks in the edge-weight model and we refer to [Mue+22] for an overview. We mainly consider problems where the output itself is also a combinatorial object that minimizes or maximizes some cost function.

The Minimum-Weight Spanning Tree problem is one of those with many applications. A **MST** can, for example, be used as a sparse representation of low-cost routes in road networks like the New York taxi cab setting. Other applications were found as a subroute in marginal selection in synthetic data generation [MMS21] and in the context of clustering algorithms [Pin18; Pin+18; Bat+17; Jay+24; LRNo9].

We also consider the problem of releasing a Minimum-Weight (Perfect) Matching (**MWPM**) analogously.

Finally, in Chapter 5 we consider the problem of privately releasing a *dendrogram* which is a tree-based data structure encoding a hierarchical clustering. Releasing hierarchical has many applications, e.g. in data visualization [SS02; HHE04] or mining [Bro+92]. We consider the rather recently proposed cost function due to

That cost function 678
recently won a Test of 679
Time Award at STOC
[ACM26].

Dasgupta [Das16]. Our main goal is to show that this problem is very hard under privacy constraints, even if the graph is very sparse.

CONTRIBUTIONS AND OVERVIEW

We now give a summary of the results in this thesis. The thesis is written as a “synopsis with manuscripts of papers or already published papers attached” in accordance with the *general rules and guidelines for the PhD program* at the Faculty of Science at the University of Copenhagen. This thesis is based on the following published results.

- P1** Optimal Bounds for Private MSTs via Input Perturbation [[Pag+24](#)]
with Rasmus Pagh, Hao Wu, and Hanwen Zhang
Principles of Database Systems (**PODS**) 2025
Also presented as a research highlight at FORC 2025
- P2** Private Lossless Multiple Release [[And+25](#)]
with Joel Daniel Andersson, Boel Nelson, and Rasmus Pagh
International Conference on Machine Learning (**ICML**) 2025
- P3** The Correlated Gaussian Sparse Histogram Mechanism [[LR25](#)],
with Christian Janos Lebeda
Foundations of Responsible Computing (**FORC**) 2025

At the time of writing, some manuscripts had been accepted for poster presentation at the annual privacy workshop *Theory and Practice of Differential Privacy* (TPDP) but had not yet published in a conference.

- T1** Lower Bounds for Private Graph Optimization Problems using Reconstruction Attacks
with Jacob Imola and Rasmus Pagh – 2026
Some of these results overlap with “Optimal Bounds for Private Graph Problems” together with Anders Aamand and Rasmus Pagh. These results were independently presented at TPDP 2025
- T2** Faster Private Minimum Spanning Trees [[PR24](#)]
with Rasmus Pagh – 2025

P1 provides strictly stronger results than **T2**. Therefore, **T2** is excluded from this thesis, but we briefly describe the main ideas below.

Joint work with Geis, Leonhardt, Meintrup, Meyer and Penschuck [[Gei+26](#)] was accepted at ALENEX 2026; another work, extending the author’s master’s thesis with Paloma Lima is under submission. These results fall outside the scope of this thesis and are therefore excluded.

SUMMARY OF THE RESULTS

We call a private mechanism $\mathcal{M}$ *input-perturbation based* if it is a composition $\mathcal{M} = \mathcal{A} \circ \mathcal{P}$ of a randomized perturbation $\mathcal{P}$ of the input and an arbitrary non-private algorithm $\mathcal{A}$. In other words, on input $\mathbf{x}$, it first perturbs the input $\tilde{\mathbf{x}} \leftarrow \mathcal{P}(\mathbf{x})$, runs $\mathcal{A}$ on it, and only then releases the final output $\mathcal{A}(\tilde{\mathbf{x}})$. Additive Noise Mechanisms are exactly those mechanisms where $\mathcal{A}$ is the identity map.

In this thesis, we investigate when input perturbation gives optimal private mechanisms. We study the question in the following three settings: private graph optimization problems, multiple releases, and sparse histograms.

(T2: FASTER PRIVATE MINIMUM SPANNING TREES) We start by considering the problem of releasing an **MST** under approximate **DP** and the ℓ_∞ neighboring relationship. The error α of a released tree $T \subseteq E$ is measured in an (expected) additive sense with respect to the underlying private weights:

Recall that ℓ_∞ allows
all weights to change
by at most one.

Definition 3.1. For a graph $G = (V, E, \mathbf{w})$, we define the error of some spanning tree $T \leftarrow \mathcal{M}(G)$ drawn from some **MST** algorithm $\mathcal{M}$ as

$$\alpha := \mathbb{E}_{T \leftarrow \mathcal{M}(G)} \left[\sum_{e \in T} w_e \right] - \sum_{e \in T^*} w_e, \quad (1)$$

where T^* is the optimal **MST** in G .

We consider the maximization version, which is equivalent to the minimization version after negating all weights. This work is not included in the thesis because **P1** provides strictly stronger results. Nevertheless, we quickly sketch the main ideas here. A privatized version of Prim–Jarník’s algorithm for finding an **MST** starts at some vertex and keeps a frontier of edges that can be added to the solution without creating a cycle. The cut property ensures that the minimum-weight edge crossing any cut must be part of an optimal **MST**. Pinot, Morvan, Yger, Gouy-Pailler, and Atif [Pin+18] proved that this selection process can be replaced by a private selection mechanism, in their case the exponential mechanism, to achieve privacy guarantees. The problem with this approach is that it needs to generate $\mathcal{O}(m)$ noise terms in each of the $\mathcal{O}(n)$ steps of the algorithm. This naively results in a running time of $\mathcal{O}(m \cdot n)$. We provide an efficient implementation of *Report-Noisy-Max* where the set of possible edges is constantly updated. *Report-Noisy-Max* adds independent exponential noise to each weight and then releases the edge with the maximum noisy weight. By the composition of ρ -zCDP over the $\mathcal{O}(n)$ rounds of Prim–Jarník’s algorithm, we need to scale the noise proportionally to $\sqrt{n}$.

For a fixed sensitivity and privacy level, the main idea is to discretize and group all weights into buckets of constant width. We then only need to consider $\mathcal{O}(\sqrt{n} \log m)$ of the largest groups by a concentration bound of the exponential distribution. The remaining edges that are further away from the maximum are extremely unlikely to get enough noise to become the maximum. We provide a simulation argument that preserves these low-probability events exactly. Furthermore, we show that a single sample is sufficient within each group by another simulation argument—essentially, one only needs the maximum noise within each group, and by a symmetry argument, we can just assign it to some uniformly drawn representative within the group. We provide a special priority queue that keeps the maximum in sublinear time $\mathcal{O}(m^{1/4})$ and supports constant-time insertions and deletions. In this way, we achieve a running time of $\mathcal{O}(m + n \cdot (m^{1/4} + \sqrt{n} \log m)) = \mathcal{O}(m + n^{3/2} \log m)$: the $\mathcal{O}(m)$ terms accounts for adding and removing elements to the frontier over the whole execution (each of the m edges is inserted and removed once in constant time). Furthermore, in each of the $\mathcal{O}(n)$ rounds, we spend $\mathcal{O}(m^{1/4})$ to find the maximum edge in the current set of candidates (frontier) but this becomes asymptotically negligible. The remaining factor of $\mathcal{O}(\sqrt{n} \cdot \log m)$ comes exactly from the running time of our private selection mechanism..

For dense graphs with $m = \Theta(n^2)$, our algorithm runs in time linear in the size of the input. Furthermore, the algorithm matches the previous error bound of $\mathcal{O}(n^{3/2} \log n)$ up to a constant factor.

PAPER P1 In Chapter 4 we improve these results. We provide an optimal algorithm for privately releasing an **MST** under the ℓ_∞ neighboring relationship and approximate **DP**. We improve the previous work by providing a simple input-perturbation based algorithm with error $\alpha \in \mathcal{O}\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot \log n \cdot \sqrt{\log(1/\delta)}\right)$. The main idea is to add carefully calibrated Gumbel noise to each edge and then release the output of *any* chosen **MST** algorithm $\mathcal{A}$. Therefore, the overall running time is given by $\mathcal{O}(m + t_{\mathcal{A}}(m, n))$, where $t_{\mathcal{A}}(m, n)$ is the running time of $\mathcal{A}$ on a graph with m edges and n vertices. This improves the previous $\mathcal{O}(m \cdot n)$ time algorithm due to [Pin+18] and our subsequent work **T2**, which gives $\mathcal{O}(m + n^{3/2} \log n)$.

We first observe that the **MST** is unique after adding noise with high probability. Hence, every **MST** algorithm returns the same tree as the one directly found by Kruskal’s algorithm. On the other hand, a privatized version of Kruskal’s algorithm can be obtained by iteratively adding an approximately heaviest edge via the *Exponential Mechanism* and then discarding edges that would create a cycle in subsequent steps. We show that the output distributions of those two mechanisms match exactly.

This is closely related to a known *one-shot* mechanism for private top- k selection, where Gumbel noise is likewise used to simulate the iterative application of the *Exponential Mechanism* (the “peeling mechanism”) [DR19]. In some sense, this is precisely what happens when Kruskal’s algorithm is run on the graph with the perturbed weights. Our setting is more involved because the edges with the largest noisy weights need not form a valid spanning tree, as some of them may close a cycle. Nevertheless, we show by a careful analysis that this topological constraint is not an obstacle at all. We show that the one-shot approach still exactly simulates the *Exponential Mechanism* even when in each step, some of the possible future candidates are removed. We complement our upper bound with a lower bound of $\mathcal{O}\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot \ln n\right)$ for $\delta \leq \mathcal{O}(1/\sqrt{n})$. This proves that the error is optimal up to a logarithmic factor. We confirm our theoretical findings with some carefully designed experiments.

PAPER T1 In Chapter 5, we provide new *reconstruction-attack* based lower bounds for private graph optimization problems. Recently, Hladík and Tetek [HT24] found a new lower bound for private **MST** under pure **DP**. Their packing-based lower bound shows that an error of $\Omega(\frac{1}{\epsilon} \cdot n \cdot \log n)$ under ℓ_1 and $\Omega(\frac{1}{\epsilon} \cdot n^2 \cdot \ln n)$ under ℓ_∞ is necessary for the complete graph. This worst-case lower bound asymptotically matches the error given by the Laplace mechanism. The authors explicitly asked for lower bounds for approximate **DP** as well. In **P1**, we already provided a nearly-optimal lower bound under ℓ_∞ .

In the manuscript **T1**, we focus on the ℓ_1 neighboring relationship. One way to get lower bounds for approximate **DP** is via reconstruction attacks from the perspective of an adversary. The overall idea is that any private mechanism cannot be *blatantly non-private*, i.e., it cannot leak a large fraction of its input (assuming sufficiently high entropy in the dataset). This motivates the idea of encoding the private dataset into the solution of the graph problem for which one wants to obtain a lower bound. Any solution that is too accurate would then allow reconstruction of the private input, which in turn contradicts privacy.

For **MST** and **MWPM**, we provide a novel reconstruction-attack based lower bound that shows that an error of $\Omega(\frac{1}{\epsilon} \cdot \frac{m}{n} \cdot \log(m/n))$ is optimal (assuming δ is polynomially small). Consequently, any private **MST** algorithm that is too accurate would violate fundamental reconstruction lower bounds of **DP**.

This exactly matches the upper bound obtained by simply adding Laplace noise to each edge and closes a remaining logarithmic gap from [Sea16]. Observe that this implies that input perturbation is asymptotically optimal for private **MST**. With minor modifications, we also show that this lower bound works for the problem of privately releasing a **MWPM** and additionally, we extend the packing-based argument in [HT24] under pure **DP** to Minimum-Weight (Perfect) Matching.

We then turn our attention to a different problem, namely privately releasing a hierarchical clustering under Dasgupta's cost function. Recently, Deng, Gao, Upadhyay, Wang, and Zhou [Den+25] showed a packing-based lower bound of $\Omega(n^2/\epsilon)$ on the complete graph topology. We give a reconstruction-based lower bound of $\Omega(\frac{1}{\epsilon} \cdot n \cdot \frac{\phi}{d_{\max}})$ that is parameterized by the balanced min-cut ϕ and the maximum degree $d_{\max}$ of the graph topology. It holds in expectation for a reasonable regime of the privacy parameters. This immediately recovers the previous bound [Den+25] on the complete graph (where $\phi = \Theta(n^2)$ and $d_{\max} = \Theta(n)$). Moreover, observe that, e.g., for d -regular expander graphs $\phi = \Omega(nd)$ and hence the quadratic lower bound still holds for extremely sparse graphs.

Again, the main idea is to encode a random binary vector into a random subset of the edges. The reconstruction attack then works by using new insights into the connection between the released clustering tree and an induced balanced min-cut. At a high level, we can reconstruct a large fraction of the edges that cross this cut if the private HC algorithm was too accurate. Finally, it matches the known (exponential-time) upper bound of $\mathcal{O}(\frac{1}{\epsilon} \cdot n^2 \cdot \log n)$ up to a logarithmic factor.

P2: PRIVATE MULTIPLE LOSSLESS RELEASES In Chapter 6, we study the setting where the same statistic is released under different privacy levels. This resembles natural settings where some parties are trusted more than others. As a simple baseline, one could just create these releases independently but this causes privacy costs in terms of composition. Our ultimate goal is to find cases where we can bypass this, so that the full sequence is jointly as private as the least private release within this sequence.

We formalize this intuition by introducing the notion of *lossless multiple release*. Informally, we say that a mechanism satisfies *lossless multiple release* if for any set of outputs, the joint distribution can be viewed as post-processing of the least private release in that subset. Additionally, we want each of those releases to be distributed as a direct release with its corresponding privacy level. A simple example where this works is the Gaussian mechanism: adding fresh Gaussian noise to an existing release yields a more private release that is obtained by post-processing.

We generalize this observation of “going from one noise term to another” and introduce a distributional property that we call *convolution preorder*. For any (possibly infinite) set of privacy parameters S , we say that a family of distributions $\mathcal{D}(\rho)$ parameterized by $\rho \in S$ admits *convolution preorder*, if there exists another distribution $\mathcal{C}(\rho_1, \rho_2)$ such that the convolution $\mathcal{D}(\rho_2) * \mathcal{C}(\rho_1, \rho_2)$ is distributed as $\mathcal{D}(\rho_1)$ for all $\rho_1 < \rho_2$. Note that the ρ 's naturally correspond to some privacy level where a higher value means less privacy. Using this, one can not only create a more private release, but (given access to an additional independent release), one can also sample a less private release by combining the noise in a particular way known as inverse variance weighting. These settings are known as the *gradual*

privacy level here is
a very general
concept and not tied
to any particular
definition.

release and the *privacy relaxation*. Our work goes one step further and allows for arbitrary release orders.

Our main result proves a meta theorem stating that Private Lossless Multiple Release is possible for every Independent Additive Noise Mechanism that uses a noise distribution satisfying *convolution preorder*. Furthermore, we show that many distributions such as Gaussian, Laplace (and the discrete counterpart) and Poisson, satisfy this property.

In unpublished follow-up work, we initiate the study of non-additive noise mechanisms. We derive a Markov chain formulation that is equivalent to our lossless multiple release definition. We then use this to enable arbitrary releases for k -ary *Randomized Response*, extending and simplifying previous work [Pan24]. Finally, we give a numerical counterexample where Private Lossless Multiple Release (PLMR) is not possible, answering a question we raised in the published paper.

P3: THE CORRELATED SPARSE GAUSSIAN HISTOGRAM MECHANISM In Chapter 7, we show how a single correlated noise sample can improve the utility when releasing a sparse histogram under DP. Recently, Lebeda [Leb25] has observed that the release of d counting queries with the Gaussian mechanism can be improved in the case of the *add/remove* relationship. His result shows that instead of scaling with $\sqrt{d}$, the standard deviation scales as $\frac{1}{2}(\sqrt{d} + 1)$, which improves the utility by up to a factor of two. This is because, in this particular neighboring relationship, it is impossible for some counts to increase while others decrease. Their main insight is that using a single correlated noise sample can exploit this structural constraint.

In that paper, we extend this approach to *sparse* histograms. One way to preserve the sparsity of a histogram is to only add noise to the non-zero entries. However, this can result in an event of infinite privacy loss when a zero changes to a one in neighboring datasets (or vice versa). In that case, an adversary can immediately infer what the underlying dataset was. One solution is to introduce a threshold $\tau > 0$ and only release those values whose noisy value is above τ .

This mechanism can still have two kinds of bad events: some privacy loss δ_{gauss} that is caused by the Gaussian mechanism and some loss δ_{inf} when the noise of a neighboring non-zero count exceeds the threshold.

An observation due to [GAT20] is that if the Gaussian mechanism satisfies $(\epsilon, \delta_{\text{gauss}})$ -DP, adding the thresholding technique gives a mechanism that is $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -DP. We use this result together with the exact analysis of the Gaussian mechanism due to Wilkins, Kifer, Zhang, and Karrer [Wil+24] to prove that a correlated sample can also give an advantage in this setting. Finally, we give an even tighter analysis using ideas of Wilkins, Kifer, Zhang, and Karrer [Wil+24], making use of the fact that each coordinate can contribute to either δ_{inf} or δ_{gauss} , but never to both at the same time. We complement these results by extending them to other settings such as aggregate queries and top- k counting queries, by giving practical considerations for discrete noise, and by running experiments that verify the utility gain of our approach.

very informative
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was written by
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Part II

903

INCLUDED PAPERS

904

We study the problem of privately releasing an approximate minimum spanning tree (MST). Given a graph $G = (V, E, \mathbf{w})$ where V is a set of n vertices, E is a set of m undirected edges, and $\mathbf{w} \in \mathbb{R}^{|E|}$ is an edge-weight vector, our goal is to publish an approximate Minimum-Weight Spanning Tree (MST) under *edge-weight* differential privacy, as introduced by Sealfon in PODS 2016, where V and E are considered public and the weight vector is private. We use the ℓ_∞ -distance on weights: for a sensitivity parameter Δ_∞ , graphs $G = (V, E, \mathbf{w})$ and $G' = (V, E, \mathbf{w}')$ are neighboring if $\|\mathbf{w} - \mathbf{w}'\|_\infty \leq \Delta_\infty$.

Existing private MST algorithms face a trade-off, sacrificing either computational efficiency or accuracy. We show that it is possible to get the best of both worlds: With a suitable random perturbation of the input that does *not* suffice to make the weight vector private, the result of *any* non-private MST algorithm will be private and achieves a state-of-the-art error guarantee. Furthermore, by establishing a connection to *Private Top-k Selection* [Steinke and Ullman, FOCS '17], we give the first privacy-utility trade-off lower bound for MST under approximate differential privacy, demonstrating that the error magnitude, $\tilde{O}(n^{3/2})$, is optimal up to logarithmic factors. That is, our approach matches the time complexity of any non-private MST algorithm and at the same time achieves optimal error. We complement our theoretical treatment with experiments that confirm the practicality of our approach.

In this section, we consider the problem of privately releasing an MST for a graph G where both the set of vertices $V = \{1, \dots, n\}$ and edges $E = \{1, \dots, m\}$ are public, but we keep the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^{|E|}$ private. We consider the ℓ_∞ neighborhood relation, where all weights can differ by at most Δ_∞ and want to protect the information encoded on the weights, which is known as *edge-weight differential privacy* [Sea16]. An application example is to model passenger data on a public city traffic network. Another setting comes from graphical modeling: consider a dataset $D = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(d)})$ of size d where each vector $\mathbf{x}^{(i)} \in \{0, 1\}^n$ represents a list of sensitive binary attributes. The *Chow-Liu Tree* [CL68] is the minimum spanning tree derived from the negated *mutual information matrix* encoding all pairwise mutual information of the attributes. Changing one $\mathbf{x}$ in D could simultaneously alter *all* weights.

Currently, there are two different algorithmic approaches: *input-privatization* [Sea16] and *in-place* [Pin+18; MMS21]. The former method adds noise (e.g., Laplace or Gaussian) to each edge weight and publishes the entire noisy weight vector. This allows anyone to compute the MST while privacy is ensured by the *post-processing* property of differential privacy. While this approach allows the design of an (expected) linear-time algorithm, it provides only a worst-case additive error of $\tilde{O}(n^2)$.

input-privatization
is equivalent to
input-perturbation
with the identity map.

The latter approach, which adds noise during the execution of an MST algorithm, achieves a better error bound of $\tilde{O}(n^{3/2})$, but its best-known running time is $\tilde{O}(n^{3/2} + m)$ for a fixed privacy parameter [PR24]. Thus, there is a gap, and whether

Reference	Additive Error	Time	Technique
[Sea16]	$O\left((1/\varepsilon) \cdot n\sqrt{m} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input privatization
[Sea16]	$\Omega(n)$	–	Lower bound
[Pin18]	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$O(nm)$	In-place noise
[PR24]	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$O\left(m + n^{3/2} \log n\right)$	In-place noise
New	$O\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n) \sqrt{\log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input perturbation
New	$\Omega\left((1/\varepsilon) \cdot n^{3/2} \cdot (\log n)\right)$	–	Lower bound

Table 4.1: Results on (ε, δ) -DP MST with ℓ_∞ neighboring relationship on edge weights, assuming sensitivity $\Delta_\infty = 1$. Sealfon [Sea16] and Pinot [Pin18] originally provided upper bounds for pure DP, but these can be adapted for approximate DP. Sealfon considered the ℓ_1 neighboring relationship, which leads to the same lower bound and a weaker upper bound for ℓ_∞ . The unpublished manuscript [PR24] was written by a subset of the authors. The “MST” in the **Time** column refers to the running time of any non-private MST algorithm.

it is possible to achieve the “best of both worlds” has remained unknown. We answer these two open questions in the affirmative:

Question 1: Can we design an (expected) linear-time private MST algorithm that matches the error guarantee of $\tilde{O}(n^{3/2})$ of the *in-place* approach?

Question 2: Is the error asymptotically optimal?

OUR CONTRIBUTION We introduce the *first* algorithmic framework that reduces the private MST problem (under the ℓ_∞ neighboring relationship) to the non-private one, achieving an additive error of $\tilde{O}(n^{3/2})$. Compatible with any non-private MST algorithm, it enables the *first* (expected) linear-time private MST solution. We also establish the *first* $\tilde{\Omega}(n^{3/2})$ error lower bound, proving the asymptotic optimality of our framework. Formally, we show the following upper and lower bounds:

Theorem 4.1 (Upper Bound). *Let $G = (V, E, \mathbf{w})$ be a graph with n vertices and m edges, and let $\varepsilon, \delta > 0$. Consider an arbitrary (non-private) algorithm that computes an MST of G within time $t(n, m)$, independent of the weight vector. Then there exists an (ε, δ) -differentially private mechanism $\mathcal{M}$ that releases a random, approximately minimum spanning tree $\mathcal{M}(G)$, such that, if T^* is a minimum spanning tree:*

- *the weight difference satisfies*

$$\mathbb{E} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] - \sum_{e \in T^*} w_e \in O\left(\frac{1}{\varepsilon} \cdot n^{3/2} \cdot (\log n) \cdot \sqrt{\log(1/\delta)}\right), \text{ and}$$

- *the running time is $t(n, m) + \mathcal{O}(m)$.*

For example, the randomized linear time algorithm by Karger, Klein, and Tarjan [KKT95] or the best-known near-linear time deterministic algorithm due to Chazelle [Cha00] can be used in Theorem 4.1. Our approach matches all known upper bounds, meanwhile it is very simple to implement. See Table 4.1 for a comprehensive summary. Moreover, the result can be easily extended to finding a maximum weight independent set in any matroid with rank n and m elements, where $t(n, m)$ is the running time for any algorithm that computes the maximum weight independent set in such a matroid.

The key technique is a novel approach (formally introduced in Section 4.2.3) to iteratively sample k items from a larger candidate set with probabilities proportional to their weights. After each selection, the chosen item and a subset of the remaining candidates—determined by the sequence of previously selected items—are removed before the next selection. Our approach generates noisy weights for all items at the beginning of the sampling process, which are reused throughout the iterative selection procedure, thereby eliminating the need to regenerate fresh randomness at each step. As discussed in Section 4.4, this technique extends a substantial body of research on top- k sampling [Yel77; Ohl90; Ros97; Coh97].

Lower Bound. Our lower bound partially resolves an open question of Hladík and Tetek [HT24], establishing a tight bound for worst-case instances under approximate differential privacy:

Theorem 4.2 (Lower Bound). *Let $\varepsilon \leq 1$ and $\delta \in O(1/\sqrt{n})$, and let $\mathcal{M}$ be an (ε, δ) -DP MST algorithm. There exists an input graph $G = (V, E, \mathbf{w})$ with n vertices such that, if T^* is the minimum spanning tree of G , the (random) spanning tree $\mathcal{M}(G)$ released by $\mathcal{M}$ satisfies:*

$$\mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in \Omega\left(\frac{1}{\varepsilon} \cdot n^{3/2} \cdot \ln n\right). \quad (1)$$

The proof technique is based on a lower bound technique for top- k selection (reporting k elements of approximately maximum value) in [SU17]. First, observe that maximization and minimization are equivalent for spanning trees by simply flipping the sign of all edge weights. A maximum spanning tree can be seen as a top- $(n-1)$ selection problem (selecting $n-1$ edges of the approximately largest value) with an additional constraint that the edge set is acyclic. Thus, it is natural to guess that the hard instances of Steinke and Ullman [SU17], embedded into the weights of a complete graph, yield a hard instance for the maximum spanning tree as well. By carefully adapting the proof in [SU17], we show that this is indeed the case. In particular, leveraging a result from the Erdős–Rényi random graph model [ER59], we show the existence of a spanning tree with weight similar to the top- $(n-1)$ edge weights.

Organization. The rest of the section is organized as follows. Section 4.1 introduces the problem formally and reviews the necessary preliminaries. Section 4.2 describes the reduction from normal non-private MST algorithm to private MST algorithm that achieve the desired utility and running time. Section 4.3 establishes the lower bound for the problem. Section 4.4 presents a detailed comparison of our techniques with previous ones and explores the broader background of the problem. Experiments in Section 4.5 show the practicality of our approach.

4.1 PRELIMINARIES

We consider a graph $G = (V, E, \mathbf{w})$ where the set of vertices $V = \{1, \dots, n\}$ and edges $E = \{1, \dots, m\}$ are public, and the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^m$ is private. For each edge $e \in E$, let w_e denote its weight. The cost of a subset $T \subseteq E$ is defined as $w(T) \doteq \sum_{e \in T} w_e$. A *spanning tree* is an acyclic subset $T \subseteq E$ that makes the graph connected. Denote $\mathcal{T}(G)$ the collection of all possible spanning trees in G . A *minimum (cost) spanning tree* (MST) is a spanning tree T^* which minimizes $w(T^*)$. We sometimes hide logarithmic factors using tilde notation for $\tilde{O}$, $\tilde{\Theta}$ and $\tilde{\Omega}$.

This section makes use several probability distributions and we refer to Part [iii](#) for their definitions.

4.1.1 Differential Privacy

Given an input graph $G = (V, E, \mathbf{w})$, the private **MST** problem aims to find a spanning tree T while preserving the privacy of the weight vector $\mathbf{w}$. In this section, we consider the ℓ_∞ neighboring relationship, restated here for convenience.

Definition 4.1 (ℓ_∞ -neighboring inputs). *Given $\Delta_\infty > 0$, two weight vectors $\mathbf{w}, \mathbf{w}' \subseteq \mathbb{R}^E$ are neighboring, denoted $\mathbf{w} \sim \mathbf{w}'$, if and only if $\|\mathbf{w} - \mathbf{w}'\|_\infty = \max_{e \in E} |w_e - w'_e| \leq \Delta_\infty$. Two graphs $G = (V, E, \mathbf{w})$ and $G' = (V', E', \mathbf{w}')$ are neighboring, denoted $G \sim G'$, if and only if $V = V'$, $E = E'$, and $\mathbf{w} \sim \mathbf{w}'$.*

For simplicity, we assume throughout the paper that $\Delta_\infty = 1$ and all bounds presented can be generalized by scaling a factor of Δ_∞ .

We will use ρ -zero concentrated DP (ρ -zCDP) as defined in Definition [2.9](#).

Exponential Mechanism. The exponential mechanism [\[MT07\]](#) can be used to differentially private release discrete outputs. It operates over an input space $\mathcal{X}$ (with neighboring datasets defined by a relation $\sim$) and a finite output space $\mathcal{Y}$. The mechanism $\mathcal{M}_{\text{EXP}} : \mathcal{X} \rightarrow \mathcal{Y}$ assigns the following probabilities for a given dataset $x \in \mathcal{X}$:

$$\mathbb{P}[\mathcal{M}_{\text{EXP}}(x) = y] \propto \exp(-\varepsilon \cdot \mathcal{E}(x, y) / (2 \cdot \Delta_{\text{EXP}})), \quad \forall y \in \mathcal{Y} \quad (2)$$

where $\mathcal{E} : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is the *loss function* quantifying the cost of selecting y given input x , and Δ_{EXP} is the *sensitivity* of $\mathcal{E}$, i.e., the maximum change in the loss function for neighboring datasets:

$$\Delta_{\text{EXP}} \doteq \max_{x \sim x', y \in \mathcal{Y}} |\mathcal{E}(x, y) - \mathcal{E}(x', y)|.$$

Fact 4.3 ([\[MT07\]](#) Properties of $\mathcal{M}_{\text{EXP}}$). *The exponential mechanism $\mathcal{M}_{\text{EXP}}$ satisfies ε -differential privacy. Moreover, for each $\beta \in (0, 1)$, and $\tau \doteq \frac{2 \cdot \Delta_{\text{EXP}}}{\varepsilon} \cdot \ln \frac{|\mathcal{Y}|}{\beta}$, it holds that:*

$$\mathbb{P}[\mathcal{E}(x, \mathcal{M}_{\text{EXP}}(x)) \geq \min_{y \in \mathcal{Y}} \mathcal{E}(x, y) + \tau] \leq \beta, \quad \forall x \in \mathcal{X}. \quad (3)$$

4.2 REDUCTION FROM PRIVATE MST TO NON-PRIVATE MST

In this section, we present a novel algorithmic framework (Algorithm [1](#)) that enables a reduction from the private **MST** problem to the non-private one, thereby proving Theorem [4.1](#). A key feature of this framework is its independence from the chosen **MST** algorithm. Since the graph's topology is public, the most suitable algorithm can always be selected in advance. Notably, this framework facilitates an (expected) linear-time private **MST** algorithm.

Overview. Figure [4.1](#) outlines the roadmap of this section. In Section [4.2.1](#), we introduce a simple algorithmic framework that fulfills Theorem [4.1](#). The framework adds certain noise to each edge weight, then applies an arbitrary non-private

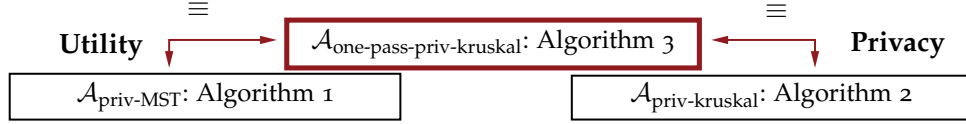


Figure 4.1: Roadmap of Section 4.2. The figure outlines the proof structure: the utility guarantee is established for Algorithm 1, and the privacy guarantee is proven for Algorithm 2. A "bridging" algorithm (Algorithm 3) is introduced to demonstrate the equivalence of Algorithms 1 and 2 by showing they share the same output distribution.

MST algorithm to publish only the set of MST edges with respect to the noisy weights. Proving its time complexity and utility is straightforward, but its privacy guarantee requires more effort. Instead of proving privacy directly, we establish it through two equivalent algorithms. Section 4.2.2 describes a private version of Kruskal's algorithm [Kru56] (Algorithm 2), which iteratively and privately selects the approximate MST edges. Its privacy analysis follows directly from the composition property of DP, though its utility analysis is more intricate. Finally, in Section 4.2.3, we present a "bridging" algorithm (Algorithm 3) that has the same output distribution as both Algorithm 1 and Algorithm 2, establishing their equivalence. This allows us to transfer privacy and utility guarantees between the algorithms, simplifying the proofs.

A detailed comparison of our algorithmic techniques with prior work, along with a broader discussion of the problem's background, is deferred to Section 4.4.

4.2.1 Reduction

The framework is outlined in Algorithm 1. It is straightforward to implement and requires only 3 lines of code using standard libraries. Given a privacy parameter ρ and an input graph $G = (V, E, \mathbf{w})$, it computes for each edge $e \in E$ a noisy weight $\tilde{w}_e \doteq w_e + (2/\epsilon') \cdot \ln \text{Exp}(1)$, where $\text{Exp}(1)$ denotes a random variable following the exponential distribution (Definition A.4). Finally, it computes the MST using an existing (non-private) MST algorithm $\mathcal{A}_{\text{MST}}$ on the noisy weights $\tilde{\mathbf{w}} = (\tilde{w}_1, \dots, \tilde{w}_m)$.

Algorithm 1 $\mathcal{A}_{\text{priv-MST}}$: Private MST Framework

Require: $G = (V, E, \mathbf{w})$, any MST algorithm $\mathcal{A}_{\text{MST}}$, privacy parameters (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: $\tilde{w}_e \leftarrow w_e + (2/\epsilon') \cdot \ln(\text{Exp}(1))$ for all $e \in E$.
 - 3: **return** $\mathcal{A}_{\text{MST}}(V, E, \tilde{\mathbf{w}})$.
-

EXTENSION TO FINDING A MAXIMUM WEIGHT INDEPENDENT SET IN MATROIDS The framework extends to general matroids by perturbing each element's weight with similar noise and computing the maximum weight independent set using any suitable algorithm on the noisy weights, see Section 4.7.1.

The properties of Algorithm 1 are outlined in Theorem 4.1.

Proof of Theorem 4.1. It remains to analyze its running time, privacy and utility guarantees.

Running Time. Adding noise to all edge weights requires only $O(m)$ time. Consequently, the overall running time of Algorithm 1 is given by $O(m + t_{\mathcal{A}_{\text{MST}}}(n, m))$, where $t_{\mathcal{A}_{\text{MST}}}(n, m)$ denotes the running time of the chosen MST algorithm $\mathcal{A}_{\text{MST}}$ for a graph with n vertices and m edges. For example, with the celebrated MST

algorithm by Karger, Klein, and Tarjan [KKT95], Algorithm 1 achieves an expected linear running time, where $t_{A_{MST}}(n, m) \in O(n + m)$. Our reduction is compatible with any **MST** algorithm, even in parallel or distributed settings.

Utility Guarantee. We analyze the utility of Algorithm 1 before establishing the privacy guarantee, as the former is more straightforward. We need the following fact, whose proof is in Appendix 4.7.3.

Fact 4.4. Let $Z_i \sim \text{ln Exp}(1)$ for $i \in [m]$ be independent. Then,

$$\mathbb{E} \left[\max_{i \in [m]} |Z_i| \right] \in O(\ln m).$$

Let T^* and T denote the MSTs with respect to the original edge weights $\mathbf{w}$ and the noisy edge weights $\tilde{\mathbf{w}}$, respectively. Note that T is precisely the spanning tree returned by Algorithm 1. We aim to prove that T satisfies the utility guarantee specified in Theorem 4.1. By Fact 4.4 and noting that both T and T^* contain fewer than n edges, it follows that

$$\mathbb{E} [\max_{e \in E} |\tilde{w}_e - w_e|] \in O(1/\epsilon' \cdot \ln m) \implies \frac{\mathbb{E} [|w(T) - \tilde{w}(T)|]}{\mathbb{E} [|w(T^*) - \tilde{w}(T^*)|]} \in O(n/\epsilon' \cdot \ln m). \quad (4)$$

As T is the **MST** on the noisy weights $\tilde{\mathbf{w}}$, we have $\tilde{w}(T) \leq \tilde{w}(T^*)$. Therefore,

$$\begin{aligned} \mathbb{E} [w(T) - w(T^*)] &\leq \mathbb{E} [w(T) - \tilde{w}(T) + \tilde{w}(T^*) - w(T^*)] \\ &\leq \mathbb{E} [|w(T) - \tilde{w}(T)|] + \mathbb{E} [|\tilde{w}(T^*) - w(T^*)|] \in O(n/\epsilon' \cdot \ln m). \end{aligned} \quad (5)$$

Finally, to obtain the utility bound in Theorem 4.1, based on Algorithm 1, Line 1, we observe that

$$\begin{aligned} \frac{1}{\epsilon'} &= \sqrt{\frac{n-1}{2\rho}} \in O\left(\frac{1}{\epsilon} \sqrt{n \log \frac{1}{\delta}}\right), \text{ since,} \\ \frac{1}{\sqrt{\rho}} &= \frac{1}{\sqrt{\epsilon + \log \frac{1}{\delta}} - \sqrt{\log \frac{1}{\delta}}} = \frac{\sqrt{\epsilon + \log \frac{1}{\delta}} + \sqrt{\log \frac{1}{\delta}}}{\epsilon} \in O\left(\frac{1}{\epsilon} \sqrt{\log \frac{1}{\delta}}\right). \end{aligned}$$

Substituting the expression for $1/\epsilon'$ into Equation (5) completes the proof.

Privacy Guarantee. As outlined in the *roadmap* and illustrated in Figure 4.1, directly proving the privacy guarantees is challenging. To address this, we introduce Algorithm 2 in Section 4.2.2, whose privacy guarantee can be easily established, and Algorithm 3 in Section 4.2.3, which shares the same output distribution as Algorithms 1 and 2, transferring the privacy guarantee to the former. $\square$

4.2.2 Private Kruskal Algorithm

Algorithm 2 presents a privatized version of Kruskal's algorithm. Compared to the non-private one [Kru56], at each iteration (Line 3-6), instead of selecting the edge with minimum weight, it selects an approximately minimum-weight edge by the exponential mechanism [MT07] which samples an edge e with probability proportional to $e^{-(\epsilon'/2) \cdot w_e}$.

Algorithm 2 $\mathcal{A}_{\text{private-Kruskal}}$: Private Kruskal's Algorithm**Require:** Graph $G = (V, E, \mathbf{w})$, privacy parameter (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2} \cdot \rho / (n - 1)$;
- 2: Initialize $T \leftarrow \emptyset, F \leftarrow E$.
- 3: **for** $i \leftarrow 1$ to $|V| - 1$ **do**
- 4: Sample $e \in F$ with probability $\propto e^{-\frac{\epsilon'}{2} \cdot w_e}$.
- 5: $T \leftarrow T \cup \{e\}$.
- 6: $F \leftarrow F \setminus \{e' \in F : \{e'\} \cup T \text{ contains a cycle}\}$.
- 7: **return** T .

Theorem 4.5 (Properties of Algorithm 2). *Given an input graph $G = (V, E, \mathbf{w})$ with n vertices and m edges, Algorithm 2 runs in $\mathcal{O}(m + n \log n)$ time, is (ϵ, δ) -DP, and returns an approximately minimum spanning tree $\mathcal{M}(G)$ such that, if T^* is a minimum spanning tree:*

$$\mathbb{E} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in O\left(\frac{1}{\epsilon} \cdot n^{3/2} \cdot (\log n) \cdot \sqrt{\log(1/\delta)}\right). \quad (6)$$

1096 *Proof of Theorem 4.5.* We analyze the running time, privacy and utility guarantees
1097 of Algorithm 2.

1098 **Running Time.** Algorithm 2 can be implemented in $\mathcal{O}(m + n \log n)$ time using a
1099 designated data structure. As it is not our primary focus, we refer for the details to
1100 Section 4.7.5. It remains to analyze the privacy-utility trade-offs of Algorithm 2.

1101 **Privacy Guarantee.** We leverage ρ -zCDP for its tighter composition bounds, so the
1102 proof involves conversion from DP to ρ -zCDP and back. We need to show that
1103 Algorithm 2 guarantees (ϵ, δ) -DP.

- 1104 • Based on the privacy guarantee of the exponential mechanism (Fact 4.3), each
1105 round of Algorithm 2 is ϵ' -DP. Using the conversion from DP to ρ -zCDP (Fact 2.2)
1106 and the initialization of ϵ' (Line 1), this corresponds to ρ' - ρ -zCDP, where $\rho' =$
1107 $(\epsilon')^2/2 = \rho/(n - 1)$.
- 1108 • Since the algorithm runs $n - 1$ rounds, by the composition property (Fact 2.5),
1109 Algorithm 2 satisfies $(n - 1) \cdot \rho' = \rho$ - ρ -zCDP.
- 1110 • Applying Fact 2.2 in reverse (from ρ -zCDP to DP), we conclude that the algorithm
1111 satisfies $(\rho + 2 \cdot \sqrt{\rho \log(1/\delta)}, \delta)$ -DP. Substituting

$$\rho \leftarrow \left(\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)} \right)^2 \text{ yields } \rho + 2 \cdot \sqrt{\rho \log(1/\delta)} = \epsilon.$$

1112 **Utility Guarantee.** The utility analysis is more involved, so we defer it to the
1113 next section, where we introduce a simplified implementation of Algorithm 2 and
1114 establish its equivalence to Algorithm 1. This allows the straightforward utility
1115 analysis of Algorithm 1 to transfer directly to Algorithm 2, demonstrating the
1116 benefit of proving the algorithms' equivalence.

1117 It is worth noting that the utility guarantee of Algorithm 2 can be proven directly,
1118 although it is much more complicated and requires a much deeper insight into
1119 Kruskal's algorithm. A proof is provided in Section 4.7.4 for reference.

1120

□

4.2.3 One-Pass Private Kruskal

Finally, we introduce the last private **MST** algorithm. It achieves the same running time as Algorithm 2, but features a significantly simplified structure, allowing it to be implemented using existing libraries for non-private Kruskal's algorithm. More importantly, it serves as a bridge to establish the equivalence between Algorithm 1 and Algorithm 2.

The algorithm is outlined in Algorithm 3. It begins by generating noisy edge weights using $\tilde{w}_e \doteq \text{Exp}(1) / e^{-(\epsilon'/2)w_e}$ and then applies the non-private Kruskal's algorithm to the noisy weights.

Algorithm 3 $\mathcal{A}_{\text{one-pass-private-Kruskal}}$

Require: Graph $G = (V, E, \mathbf{w})$, privacy parameter (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: Initialize $T \leftarrow \emptyset$, $F \leftarrow E$.
 - 3: $\tilde{w}_e \leftarrow \text{Exp}(1) / e^{-\frac{\epsilon'}{2} \cdot w_e}$. $\triangleright \text{Exp}(1)$ represents exponential noise
 - 4: **for** $i \leftarrow 1$ to $|V|-1$ **do**
 - 5: $e \leftarrow \arg \min_{e \in F} \tilde{w}_e$.
 - 6: $T \leftarrow T \cup \{e\}$.
 - 7: $F \leftarrow F \setminus \{e' \in F : \{e'\} \cup T \text{ contains a cycle}\}$.
 - 8: **return** T .
-

Theorem 4.6. *Algorithm 3 shares the properties of Algorithm 2 as stated in Theorem 4.5.*

Proof of Theorem 4.6. We analyze the runtime, utility, and privacy guarantees of Algorithm 3. In the process, we establish the equivalence among Algorithm 1, Algorithm 2, and Algorithm 3, i.e., they have the same output distributions.

Running Time. The running time is dominated by Kruskal's algorithm, which is $O(n + m \log n)$.

Utility Guarantee. The utility guarantee is established through the equivalence between Algorithm 1 and Algorithm 3, allowing the utility guarantee of the former to transfer to the latter.

First, observe that in line 5 of Algorithm 3, the returned edge satisfies

$$\arg \min_{e \in F} \tilde{w}_e = \arg \min_{e \in F} (w_e + \frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)). \quad (7)$$

Hence, we can replace the noisy weight $\tilde{w}_e \doteq \text{Exp}(1) / e^{-\frac{\epsilon'}{2} \cdot w_e}$ in Algorithm 3 by $\tilde{w}_e \doteq w_e + \frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)$. Second, since $\frac{2}{\epsilon'} \cdot \ln \text{Exp}(1)$ is a continuous random variable, the **MST** on the noisy graph is unique with probability 1. Consequently, *any* **MST** algorithm computes the same solution, thereby establishing the equivalence between Algorithm 1 and Algorithm 3.

Privacy Guarantee. The privacy guarantee is established through the equivalence between Algorithms 2 and 3, allowing the privacy guarantee of the former to transfer to the latter.

To establish the equivalence, we introduce a generalized model, *Probability Proportional to Sizes with Adaptive Candidate Removal* (Probability Proportional to Sizes with Adaptive Candidate Removal (**PPSACR**)), for top- k sampling, along with

a new sampling technique for this model. As will be shown, Algorithm 2 can be viewed as a special case of PPSACR, and Algorithm 3 directly implements the new sampling technique, thereby demonstrating its equivalence to Algorithm 2. It is worth noting that the equivalence between Algorithms 2 and 3 can also be established directly, without relying on PPSACR. However, we choose to present PPSACR here, as it is of independent interest and may have broader applications to other problems.

PPSACR MODEL: The model is described in Algorithm 4. Let U be a finite set of items, indexed from 1 to $|U|$. Each item $j \in U$ is associated with a weight $s(j) \in \mathbb{R}^+$. PPSACR iteratively samples k items. It maintains a candidate set $\mathcal{C}$, initialized as U , and a selected item list $\mathcal{I}$, which is initially empty. At each step, an item $j \in \mathcal{C}$ is sampled with probability proportional to $s(j)$ and appended to $\mathcal{I}$. Afterward, the sampled item j and an additional subset of items, denoted by $f(\mathcal{I})$, are removed from $\mathcal{C}$. Here, $f : (\bigcup_{i \in [k]} U^i) \rightarrow 2^U$ is a function that determines the additional items to be removed based on the sequence of selected items so far. Since the set $U \setminus \mathcal{C}$ at each step is fully determined by $\mathcal{I}$ and f , the function f , given $\mathcal{I}$ as input, effectively encodes the information about $U \setminus \mathcal{C}$.

Example 4.7. Algorithm 2 can be interpreted as a special case of this sampling model, where U represents the set of edges E , $s(e)$ is the exponential mechanism sampling weight $e^{-\lambda \cdot w_e^*}$ for each $e \in E$, k is $n - 1$ (the number of edges in a spanning tree), and f is the function that removes edges forming cycles based on the previously selected edges.

Algorithm 4 PPSACR

```

1:  $\mathcal{C} \leftarrow U, \mathcal{I} \leftarrow$  An empty list  $\triangleright$  possible candidates and sampled items so far.
2: while  $|\mathcal{I}| \leq k$  and  $\mathcal{C} \neq \emptyset$  do
3:   Sample  $j \in \mathcal{C}$  with probability proportional to  $s(j)$ .
4:   Add  $j$  to the back of  $\mathcal{I}$ .
5:    $\mathcal{C} \leftarrow \mathcal{C} \setminus (\{j\} \cup f(\mathcal{I}))$ .
6: return  $\mathcal{I}$ 

```

Theorem 4.8 (One-Shot-PPSACR, Algorithm 6). *Algorithm 4 can be equivalently implemented with the following modifications:*

1. Add a noisy weights generating step before line 1: $\tilde{s}(j) \leftarrow \text{Exp}(1) / s(j), \forall j \in U$.
2. Replace line 3 with $j \leftarrow \arg \min_{j' \in \mathcal{C}} \tilde{s}(j')$.

The modified algorithm is referred to as One-Shot-PPSACR.

Due to a cleaner exposition, the pseudo codes of the modified algorithm (Algorithm 6) is given in Section 4.7.2. Note that if Algorithm 2 is viewed as a special case of PPSACR, as described in Example 4.7, then Algorithm 3 directly implements the modifications proposed in Theorem 4.8. Therefore, Theorem 4.8 immediately establishes the equivalence between Algorithms 2 and 3.

To proceed, we require the following properties of the exponential distribution, the proofs of which are included in Appendix 4.7.3.

Fact 4.9 (Scaling). *If $X \sim \text{Exp}(1)$, then for all $\lambda > 0$, we have that $Y \doteq X/\lambda$ has distribution $\text{Exp}(\lambda)$.*

Fact 4.10 (Minimum). *If $X_1 \sim \text{Exp}(\lambda_1), \dots, X_d \sim \text{Exp}(\lambda_d)$, then*

$$\mathbf{P} \left[X_i = \min_{j \in [d]} X_j \right] = \frac{\lambda_i}{\sum_{j \in [d]} \lambda_j}.$$

Fact 4.11 (Memoryless property of exponential distribution). *If $X \sim \text{Exp}(\lambda)$, then*

$$\mathbf{P}[X \geq x + y \mid X \geq x] = \mathbf{P}[X \geq y], \forall x, y \geq 0.$$

Proof of Theorem 4.8. To simplify the discussion, we assume that k sampled items are returned by **PPSACR**. The proof for the case when less than k items are returned is similar. Let $\mathcal{J} \doteq (j_1, j_2, \dots, j_k) \in U^k$ be a feasible output sequence of **PPSACR**. We will prove that **One-Shot-PPSACR** outputs $\mathcal{J}$ with the same probability as **PPSACR**. **PPSACR**: For each $i \in [k]$, denote $\mathcal{C}_i$ the set of candidates to be sampled in Algorithm 4, Line 3 during the i -th iteration. The probability of selecting j_i is $s(j_i)/(\sum_{j \in \mathcal{C}_i} s(j))$. Thus, the probability that **PPSACR** outputs $\mathcal{J}$ is

$$\frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)} \cdot \frac{s(j_2)}{\sum_{j \in \mathcal{C}_2} s(j)} \cdots \frac{s(j_k)}{\sum_{j \in \mathcal{C}_k} s(j)}. \quad (8)$$

One-Shot-PPSACR: For each $i \in [k]$, let $\mathcal{E}_i$ be the event that $\tilde{s}(j_i) = \min_{j \in \mathcal{C}_i} \tilde{s}(j)$, and define $\mathcal{E}_{1:i} \doteq \mathcal{E}_1 \wedge \dots \wedge \mathcal{E}_i$. First, by Fact 4.9, for each $j \in U$, $\tilde{s}(j) \doteq \text{Exp}(1)/s(j)$ follows distribution $\text{Exp}(s(j))$. Based on Fact 4.10, the minimum property of a collection of exponential random variables, it holds that

$$\mathbf{P}[\mathcal{E}_1] = \frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)}. \quad (9)$$

Second, let $z_1 \leq z_2 \leq \dots \leq z_k$ be a feasible realization of $\tilde{s}(j_1), \dots, \tilde{s}(j_k)$. For each $i \in [2 \dots k]$, define $\tilde{s}(j_{1:i-1}) \doteq (\tilde{s}(j_1), \dots, \tilde{s}(j_{i-1}))$ and $z_{1:i-1} = (z_1, \dots, z_{i-1})$. Conditioned on the events $\mathcal{E}_{1:i-1}$ and $\tilde{s}(j_{1:i-1}) = z_{1:i-1}$, we know that for each $j \in \mathcal{C}_i$, $\tilde{s}(j) \geq (\max z_{1:i-1}) = z_{i-1}$. Since the $\tilde{s}(j)$ are independent, by the memoryless property of the exponential distribution (Fact 4.11), $\tilde{s}(j) - z_{i-1}$ still follows the distribution $\text{Exp}(s(j))$. Together with the fact that the minimum is invariant under a constant shift, the event $\mathcal{E}_i$ is now equivalent to $\tilde{s}(j_i) - z_{i-1} = \min_{j \in \mathcal{C}_i} (\tilde{s}(j) - z_{i-1})$, applying Fact 4.10 again yields:

$$\begin{aligned} & \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1}) = z_{1:i-1}] \\ &= \mathbf{P} \left[\tilde{s}(j_i) - z_{i-1} = \min_{j \in \mathcal{C}_i} (\tilde{s}(j) - z_{i-1}) \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1}) = z_{1:i-1} \right] \\ &= \frac{s(j_i)}{\sum_{j \in \mathcal{C}_i} s(j)}. \end{aligned}$$

1198 Taking expectation over $\tilde{s}(j_{1:i-1})$ gives

$$\mathbb{E}_{\tilde{s}(j_{1:i-1})} [\mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}, \tilde{s}(j_{1:i-1})]] = \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}] = \frac{s(j_i)}{\sum_{j \in \mathcal{C}_i} s(j)}.$$

1199 Finally, based on chain rule of probability, $\mathcal{E}_{1:k}$ has exactly the same probability
1200 as Equation (8):

$$\mathbf{P}[\mathcal{E}_{1:k}] = \mathbf{P}[\mathcal{E}_1] \cdot \prod_{i=2}^k \mathbf{P}[\mathcal{E}_i \mid \mathcal{E}_{1:i-1}] = \frac{s(j_1)}{\sum_{j \in \mathcal{C}_1} s(j)} \cdot \frac{s(j_2)}{\sum_{j \in \mathcal{C}_2} s(j)} \cdots \frac{s(j_k)}{\sum_{j \in \mathcal{C}_k} s(j)}. \quad (10)$$

1201 □

1202 4.3 A LOWER BOUND FOR APPROXIMATE DP WITH ℓ_∞

1203 In this section, we prove the lower bound stated in Theorem 4.2. Our proof consists
1204 of two steps: in Section 4.3.1, we present a simple reduction from (ϵ, δ) -DP **MST**
1205 algorithms to $(1, \delta)$ -DP **MST** algorithms to facilitate the derivation of lower bounds;
1206 in Section 4.3.2, we present a lower bound for all $(1, \delta)$ -DP **MST** algorithms, which
1207 can be extended to (ϵ, δ) -DP lower bound using the previous reduction.

1208 4.3.1 Reduction from (ϵ, δ) -DP to $(1, \delta)$ -DP

Lemma 4.12. Assume that $\epsilon < 1$. Suppose that there is an $(\epsilon, \frac{\epsilon^\epsilon - 1}{\epsilon - 1} \cdot \delta)$ -DP **MST** algorithm $\mathcal{M}$ under the ℓ_∞ neighboring relationship, such that for every input graph $G = (V, E, \mathbf{w})$ (with the **MST** denoted by T^*), it holds that

$$\mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] - \sum_{e \in T^*} w_e \in o\left((1/\epsilon) \cdot n^{3/2} \cdot \ln n\right). \quad (11)$$

Then there exists a $(1, \delta)$ -DP **MST** algorithm $\mathcal{M}'$, such that for every input graph $G' = (V', E', W')$ (with the **MST** denoted by T'), it holds that

$$\mathbb{E}_{\mathcal{M}'} \left[\sum_{e \in \mathcal{M}'(G')} w'_e \right] - \sum_{e \in T'} w'_e \in o\left(n^{3/2} \cdot \ln n\right). \quad (12)$$

1209 *Proof.* We demonstrate how to construct $\mathcal{M}'$ based on $\mathcal{M}$. Given a graph $G' =$
1210 (V', E', W') , we create a new graph $G = (V, E, W)$ as follows: $V = V'$, $E = E'$, and
1211 $W = (1/\epsilon) \cdot W'$. To simplify the discussion, we assume that $1/\epsilon \in \mathbb{N}^+$; otherwise,
1212 we replace $1/\epsilon$ with $\lceil 1/\epsilon \rceil$, which affects the privacy and utility guarantees only by
1213 a constant factor, as $\epsilon \leq 1$. Based on this construction, we have $T' = T^*$. Finally, we
1214 define $\mathcal{M}'$ as: $\mathcal{M}'(G') \doteq \mathcal{M}(G)$. Then based on group privacy property (Fact 2.7),
1215 $\mathcal{M}'$ is $\left(\epsilon \cdot \frac{1}{\epsilon}, \frac{\epsilon^{\frac{1}{\epsilon} \cdot \epsilon} - 1}{\epsilon^\epsilon - 1} \cdot \frac{\epsilon^\epsilon - 1}{\epsilon - 1} \cdot \delta\right) = (1, \delta)$ -DP. We conclude the proof by observing that

$$\begin{aligned} \mathbb{E}_{\mathcal{M}'} [\sum_{e \in \mathcal{M}'(G')} w'_e] &= \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w'_e] = \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} \epsilon \cdot w_e] \\ &= \epsilon \cdot \mathbb{E}_{\mathcal{M}} [\sum_{e \in \mathcal{M}(G)} w_e] \\ &= \epsilon \cdot \left(\sum_{e \in T^*} w_e + o\left((1/\epsilon) \cdot n^{3/2} \cdot \ln n\right) \right) \\ &= \sum_{e \in T'} w'_e + o\left(n^{3/2} \cdot \ln n\right). \end{aligned} \quad (13)$$

1216 □

4.3.2 $(1, \delta)$ -DP Lower Bound

In this subsection, we show that no $(1, \delta)$ -DP algorithm $\mathcal{M}$ satisfies Equation (12) for $\delta = O(1/\sqrt{n})$, thereby proving Theorem 4.2. Our proof heavily relies on a technique originally developed by Steinke and Ullman [SU17] for establishing lower bounds for the private maximum top- k selection problem. To apply this technique, we negate the weights of the input graph. Instead of proving a lower bound for the minimum spanning tree problem, we demonstrate that any $(1, \delta)$ -DP maximum spanning tree algorithm $\mathcal{M}$ incurs an error of $\Omega(n^{3/2} \log n)$ for $\delta = O(1/\sqrt{n})$.

Theorem 4.13 (Theorem 3 in [SU17]). *Let $\beta, \gamma, \Delta, k > 0$ and $s, d \in \mathbb{N}^+$ be a fixed set of parameters. Let $P = (P_1, \dots, P_d)$ be independent samples from the beta distribution $\text{Beta}(\beta, \beta)$, and let $X \in \{0, 1\}^{s \times d}$ be a random dataset such that every $X_{i,j}$ is an independent sample from Bernoulli distribution with mean P_j for every $i \in [s]$ and $j \in [d]$.*

Let $\mathcal{A} : \{0, 1\}^{s \times d} \rightarrow \mathbb{R}^d$ be a $(1, \beta\gamma k/(s\Delta))$ -differentially private algorithm (where two datasets X and X' are neighboring if and only if they differ by at most one row).

Assume $\mathbb{E}_{P,X,\mathcal{A}} [\|\mathcal{A}(X)\|_2^2] = k$ and $\mathbf{P} [\|\mathcal{A}(X)\|_1 \leq \Delta] = 1$ and

$$\mathbb{E}_{P,X,\mathcal{A}} \left[\sum_{j \in [d]} \mathcal{A}(X)_j \cdot \left(P_j - \frac{1}{2} \right) \right] \geq \gamma k, \quad (14)$$

then $s \geq \beta\gamma\sqrt{k}$.

Informally speaking, from Theorem 4.13, any DP algorithm with a good utility requires a lot of samples. To translate the lower bound on samples to a lower bound on the utility for MST, our goal is to construct a hard distribution of graphs, where the weight vector simulates the random dataset X and the algorithm's utility ties to the same quantity in the theorem. The proof is by contradiction. When the weight vector, or equivalently, the number of samples, is fixed, Theorem 4.13 shows that any DP algorithm with a "good" utility requires more samples than this fixed number, leading to a contradiction. We will present the proof formally in the rest of this section.

Definition 4.2 (Hard Distribution). *Consider a complete graph G with n -vertices and let $m = n(n-1)/2$ be the number of edges. Let $\beta, \gamma > 0$ and $s \in \mathbb{N}^+$ be parameters to be determined later. For each edge e , we first sample P_e from $\text{Beta}(\beta, \beta)$ independently. Then let $X \in \{0, 1\}^{s \times m}$ be a random dataset such that $X_{i,e}$ is an independent sample from Bernoulli distribution with mean P_e , $\forall i \in [s], e \in [m]$. Finally, let $w_e \doteq \sum_{i \in [s]} X_{i,e}$ for each $e \in [m]$. Hence, w_e follows binomial distribution $\text{Bin}(s, P_e)$.*

Since a spanning tree algorithm $\mathcal{M}$ has a different input and output format from the algorithm in Theorem 4.13, a conversion is required.

Definition 4.3 (Conversion). *Let G be the random graph and X the random dataset*

generated as described in Definition 4.2. Given an algorithm $\mathcal{M}$ that takes G as input and outputs a set of spanning tree edges, define $\mathcal{A}_{\mathcal{M}}$ as the algorithm that, given input X , outputs an indicator vector $\mathcal{A}_{\mathcal{M}}(X) \in \{0, 1\}^{|E|}$, where $\mathcal{A}_{\mathcal{M}}(X)_e = 1$ if and only if $e \in \mathcal{M}(G)$.

It is straightforward to see that if $\mathcal{M}$ is $(1, \delta)$ -DP (where two input graphs are considered neighboring if the weights of each edge differ by at most 1), then $\mathcal{A}_{\mathcal{M}}$ is also $(1, \delta)$ -DP (where two input datasets are neighboring if they differ in at most one row). Therefore, under the hard distribution, the sample complexity lower bound in Theorem 4.13 for the algorithm $\mathcal{A}_{\mathcal{M}}$ transfer to $\mathcal{M}$.

Before presenting our main lower bound, we state a lemma that is critical to our proof.

Lemma 4.14. *For $s \geq 10$, $\beta = \frac{1}{2} \ln n$ and $n \geq 2 \times 10^7$, let G be a random graph sampled from the hard distribution in Definition 4.2. With probability at least 0.99, there exists a spanning tree of G where each edge has weight at least $3/4 \cdot s$.*

The complete proof of Lemma 4.14 is included in Appendix 4.7.6. Intuitively, the result follows from the properties of the beta distribution and the Erdős–Rényi random graph model [ER59]. Let H be the subgraph of G that includes all vertices and all edges e such that $w_e \geq \frac{3}{4} \cdot s$. Since the edge weights are independent and due to our choice of β , the graph H is an instance of the Erdős–Rényi model with edge sampling probability $p = \mathbf{P}[w_e \geq s \cdot 3/4] \in \Omega(\ln(n)/n)$. Finally, G has a spanning tree where all edges e satisfy $w_e \geq s \cdot 3/4$ if and only if H is connected, which occurs almost surely when $p \in \Omega(\ln(n)/n)$ [ER59].

Assume Lemma 4.14 holds, we directly have

$$\mathbb{E}_G [\sum_{e \in T^*} w_e] \geq 0.99 \cdot \frac{3}{4} \cdot s \cdot (n - 1). \quad (15)$$

where T^* be the maximum spanning tree in G . Next, we will present the main theorem in this section that captures the utility lower bound.

Theorem 4.15 (MST lower-bound). *Let $n \geq 9 \times 10^8$, $\beta = \frac{1}{2} \ln n$ and $s = \frac{1}{100} \sqrt{n} \ln n$, G be a random graph sampled from the hard distribution in Definition 4.2. Let $\gamma = 0.04$ and $\delta = \beta\gamma/s = 2/\sqrt{n}$. For any $(1, \delta)$ -differentially private maximum spanning tree algorithm $\mathcal{M}$, it holds that*

$$\mathbb{E}_{G, \mathcal{M}} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] \leq \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000}, \quad (16)$$

where T_G^* be the maximum spanning tree given G .

Proof. The proof proceeds by contradiction. Assume there exists a $(1, \delta)$ -DP maximum spanning tree algorithm $\mathcal{M}$, s.t.

$$\mathbb{E}_{G, \mathcal{M}} \left[\sum_{e \in \mathcal{M}(G)} w_e \right] > \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000}, \text{ and}$$

let X be the random dataset generated when G is constructed, as described in Definition 4.2, and let $\mathcal{A}_M$ denote the algorithm converted from $\mathcal{M}$, as described in Definition 4.3. We will prove that

$$\mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (P_e - \frac{1}{2}) \right] > \gamma(n-1), \quad (17)$$

Since $\mathcal{M}$ outputs $(n-1)$ edges, we always have $\|\mathcal{A}_M(X)\|_2^2 = \|\mathcal{A}_M(X)\|_1 = n-1$. Hence, $\mathcal{A}_M$ satisfies all conditions in Theorem 4.13, where we set $k = \Delta = (n-1)$ and $\delta = \beta\gamma k/(s\Delta) = \beta\gamma/s$. Therefore, Theorem 4.13 implies that $s \geq \beta\gamma\sqrt{n-1}$, leading to a contradiction since the assumption in Theorem 4.15 implies $s = 0.01\sqrt{n} \ln n < \beta\gamma\sqrt{n-1}$.

Proving Equation (17): Since $w_e \sim \text{Bin}(s, P_e)$, we have $\mathbb{E}[w_e] = s \cdot P_e$. Further, the following lemma holds, whose proof employs a standard technique using Hoeffding's inequality.

Lemma 4.16. $\mathbb{E}[\max_{e \in E} |w_e - s \cdot P_e|] \leq \sqrt{3s \ln n}$.

Proof. Since w_e is a sum of s independent random variables between $[0, 1]$, and has expectation $s \cdot P_e$, by Hoeffding's inequality, it holds that $\mathbf{P}[|w_e - s \cdot P_e| \geq t] \leq 2 \cdot \exp\left(-\frac{2t^2}{s}\right)$ for all $t \geq 0$.

By union bound, it holds that for all $t \geq 0$, $s \geq 1$ and $n \geq 100$:

$$\begin{aligned} \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t + \sqrt{s \cdot \ln n}\right] &\leq 2 \cdot \frac{n(n-1)}{2} \cdot \exp\left(-2 \cdot \left(t + \sqrt{s \cdot \ln n}\right)^2 / s\right) \\ &\leq n^2 \cdot \exp\left(-2 \cdot t^2 / s\right) \cdot \exp\left(-2 \cdot s \cdot (\ln n) / s\right) = \exp\left(-2 \cdot t^2 / s\right). \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{E}\left[\max_{e \in E} |w_e - s P_e|\right] &= \int_0^{\sqrt{s \ln n}} \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t\right] dt \\ &\quad + \int_{\sqrt{s \ln n}}^\infty \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t\right] dt \\ &\leq \sqrt{s \cdot \ln n} + \int_0^\infty \mathbf{P}\left[\max_{e \in E} |w_e - s \cdot P_e| \geq t + \sqrt{s \cdot \ln n}\right] dt \\ &\leq \sqrt{s \cdot \ln n} + \int_0^\infty \exp\left(-2 \cdot t^2 / s\right) dt = \sqrt{s \cdot \ln n} + \frac{1}{2} \cdot \int_{-\infty}^\infty \exp\left(-\frac{t^2}{2 \cdot (s/4)}\right) dt \\ &= \sqrt{s \cdot \ln n} + \frac{\sqrt{2 \cdot \pi \cdot (s/4)}}{2} < \sqrt{3s \ln n}. \quad \square \end{aligned}$$

Combing Lemma 4.16 and that $\|\mathcal{A}_M(X)\|_1 \equiv n-1$, we see

$$\begin{aligned} \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (s \cdot P_e - \frac{s}{2}) \right] &\geq \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e w_e \right] \\ &\quad - (n-1) \cdot \left(\sqrt{3s \ln n} + \frac{s}{2} \right). \end{aligned}$$

By the construction of $\mathcal{A}_M$ (Definition 4.3), along with the assumption that $\mathcal{M}$ violates Equation (16),

$$\mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e w_e \right] = \mathbb{E}_{G, M} \left[\sum_{e \in M(G)} w_e \right] > \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] \frac{n^{3/2} \ln n}{1000}.$$

1277 Hence, we can bound the expectation $\mu_E \doteq \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (s \cdot P_e - \frac{s}{2}) \right]$,

$$\begin{aligned} \mu_E &> \mathbb{E}_G \left[\sum_{e \in T_G^*} w_e \right] - \frac{n^{3/2} \ln n}{1000} - (n-1)(\sqrt{3s \ln n} + \frac{s}{2}) \\ &> 0.99 \cdot \frac{3}{4} s(n-1) - \frac{n^{3/2} \ln n}{1000} - (n-1)(\sqrt{3s \ln n} + \frac{s}{2}) \\ &= 0.2425s(n-1) - \frac{n^{3/2} \ln n}{1000} - (n-1)\sqrt{3s \ln n}. \end{aligned}$$

1278 Since $n \geq 9 \times 10^8$, $s = \frac{1}{100} \sqrt{n \ln n}$ and $\gamma = 0.04$, it follows that

$$\begin{aligned} \mathbb{E}_{P, X, \mathcal{A}_M} \left[\sum_{e \in [m]} (\mathcal{A}_M(X))_e (P_e - \frac{1}{2}) \right] &> 0.2424(n-1) - \left(\frac{(n-1)}{s} \sqrt{3s \ln n} + \frac{n^{3/2} \ln n}{1000} \right) \\ &> \gamma(n-1). \end{aligned}$$

1279

□

1280 4.4 RELATED WORK

1281 Releasing various graph statistics under differential privacy constraints is a funda-
1282 mental and well-studied task. For more background, see [Li+23; Mue+22].

1283 4.4.1 Top- k Selection.

1284 The private **MST** problem is closely related to the private top- k selection problem.
1285 Formulated in the framework of this paper, the problem involves m items with
1286 weights $w_1, \dots, w_m$, and the goal of private top- k selection is to privately sample k
1287 items with approximately maximum or minimum weights. Notably, maximization
1288 and minimization are equivalent by flipping the sign of all weights. The private
1289 **MST** problem can be seen as a variant of top- $(n-1)$ selection with the additional
1290 topological constraint that the selected items must form a spanning tree.

1291 Durfee and Rogers [DR19] introduced the first linear-time algorithm for private
1292 top- k selection with asymptotically optimal privacy-utility trade-off under approxi-
1293 mate DP. They showed that iteratively applying the exponential mechanism [MT07]
1294 to select k items with an approximately maximum weight, is equivalent to adding
1295 random noise following the Gumbel distribution (see Definition A.5) to each item
1296 weight and then returning the k items with maximum noisy weights. This technique
1297 has a rich research history in the context of the non-private top- k problem [Yel77;
1298 Ohl90; Ros97; Coh97], appearing under various names. For the top-1 maximum
1299 selection problem, Yellott [Yel77] showed very early that adding Gumbel noise to
1300 the $(\ln w_i)$'s and selecting the maximum allows sampling an item with probability
1301 proportional to their weights. Rosén [Ros97] studied the top- k maximum selection
1302 problem under the name of "order sampling" and proposed a method to sample k
1303 items with probability proportional to their weights, by generating noisy scores
1304 $\text{Exp}(w_i) = \text{Exp}(1)/w_i$ for $i \in [m]$ and selecting k items with the minimum scores.

Reference	PN	NH	Error	Time	Technique
[Sea16]	ε -DP	ℓ_1	$O((1/\varepsilon) \cdot n \log n)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	ε -DP	ℓ_1	$O((1/\varepsilon) \cdot n \log n)$	$O(nm)$	In-place noise
[HT24]	ε -DP	ℓ_1	$\Omega((1/\varepsilon) \cdot n \log n)$	–	Lower bound
[Sea16]	(ε, δ) -DP	ℓ_1	$O\left((1/\varepsilon) \cdot n \sqrt{(\log n) \cdot \log(1/\delta)}\right)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	(ε, δ) -DP	ℓ_1	$O\left((1/\varepsilon) \cdot n \sqrt{(\log n) \cdot \log(1/\delta)}\right)$	$O(nm)$	In-place noise
[Sea16]	(ε, δ) -DP	ℓ_1	$\Omega(n)$	–	Lower bound
[Sea16]	ε -DP	ℓ_∞	$O((1/\varepsilon) \cdot nm \log n)$	$\text{MST} + O(m)$	Input privatization
[Pin18]	ε -DP	ℓ_∞	$O((1/\varepsilon) \cdot n^2 \log n)$	$O(nm)$	In-place noise
[HT24]	ε -DP	ℓ_∞	$\Omega((1/\varepsilon) \cdot n^2 \log n)$	–	Lower bound

Table 4.2: Extended landscape of results and complementing Table 4.1 with further known results for a graph with n vertices and m edges. PN and NH denote the Privacy Notation and the Neighboring Relationship respectively. The table compares the previous works [Sea16; Pin18; HT24] for both the ℓ_1 and ℓ_∞ neighborhood relation. “MST” in the Time column is the running time of any non-private MST algorithm.

This is equivalent to finding the k items maximizing $-\ln \text{Exp}(1) + \ln w_i$, where $-\ln \text{Exp}(1)$ follows exactly a Gumbel distribution (Fact 4.17).

Our sampling model, PPSACR, and the corresponding technique in Algorithm 4, extend this line of research [Coh97; Yel77; Ros97] by enabling the adaptive removal of candidate items after each sampling step, based on previously selected items.

Qiao, Su, and Zhang [QSZ21] proposed a linear-time private top- k selection algorithm that achieves an asymptotically optimal privacy-utility trade-off (up to a logarithmic factor) under approximate differential privacy. Their approach involves adding independent Laplace noise to each item weight and returning the set of top- k items with the smallest noisy weights. However, their current privacy-utility trade-off analysis heavily depends on the returned items corresponding to the true k smallest noisy weights. Extending their approach and analysis to the private MST problem remains open, as the selected items or edges may not necessarily form a spanning tree.

4.4.2 Releasing an MST under DP

We supplement results for approximate DP (Table 4.1) with results for pure DP in Table 4.2. Before our work there were two approaches to private MST: input privatization and in-place algorithms [Pin+18] that we review next.

IN-PLACE Both Prim-Jarník’s and Kruskal’s algorithm [Kru56; Pri57; Jar30] start with an empty set of edges and then iteratively grow it while guaranteeing that it is still a subset of an MST. In each step, they greedily select the lightest new edge between cuts that respect the edges already chosen. One can privatize them by injecting noise whenever a weight is accessed during the computation of an MST. We can replace the selection step using any differentially private selection mechanism, for instance, Report-Noisy-Max [DR14], Permute-and-Flip [MS20], or the Exponential Mechanism [MT07] and get overall privacy by composition. PAMST [Pin18] is based on the Prim-Jarník algorithm and gives the same asymptotic utility as our approach. Furthermore, a private version of Kruskal’s algorithm has been suggested by McKenna, Miklau, and Sheldon [MMS21] as a subroutine in privately generating synthetic data using a probabilistic graphical model, but its utility has never been discussed. We provide such an analysis in Section 4.2.2. Compared to

the *input privatization* in the next paragraph [Sea16], this approach gives strictly better utility under the ℓ_∞ neighboring relationship, assuming the graph is not too sparse. Unfortunately, the caveat is the running time because of the cost of the noisy selection. In an unpublished manuscript, Pagh and Retschmeier [2024] brought down the running time to $O(m + n^{3/2} \log(n) / \sqrt{\rho})$ by designing a special priority queue for Prim–Jarník’s algorithm, which simulates *Report-Noisy-Max* in sublinear time $O(\sqrt{n/\rho} \log n)$, where ρ can be chosen as in Algorithm 1 to ensure (ϵ, δ) -DP.

INPUT PRIVATIZATION One simple idea is to release a private synthetic graph by adding noise to *all* the edge weights and obtain privacy by post-processing. Sealfon [Sea16] was the first to analyze *input perturbation* using Laplace noise to achieve ϵ -DP under the ℓ_1 neighborhood and gave error of $O(n \log n)$ which is known to be asymptotically optimal [HT24]. However, under ϵ -DP with ℓ_∞ neighboring relationship, this technique gives an additive error of $O(nm \log n)$. For dense graphs, this can leave a gap of up to a factor of $O(n)$ to the known lower bound of $\Omega(n^2 \log n)$ under ϵ -DP. The advantage is that it allows flexibility in choosing any (non-private) **MST** algorithm e.g. the expected linear time algorithm by Karger, Klein, and Tarjan [KKT95], the deterministic near-linear time algorithm by Chazelle [Chao0], or the deterministic linear time for dense graphs by Fredman and Tarjan [FT87]. The overall running time is compounded by the $O(m)$ time it takes to add fresh noise to each edge and the running time of the chosen **MST** algorithm. Another advantage is that releasing a single private synthetic graph simultaneously allows the computation of other graph statistics, such as finding *shortest paths* or *minimum-weight (perfect) matchings* [Sea16].

LOWER BOUNDS Recent work by Hladík and Tetek [HT24] showed tight asymptotical worst-case bounds for ϵ -DP using a packing argument: $\Omega(n \log n / \epsilon)$ for the ℓ_1 neighborhood and $\Omega(n^2 \log n / \epsilon)$ for ℓ_∞ . They improved Sealfon’s ℓ_1 bound of $\Omega(n)$. Under the ℓ_∞ neighborhood, we show in Theorem 4.1 together with Theorem 4.2, that an expected error of $\tilde{\Theta}(n^{3/2})$ is asymptotically tight.

4.5 EMPIRICAL EVALUATION

To confirm our theoretical claims, we implemented PAMST [Pin18], Sealfon’s input privatization [Sea16], and our Algorithm 1 (instantiated with Prim–Jarník) in *Python* 3.9. The implementation relies on version 3.2.1 of the *NetworkX* library [HSSo8]. All experiments ran locally on a MacBook Pro with an Apple M2 Pro processor (10 Cores, up to 3.7GHz) and 16GB of RAM. We present two experiments that support our theoretical claims. The first experiment resembles a natural setting in the context of synthetic data generation, and the second explores the influence of the graph’s density. The experiments indicate that the output distribution of our algorithm indeed matches PAMST and, hence, outperforms the *input privatization* approach if the graph is not too sparse. The results are shown in Figure 4.2, and we refer for more details to Section 4.7.7.

MUTUAL INFORMATION A natural setting is finding the Chow-Liu tree [CL68], which is the **MST** on the graph encoding the negated mutual information matrix on all pairwise attributes. Let $X_1, \dots, X_n \in \{0, 1\}$ be random bits. We draw $X_1 \sim \text{Ber}(1/2)$, and then recursively X_i by flipping the bit X_{i-1} with probability $0 < p < 1/2$. This simulates a natural scenario where there is mutual information

Note that in the Chow-Liu setting, we want to find the maximum spanning tree, which is the same as the **MST** on the negated weights.

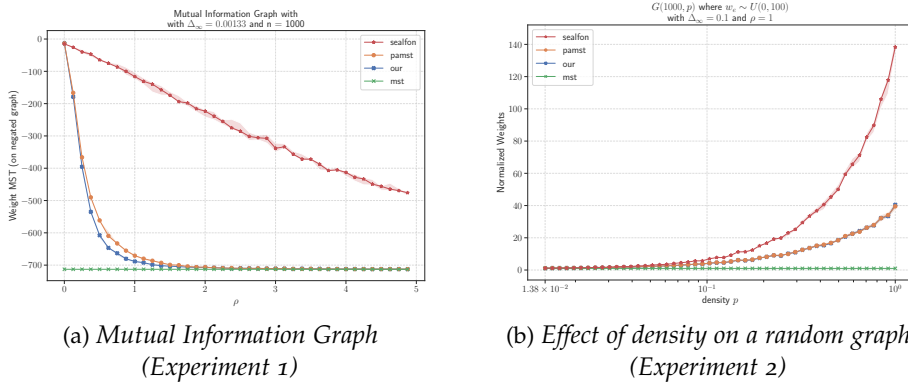


Figure 4.2: The results of our experiment. **a)** Shows the instance described in Experiment 1. Note that we have to negate all weights to find the maximum spanning tree on the mutual information graph. **b)** Shows the impact of the graph’s density on random graphs with $n = 1000$ vertices for a fixed privacy level $\rho = 1$: The figure shows the ratio between the real **MST** and the private one, where each edge weight is uniformly drawn from the interval $[0, 100]$. Because the noise scale of Sealfon’s input perturbation scales with the number of edges in the graph, we see a larger gap for denser graphs. Each data point shows the median of ten runs.

between X_i and X_{i-1} controlled by p . The **MST** is formed by the edges on the path $P = (X_1, \dots, X_n)$ (visualized in Figure 4.3). A potential underlying dataset of size d has the sensitivity of mutual information, which is $\Delta_{mi} = \log_2(d)/d$. As later shown in Section 4.7.7, for this process the pairwise mutual information between all X_i and X_j define a complete graph $G = (V, E)$ with weights

$$\forall i, j \in [n], i \neq j : w_{ij} = \frac{1}{2} (p_1 \log_2(p_1) + p_2 \log_2(p_2)) ,$$

where $p_1 = 1 + (1 - 2p)^k$ and $p_2 = 1 - (1 - 2p)^k$. In the experiment, we used $n = 1000$ with flip probability of $p = 0.05$. We set the underlying dataset size to $d = 10^5$, thus $\Delta_\infty = \Delta_{mi} \approx 0.00133$. The results are shown in Figure 4.2-a). We can see that the error of our approach closely resembles PAMST, outperforming the input privatization approach.

THE EFFECT OF THE DENSITY The second experiment follows the setup proposed in [Pin18]. We sample an Erdős–Rényi random graph $\mathcal{G}(n, p)$ where we draw the weights $w_e \sim U(0, 100)$ uniformly. The experiments shown in Figure 4.2-b) were run on graphs of size $n = 1000$ using $\Delta_\infty = 0.1$. As shown, input privatization does significantly worse as the graph’s density increases, whereas our approach resembles the output distribution of PAMST.

4.6 CONCLUSION AND OPEN PROBLEMS

Our work shows that a simple input perturbation yields a privacy guarantee for the output of any **MST** algorithm far better than what is implied by the post-processing property of DP. It is natural to wonder if something similar holds for other problems where the best existing in-place private algorithms add noise during the computation.

There remains a small gap of $O(\sqrt{\log(1/\delta)})$ between known upper and lower utility bounds. As discussed in Section 4.4.1, extending the Laplace noise-based

approach for top- k selection by Qiao, Su, and Zhang [QSZ21] to the private **MST** problem remains an open question.

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4.7 APPENDIX

4.7.1 Algorithm for Maximum Weight Independent Set in a Matroid

A matroid $(U, \mathcal{I})$ is defined by a finite ground set U and a family of *independent sets* $\mathcal{I} \subseteq 2^U$. The family $\mathcal{I}$ satisfies the following three properties:

- $\emptyset \in \mathcal{I}$;
- $\forall S \in \mathcal{I}$ and $T \subseteq S, T \in \mathcal{I}$;
- $\forall S, T \in \mathcal{I}$ and $|S| < |T|$, there exists $t \in T \setminus S$ such that $S \cup \{t\} \in \mathcal{I}$.

As an example, let U be the set of edges in a graph and $\mathcal{I}$ be the family of all subsets of edges that form a forest. It's easy to verify that $(U, \mathcal{I})$ is indeed a matroid, more concretely a *graphic matroid*. Given a weight function $w : U \rightarrow \mathbb{R}^+$, the weight of an independent set S , $w(S)$, is defined as the sum of the weight of its elements. Therefore, **MST** problem is a special case of finding a maximum weight independent set in this matroid. Kruskal's [Kru56] algorithm was generalized to the problem of finding a maximum weight independent set in a matroid by Edmonds [Edm71]. The algorithm first sorts all the elements in the matroid by weight in decreasing order and then tries to insert elements one by one into the independent set if possible. Since the analysis for Algorithm 1 is based on Kruskal's algorithm, it naturally generalizes to finding a maximum weight independent set in general matroids. For matroids with rank n and m elements and since there is no upper bound for m on n , the expected error we have will be $O(n^{3/2} \log m / \sqrt{\rho})$.

Algorithm 5 Private Maximum-Weight-Independent-Set-in-Matroid Framework

Require: A matroid $(U, \mathcal{I})$ and the weights $\mathbf{w}$, algorithm $\mathcal{A}_{\text{MWIS}(\text{matroid})}$, privacy parameters (ϵ, δ)

- 1: $\rho \leftarrow (\sqrt{\epsilon + \log(1/\delta)} - \sqrt{\log(1/\delta)})^2$; $\epsilon' \leftarrow \sqrt{2 \cdot \rho / (n-1)}$;
 - 2: $\tilde{w}_e \leftarrow w_e + (2/\epsilon') \cdot \ln(\text{Exp}(1))$ for all $e \in U$.
 - 3: **return** $\mathcal{A}_{\text{MWIS}(\text{matroid})}(U, \mathcal{I}, \tilde{\mathbf{w}})$.
-

1437 4.7.2 One-Shot-PPSACR

Algorithm 6 One-Shot-PPSACR

```

1:  $\tilde{s}(j) \leftrightarrow \mathbb{E} \exp(1) / s(j), \forall j \in U.$ 
2:  $\mathcal{C} \leftarrow U, \mathcal{I} \leftarrow$  An empty list.  $\triangleright$  possible candidates and sampled items so far
3: while  $|\mathcal{I}| \leq k$  and  $\mathcal{C} \neq \emptyset$  do
4:    $j \leftarrow \arg \min_{j' \in \mathcal{C}} \tilde{s}(j').$ 
5:   Add  $j$  to the back of  $\mathcal{I}$ 
6:    $\mathcal{C} \leftarrow \mathcal{C} \setminus (\{j\} \cup f(\mathcal{I})).$ 
7: return  $\mathcal{I}.$ 

```

1438 4.7.3 Further Background

1439 The distribution of the logarithm of an exponential random variable is closely
 1440 connected to the *Gumbel distribution*.

Fact 4.17. If $Z \sim \mathbb{E} \exp(1)$, then $-\ln Z \sim \text{Gumbel}(1)$.

1441 *Proof.* $\mathbf{P}[-\ln X \leq z] = \mathbf{P}[X \leq \exp(-z)] = \exp(-e^{-z})$ for all $z \in \mathbb{R}$. □

1442 **Proof of Fact 4.4.** For our proof, we require the following fact, which we will prove
 1443 later.

Fact 4.18. If $Z_i \sim \ln \mathbb{E} \exp(1)$, then for each $\beta \in (0, 1)$,

$$\mathbf{P}\left[Z_i > \ln \ln \frac{1}{\beta}\right] = \mathbf{P}\left[Z_i < \ln \ln \frac{1}{1-\beta}\right] = \beta.$$

1444 Since $\ln(1+x) \leq x, \forall x > -1$, it holds that

$$\ln \ln \frac{1}{1-\beta} = \ln(-\ln(1-\beta)) \geq \ln \beta = -\ln \frac{1}{\beta}.$$

1445 Then it follows that

$$\mathbf{P}\left[Z_i < -\ln \frac{1}{\beta}\right] \leq \mathbf{P}\left[Z_i > \ln \ln \frac{1}{\beta}\right] = \beta.$$

1446 Define $X \doteq \max_{i \in [m]} |Z_i|$. By a union bound, we have

$$\mathbf{P}\left[|X| \geq \ln \frac{2 \cdot m}{\beta}\right] \leq \beta, \quad \forall \beta \in (0, 1).$$

1447 Define $t \doteq \ln \frac{2 \cdot m}{\beta} \geq \ln(2 \cdot m)$. Then $\beta = (2 \cdot m) \cdot e^{-t}$. Therefore,

$$\begin{aligned}
 \mathbb{E}[|X|] &= \int_0^{\ln(2 \cdot m)} \mathbf{P}[|X| \geq t] dt + \int_{\ln(2 \cdot m)}^{\infty} \mathbf{P}[|X| \geq t] dt \\
 &\leq \ln(2 \cdot m) + \int_{\ln(2 \cdot m)}^{\infty} (2 \cdot m) \cdot e^{-t} dt \leq \ln(2 \cdot m) + (2 \cdot m) \cdot e^{-\ln(2 \cdot m)} \\
 &= \ln(2e \cdot m).
 \end{aligned}$$

1448 *Proof of Fact 4.18.* If $Y \sim \text{Exp}(1)$ where $\mathbf{P}[Y \leq y] = e^{-y}$ we have for each $\beta \in (0, 1)$,

$$\begin{aligned} \mathbf{P}\left[Y > \ln \frac{1}{\beta}\right] &= \exp\left(-\ln \frac{1}{\beta}\right) = \beta, \text{ and } \mathbf{P}\left[Y < \ln \frac{1}{1-\beta}\right] \\ &= 1 - \exp\left(-\ln \frac{1}{1-\beta}\right) = \beta. \end{aligned}$$

1449 It follows that

$$\mathbf{P}\left[\ln Y > \ln \ln \frac{1}{\beta}\right] = \mathbf{P}\left[\ln Y < \ln \ln \frac{1}{1-\beta}\right] = \beta.$$

1450

□

1451 **Proof of Fact 4.9.**

$$p_Y(y) = p_X(\lambda \cdot y) \cdot \frac{dX}{dY} = \lambda \cdot e^{-\lambda \cdot y}, \forall y \geq 0.$$

1452

□

Fact 4.19 (Order of Exponentials). *If $X \sim \text{Exp}(\lambda_1)$, and $Y \sim \text{Exp}(\lambda_2)$, then*

$$\mathbf{P}[X \leq Y] = \int_0^\infty \lambda_1 e^{-\lambda_1 x} \cdot \mathbf{P}[Y \geq x] dx = \int_0^\infty \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}.$$

1453 **Proof of Fact 4.10.** Let $Y \doteq \min_{j \in [d] \setminus \{i\}} X_j$. Then

$$\mathbf{P}[Y \geq y] = \prod_{j \in [d] \setminus \{i\}} \mathbf{P}[X_j \geq y] = \exp\left(-y \sum_{j \in [d] \setminus \{i\}} \lambda_j\right). \quad (18)$$

1454 Therefore, $Y \sim \text{Exp}\left(\sum_{j \in [d] \setminus \{i\}} \lambda_j\right)$.

1455 Applying Fact 4.19 gives $\mathbf{P}\left[X_i = \min_{j \in [d]} X_j\right] = \frac{\lambda_i}{\sum_{j \in [d]} \lambda_j}$. □

1456 **Proof of Fact 4.11.** $\mathbf{P}[X \geq x + y \mid X \geq x] = \frac{\exp(x+y)}{\exp(x)} = \exp(y) = \mathbf{P}[X \geq y]$. □

1457 4.7.4 Utility Guarantee of Private Kruskal

1458 We prove the utility of private Kruskal's algorithm (Algorithm 2) directly. Suppose
 1459 we use Kruskal's algorithm to find the **MST** on the noisy and original graphs in
 1460 parallel. When we try to add the k -th edge in noisy graph, we are actually finding
 1461 the noisy minimum-weighted edge among all the cycle-free candidates. Also, there
 1462 must be a candidate from the first k added edges when we run Kruskal's algorithm
 1463 in the original graph, as all forests form a matroid. Therefore, the real minimum
 1464 weight is not greater than the payment for the k -th step on the original graph,
 1465 so the extra cost induced in this step only comes from selecting the noisy-min.
 1466 Accumulate the error over all steps, we show the desired utility guarantee of the
 1467 algorithm.

Proof. For each $i \in \{1, \dots, n-1\}$, let e_i be the edge added to T at the i^{th} iteration of Algorithm 2. Let $T_i \doteq \bigcup_{j \in [i]} \{e_j\}$ be the partial spanning forest after iteration i and for convenience, let $T_0 = \emptyset$. Further, initially set $F_0 = E$, and denote with F_i the set F immediately after the i^{th} iteration containing all edges that still could be added to T without creating a cycle. Assume that also run a non-private Kruskal's algorithm on the same graph with the true edge weights in parallel, to obtain the real MST $T_i^* = \bigcup_{j \in [i]} \{e_j^*\}$, where e_j^* is the edge added at iteration j .

To prove the utility guarantee, we want to show that for all steps, with probability $1 - \beta$ and for some universal constant $c \in \mathbb{R}^+$,

$$w_{e_i} \leq w_{e_i^*} + \frac{c}{\epsilon'} \cdot \ln \frac{n}{\beta}. \quad (19)$$

Note that the utility guarantee of the exponential mechanism gives us with probability $1 - \beta$

$$w_{e_i} \leq \min_{e \in F_{i-1}} w_e + \frac{c}{\epsilon'} \cdot \ln \frac{n}{\beta} \quad (20)$$

because e_i is exactly the noisy minimum of the set F_{i-1} chosen by the algorithm. Therefore, it suffices to show that

$$\min_{e \in F_{i-1}} w_e \leq w_{e_i^*}. \quad (21)$$

Note that $w_{e_{i-1}^*} \leq w_{e_i^*}$ by the way Kruskal adds new edges to T^* . Using this, we claim that F_{i-1} contains at least one edge which belongs to T_i^* and can be added to T_{i-1} without forming a cycle.

First, note that edges in T_{i-1} induce a forest. Let $\mathcal{T}$ be an arbitrary tree in this forest, and assume that it has t edges. Then, the number of edges $e = (u, v) \in T_i^*$, such that both $u, v \in \mathcal{T}$ is at most t without inducing a cycle.

There are $i-1$ edges in T_{i-1} , but i edges in T_i^* . Therefore, there must be *at least one* edge from T^* , which connects two trees in the forest induced by T_{i-1} . It follows that this edge belongs to F_{i-1} , proving Equation (21). Summing over all possible edges gives

$$\sum_{i \in [n-1]} w_{e_i} \leq \sum_{i \in [n-1]} w_{e_i^*} + \frac{c(n-1)}{\epsilon'} \ln \frac{n}{\beta}.$$

By a standard integration technique, this implies an expectation of $\mathcal{O}\left(\frac{n}{\epsilon'} \ln n\right)$. $\square$

4.7.5 Efficiently Implementing $\mathcal{A}_{\text{priv-kruskal}}$

This section describes how $\mathcal{A}_{\text{priv-kruskal}}$ can be efficiently implemented. Although this result is, of course, overshadowed by our main result, we believe it is interesting enough to be stated here.

The algorithm can be implemented in $O(n + m \log n)$ time. In the initialization phase, we construct a complete binary tree where every level, except possibly the last, is fully filled, and all nodes in the last level are as far left as possible—with m leaf nodes. Each leaf node represents an edge and is assigned a weight $e^{-\epsilon' w_e}$. Each internal node of the binary tree is assigned a weight equal to the sum of the weights of the leaf nodes in the subtree rooted at that node.

This binary tree is used to manage the nodes in F efficiently: it supports both the sampling operation (Algorithm 2, Line 4) and the update operation (Algorithm 2, Line 6) in $O(\log n)$ amortized time per edge.

1505 *Sampling.* We apply a top-down approach starting from the tree's root to sample
 1506 an edge (i.e., a leaf node). At each step, we move to the left child with probability
 1507 equal to the ratio of the left child's weight to the current node's weight, and move
 1508 to the right child otherwise. The sampling procedure terminates when the current
 1509 node is a leaf. It is straightforward to verify via an induction on the number of
 1510 levels that this method samples a leaf node with probability proportional to $e^{-\epsilon' w_e}$
 1511 and that the sampling process completes in $O(\log n)$ time.

1512 *Update.* We introduce two auxiliary data structures for this step. First, we main-
 1513 tain a union-find data structure [Tar75], which allows us to determine the connected
 1514 component (induced by the edges in T) to which a vertex belongs in $O(\alpha(n))$ amor-
 1515 tized time, where $\alpha(n)$ is the inverse Ackermann function. Additionally, for each
 1516 connected component, we maintain a linked list to track the vertices it contains.
 1517 When a new edge $e = (u, v)$ is added to the set T , let C_u and C_v denote the con-
 1518 nected components containing u and v , respectively. To merge C_u and C_v , we
 1519 update both the union-find structure and the linked list, which can be performed
 1520 in $O(\alpha(n) + 1)$ time. Next, we need to remove from F all edges connecting C_u and
 1521 C_v . To do so efficiently, we choose the smaller of C_u and C_v (in term of the number
 1522 of vertices)—assume, without loss of generality, it is C_u —and check all edges incident
 1523 to it. For each edge, if its other endpoint belongs to C_v , we remove it from F .

1524 We now analyze the number of times an edge is checked and the cost of its
 1525 removal from F . For the former, define the rank of a vertex as the number of
 1526 vertices in the connected component to which it belongs. Each time an edge is
 1527 checked, the rank of one of its endpoints at least doubles. Since a vertex can have a
 1528 rank of at most n , an edge can be checked at most $O(\log n)$ times.

1529 For the latter, removing an edge from F is handled by removing the correspond-
 1530 ing leaf node from the binary tree and then updating the weights of the nodes
 1531 along the leaf-to-root path. This operation can be completed in $O(\log n)$ time.

1532 4.7.6 Proof of Lemma 4.14

1533 Our proof relies on a standard result from the Erdős–Rényi model $\mathcal{G}(n, p)$, where
 1534 $\mathcal{G}$ is a random graph with n vertices such that each edge is independently included
 1535 with probability p . The following lemma is stated with explicit constants, as our
 1536 subsequent proof relies on them. Despite extensive searching, we could not find a
 1537 version of the lemma with explicit constants. For completeness, we provide a proof
 1538 of this lemma with explicit constants in the complete version [Pag+24].

Lemma 4.20 ([ER59] Connectivity). *For any real $\epsilon > 0$, if $p > \frac{(1+\epsilon)\ln n}{n}$, then a random graph in the Erdős–Rényi model $\mathcal{G}(n, p)$ with n vertices is almost surely connected. In particular, if $p \geq \frac{2\ln n}{n-1}$ and $n \geq 1700$, the probability that the random graph is not connected is at most $\frac{6}{n}$.*

1539 Recall that the random graph G considered in Lemma 4.14 is generated as
 1540 follows: G consists of n vertices and $m = \frac{n(n-1)}{2}$ edges. For each edge e , we first
 1541 sample $P_e \sim \text{Beta}(\beta, \beta)$ and then sample $w_e \sim \text{Bin}(s, P_e)$.

1542 Let H be the subgraph that includes all vertices in G and all edges e such that
 1543 $w_e \geq 3/4 \cdot s$. Since the edge weights are independent of each other, the graph H
 1544 is an instance of Erdős–Rényi model with $p = \mathbf{P}[w_e \geq 3/4 \cdot s]$. Note that G has a
 1545 spanning tree where all edges e have $w_e \geq 3/4 \cdot s$ if and only if H is connected.

Given Lemma 4.20, to prove Lemma 4.14, it just remain to show that for each edge e , we have

$$\mathbf{P} \left[w_e \geq \frac{3}{4} \cdot s \right] > \frac{2 \ln n}{n-1}. \quad (22)$$

Fact 4.21 (Steinke and Ullman [SU17]). *Let $\beta \geq 1$, and X be a random variable following beta distribution $\text{Beta}(\beta, \beta)$. Then for all $p \in [0, 1/2]$, it holds that*

$$\mathbf{P}[X \leq p] = \mathbf{P}[X \geq 1-p] \geq (4 \cdot p \cdot (1-p))^{\beta-1} \cdot \frac{p}{\beta}. \quad (23)$$

According to Fact 4.21, we have $\mathbf{P}[P_e \geq 0.9] \geq \frac{0.36^{\beta-1}}{10\beta}$ and since $\mathbb{E}[w_e] = s \cdot P_e$, by Hoeffding's inequality,

$$\mathbf{P}[w_e > s \cdot P_e - 0.15s \mid P_e \geq 0.9] < \exp(-0.045s). \quad (24)$$

Therefore,

$$\begin{aligned} \mathbf{P} \left[w_e \geq \frac{3}{4}s \right] &\geq \mathbf{P}[P_e \geq 0.9] \cdot \mathbf{P}[w_e \leq s \cdot P_e - 0.15s \mid P_e \geq 0.9] \\ &\geq \frac{0.36^{\beta-1}}{10\beta} \cdot (1 - \exp(-0.045s)). \end{aligned}$$

For $s \geq 10$ and $\beta = \frac{1}{2} \ln n$ and $n \geq 2 \times 10^7$, we have

$$\mathbf{P} \left[w_e \geq \frac{3}{4}s \right] \geq \frac{0.36^{\frac{1}{2} \ln n}}{1.8 \ln n} \cdot (1 - \exp(-0.45)) > \frac{0.36}{1.8 n^{0.52} \ln n} > \frac{2 \ln n}{n-1}. \quad (25)$$

□

4.7.7 Proofs for the Experiments

Assume $0 < p < \frac{1}{2}$. The mutual information $I(X; Y)$ between two discrete random variables X and Y quantifies the amount of information that X contains about Y (or vice versa).

Definition 4.4 (Mutual Information [Ash90]).

$$I(X; Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \mathbf{P}[X = x, Y = y] \log_2 \left(\frac{\mathbf{P}[X = x, Y = y]}{\mathbf{P}[X = x] \mathbf{P}[Y = y]} \right). \quad (26)$$

We now prove the validity of the weight function given in Section 4.5.

Definition 4.5. We define the random variables $X_1, \dots, X_n \in \{0, 1\}$ recursively by

setting

$$X_1 \sim \text{Ber}\left(\frac{1}{2}\right) \quad \text{and} \quad X_i = \begin{cases} 1 - X_{i-1} & \text{with probability } p \\ X & \text{else} \end{cases}.$$

1560

1561 By definition, $\mathbf{P}[X_i = 0] = \mathbf{P}[X_i] = 1 = \frac{1}{2}$. Note that mutual information is min-
 1562 imized for $p \approx \frac{1}{2}$ and note that for all X_i : $\mathbf{P}[X_i = 1] = \mathbf{P}[X_i = 0] = \frac{1}{2}$. We need
 1563 another short lemma for the parity of a binomial random variable:

Lemma 4.22 (Parity of Binomials.). Denote $Z \sim \text{Bin}(n, p)$, then

$$\Pr[Z \text{ is even}] = 1 - \mathbf{P}[Z \text{ is odd}] = \frac{1}{2} + \frac{1}{2}(1 - 2p)^n. \quad (27)$$

1564 *Proof.* By direct computation and application of the binomial theorem in the last
 1565 step:

$$\begin{aligned} \mathbf{P}_{Z \sim \text{Bin}(n, p)}[Z \text{ even}] &= \sum_{k=0}^n \frac{(-1)^k + 1}{2} \mathbf{P}[Z = k] \\ &= \frac{1}{2} \left(\sum_{k=0}^n (-1)^k \mathbf{P}[Z = k] + \sum_{k=0}^n \mathbf{P}[Z = k] \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n ((-1)^k \mathbf{P}[Z = k]) + 1 \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n ((-1)^k \binom{n}{k} p^k (1-p)^{n-k}) + 1 \right) \\ &= \frac{1}{2} \left(\sum_{k=0}^n \left(\binom{n}{k} (-p)^k (1-p)^{n-k} \right) + 1 \right) = \frac{1}{2}(1 - 2p)^n + \frac{1}{2}. \end{aligned}$$

1566

□

Claim 4.23. Assume $0 < p < \frac{1}{2}$, the mutual information score between X_i and X_j that are $k = |i - j|$ steps apart can directly be computed by

$$\begin{aligned} I(X_i; X_j) &= \left(\frac{1}{2} + \frac{1}{2}(1 - 2p)^k \right) \log_2 \left(1 + (1 - 2p)^k \right) \\ &\quad + \left(\frac{1}{2} - \frac{1}{2}(1 - 2p)^k \right) \log_2 \left(1 - (1 - 2p)^k \right). \end{aligned} \quad (28)$$

1567 *Proof.* We shortly write $p_{00}(k) = \mathbf{P}[X_i = 0, X_j = 0]$ and $p_0 = \mathbf{P}[X_i = 0]$ (resp $p_{11}(k)$,
 1568 $p_{10}(k)$ and $p_{01}(k)$, p_1). Furthermore, denote $p_{\text{even}}(k) = \mathbf{P}_{Z \sim \text{Bin}(k, p)}[Z \text{ is even}]$ and
 1569 $p_{\text{odd}}(k)$ as proven above. Note that we can compute these probabilities:

$$\begin{aligned} p_{00}(k) &= \mathbf{P}[X_i = 0 | X_j = 0] \cdot \mathbf{P}[X_j = 0] = p_{\text{even}}(k) \cdot \frac{1}{2} = \frac{1}{4} + \frac{1}{4} \cdot (1 - 2p)^k, \\ p_{10}(k) &= \mathbf{P}[X_i = 1 | X_j = 0] \cdot \mathbf{P}[X_j = 0] = p_{\text{odd}}(k) \cdot \frac{1}{2} = \frac{1}{4} - \frac{1}{4} (1 - 2p)^k. \end{aligned}$$

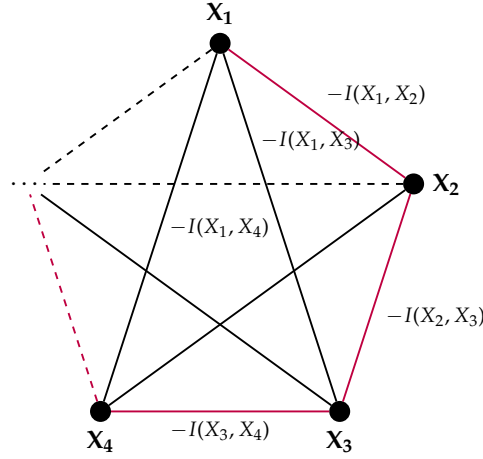


Figure 4.3: An extract of the complete graph encoding the mutual information between the random variables $X_1, \dots, X_n$ described in Definition 4.5 and used in Section 4.5. The weights encode the negated mutual information corresponding to the described process. The **MST** is formed by the vertices on the path $P(X_1, X_2, \dots)$. In our experiment with $n = 1000$ vertices and the flip probability $p = 0.05$, we have $-I(X_1, X_2) = I(X_2, X_3) = \dots \approx -0.7136$, $-I(X_1, X_3) = I(X_2, X_4) = \dots \approx -0.5471$ and $-I(X_1, X_4) \dots \approx -0.4277$.

1570 We trivially get $p_{11}(k) = p_{00}(k)$ and $p_{01}(k) = p_{10}(k)$ by symmetry. Then, we can
 1571 compute the mutual information directly.

$$\begin{aligned}
 I(X_i; X_j) &= p_{00}(k) \log_2 \frac{p_{00}(k)}{p_0 p_0} + p_{01}(k) \log_2 \frac{p_{01}(k)}{p_0 p_1} \\
 &\quad + p_{10}(k) \log_2 \frac{p_{10}(k)}{p_1 p_0} + p_{11}(k) \log_2 \frac{p_{11}(k)}{p_1 p_1} \\
 &= p_{00}(k) \log_2(4p_{00}(k)) + p_{01}(k) \log_2(4p_{01}(k)) \\
 &\quad + p_{10}(k) \log_2(4p_{10}(k)) + p_{11}(k) \log_2(4p_{11}(k)) \\
 &= 2p_{00}(k) \log_2(4p_{00}(k)) + 2p_{01}(k) \log_2(4p_{01}(k)) \\
 &= \left(\frac{1}{2} + \frac{1}{2}(1-2p)^k \right) \log_2 \left(1 + (1-2p)^k \right) \\
 &\quad + \left(\frac{1}{2} - \frac{1}{2}(1-2p)^k \right) \log_2 \left(1 - (1-2p)^k \right).
 \end{aligned}$$

1572

□

LOWER BOUNDS FOR PRIVATE GRAPH OPTIMIZATION PROBLEMS USING RECONSTRUCTION ATTACKS

Under submission, partial results appeared at Theory and Practice on Differential Privacy (TPDP) 2025 & 2026

Joint work with Jacob Imola and Rasmus Pagh

This section studies fundamental graph optimization problems under differential privacy (DP) and shows new, reconstruction-based lower bounds. We consider a graph $G = (V, E, \mathbf{w})$ where the vertex set V and edges E are public and the weights $\mathbf{w} : E \rightarrow \mathbb{R}$ must be kept differentially private under an ℓ_1 neighboring relation. For the problems of releasing a minimum-weight spanning tree and a minimum-weight perfect matching, we show new, tight error bounds of $\Omega(n \cdot \log(m/n)/\epsilon)$ on worst-case graphs with n vertices and $m > 2n$ edges. The upper bounds are known pure DP algorithms while the new lower bound holds even under approximate (ϵ, δ) -DP as long as $\delta \leq (n/m)^{\Omega(1)}$. Our lower bounds improve the $\Omega(n/\epsilon)$ lower bounds of Sealfon (PODS '16).

The fact that approximate DP does not reduce error for Minimum-Weight Spanning Tree (MST) under the ℓ_1 neighboring relation contrasts with the recent upper bound of Pagh et al. (PODS '25) which shows that approximate DP allows much better error under the ℓ_∞ neighboring relation. Going beyond worst-case graphs, we give lower bounds for large families of sparse graphs with expansion properties. We show a lower bound of $\Omega(n/\epsilon)$ for the minimum spanning tree for any graph where the minimum cut is at least $\Omega(\log(n))$. Finally, we consider the problem of private hierarchical clustering under Dasgupta's cost function (STOC '16) and show the first approximate DP lower bound parameterized by the minimum weight of a balanced cut. This extends lower bounds of Deng et al. (ICLR '25) to general graphs and to approximate DP.

Graph-structured data is a fundamental abstraction in modern data management. Weighted graphs naturally arise, for example, from relational data describing transportation networks, communication infrastructure, recommendation systems, similarity relations, and interactions between entities. Many important analysis tasks on such data reduce to classical graph optimization problems, including computing sparse connectivity structures such as minimum spanning trees, assignment structures such as minimum-weight perfect matchings, and multiscale summaries such as hierarchical clusterings. These structures serve as compact summaries that support downstream tasks such as routing, indexing, similarity search, visualization, and unsupervised learning.

In many applications, however, the edge weights encode sensitive aggregate information derived from individuals. For example, edge weights may represent traffic intensities collected from users, communication frequencies between populations, similarity scores derived from behavioral data, or compatibility measures computed from medical records. In such settings, a single individual may affect multiple edge weights simultaneously, but only by a small amount in each coordinate. Compared to edge-level privacy models that treat edges as discrete records, the ℓ_1 model naturally captures continuous-valued and aggregate data

contributions, making it particularly well suited for weighted graph optimization problems. This motivated Sealfon [Sea16] to study differential privacy under an ℓ_1 neighboring relation on *edge weights*, where neighboring weighted graphs differ by bounded total change in their edge-weight vectors. Other researchers have applied this appealing privacy model to other graph problems, including hierarchical clustering [Den+25] and all pairs shortest paths [Che+23].

Examples. In a transportation network, an **MST** can represent a sparse backbone connecting cities or distribution centers using low-cost routes inferred from aggregate traffic data. Publishing such a structure enables routing, infrastructure planning, and resilience analysis while dramatically compressing the underlying weighted network. Similarly, in a recommendation or similarity graph, a hierarchical clustering, also known as a dendrogram, provides a navigable taxonomy of users, products, or documents derived from sensitive behavioral correlations. These structures often involve sensitive data, such as road traffic data or user browsing activity, and thus are good candidates for private data release.

5.0.1 Our Contributions

We study the Minimum-Weight Spanning Tree, Minimum-Weight (Perfect) Matching (**MWPM**), and Minimum-Cost Hierarchical Clustering (**HC**) problems in the edge-weight model of differential privacy. We derive *two* types of lower bounds. For **MST** and **MWPM**, we provide lower bounds that the additive error must grow with a $\Omega(n \log(\frac{m}{n}))$ factor, where m is the number of edges in the graph. These lower bounds work by demonstrating effective reconstruction attacks for *worst-case* graph topologies, ruling out differential privacy. Additionally, they improve over prior lower bounds of Sealfon [Sea16] and match the error of simple input perturbation algorithm [Aam26]. The fact that input perturbation is an optimal approach for these problems is additional evidence that input perturbation is often a good approach to graph optimization problems, as seen recently in the paper of Pagh, Retschmeier, Wu, and Zhang [Pag+24].

We also investigate a novel generalization of lower bounds in edge-weight DP: the question of establishing lower bounds for a *fixed* graph topology that is not necessarily worst-case. This is inspired by the fact that the graph topology in edge-weight DP is information that is known beforehand that can be leveraged by the algorithm. We make progress towards answering this question on **MST** and **HC**. For **MST**, we show that topology with *minimum cut* at least $\Omega(\log(n))$ must incur error $\Omega(n)$, differing from the lower bound of the worst-case topology by a $\log(\frac{m}{n})$ factor. For **HC** under Dasgupta’s cost function [Das16], we obtain a fully-parameterized lower bound that grows with the expansion (in the sense of expander graphs), and decreases with the maximum degree of the graph. For d -regular expander graphs, our lower bound is $\Omega(n^2)$, which matches the state-of-the-art lower bound of [Den+25; Imo+23] but generalizes it to approximate DP and beyond the worst-case topology.

A summary of all our new results together with known bounds appears in Figure 5.1. Note that the error bounds depend only on graph parameters, so the error is independent of the weight (or cost) of a solution. Thus, these problems can be solved privately with good error bounds whenever the weight (or cost) of an optimal solution is sufficiently large compared to the error bound.

For **MST** and **MWPM**, the fact that upper bounds under pure differential privacy ($\delta = 0$) can be matched by lower bounds shown under approximate differential privacy (with polynomially small δ) is somewhat surprising given the polynomial

Problem	Error Bound	Reference	Graph type	Parameters
MST	$\mathcal{O}(n \cdot \log n)$	[Sea16]	Any	$\delta = 0$
	$\mathcal{O}(n \cdot \log(m/n))$	[Aam26]	Any	$\delta = 0$
	$\Omega(n)$	[Sea16]	Star graph	$\delta < c_\epsilon$
	$\Omega(n \cdot \log(m/n))$	Theorem 5.3	Worst-case	$\delta < (n/m)^{\Omega(1)}$
	$\Omega(n)$	Corollary 5.8	Min-cut $\geq 5 \log(n)$	$\delta < \frac{c_\epsilon}{m}$
MWPM	$\mathcal{O}(n \cdot \log n)$	[Sea16]	Any	$\delta = 0$
	$\mathcal{O}(n \cdot \log(m/n))$	[Aam26]	Any	$\delta = 0$
	$\Omega(n)$	[Sea16]	Collection of 4-cycles	$\delta < c_\epsilon$
	$\Omega(n \cdot \log(m/n))$	Theorem 5.10	Worst-case	$\delta < (n/m)^{\Omega(1)}$
HC	$\mathcal{O}(n^2 \log n)$	[Imo+23]	Any	$\delta = 0$
	$\Omega(n^2)$	[Den+25]	Complete graph	$\delta = 0$
	$\Omega(n^2)$	Theorem 5.18	Degree- $\mathcal{O}(1)$ expander	$\delta < \frac{c_\epsilon}{m}$

Figure 5.1: Upper and lower bounds for the error of differentially private graph optimization problems: MST, MWPM, and HC under the ℓ_1 neighboring relationship. Upper bounds hold for pure differential privacy ($\delta = 0$), while lower bounds are for (ϵ, δ) -DP. All bounds omit a multiplicative factor $1/\epsilon$ for readability; $c_\epsilon > 0$ is a constant that depends only on ϵ .

separation between pure and approximate differential privacy for MINIMUM SPANNING TREE under the ℓ_∞ neighboring relation [Pag+24]. It remains an interesting open question to show tight bounds for larger values of δ .

TECHNIQUES Our lower bounds work via *reconstruction attacks*, where a private dataset is encoded into the weights of the graph topology, and based on the solution to the graph problem, a decoding process attempts to reconstruct the dataset. If the error of the graph algorithm is low enough, we show that the reconstruction will be successful, contradicting (ϵ, δ) -DP. For MINIMUM SPANNING TREE and Minimum-Weight (Perfect) Matching, we give new encoding and decoding procedures for datasets in a bigger domain $[m/n]^d$. This allows us to encode more information into the weights of the graph than simply using a binary dataset, lifting the existing lower bound by a $\log(m/n)$ factor.

For our second type of lower bound, we focus on reconstructing binary datasets and encode them *randomly* into the edge weights. This requires analyzing the behavior of the encoded dataset using ideas from random graph theory. The benefit of this less structured encoding is that it can apply to general classes of graph topologies. For MINIMUM SPANNING TREE, a randomized encoding and a simple decoding procedure is enough to give lower bounds for all graphs with a sufficient minimum cut. For HIERARCHICAL CLUSTERING, we have to further subsample the edges to which the dataset is encoded and utilize the fact that the graph algorithm does not know these subsampled edges to show proper reconstruction.

Our lower bounds also use generalized reductions from differential privacy to reconstruction rates (Section 5.1), which may find more application in the edge weight setting. As recently observed by Aamand [Aam26] the lower bounds for MST and Minimum-Weight (Perfect) Matching are matched by Sealfon’s input perturbation algorithms that simply add Laplace noise to each edge weight and run the non-private optimization algorithm as post-processing. For completeness, we include a simple proof of the improved bound of [Aam26].

5.0.2 Related Work

GRAPH OPTIMIZATION PROBLEMS. MINIMUM SPANNING TREE and Minimum-Weight (Perfect) Matching are classic graph optimization problems, with applications including tree-structured graphical models, synthetic data generation, and market or network matching. HIERARCHICAL CLUSTERING is an unsupervised learning technique that - unlike *flat* clustering methods such as k -means - organizes data into clusters at multiple levels of granularity. The problem dates back more than half a century [War63] and has applications across many different domains, including network analysis [LRU14], text analysis [SKK00], and biology.

RECONSTRUCTION ATTACKS. Our lower-bound proofs are part of a long line of reconstruction attacks in differential privacy. The seminal work of Dinur and Nissim [DN03] showed that answering too many queries too accurately can allow reconstruction of most of a private database. Subsequent work developed this connection into general lower-bound methods for private data release, including iterative constructions [GRU12], lower bounds based on reconstruction and fingerprinting ideas [De12], and robust traceability arguments [Dwo+15]. In the graph setting, Sealfon [Sea16] and Eden, Liu, Raskhodnikova, and Smith [Ede+25] achieve lower bounds via reconstruction attacks; we improve on the results of Sealfon, while the latter result applies to a different privacy model.

PRIVATE GRAPH ALGORITHMS. Early graph-DP work studied edge-level statistics such as degree distributions [Hay+09], while node-DP was formalized and studied by Kasiviswanathan, Nissim, Raskhodnikova, and Smith [Kas+13]. Under edge-level DP, Eliás, Kapralov, Kulkarni, and Lee [Eli+20] gave near-optimal algorithms for approximating cuts via synthetic graphs, and Liu, Upadhyay, and Zou [LUZ24] obtained optimal bounds for several other cut-related private graph approximation tasks. In a *local* version of edge-DP where each node releases its own perturbed view of the graph, Eden, Liu, Raskhodnikova, and Smith [Ede+25] studied triangle counting and used reconstruction-style lower-bound techniques. Private matching in a different privacy model and objective, has been studied in discrete allocation and graph models [Hsu+14; Din+25]. These privacy models are distinct from the edge-weight model, and the techniques do not easily carry over.

EDGE-WEIGHT PRIVATE GRAPH OPTIMIZATION. The edge-weight model was introduced by Sealfon [Sea16], modeling the situation where graph structure is public but weights are sensitive. He studied differentially private release of approximate shortest paths and distances, and also gave algorithms and lower bounds for releasing MINIMUM SPANNING TREE, SINGLE-SOURCE SHORTEST-PATH TREES, and Minimum-Weight (Perfect) Matching under the ℓ_1 neighboring relation on weights. A recent line of work has studied private release of *all-pairs shortest-path distances* under the ℓ_1 neighboring relation, with lower bounds shown using discrepancy-based methods [FLL22; Che+23; Bod+24]. MST was studied under the ℓ_∞ neighboring relation in Hladík and Tetek [HT24] and Pagh, Retschmeier, Wu, and Zhang [Pag+24], showing tight error bounds in pure and approximate DP settings, respectively.

PRIVATE HIERARCHICAL CLUSTERING. The non-private objective framework for hierarchical clustering was introduced by Dasgupta [Das16] and recently won a STOC Test of Time Award [ACM26]. It was further developed by Cohen-Addad,

1744 Kanade, Mallmann-Trenn, and Mathieu [Coh+19] and the differentially private
 1745 version was first studied by Imola, Epasto, Mahdian, Cohen-Addad, and Mirrokni
 1746 [Imo+23], who gave algorithms and proved a packing-based $\Omega(n^2/\epsilon)$ additive-error
 1747 lower bound for pure edge DP. Deng, Gao, Upadhyay, Wang, and Zhou [Den+25]
 1748 subsequently studied hierarchical clustering in the edge-weight model and showed
 1749 that hard instances can be embedded into the weights of the complete graph,
 1750 implying an $\Omega(n^2/\epsilon)$ lower bound under pure DP. Our hierarchical-clustering
 1751 lower bound follows the edge-weight line of Deng, Gao, Upadhyay, Wang, and
 1752 Zhou [Den+25], but strengthens the picture by applying to approximate DP and to
 1753 broader graph topologies.

1754 5.1 PRELIMINARIES

1755 For two vectors $\mathbf{w}, \mathbf{w}' \in \mathbb{R}^m$, we define $d_H(\mathbf{w}, \mathbf{w}') := \sum_{i=1}^m \mathbb{1}[w_i \neq w'_i]$ to be the
 1756 *Hamming distance*, i.e., the number of coordinates in which they differ. For sets S, S'
 1757 we extend the definition by the symmetric difference $d_H(S, S') := |S \setminus S'| + |S' \setminus S|$.
 1758 We write $\mathbf{x} \sim_H \mathbf{x}'$ if $d_H(\mathbf{x}, \mathbf{x}') \leq 1$.

1759 **GRAPH TERMINOLOGY** We consider a finite, simple and undirected graph
 1760 $G = (V, E, \mathbf{w})$, where the set of n vertices V and the set of m edges E are public
 1761 and the weight vector $\mathbf{w} = (w_1, \dots, w_m) \in \mathbb{R}^m$ is private. We use $\mathbf{w} \in \mathbb{R}^m$ and
 1762 $w : E \rightarrow \mathbb{R}$ interchangeably to denote the edge weights. We refer to $F = (V, E)$ as
 1763 the *topology* of G . Throughout the paper, we assume $m > n$ and that F is connected.
 1764 We denote by $\mathcal{G}$ (respectively $\mathcal{G}_w$) the family of all unweighted (weighted) graphs.

1765 **WEIGHTS OF SETS AND CUTS** For a subset of edges $S \subseteq E$ of edges, we denote
 1766 the *weight* as $w(S) = \sum_{e \in S} w_e$. We refer to a partition A, B of V as a *cut* of G . The
 1767 *weight* of the cut is given by $w(A, B) := \sum_{uv \in E \cap (A \times B)} w_{uv}$. Similarly, for any edge set
 1768 $E' \subseteq E$, we denote $E'(A, B) = |\{(u, v) \in E' : u \in A, v \in B\}|$ to be the size of the cut,
 1769 or the number of edges crossing between A, B . We refer to any cut A, B satisfying
 1770 $\frac{1}{3}n \leq |A|, |B| \leq \frac{2}{3}n$ as a *balanced cut*.

1771 **TREES AND PERFECT MATCHINGS** A *spanning tree* is an acyclic subset $T \subseteq E$
 1772 of size $n - 1$ making the graph connected. A *matching* $M \subseteq E$ is a set of pairwise
 1773 vertex-disjoint edges, i.e., $e \cap e' = \emptyset$ for any two distinct $e, e' \in M$. For a matching
 1774 M and a subset $S \subseteq V$, we let $S(M)$ denote those vertices in S that are incident to
 1775 an edge of M . A *perfect matching* is a matching that covers every vertex of V . We
 1776 denote the set of all spanning trees and perfect matchings of G as $\mathcal{S}(G)$ and $\mathcal{P}(G)$,
 1777 respectively. A *minimum spanning tree* (MST) is a spanning tree T^* that minimizes
 1778 $w(T)$ among all spanning trees $T \in \mathcal{S}(G)$; analogously, a *minimum weight perfect*
 1779 *matching* (MWPM) is a perfect matching $M^* \in \mathcal{P}(G)$ that minimizes $w(M)$ among
 1780 all $M \in \mathcal{P}(G)$.

1781 **HIERARCHICAL CLUSTERING** A *hierarchical clustering* (HC) for a graph G is
 1782 represented by a rooted tree T whose leaves correspond to V , and where each
 1783 internal node corresponds to the union of the leaves in its subtree. Traversing the
 1784 tree from the root to the leaves corresponds to successively refining the clustering.
 1785 In seminal work [Das16], Dasgupta introduced a cost function $\text{cost}_w(T)$ evaluating
 1786 the quality of a hierarchical clustering, setting the stage for optimization. Intuitively,
 1787 it charges the total weight separated at each merge times the size of the subtree at
 1788 that merge, encouraging less weight to cross higher merges. Formally,

Definition 5.1 ([Das16] Dasgupta’s cost objective). *Given an input graph $G = (V, E)$ together with weights $w : E \rightarrow \mathbb{R}_{\geq 0}$ and a hierarchical clustering tree T , the Dasgupta cost function cost_w is given by*

$$\text{cost}_w(T) := \sum_{\{u,v\} \in E} |T[u \vee v]| \cdot w(\{u, v\}), \quad (1)$$

where $|T[u \vee v]|$ denotes the number of leaves in the subtree induced by the lowest common ancestor of u and v in T . As noted by Dasgupta, we may assume without loss of generality that T is a rooted *binary* tree, because every non-binary hierarchical clustering tree can be transformed into a binary one without increasing $\text{cost}_w(T)$. We illustrate this cost function on a small example in Appendix 5.7.

RECONSTRUCTION ATTACKS. Our DP lower bounds will be shown via reconstruction attacks, which are algorithms that try to reconstruct the private dataset $\mathbf{x}$ based on the algorithm output. If the attack reconstructs coordinates of $\mathbf{x}$, they demonstrate an impossibility for DP, and any reasonable notion of privacy (see [Vad17] for an overview of reconstruction attacks).

We will use two types of reconstruction attacks in our lower bounds. In the first type, we consider an algorithm that attempts to reconstruct a fixed coordinate i of $\mathbf{x}$, and show that any DP algorithm cannot be too successful at doing so. This is a slight generalization of Sealfon [Sea16, Lemma 5.3] to general data domains $\mathcal{X}$.

Lemma 5.1. *Let $\mathcal{B} : \mathcal{X}^d \rightarrow \mathcal{X}^d$ be any mechanism that is (ϵ, δ) -differentially private under the Hamming neighborhood relationship. Then for a random dataset $\mathbf{x} \leftarrow \mathcal{X}^d$ drawn uniformly at random, we have for each $i \in [d]$, we have $\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] \leq \frac{\exp(\epsilon)}{|\mathcal{X}|} + \delta$.*

As we will see, using a data domain of size $|\mathcal{X}| = \frac{m}{n}$ will be critical to obtaining tight lower bounds. We also consider reconstructing an entire set $I \subseteq [d]$ of indices of a binary dataset $\mathbf{x}$. An attack is then a mechanism $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$, and its success is measured by

$$\text{Succ}(\mathcal{B}) = \frac{1}{2} - \frac{\mathbb{E}[d_H(\mathbf{x}|_I, \mathbf{y}|_I)]}{\mathbb{E}[|I|]}, \quad (2)$$

where $\mathbf{x} \sim \{0, 1\}^d$ and $(I, \mathbf{y}) := \mathcal{B}(\mathbf{x})$, and $\mathbf{x}|_I$ indicate the subvector of $\mathbf{x}$ on indices I . Intuitively, success is the normalized error that randomly guessing I would yield (namely $\frac{1}{2}$) minus the normalized error made by $\mathcal{B}$ in the set I . DP limits the possible success as follows.

Theorem 5.2. *Let $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$ be any (ϵ, δ) -DP algorithm and let $\mathbf{x} \sim \{0, 1\}^d$ be drawn uniformly at random. Then, $\text{Succ}(\mathcal{B}) \leq \frac{e^\epsilon - 1}{2(e^\epsilon + 1)} + \frac{d\delta}{e^\epsilon + 1}$.*

The proof appears in Section 5.7.1. It differs from prior reconstruction attacks because I is itself a differentially private output. When $\delta = o(\frac{1}{d})$, the second term becomes negligible, and when $\epsilon < 1$, we have that $\text{Succ}(\mathcal{B}) = O(\epsilon)$.

5.2 PRIVATE MINIMUM-WEIGHT SPANNING TREES

In Section 5.2.1, we show that there is a worst-case topology F on which the additive error of **MST** under DP is tight in $\Theta((n/\epsilon) \cdot \log(m/n))$ (for small enough δ) achieved

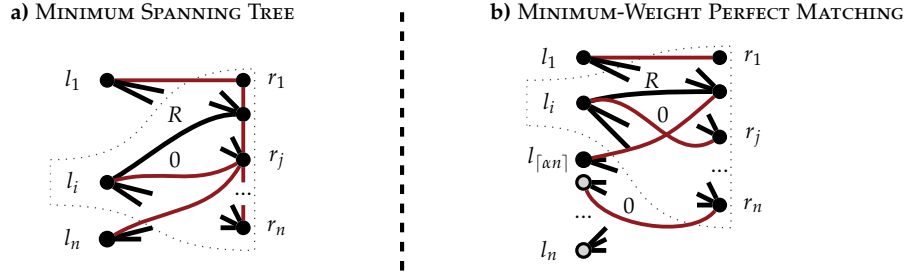


Figure 5.2: **a)** Encoding the vector $\mathbf{x} = (x_1, \dots, x_n)$ into the minimum-weight spanning tree. **b)** The encoding for min-weight perfect matching for a vector of length $\lceil \alpha n \rceil$. We pad the left side with the gray vertices. In both cases, each vertex r_j represents the integer j , and we set the weight of the edge from l_i to r_j to 0, if $x_i = j$ and otherwise to $R = \mathcal{O}(\frac{1}{\epsilon} \ln n)$.

by the Laplace mechanism. Going beyond worst-case graphs, in Section 5.2.2, we show that the lower bound of $\Omega(n/\epsilon)$ due to [Sea16] holds on all graph topologies with minimum cut $\Omega(\log(n))$.

5.2.1 Tight Lower Bounds on Worst-Case Topologies

Assuming $\delta \leq n^{-\Omega(1)}$, we now close the gaps between the known upper and the lower bounds. The idea is to encode a random vector $\mathbf{x} \in [n]^d$ into the **MST** of a dense graph G so that releasing an overly accurate **MST** by a differentially private mechanism would contradict the reconstruction rate in Lemma 5.1. As a by-product, we also get a lower bound for ϵ -DP, obtained by taking the limit of δ towards 0 recovering the packing-based lower bound obtained by Hladík and Tetek [HT24].

Theorem 5.3 (Worst-Case Lower Bound MST). *Let $\epsilon > 0$ and $\delta \leq (n/m)^{\Omega(1)}$ where $m > 2n$. There exists a graph topology $F = (V, E)$ and a distribution $\mathcal{D}_\omega$ of weights such that for any (ϵ, δ) -DP algorithm $\mathcal{A}$ that outputs an approximate **MST** T under the ℓ_1 neighboring relationship, if the weights $\mathbf{w} \sim \mathcal{D}_\omega$ then the expected additive error satisfies*

$$w(T) - w(T^*) \geq \Omega((n/\epsilon) \cdot \min(\ln(1/\delta), \ln(m/n))),$$

where T^* is the optimal MST.

We now describe how to encode a vector $\mathbf{x} \in [n]^n$ (see Figure 5.2). The first construction gives a lower bound that depends solely on n and we will later generalize it to include the number of edges m as well. Throughout the construction, we assume the topology $F \in \mathcal{G}$ to be public.

ENCODING Let $\text{Enc}_R : [n]^n \rightarrow \mathcal{G}_\omega$ be the encoding function that takes some dataset $\mathbf{x} \in [n]^n$ and returns a weighted graph that encodes the dataset into the **MST**. The parameter $R > 1$ is used to control the edge weights. We construct a new connected graph $G = (V_1 \cup V_2, E, \mathbf{w})$ with $2n$ vertices and $n^2 + (n - 1)$ edges in the following way: Let $V_1 = \{\ell_1, \dots, \ell_n\}$ and $V_2 = \{r_1, \dots, r_n\}$ be two vertex sets each of size n . Each vertex $\ell_i \in V_1$ represents one coordinate of $\mathbf{x}$ and has edges to each vertex in V_2 . Set $w(\{\ell_i, r_{x_i}\}) := 0$ and $w(\{\ell_i, r_j\}) := R$ for $j \neq x_i$. We ensure

connectivity by adding a path through the vertices in V_2 with zero weighted edges (contributing the $n - 1$ extra edges).

DECODING Let $\text{Dec}_G : \mathcal{S}(G) \rightarrow [n]^n$ be a decoder and assume that the topology G was produced by the encoder (to simplify notation we assume F is public as soon as the instance has been created).

Given a spanning tree $T \in \mathcal{S}(G)$, we reconstruct the vector $\mathbf{y} \in [n]^n$ as follows: for each $i \in [n]$, choose any $r_j \in V_2$ with $\{\ell_i, r_j\} \in T$ and set $y_i := j$. Note that any vertex in V_1 must have at least one edge in T by construction, and therefore, the reconstruction for each coordinate is well-defined. In case there are multiple, we break ties arbitrarily.

Now define the attack $\text{Rec} : [n]^n \rightarrow [n]^n$ as $\text{Rec}_{\mathcal{A}_{\text{MST}}, R}(\mathbf{x}) := \text{Dec}_G(\mathcal{A}_{\text{MST}}(\text{Enc}_R(\mathbf{x})))$ where $\mathcal{A}_{\text{MST}}$ is any (possibly randomized) **MST** algorithm. By the construction above, it is clear that an optimal **MST** algorithm $\mathcal{A}_{\text{OPT}}$ recovers all coordinates perfectly, i.e. $\text{Rec}_{\mathcal{A}_{\text{OPT}}, R}(\mathbf{x}) = \mathbf{x}$. This leads to the following two observations.

Observation 5.4. Let $R > 0$ and $G = (F, \mathbf{w})$ be obtained from $\text{Enc}_R(\mathbf{x})$. If $T \in \mathcal{S}(F)$ is any and T^* the optimal spanning tree of G , then

$$d_H(\text{Dec}_F(T), \mathbf{x}) \leq \frac{w(T) - w(T^*)}{R} = \frac{w(T)}{R},$$

Indeed, $w(T^*) = 0$, and whenever $\text{Dec}_F(T)_i \neq x_i$, the tree T contains an edge $\{\ell_i, r_j\}$ with $(j \neq x_i)$, which has weight R . These bad edges are distinct for different i , so the number of decoding errors is at most $w(T)/R$.

Observation 5.5 (Induced ℓ_1). For every $\mathbf{x} \sim_H \mathbf{x}'$ and $R > 0$, setting $k := 2\lceil R \rceil$, there is a chain of ℓ_1 -neighboring graphs, i.e. $\text{Enc}_R(\mathbf{x}) := G_0 \sim_1 G_1 \sim_1 \dots \sim_1 G_k =: \text{Enc}_R(\mathbf{x}')$.

Each consecutive G_i is obtained by successively changing a single weight by one. By group privacy (see e.g. [Vad17, Lemma 7.2.2]), this means there is a $(2\lceil R \rceil\epsilon, 2\lceil R \rceil e^{2\lceil R \rceil\epsilon}\delta)$ -DP guarantee on $\text{Enc}_R(\mathbf{x})$ under the *Hamming* neighboring relationship. We are now ready to prove Theorem 5.3.

Proof of Theorem 5.3. Suppose for contradiction, that there exists an (ϵ, δ) -DP **MST** algorithm $\mathcal{A}_{\text{MST}}$ with expected additive error less than $\frac{n}{100} \cdot R$. Assume a random dataset $\mathbf{x} \sim \text{Uni}([n]^n)$. Note that the decoding step is simply post-processing, and hence $\text{Rec}_{\mathcal{A}_{\text{MST}}, R}$ parameterized by $\mathcal{A}_{\text{MST}}$ and R is $(2\lceil R \rceil\epsilon, 2\lceil R \rceil e^{2\lceil R \rceil\epsilon}\delta)$ -DP. Now set $R := \min\left(\frac{\ln n}{2c\epsilon}, \frac{\ln(1/\delta)}{2c\epsilon}\right)$ for some $c > 1$. Combining with Observation 5.4 and by the utility guarantee of $\mathcal{A}_{\text{MST}}$, we would leak at least $\frac{99}{100} \cdot n$ of the input coordinates in expectation. Let $\mathbf{y} := \text{Rec}_{\mathcal{A}_{\text{MST}}, R}(\mathbf{x})$ and by Lemma 5.1,

$$\begin{aligned} \mathbf{P}[y_i = x_i] &\leq \frac{e^{2\lceil R \rceil\epsilon}}{n} + 2\lceil R \rceil e^{2\lceil R \rceil\epsilon}\delta \\ &\leq \frac{1}{n} \exp\left(\ln(n^{1/c})\right) + \frac{\ln(n)}{\epsilon c} \exp\left(\ln n^{1/c}\right) \delta \\ &= n^{(1-c)/c} + \frac{\ln(n)}{\epsilon c} n^{1/c} \delta. \end{aligned}$$

By our assumption on $\delta = n^{-\Omega(1)}$, the second term vanishes. Linearity of expectation gives

$$\mathbf{E}\left[\sum_{i \in [n]} (\mathbb{1}[y_i = x_i])\right] \leq n \cdot n^{(1-c)/c} = n^{1/c}.$$

Algorithm 7 Subset Reconstruction Attack Rec^{MST} for MST

Require: Public graph topology $F = (V, E)$, dataset $\mathbf{x} \in \{0, 1\}^m$,
parameter $R > 0$, algorithm $\mathcal{A}_{\text{MST}}$.

- 1: Initialize weights $\mathbf{w} \in \{0, R\}^m$.
- 2: Let $\ell : E \rightarrow [m]$ be any bijection from E to $[m]$ labels.
- 3: Set $w(e) := Rx_{\ell(e)}$ for all $e \in E$. $\triangleright$ Encode the dataset into the weights
- 4: Let $G = (F, \mathbf{w})$.
- 5: Compute minimum spanning tree $T := \mathcal{A}_{\text{MST}}(G)$.
- 6: Initialize reconstruction set $I = \{\ell(e) : e \in T\} \subseteq [m]$.
- 7: **for each** $i \in I$, guess $y_i = 0$.
- 8: **return** Reconstruction set I , reconstructed vector $\mathbf{y} = \{y_i : i \in I\}$.

1878 In particular, for a sufficiently large constant c , we contradict the utility guarantee
1879 of $\mathcal{A}_{\text{MST}}$. Note also that we require $R \leq \frac{\ln n}{2c\epsilon}$ to get any meaningful probability.
1880 Thus, we have shown a lower bound of $\Omega((n/\epsilon) \ln n)$ in the dense case ($m = \Theta(n^2)$),
1881 for sufficiently small δ .

1882 It remains to show that this technique can be parameterized by the number
1883 of edges m to yield a bound of $\Omega((n/\epsilon) \cdot \ln(m/n))$ for any $n \leq m \leq n^2$. Suppose,
1884 we want a lower bound for graphs with m edges and $2n$ vertices. Without loss
1885 of generality, we treat m/n as an integer. We can build n^2/m many subgraphs
1886 on $2(m/n)$ vertices (constructed as described above for the dense case with n
1887 replaced by m/n). To make the full graph connected, we add a path through one
1888 representative vertex from each subgraph, adding only $\mathcal{O}(n^2/m)$ low-cost edges
1889 that would only affect the bound by a constant. Ignoring this lower-order term,
1890 observe that the number of edges is exactly $(m/n)^2 \cdot (n^2/m) = m$ (we ignore edges
1891 on the path of length $\leq n$).

1892 For $i \in n^2/m$, let each subgraph encode an i.i.d. random vector $\mathbf{x}^{(i)} \in [m/n]^{m/n}$
1893 of length m/n . Since an (ϵ, δ) -DP algorithm satisfies the same privacy guarantees
1894 on the weights of each individual subgraph, the previous lower bound applied
1895 to a single subgraph states that we cannot reconstruct a single $\mathbf{x}^{(k)}$ with expected
1896 error better than $\Omega(\frac{m}{n\epsilon} \cdot \ln \frac{m}{n})$. By linearity of expectation, we get a lower bound on
1897 the expected total error of $\Omega(\frac{m}{n\epsilon} \cdot \ln(\frac{m}{n}) \cdot \frac{n^2}{m}) = \Omega(\frac{n}{\epsilon} \cdot \ln(\frac{m}{n}))$. $\square$

1898 5.2.2 Lower Bounds for General Topology Classes

1899 We will design a subset reconstruction attack that uses any given approximate
1900 DP **MST** algorithm $\mathcal{A}_{\text{MST}}$. Assuming the additive cost of $\mathcal{A}_{\text{MST}}$ is sufficiently low,
1901 the reconstruction success will be high enough to form a contradiction with
1902 Theorem 5.2 and establish an error lower bound for any DP algorithm.

1903 Our reconstruction attack works as follows: the dataset is drawn from $\mathbf{x} \sim$
1904 $\text{Uni}(\{0, 1\}^m)$, and then arbitrarily encoded into E with a labeling function $\ell : E \rightarrow$
1905 $[m]$. The weights are set as $w(e) = Rx_{\ell(e)}$, where R is a factor used to amplify the
1906 weight that we will choose later. Then, $\mathcal{A}_{\text{MST}}$ is run on $(V, E, \mathbf{w})$ to produce a tree
1907 $T \subseteq E$. The returned tree T contains edges whose weight is more likely to be 0, and
1908 thus the reconstructed set is set to be $I = \{\ell(e) : e \in T\}$, and all guesses $\{y_i : i \in I\}$
1909 are $y_i = 0$. This procedure appears in Algorithm 7.

1910 We will show that if $\mathcal{A}_{\text{MST}}$ has sufficiently low additive error, then this attack has
1911 a success rate much higher than 0. Specifically, we will show $\mathbb{E}[\sum_{i \in I} x_i] \leq 0.1 \mathbb{E}[|I|]$.
1912 By construction, $w(T) = R \sum_{i \in I} x_i$. Using the additive error guarantee of $\mathcal{A}_{\text{MST}}$,

we have $\mathbb{E}[\sum_{i \in I} x_i] = \frac{1}{R} \mathbb{E}[w(T)] \leq \frac{1}{R} w(T^*) + \frac{1}{10} n$. Thus, we need to show that the optimal **MST** in $(V, E, \mathbf{w})$ is likely to have low weight. This is where we assume that the minimum cut of F is not too small. Recall that the minimum cut is given by $\min_{A \sqcup B = V} E(A, B)$. We use a concentration inequality, and a union bound over all cuts, to argue that if the mincut of F is at least $5 \log n$, then with high probability all cuts have an edge labeled 0. This immediately implies that an **MST** of weight 0 exists. In order for the union bound to work, we use a result of Karger [Kar93] bounding the number of approximate minimum cuts, as these cuts have higher failure probability. Formally,

Lemma 5.6. *Suppose the graph topology $F = (V, E)$ has minimum cut $\lambda \geq 5 \log(n)$, and the random dataset $\mathbf{x} \sim \text{Uni}(\{0, 1\}^m)$ is embedded into $\mathbf{w}$ via an arbitrary bijection. Then, with probability at least $1 - \frac{3}{n^2}$, there exists a zero-cost spanning tree $T \subseteq E$, i.e. $w(T) = 0$.*

The proof appears in Section 5.7.2. Note that $\mathbb{E}[|I|] = |I| = n - 1$ always. Having established bounds on both $\mathbb{E}[\sum_{i \in I} x_i]$ and $\mathbb{E}[|I|]$, we can show a reconstruction attack lower bound:

Theorem 5.7. *Let (V, E) be a public graph topology with minimum cut at least $5 \log(n)$. Suppose that $\mathcal{A}_{\text{MST}}$ attains expected additive error $\leq \frac{R}{10} n$ for some $R > 0$. Then $\text{Succ}(\text{Rec}^{\text{MST}}) \geq 0.4$.*

Proof. Let $\mathcal{E}$ denote the event that the optimal **MST** has weight that is not 0. We have that

$$\begin{aligned} \mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] &= \mathbb{E} \left[\sum_{i \in I} x_i \right] \leq \mathbb{E} \left[\sum_{i \in I} x_i \mid \neg \mathcal{E} \right] + \mathbf{P}[\mathcal{E}] \mathbb{E} \left[\sum_{i \in I} x_i \mid \mathcal{E} \right] \\ &\leq \mathbb{E} \left[\sum_{i \in I} x_i \mid \neg \mathcal{E} \right] + \mathbf{P}[\mathcal{E}] \cdot n \cdot R. \end{aligned}$$

By Lemma 5.6, we know $\mathbf{P}[\mathcal{E}] \leq \frac{3}{n^2}$. Furthermore, by the construction of I , we know $R \sum_{i \in I} x_i = w(T)$. Given that $\neg \mathcal{E}$ occurs, we have $\mathbb{E}[w(T)] \leq \frac{R}{10} n$ by the error guarantee of $\mathcal{A}_{\text{MST}}$. Thus, the above bound is $\frac{n}{10} + \mathcal{O}(\frac{R}{n})$. $\square$

Theorem 5.7 and Lemma 5.2 immediately imply the following corollary:

Corollary 5.8. *For any topology $F = (V, E)$ with minimum cut at least $5 \log(n)$, there is no (ϵ, δ) -DP algorithm $\mathcal{A}_{\text{MST}}$ attaining $\frac{1}{10\epsilon} n$ additive error for any $\epsilon \leq 1, \delta \leq \frac{\epsilon 0.01}{m}$.*

Proof. Suppose the contrary, and instantiate Rec^{MST} with $\mathcal{A}_{\text{MST}}$ and $R = \frac{1}{\epsilon}$. By group privacy, Rec^{MST} satisfies $(\epsilon \lceil R \rceil, \delta \lceil R \rceil e^{\epsilon \lceil R \rceil})$ -DP, which means it satisfies $(2, \frac{10}{\epsilon} \delta)$ -DP as $\epsilon \lceil R \rceil < 2$. By Lemma 5.2, this means it can only be a $\frac{1}{2} - \frac{1}{1+e^2} + m\delta < 0.39$ -successful reconstruction attack, yet this contradicts Theorem 5.7. $\square$

While looser than Theorem 5.3 by a $\log(\frac{m}{n})$ factor, this result establishes that the error of **MST** must still grow linearly under (ϵ, δ) -DP for many topologies. We note that this proof would still work if F had any linear-sized subgraph with large enough min-cut.

1941 5.3 MINIMUM-WEIGHT PERFECT MATCHINGS

1942 We will now show multiple lower bounds for the problem of **MWPM**.

1943 5.3.1 Reconstruction-Based Lower Bound

1944 As any perfect matching has size $n/2$, the upper bound in Theorem 5.25 immedi-
 1945 ately carries over to Minimum-Weight (Perfect) Matching as well. We now show
 1946 that the resulting additive error of $\Theta((n/\epsilon) \cdot \ln(\frac{m}{n}))$ is tight for this problem as well
 1947 (for sufficiently small δ). The encoding will be slightly different for **MST** because
 1948 all coordinates with the same value would be encoded by the same vertex, and a
 1949 matching can only recover one of those. To fix this, we prove a standard balls and
 1950 bins result stating that if x has length $n' = \alpha n$ for some small $0 < \alpha \leq 1$, only a
 1951 α^2 fraction of a random $\mathbf{x} \sim \text{Uni}([n]^{\lceil \alpha n \rceil})$ has collisions with high probability. The
 1952 proof to the following statement uses a standard concentration inequality.

1953 **Lemma 5.9** (Collision-free). *Let $(X_1, \dots, X_d) \sim \text{Uni}([n]^d)$ with $d = \lceil \alpha n \rceil$ for a constant*
 1954 *$0 < \alpha \leq 1$. Then, with probability at least $1 - \exp(-\Omega(\alpha^3 n))$, the number of indices i for*
 1955 *which there exists some $j \neq i$ with $X_i = X_j$ is $\mathcal{O}(\alpha^2 n)$.*

1956 *Proof.* Define $\tilde{X}_i = \mathbb{1}[\exists j \neq i : X_j = X_i]$ and let $Y = \sum_{i=1}^d \tilde{X}_i$. By a union bound,
 1957 $\mathbb{E}[\tilde{X}_i] \leq \sum_{j \neq i} \mathbb{P}[X_j = X_i] = (d-1)/n$, so that $\mathbb{E}[Y] \leq d(d-1)/n = \alpha^2 n - \alpha \leq \alpha^2 n$.
 1958 Since changing a single X_i affects at most two of the $\tilde{X}_i$'s, Y is 2-Lipschitz. By
 1959 McDiarmid's inequality [McD89], for any $t > 0$, $\mathbb{P}[Y \geq \mathbb{E}[Y] + t] \leq \exp(-2t^2/(4d))$.
 1960 Setting $t = \mathbb{E}[Y]$ gives $\mathbb{P}[Y \geq 2\mathbb{E}[Y]] \leq \exp(-\mathbb{E}[Y]^2/(2d)) = \exp(-\Omega(\alpha^3 n))$. Thus,
 1961 with probability at least $1 - \exp(-\Omega(\alpha^3 n))$, we have $Y \leq 2\mathbb{E}[Y] = \mathcal{O}(\alpha^2 n)$. $\square$

1962 **Theorem 5.10** (Lower bound MWPM). *Let $\epsilon > 0$, $\delta \leq (n/m)^{\Omega(1)}$, and $m > 2n$. Then*
 1963 *there exists a graph topology $F = (V, E)$ and a distribution of weights $\mathcal{D}_\omega$ on weights of*
 1964 *E such that for any (ϵ, δ) -DP protocol $\mathcal{B}$ that outputs an approximate minimum-weight*
 1965 *perfect matching M under the ℓ_1 neighboring relationship, if the weights $\mathbf{w} \sim \mathcal{D}_\omega$, then*
 1966 *the expected additive error satisfies*

$$w(M) - w(M^*) \geq \Omega((n/\epsilon) \cdot \min(\ln(1/\delta), \ln(m/n))),$$

1967 *where M^* is an optimal minimum-weight perfect matching.*

1968 Fix some constant $0 < \alpha \leq 1$ and let R again be a parameter that controls the
 1969 weights. Now let $\text{Enc}_R : [n]^d \rightarrow \mathcal{G}_\omega$ be constructed as follows. Let $\mathbf{x} \in [n]^d$ be some
 1970 vector of length $d = \alpha n$ and $0 < \alpha \leq 1$. Then the encoder creates a complete bipartite
 1971 graph $G = (V_1 \cup V_2, E, \mathbf{w})$ with bipartitions $V_1 = \{\ell_1, \dots, \ell_n\}$ and $V_2 = \{r_1, \dots, r_n\}$.
 1972 Each of the first d vertices $\ell_i \in V_1$ represents a single coordinate of $\mathbf{x}$. For each
 1973 $i \in [d]$, we set $w(\{\ell_i, r_{x_i}\}) := 0$ and $w(\{\ell_i, r_j\}) := R$ for $j \neq x_i$. We treat the remaining
 1974 $(1 - \alpha)n$ vertices in V_1 as dummy nodes that simply get zero-weight edges to all
 1975 vertices in V_2 . Furthermore, let $\text{Dec}_G : \mathcal{P}(G) \rightarrow [n]^d$ and $\text{Rec}_G : [n]^d \rightarrow [n]^d$ be
 1976 analogously defined as in Section 5.2. Find this construction also in Figure 5.2.

1977 Using this encoding, the optimal solution is determined by the number of
 1978 collisions C in $\mathbf{x}$ and is not necessarily 0 as before. By Lemma 5.9, we can make C
 1979 arbitrarily small with probability exponentially small in n . The key observation is
 1980 that there are still $\Theta(n)$ many coordinates that do not collide. Every non-colliding
 1981 coordinate can be recovered from a correct edge in some perfect matching M .
 1982 Hence, we get this observation.

Observation 5.11. Let $G = (F, \mathbf{w})$ be obtained from $\text{Enc}_R(\mathbf{x})$ and let

$$C = |\{i \in [d] \mid \exists j \neq i, x_i = x_j\}|$$

be the number of colliding coordinates. If $M \in \mathcal{P}(F)$ is any perfect matching, then

$$d_H(\text{Dec}_F(M), \mathbf{x}) \leq \frac{w(M) - w(M^*)}{R} + C,$$

where M^* is an optimal Minimum-Weight (Perfect) Matching of G .

We now have all the ingredients ready to prove the theorem. The proof is similar to Theorem 5.3. First, draw $\mathbf{x} \sim \text{Uni}([n]^{\alpha n})$ and fix a small constant $0 < \alpha \leq 1$. WLOG, assume that αn is an integer. Then $C \leq \alpha^2 n$ with probability $1 - \exp(-\Omega(\alpha^3 n))$ by Lemma 5.9.

Suppose for contradiction that there exists an (ε, δ) -DPMMinimum-Weight (Perfect) Matching algorithm $\mathcal{B}$ with expected additive error $\mathbb{E}[w(M) - w(M^*)] < 0.01 \alpha n R$. Then by Observation 5.11, $\mathbb{E}[d_H(\text{Dec}_G(M), \mathbf{x})] \leq 0.01 \alpha n + \alpha^2 n$. So $\mathcal{B}$ correctly recovers at least $\alpha n - 0.01 \alpha n - \alpha^2 n = \alpha n(0.99 - \alpha)$ coordinates in expectation.

Observe that for some fixed $R > 1$, Observation 5.5 holds again, and the induced ℓ_1 -distance of two encoded graphs differs by at most $2\lceil R \rceil$ if the encoded vectors are Hamming neighbors. Hence, by Lemma 5.1,

$$\begin{aligned} \mathbf{P}[\text{Rec}_{\mathcal{A}, R}(\mathbf{x})_i = x_i] &\leq \frac{e^{2\lceil R \rceil \varepsilon}}{n} + 2\lceil R \rceil e^{2\lceil R \rceil \varepsilon} \delta \\ &\leq \frac{1}{n} \exp(\ln(n^{1/c})) + \frac{\ln(n)}{\varepsilon c} \exp(\ln n^{1/c}) \delta \\ &= n^{(1-c)/c} + \frac{\ln(n)}{\varepsilon c} n^{1/c} \delta. \end{aligned}$$

Assuming $\delta = n^{-\Omega(1)}$, the second term is $o(1/n)$ and vanishes. Using linearity of expectation over all $i \in [\alpha n]$,

$$\mathbb{E} \left[\sum_{i \in [\alpha n]} \mathbf{1}[\text{Rec}_{\mathcal{A}, R}(\mathbf{x})_i = x_i] \right] \leq \left(n^{(1-c)/c} \cdot \alpha n \right) = \alpha \cdot n^{1/c},$$

For sufficiently large $c > 1$, this is sublinear in n , contradicting the assumed utility guarantee of $\Theta(\alpha n)$. Finally, to obtain a bound that also depends on the number of edges, we use the same decomposition technique as in the proof of Theorem 5.3 with simplification that we don't need to connect the subgraphs.

5.3.2 Packing-Based Lower Bound

We now show tight error bounds for MWPM under pure-DP for both the ℓ_1 and ℓ_∞ neighboring relationships using a similar argument to the one proposed by Hladík and Tetek [HT24]. The authors explicitly asked whether their technique can be used for other problems as well. The presented lower bound of $\Omega(\frac{n^2}{\varepsilon} \ln n)$ matches the one the authors gave for private MST. The proof also works for ℓ_1 and recovers the reconstruction-based lower bound we gave in Theorem 5.10.

For the following, let $\mathcal{P}(G)$ be the set of all perfect matchings in some graph G . Furthermore, define the Hamming distance of two perfect matchings M, M' to be $d_H(M, M') := |M \setminus M'|$. The key observation is that similar to their setting, we can again greedily construct an exponential-sized set of dissimilar perfect matchings where each pair has a large Hamming distance of $d = \Theta(n)$. While [HT24] used

a complete graph for constructing their family of hard instances, we will use the complete bipartite graph because it makes it easier to argue about perfect matchings. Define M^* to be the optimal minimum-weight perfect matching on $(G, \mathbf{w})$ (the graph will be clear from the context), and $w(M) := \sum_{e \in M} \mathbf{w}_e$ the weight of a matching, so that the additive error of M is $w(M) - w(M^*)$.

We denote by $\mathcal{L}_{(G, \mathbf{w})}(\alpha)$ the set of all perfect matchings with additive error less than some α on some graph $(G, \mathbf{w})$.

Lemma 5.12 ([Vad17; HT24] Packing Argument for Matchings). *Given a collection of weights $\mathcal{W} \subseteq \mathbb{R}^m$ each at induced distance at most r from some fixed weight vector $\mathbf{w}_0 \in \mathbb{R}^m$. Fix some α for which all $\mathcal{L}_{(G, \mathbf{w})}(\alpha)$ are disjoint. Then for any ε -DP **MWPM** mechanism $\mathcal{M}$, there exist weights $\mathbf{w} \in \mathcal{W}$ such that, drawing $M \leftarrow \mathcal{M}(G, \mathbf{w})$,*

$$\mathbf{P}[M \in \mathcal{L}_{(G, \mathbf{w})}(\alpha)] = \mathbf{P}[w(M) \leq w(M^*) + \alpha] \leq \frac{\exp(r \cdot \varepsilon)}{|\mathcal{W}|},$$

where M^* is the optimal **MWPM** on $(G, \mathbf{w})$.

In our cases, we will use the ℓ_1 and ℓ_∞ neighboring relationships on the weight vectors. We are now ready to prove our lower bound.

Theorem 5.13 (Lower Bound ε -DP). *Let G be the complete bipartite graph $K_{n,n}$ on $2n$ vertices. Let $S \subseteq \mathcal{P}(G)$ be a set of perfect matchings with $d_H(M, M') \geq n/4$ for all distinct $M, M' \in S$ and size at least $|S| \geq e^{(n/4) \cdot \ln n}$. Then for any ε -DP algorithm $\mathcal{M}$ that privately releases a **MWPM** $M \leftarrow \mathcal{M}(G, \mathbf{w})$, there exist weights $\mathbf{w}$ such that*

$$\frac{1}{\sqrt{|S|}} \geq \begin{cases} \mathbf{P}\left[w(M) \leq w(M^*) + \frac{1}{128} \cdot \left(\frac{n}{\varepsilon} \cdot \ln(n) - 8\right)\right] & (\text{wrt. } \ell_1) \\ \mathbf{P}\left[w(M) \leq w(M^*) + \frac{1}{64} \cdot \left(\frac{n^2}{\varepsilon} \cdot \ln(n) - 8n\right)\right] & (\text{wrt. } \ell_\infty), \end{cases}$$

where M^* is the optimal **MWPM** on $(G, \mathbf{w})$.

Proof. Let $G = (V, E)$ be the complete bipartite graph on $2n$ vertices and let $S \subseteq \mathcal{P}(G)$ be such that each pair of perfect matchings is at least d apart in Hamming distance. For each $S_i \in S$, we create a corresponding weight function $\mathbf{w}_i$ where we set the weight of an edge to some value 0, if it is part of this matching and $R \in \mathbb{R}_+$ otherwise.

Now for each $S_i \in S$ we create a corresponding weight function $\mathbf{w}_i$ where we set a weight to 0, if the corresponding edge was part of the matching and to some $R > 0$ otherwise (with value to be chosen later). We first prove the claim for the ℓ_∞ neighboring relationship. Fix some arbitrary $\mathbf{w}_0$ and note that all other weight functions have $\|\mathbf{w} - \mathbf{w}_0\|_\infty \leq \lceil R \rceil \leq R + 1$. We can show that the sets $\mathcal{L}_{(G, \mathbf{w}_i)}(\frac{1}{2}Rd)$ of matchings with error at most $\frac{1}{2}dR$ are pairwise disjoint.

To see this, fix two distinct hard instances with weights $\mathbf{w}$ and $\mathbf{w}'$ where the optimal perfect matchings $d_H(M_1, M_2) > d$. Now assume for the sake of contradiction that there exists some matching $B \in \mathcal{L}_{(G, \mathbf{w})}(\frac{1}{2}Rd) \cap \mathcal{L}_{(G, \mathbf{w}')}(\frac{1}{2}Rd)$. Then for $\alpha = \frac{1}{2} \cdot Rd$ we have both $d_H(M_1, B) < \frac{d}{2}$ and $d_H(M_2, B) < \frac{d}{2}$ by the way the weights are chosen. This would imply $d_H(M_1, M_2) < d$, immediately contradicting the

assumed Hamming distance between M_1 and M_2 . We can now apply Lemma 5.12. Let $\mathcal{M}$ be any MWPM algorithm and draw $T \leftarrow \mathcal{M}(G, \mathbf{w})$. Then,

$$\mathbf{P} \left[w(T) \leq w(T^*) + \frac{d \cdot R}{2} \right] \leq \exp((R+1) \cdot \varepsilon) \cdot \frac{1}{|S|} := \alpha, \quad (3)$$

Now set $R := \frac{1}{2\varepsilon} \cdot \ln|S| - 1$ so that $\exp((R+1)\varepsilon) = \sqrt{|S|}$ and hence $\alpha = 1/\sqrt{|S|}$. Using $d = n/4$, Lemma 5.14 implies the existence of a set of size at least $e^{(n/4) \ln n}$ with pairwise Hamming distance at least $n/4$. Hence, we can match the constants given in the theorem statement:

$$\begin{aligned} \frac{dR}{2} &= \frac{n}{8} \cdot \left(\frac{n \cdot \ln n}{8\varepsilon} - 1 \right) \\ &= \frac{1}{64} \cdot \left(\frac{n^2}{\varepsilon} \ln n - 8n \right) = \Theta\left(\frac{n^2}{\varepsilon} \ln n\right). \end{aligned}$$

For the ℓ_1 neighboring relationship, first observe that all weights have $\|\mathbf{w} - \mathbf{w}_0\|_1 \leq [2 \cdot R \cdot n]$. Then setting $\alpha = \frac{\exp((2Rn+1)\varepsilon)}{|S|}$ allows us to prove the claimed lower bound we already showed in Chapter 5. It can be easily verified that setting $R := \frac{1}{4\varepsilon n} \ln|S| - \frac{1}{2n}$ gives the desired result. $\square$

Chapter 5 gives a (ε, δ) -DP bound that recovers this pure-DP LB by taking the limit of $\delta \rightarrow 0$.

We will now show that such a set S exists and can even be constructed.

Lemma 5.14 (Volume of a d -ball around perfect matching). *For the complete bipartite graph G , any fixed perfect matching M and any integer $d \leq n$, we have*

$$|\{M' \mid d_H(M, M') \leq d\}| \leq n^{d+1}.$$

Proof. We can give a direct combinatorial proof by observing that we first pick some subset of one of the bipartitions of size at most d that we want to rewired. Then there are at most $d!$ many ways to remap those selected vertices. Hence,

$$\begin{aligned} |\{M' \in \mathcal{P}(G) \mid d_H(M, M') \leq d\}| &\leq \sum_{i=1}^{\lfloor d \rfloor} \binom{n}{i} i! \\ &\leq d \cdot \frac{n!}{(n-d)!} \leq n^{d+1}. \end{aligned} \quad \square$$

Lemma 5.15 (Large set of dissimilar matchings). *Let G be the complete bipartite graph G . Then we can construct a set $S \subseteq \mathcal{P}(G)$ of size $e^{(n/4) \cdot \ln n}$, such that $d_H(M, M') \geq n/4$ for all distinct $M, M' \in S$.*

Proof. We provide a greedy argument similar to [HT24]. We can generate a set $S \subseteq \mathcal{P}(G)$ of size at least $n! / n^{d+1}$ with pairwise Hamming distance at least d and $|S| \geq \frac{n!}{n^{d+1}}$ for some integer $d \leq n$.

Therefore, first initialize a set $S := \emptyset$ and let U_i be the set of matchings that might still be added to S . Set $U_0 := \mathcal{P}(G)$ and repeat the following process as long as U_{i-1} is non-empty: In each step i , select any $M \in U_{i-1}$ and add it to S . Then, set U_i to be U_{i-1} without those matchings that are closer than d in Hamming distance to M . When this process stops, we necessarily have $|S| \geq \frac{n!}{n^{d+1}}$, because in each step, we removed at most n^{d+1} elements by Lemma 5.14 and the number of perfect

matchings on a complete bipartite graph with $2n$ vertices is exactly n factorial. Now, we can set $d = \frac{1}{4}n$ (or any other small enough constant) and we get

$$\begin{aligned} |S| &\geq \frac{n!}{n^{d+1}} \geq \frac{(n/e)^n}{n^{n/4+1}} = e^{(3n/4-1)\ln n - n} \\ &\geq e^{(n/4) \cdot \ln n} = e^{\Omega(n \cdot \ln n)}, \end{aligned}$$

where we used the trivial lower bound of $n! \geq (n/e)^n$ by a Stirling approximation and we assume sufficiently large n for convenience where the linear n lower-order term is dominated. $\square$

5.4 HIERARCHICAL CLUSTERING UNDER DASGUPTA'S COST FUNCTION

In this section, we design the first (subset) reconstruction-attack based lower bound for hierarchical clustering. Observe that our approach for MST, where the dataset is encoded into every edge of the topology, does not work here. This is because, by a simple concentration argument, every cut in the graph will have weight approximately half the total size of the cut, preventing any cheap hierarchical clustering from existing. Instead, we use a sparse dataset encoding which requires new technical ideas to work.

Before describing the attack, we introduce new graph notation. For a topology $F = (V, E)$, let $d_{\max}(F)$ denote the maximum degree vertex in F . Also, let $\phi(F)$ denote the minimum-size balanced cut in F ; i.e. $\phi(F) = \min_{A,B \text{ balanced cut}} |E(A, B)|$ where $|E(A, B)|$ denotes the number of edges crossing the cut. For any n and $d \geq 3$, there exist *spectral expander graphs* where every node has degree d and $\phi(F) \geq \Omega(nd)$ [Vad12]. These are the graphs where our bounds will be tightest.

To ensure the existence of a low-cost hierarchical clustering, we will encode a random dataset into a random *sparse* subset, where each edge is sampled with probability $\leq 1/d_{\max}(F)$. After privately computing the hierarchical clustering, we use the *induced balanced-cut* (Lemma 5.16) to try to reconstruct edges carrying encoded data that cross this cut; these edges will form the set I . The error guarantee is sufficient to establish that $\mathbb{E}[\sum_{i \in I} x_i]$ is low. However, because the encoding is now sparse, it becomes harder to argue that $\mathbb{E}[|I|]$ is sufficiently large. To do this, we make the observation that the clustering algorithm cannot distinguish encoded vs. non-encoded edges among the edges of weight 0, and thus the number of encoded edges in the cut is not much lower than its expectation by another concentration argument. This enables us to prove a lower bound of $\Omega(\frac{\phi(F)}{\varepsilon d_{\max}(F)} n)$ that grows with the *minimum* balanced cut size. Somewhat counterintuitively, the lower bound shrinks, rather than grows, with the maximum degree, which is a byproduct of our sparse encoding. Nonetheless, this bound applies to many topologies, and is tight up to a logarithmic factor for sparse expanders (see discussion in Section 5.4.2).

5.4.1 Reconstruction Attack Outline

Our reconstruction attack encodes the dataset into a random subsample of edges $E' \subseteq E$, where each edge $e \in E$ is selected i.i.d. with probability $p = \frac{0.5}{d_{\max}(F)}$. The dataset is encoded by taking an arbitrary bijection $\ell : E' \rightarrow [k]$ (a label function) where $k = |E'|$, and setting $w(e) := R \cdot x_{\ell(e)}$ for all $e \in E'$, and $w(e) := 0$ for all $e \in E \setminus E'$. This sparse encoding ensures that there is a low-cost hierarchical clustering. Our reconstruction attack will not use the entire returned hierarchical clustering T to reconstruct data. Instead, it will use only the edges in the induced

Algorithm 8 Reconstruction Attack Rec^{HC} for Hierarchical Clustering

Require: Public graph topology $F = (V, E)$, dataset $\mathbf{x} \in \{0, 1\}^m$, parameter R , HC algorithm $\mathcal{A}_{\text{HC}}$

- 1: Initialize weight function $w : E \rightarrow \{0, R\}$.
- 2: Subsample $E' \subseteq E$ by selecting each edge from E with $p = 0.5 \frac{1}{d_{\max}}$. Let $k = |E'|$.
- 3: Let $\ell : E' \rightarrow [k]$ be any bijection from E' to $[k]$ labels.
- 4: Set $w(e) := R \cdot x_{\ell(e)}$ for all $e \in E'$ and $w(e) := 0$ for all $e \in E \setminus E'$. $\triangleright$ Encode $\mathbf{x}$ into the weights
- 5: Compute hierarchical clustering $T := \mathcal{A}_{\text{HC}}((F, w))$.
- 6: Compute induced balanced cut $(A, B) := \text{HCToBC}(T)$. $\triangleright$ See Lemma 5.16
- 7: Initialize reconstruction set $I = \{\ell(e) : e \in E'(A, B)\} \subseteq [n]$.
- 8: **for each** $i \in I$, guess $y_i = 0$.
- 9: **return** Reconstruction set I , reconstructed vector $\mathbf{y} = \{y_i : i \in I\}$.

balanced cut of T , which is defined in the following lemma from [Das16]. Their proof is constructive and forms the balanced cut with a procedure similar to a heavy-light decomposition. We refer to the balanced cut guaranteed by Lemma 5.16 as the *induced* balanced cut of T .

Lemma 5.16 ([Das16, Lemma 11]). *For any hierarchical clustering T , there exists a balanced cut A, B with $|A|, |B| \in (\frac{n}{3}, \frac{2n}{3})$ such that $\frac{w(A, B)}{|A| \cdot |B|} \leq \frac{27}{4n^3} \cdot \text{cost}_w(T)$. Moreover, there exists a procedure $\text{HCToBC}(T)$ which outputs the cut in linear time.*

The attack proceeds by running a hierarchical clustering algorithm $\mathcal{A}_{\text{HC}}$ on the graph (V, E, w) , obtaining a low-cost hierarchical clustering T , and computing the balanced cut A, B of T (which, by Lemma 5.16, is also guaranteed to have low cost). Then, it uses the reconstruction set $I = \{\ell(e) : e \in E'(A, B)\}$ (i.e. those data elements which are encoded to edges that cross A, B), and guesses $y_i = 0$ for every $i \in I$. This reconstruction procedure, Rec^{HC} , appears in Algorithm 8.

An important observation for our analysis is the following: The edges $e \in E'$ such that $x_{\ell(e)} = 0$ appear the same as the edges $e \in E \setminus E'$ in the weight function (both appear as 0), and thus the returned tree T gives us no further information to which of these two sets an edge of weight 0 belongs. Formally, define $E'_0 = \{e \in E' : x_{\ell(e)} = 0\}$ and $E'_1 = \{e \in E' : x_{\ell(e)} = 1\}$. The observation can be stated as:

Observation 5.17. *In Algorithm 8, the tree T , and subsequent post-processing of it, are conditionally independent of E'_0 given E'_1 .*

We will use this observation to show that there are many edges in $E'_0(A, B)$, just due to the randomness of E' , and thus I will be sufficiently large.

5.4.2 Reconstruction Attack Lower Bound

We will now prove the following reconstruction attack.

Theorem 5.18 (Lower Bounds HC). *Let $F = (V, E)$ be a public graph topology such that $d_{\max}(F) \geq 8$, and $\phi(F) \geq 10000d_{\max}(F) \log n$. Suppose HC is a hierarchical clustering*

algorithm with additive error $\leq R \cdot \frac{n\phi(F)}{400d_{\max}(F)}$ for some $R > 0$. Then, $\text{Succ}(\text{Rec}^{\text{HC}}) \geq 0.4$.
 Furthermore, the expected size of the reconstructed set is at least $\frac{\phi(F)}{8d_{\max}(F)}$.

This yields the following immediate corollary whose proof is identical to Corollary 5.8:

Corollary 5.19. *Let $F = (V, E)$ be a public graph topology such that $d_{\max}(F) \geq 8$ and $\phi(F) \geq 8n$, and $\frac{\phi(F)}{d_{\max}(F)} \geq 10000 \log(n)$. Then, for any $\varepsilon \leq 1, \delta \leq \frac{0.01\varepsilon}{m}$, there is no (ε, δ) -DP algorithm for hierarchical clustering with error less than $\frac{n\phi(F)}{400\varepsilon d_{\max}(F)}$.*

Corollary 5.19 is tightest when F is a d -regular spectral expander graph where $\phi(F) \geq \Omega(nd)$. In this case, the lower bound of $\Omega(\frac{n^2}{\varepsilon})$ is the same as that for the complete graph and tight with the upper bound in [Imo+23] up to $\log(n)$ factors. Observe that our lower bound argument can be applied to any *subgraph* of F by just ignoring edges and vertices, and thus it is possible to remove a small number of high-degree nodes and derive a stronger lower bound on a subgraph.

We now prove Theorem 5.18 and therefore need some supporting lemmas. First, we will show the optimal cost of clustering the encoded graph is just $\mathcal{O}(n \cdot \log(n))$ with high probability. This comes from the fact that the components in E' w.h.p. will have size $\mathcal{O}(\log(n))$.

First, we will show that the connected components in the sampled edges E' have size $\mathcal{O}(\log n)$.

Lemma 5.20 (Largest components). *Suppose $F = (V, E)$ is a graph and let H be the subgraph obtained by keeping each edge independently with probability $p \leq \frac{1-\alpha}{d_{\max}(F)}$ for some $\alpha \in (0, 1)$. Then the maximal size of a connected component in H is $\frac{2-\alpha}{\alpha^2} 4 \log n$ with probability at least $1 - n^{-3}$.*

The proof uses standard techniques and is deferred to Section 5.7.3.

Lemma 5.21 (Existence of a good Dasgupta cluster). *Suppose $F = (V, E)$ is a graph and let $\alpha \in (0, 1)$ be a constant. Now, assign weights $w(e) := 1$ with probability $\frac{1-\alpha}{d_{\max}(F)}$ and $w(e) := 0$ otherwise for each edge $e \in E$ independently. Then with probability at least $1 - 2n^{-3}$, there exists a clustering tree T , such that $\text{cost}_w(T) \leq \frac{2-\alpha}{\alpha^2} \cdot 4n \log(n)$.*

Proof. Let $H = (V, E_H)$ be the random subgraph consisting of the sampled edges, i.e. $E_H = \{e \in E \mid w(e) = 1\}$ and note that only edges in E_H contribute to the Dasgupta cost in G . This is equivalent to analyzing the connected components $C_1, \dots, C_k \subseteq V$ of H separately because the overall Dasgupta cost decomposes to a sum of the cost within each C_i .

Now, observe that any C_i can only contribute an additive term of at most $|C_i| \cdot m_i$ where m_i is the number of edges in component C_i . The worst case can be achieved if each of the m_i edges has to cross the balanced cut, thus adding cost of at most $|C_i|$ each.

Therefore, we have for the optimal Dasgupta clustering tree $\mathcal{T}$,

$$\text{cost}_w(\mathcal{T}) \leq \sum_{i \in [k]} |C_i| \cdot m_i \leq \max_{i \in [k]} |C_i| \cdot \sum_{i \in [k]} m_i = \max_{i \in [k]} |C_i| \cdot |E_H|.$$

By Lemma 5.20, with probability at least $1 - n^{-3}$, each $|C_i| \leq \frac{2-\alpha}{\alpha^2} \cdot 4 \log n$. Furthermore, $|E_H| \sim \text{Bin}(\frac{nd}{2}, \frac{1-\alpha}{d})$ and by a Chernoff bound with probability at least $1 - \exp(-\Omega(n))$, we have $|E_H| \leq n$. A union bound over both high probability events yields the desired result. $\square$

The number of mistakes made in the set I is given by $\sum_{i \in I} x_i = |E'_1(A, B)|$. We use the previous bound on the optimal clustering cost to convert the additive error of HC into an absolute bound on $|E'_1(A, B)|$.

Lemma 5.22. *Suppose Algorithm 8 is instantiated with an algorithm $\mathcal{A}_{\text{HC}}$ with expected additive error at most $C_F R n$ for a constant C_F (that may depend on n, F). Then, the balanced cut (A, B) returned by $\text{HCToBC}(T)$ has expected size $\mathbb{E}[|E'_1(A, B)|] \leq 2C_F + 50 \log(n)$.*

Proof. By the additive error guarantee of HC, we have for fixed weights w ,

$$\mathbb{E}[\text{cost}_w(T)|w] \leq \text{cost}_w(T^*) + C_F R n.$$

Recall that $R > 0$ is the weight, that we put on the subsampled edges that encode an one of the dataset. Note that in Algorithm 8, we use $\alpha = 0.5$ and then $\frac{2-\alpha}{\alpha^2} = 6$. By Lemma 5.21, with probability at least $1 - 2n^{-3}$, we have that w has an optimal clustering cost of $(6 \cdot 4) \cdot R n \log n$. In the worst case, the cost of clustering a complete graph is $n^3 R$. Thus, we can use conditional expectation to show $\mathbb{E}[\text{cost}_w(T)] \leq 24n \log(n) + 2n^{-3}n^3 R + C_F R n \leq 25R n \log(n) + C_F R n$. Furthermore, by Lemma 5.16, the balanced cut A, B satisfies

$$w(A, B) \leq \frac{27}{4n^3} |A| |B| \text{cost}_w(T) \leq \frac{27}{16n} \text{cost}_w(T),$$

because $|A| |B|$ maximizes in the interval $(\frac{n}{3}, \frac{2n}{3})$ exactly if both have size exactly $\frac{n}{2}$. The last step is to observe that $w(A, B) = R \cdot |E'_1(A, B)|$ due to the way the data is embedded. Hence, we get the following in expectation,

$$\begin{aligned} R \cdot \mathbb{E}[|E'_1(A, B)|] &= \mathbb{E}[w(A, B)] \leq \frac{27}{16n} \mathbb{E}[\text{cost}_w(T)] \\ &\leq \frac{27}{16n} (C_F R n + 25R n \log n) \\ &\leq R(2C_F + 50 \log n). \end{aligned}$$

Dividing by R yields the result. $\square$

Next, we need to show that $|I|$ is sufficiently large, as this will show that I is a successful reconstruction set. As $|I| = |E'_0(A, B)| + |E'_1(A, B)|$, and we have already upper bounded $|E'_1(A, B)|$, we will lower bound $|E'_0(A, B)|$. Precisely, we lower bound $\mathbb{E}[|E'_0(A, B)| | E'_1]$ since conditioning on E'_1 allows us to consider A, B as a constant per Observation 5.17. Under this conditioning, the edges in E'_0 are distributed multinomially, allowing us to do a straightforward Chernoff bound.

Lemma 5.23. *Suppose $\mathbf{x}$ and E' are sampled as they are in Algorithm 8. Then, for any balanced cut A_0, B_0 , we have that $\mathbb{E}[|E'_0(A_0, B_0)| | E'_1] \geq \frac{p}{2} |\tilde{E}(A_0, B_0)|$ where $\tilde{E} = E \setminus E'_1$*

Proof. (Of Lemma 5.23) By construction, each edge $e \in E$ is independently assigned to $E \setminus E'$, E'_0 and E'_1 with probabilities $(1 - p, \frac{p}{2}, \frac{p}{2})$. Thus, conditioned on E'_1 , each

2191 remaining edge in $e \in E \setminus E'_1$ is independently assigned to $E \setminus E'$ or E'_0 with
 2192 probability $(\frac{1-p}{1-p/2}, \frac{p/2}{1-p/2})$. Thus,

$$\begin{aligned} \mathbb{E}[E'_0(A_0, B_0) \mid E'_1] &= \frac{p/2}{1-p/2} |(E \setminus E'_1)(A_0, B_0)| \\ &\geq \frac{p}{2} |(E \setminus E'_1)(A_0, B_0)|. \end{aligned} \quad \square$$

2193 The prior bound of $|\tilde{E}(A_0, B_0)|$ is not completely independent of E'_1 ; the following
 2194 result shows it is almost always bigger than $\frac{2}{3}|E(A_0, B_0)|$, giving us a simple bound
 2195 to work with.

Lemma 5.24. *Suppose $d_{\max} \geq 8$ and $\phi(F) \geq 8n$. With probability $1 - 2^{-0.2n}$, for all balanced cuts A_0, B_0 , we have $|E'_1(A_0, B_0)| \leq \frac{1}{3}|E(A_0, B_0)|$.*

2196 *Proof.* (Of Lemma 5.24) $E'_1(A_0, B_0)$ is a binomial distribution on $E(A_0, B_0)$ with
 2197 probability $\frac{p}{2}$. Thus, it suffices to compute a tail bound $\mathbf{P}[X \geq \frac{m}{3}]$, where $m =$
 2198 $|E(A_0, B_0)|$ and $X \sim \text{Bin}(m, \frac{p}{2})$. By Chernoff's bound, we know $\mathbf{P}[X \geq \frac{mp}{2} + t] \leq$
 2199 $\exp(-\frac{t^2}{mp+2t/3})$ for all $t \geq 0$. By assumption we know $p = 0.5 \frac{1}{d_{\max}} \leq \frac{1}{8}$. Taking
 2200 $t = \frac{m}{4}$ the above bound is $\mathbf{P}[X \geq \frac{m}{16} + \frac{m}{4}] \leq \exp(-\frac{m^2/16}{m/4+m/6}) < \exp(-\frac{m}{8})$. Because
 2201 $m \geq \phi(F) \geq 8n$, we have $\exp(-\frac{m}{8}) \leq \exp(-n)$, the failure probability over all 2^n
 2202 balanced cuts is at most $2^n e^{-n} \leq 2^{-0.2n}$. $\square$

2203 We are now ready to put it all together and prove our main reconstruction lower
 2204 bound for HC.

2205 **Theorem 5.18** (Lower Bounds HC). *Let $F = (V, E)$ be a public graph topology such that*
 2206 *$d_{\max}(F) \geq 8$, and $\phi(F) \geq 10000d_{\max}(F) \log n$. Suppose HC is a hierarchical clustering*
 2207 *algorithm with additive error $\leq R \cdot \frac{n\phi(F)}{400d_{\max}(F)}$ for some $R > 0$. Then, $\text{Succ}(\text{Rec}^{\text{HC}}) \geq 0.4$.*
 2208 *Furthermore, the expected size of the reconstructed set is at least $\frac{\phi(F)}{8d_{\max}(F)}$.*

2209 *Proof.* Observe that, based on the construction of the set I and y , we have,

$$\sum_{i \in I} \mathbb{1}[x_i \neq y_i] = E'_1(A, B), \text{ and } |I| = E'_1(A, B) + E'_0(A, B).$$

2210 We will prove that $9 \mathbb{E}[E'_1(A, B)] \leq \mathbb{E}[E'_0(A, B)]$, as this implies

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] \leq 0.1 \mathbb{E}[|I|],$$

2211 meaning the attack is 0.4 successful. Because HC has additive error $\frac{\phi(F)R}{400d_{\max}(F)}n$,
 2212 by Lemma 5.22, we have $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{200d_{\max}(F)} + 50 \log(n)$. By assumption, we
 2213 know that

$$50 \log(n) \leq \frac{\phi(F)}{200d_{\max}(F)}.$$

2214 Therefore, $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{100d_{\max}(F)}$. By the law of total expectation, we have
 2215 $\mathbb{E}[|E'_0(A, B)|] = \mathbb{E}[\mathbb{E}[|E'_0(A, B)| | E'_1]]$. We can bound the inner quantity,

$$\begin{aligned} \mathbb{E}[|E'_0(A, B)| | E'_1] &= \mathbb{E}_{A,B} [\mathbb{E}[|E'_0(A, B)| | A, B, E'_1]] \\ &\geq \min_{A_0, B_0 \text{ balanced cut}} \mathbb{E}[|E'_0(A_0, B_0)| | A = A_0, B = B_0, E'_1] \\ &= \min_{A_0, B_0 \text{ balanced cut}} \mathbb{E}[|E'_0(A_0, B_0)| | E'_1] \quad (\text{by Observation 5.17}) \\ &\geq \frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)|. \quad (\text{by Lemma 5.23}). \end{aligned}$$

2216 Now, we will expand the whole expectation. In the following, let $\mathcal{E}$ be the event in
 2217 Lemma 5.24; i.e. that for all balanced cuts A_0, B_0 , we have

$$|E'_1(A_0, B_0)| \leq \frac{1}{3} |E(A_0, B_0)|.$$

2218 Hence,

$$\begin{aligned} \mathbb{E}[|E'_0(A, B)|] &\geq \mathbb{E} \left[\frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)| \right] \\ &\geq \mathbb{P}[\mathcal{E}] \mathbb{E} \left[\frac{p}{2} \min_{A_0, B_0 \text{ balanced cut}} |(E \setminus E'_1)(A_0, B_0)| \middle| \mathcal{E} \right] \\ &\geq (1 - 2^{-0.2n}) \frac{p}{2} \mathbb{E} \left[\min_{A_0, B_0 \text{ balanced cut}} \frac{2}{3} |E(A_0, B_0)| \middle| \mathcal{E} \right] \quad (\text{by Lemma 5.24}) \\ &> \frac{p}{4} \phi(G) = \frac{\phi(F)}{8d_{\max}}. \end{aligned}$$

2219 Recalling that $\mathbb{E}[|E'_1(A, B)|] \leq \frac{\phi(F)}{100d_{\max}(F)}$ and $\mathbb{E}[|E'_0(A, B)|] > \frac{\phi(F)}{8d_{\max}(F)}$. Putting this
 2220 together, we can obtain

$$9 \mathbb{E}[|E'_1(A, B)|] \leq \frac{9\phi(F)}{100d_{\max}(F)} < \frac{\phi(F)}{8d_{\max}(F)} < \mathbb{E}[|E'_0(A, B)|].$$

2221 □

2222 5.5 PRIVATE ALGORITHMS WITH TIGHT ERROR

2223 We now detail algorithms with superior error on **MST** and Minimum-Weight (Per-
 2224 fect) Matching that appear in ongoing work [Aam26] demonstrating that the lower
 2225 bounds of Sections 5.2.1 and 5.3 are tight. The algorithm is a simple input privatiza-
 2226 tion, adding Laplace noise to each of the edge weights independently. The novelty
 2227 lies in the analysis: instead of bounding the maximum noise added to individual
 2228 edge weights as in prior work [Sea16; HT24; Pag+24], it uses a concentration bound
 2229 on the total noise of any single spanning tree and then applies a union bound.
 2230 We can phrase it even more general: any graph problem that outputs k edges can
 2231 achieve additive error $\mathcal{O}(\frac{k}{\varepsilon} \cdot \ln(\frac{m}{k}))$, because one can bound $\binom{m}{k} \leq (\frac{em}{k})^k$. In the
 2232 cases of MSTs and MWPMs where $k = \Theta(n)$, this gives an upper bound with a
 2233 logarithmic dependence of $\ln(\frac{m}{n})$. The proof appears in Section 5.7.4.

2234 **Theorem 5.25** ([Aam26]). *Let $G = (V, E, \mathbf{w})$ be a graph and let $\mathcal{S}(G)$ be a family of*
 2235 *solutions such that every $S \in \mathcal{S}(G)$ satisfies $S \subseteq E$ and $|S| = k$. Consider the problem*
 2236 *of minimizing $w(S)$ among all $S \in \mathcal{S}(G)$. Then there exists an ε -DP algorithm $\mathcal{A}$ that*
 2237 *outputs a feasible solution $S \in \mathcal{S}(G)$, such that with probability at least $1 - n^{-\Omega(1)}$,*

$$w(S) - w(S^*) \leq \mathcal{O}\left(\frac{k}{\varepsilon} \cdot \ln\left(\frac{m}{k}\right)\right),$$

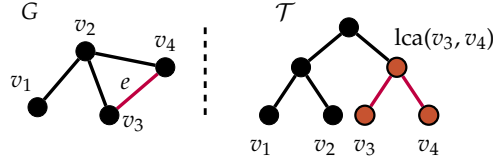


Figure 5.3: An example of a hierarchical clustering T with $\text{cost}_w(T) = 12$ for a graph $G = (V, E)$ assuming unit weights. The edge e has cost 2 as the lowest common ancestor $\text{lca}(v_3, v_4)$ has exactly two leaves in its subtree of T . Furthermore, cutting the tree at $\text{lca}(v_3, v_4)$ would give a balanced cut.

where $S^* = \arg \min_{\tilde{S} \in \mathcal{S}(G)} w(\tilde{S})$ is any optimal solution.

5.6 CONCLUSION AND FUTURE WORK

We first showed that for **MST** and Minimum-Weight (Perfect) Matching there exist worst-case topologies where the error is tight with the upper bounds. We also proved lower bounds for **MST** and HC on large classes of topologies which are close to the upper bounds, off by just a $\log(n)$ factor.

For MWPM, it remains open whether lower bounds can be derived for a large class of topologies, and what specific property that class should satisfy. For HC, closing the gap between the best lower bound of $\Omega(\frac{1}{\epsilon} \cdot n^2)$ and upper bound of $\mathcal{O}(\frac{1}{\epsilon} \cdot n^2 \cdot \log(n))$ is an interesting question, as well as deriving a polynomial-time algorithm that achieves this additive error (rather than using the exponential mechanism). Finally, we believe that both our lower bound techniques for obtaining tight bounds and bounds on general topologies are likely to carry over to other graph problems as well.

5.7 APPENDIX: RECONSTRUCTION-LOWER BOUNDS

In Figure 5.3, we illustrate the Dasgupta cost function, $\text{cost}_w(T)$, for a given hierarchical clustering tree T . We are now ready to prove the proofs we removed from the main part.

Lemma 5.1. *Let $\mathcal{B} : \mathcal{X}^d \rightarrow \mathcal{X}^d$ be any mechanism that is (ϵ, δ) -differentially private under the Hamming neighborhood relationship. Then for a random dataset $\mathbf{x} \leftarrow \mathcal{X}^d$ drawn uniformly at random, we have for each $i \in [d]$, we have $\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] \leq \frac{\exp(\epsilon)}{|\mathcal{X}|} + \delta$.*

Proof. Let $n \in \mathbb{N}$. We index the elements in $\mathcal{X}$ by $\{1, \dots, n\}$. For some arbitrarily chosen $i \in [d]$, assume a random vector $\mathbf{x} = \mathbf{x}_{<i} \oplus X_i \oplus \mathbf{x}_{>i} \sim \text{Uni}([n]^d)$ where $\oplus$ denotes vector concatenation, and we use the subscript $\mathbf{x}_{<i}$ and $\mathbf{x}_{>i}$ to split the vector at index i . Now we can bound the probability that the i 'th coordinate of the output of $\mathcal{B}(\mathbf{x})$ leaks its corresponding input x_i :

$$\begin{aligned}
\mathbf{P}[\mathcal{B}(\mathbf{x})_i = x_i] &= \frac{1}{|\mathcal{X}|^{d-1}} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(\mathbf{P}_{X_i \sim \text{Uni}([n])}[\mathcal{B}(\mathbf{x}_{<i} \oplus x_i \oplus \mathbf{x}_{>i})_i = x_i] \right) \\
&\leq |\mathcal{X}|^{1-d} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(e^\epsilon \mathbf{P}_{x_i \sim \text{Uni}([n])}[\mathcal{B}(\mathbf{x}_{<i} \oplus (1) \oplus \mathbf{x}_{>i})_i = x_i] + \delta \right) \quad (4) \\
&\leq |\mathcal{X}|^{1-d} \sum_{\mathbf{x}_{<i} \in [n]^{i-1}} \sum_{\mathbf{x}_{>i} \in [n]^{d-i}} \left(e^\epsilon \cdot \frac{1}{n} + \delta \right) \quad (5) \\
&\leq \frac{e^\epsilon}{|\mathcal{X}|} + \delta.
\end{aligned}$$

In line 4, we use the fact that each X_i is i.i.d. and in step 5, we use the privacy guarantees of the mechanism B and flip to a neighboring dataset where we explicitly set $X_i = 1$ under the hamming adjacency relation. $\square$

5.7.1 Proof of Lemma 5.2

First, we start with a supporting lemma.

Lemma 5.26. Suppose P, Q are distributions on a discrete set $\mathcal{X}$ satisfying

$$\mathbf{P}_{X \sim P}[X \in S] \leq e^\epsilon \mathbf{P}_{Y \sim Q}[Y \in S] + \delta \text{ for all } S \subseteq \mathcal{X}.$$

Then, there exists a function $\delta(x) : \mathcal{X} \rightarrow [0, \infty)$ satisfying

1. $\mathbf{P}_{X \sim P}[X = x] \leq e^\epsilon \mathbf{P}_{X \sim Q}[X = x] + \delta(x)$, and
2. $\sum_{x \in \mathcal{X}} \delta(x) \leq \delta$.

Proof. We denote $P(x) = \mathbf{P}_{X \sim P}[X = x]$ (and similar for Q).

We take the function $\delta(x) = \max\{P(x) - e^\epsilon Q(x), 0\}$. Condition 1 is satisfied because $P(x) = e^\epsilon Q(x) + (P(x) - e^\epsilon Q(x)) \leq e^\epsilon Q(x) + \delta(x)$.

For condition 2, let $N \subseteq \mathcal{X}$ denote the set where $P(x) - e^\epsilon Q(x) \geq 0$. On $x \in N$, we have $\delta(x) = P(x) - e^\epsilon Q(x)$, and on $x \in \mathcal{X} \setminus N$, we have $\delta(x) = 0$. Thus,

$$\sum_{x \in \mathcal{X}} \delta(x) = \sum_{x \in N} \delta(x) = \sum_{x \in N} P(x) - e^\epsilon Q(x) = \mathbf{P}_{X \sim P}[X \in N] - e^\epsilon \mathbf{P}_{Y \sim Q}[Y \in N] \leq \delta.$$

Theorem 5.2. Let $\mathcal{B} : \{0, 1\}^d \rightarrow 2^{[d]} \times \{0, 1\}^d$ be any (ϵ, δ) -DP algorithm and let $\mathbf{x} \sim \{0, 1\}^d$ be drawn uniformly at random. Then, $\text{Succ}(\mathcal{B}) \leq \frac{e^\epsilon - 1}{2(e^\epsilon + 1)} + \frac{d\delta}{e^\epsilon + 1}$.

Proof. It suffices to show that

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] \geq \frac{1}{1 + e^\epsilon} \mathbb{E}[|I|] - \frac{1}{1 + e^\epsilon} d\delta.$$

We use conditional expectation to write

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{1}[x_i \neq y_i] \right] = \mathbb{E} \left[\mathbb{E} \left[\sum_{i \in I} |y_i - x_i| \middle| I \right] \right] = \mathbb{E} \left[\sum_{i \in I} \mathbb{E}[|y_i - x_i| \mid I] \right].$$

2279 We will form a bound $\mathbb{E}[|y_i - x_i| | I]$ for each i . To do this, we expand it as follows
 2280 by conditional probability,

$$\begin{aligned}\mathbb{E}[|y_i - x_i| | I] &= \mathbf{P}[x_i = 1, y_i = 0 | I] + \mathbf{P}[x_i = 0, y_i = 1 | I] \\ &= \frac{\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]}{\mathbf{P}[I]}.\end{aligned}$$

2281 In the following, a sum over the variable x implicitly ranges over $\{0, 1\}^n$, and a
 2282 sum over y implicitly ranges over $\{0, 1\}^I$. We can write

$$\mathbf{P}[x_i = a, y_i = b, I] = \sum_{\substack{x: x_i = a \\ y: y_i = b}} 2^{-n} \mathbf{P}[\mathcal{A}(x) = (I, y)] = 2^{-n} \sum_{x: x_i = a} \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,b})],$$

2283 where $\mathcal{Y}_{i,b} = \{y \in \{0, 1\}^I : y_i = b\}$. For a vector $x \in \{0, 1\}^n$, let $x^{(i \rightarrow 0)}$ and $x^{(i \rightarrow 1)}$
 2284 denote x with its i th bit set to 0 (resp. 1). Because each $\mathcal{A}(x^{(i \rightarrow 0)})$ and $\mathcal{A}(x^{(i \rightarrow 1)})$
 2285 satisfy (ϵ, δ) -DP, by Lemma 5.26, there exists a non-negative function $\delta_{x \uparrow i}(I, b)$ such
 2286 that

$$\mathbf{P}[\mathcal{A}(x^{(i \rightarrow 0)}) \in (I, \mathcal{Y}_{i,b})] \leq e^\epsilon \mathbf{P}[\mathcal{A}(x^{(i \rightarrow 1)}) \in (I, \mathcal{Y}_{i,b})] + \delta_{x \uparrow i}(I, b),$$

2287 and $\sum_{I \subseteq [n]} \sum_{b \in \{0,1\}} \delta_{x \uparrow i}(I, b) \leq \delta$. Similarly, there exists a non-negative function
 2288 $\delta_{x \downarrow i}(I, b)$ such that

$$\mathbf{P}[\mathcal{A}(x^{(i \rightarrow 1)}) \in (I, \mathcal{Y}_{i,b})] \leq e^\epsilon \mathbf{P}[\mathcal{A}(x^{(i \rightarrow 0)}) \in (I, \mathcal{Y}_{i,b})] + \delta_{x \downarrow i}(I, b)$$

2289 and $\sum_{I \subseteq [n]} \sum_{b \in \{0,1\}} \delta_{x \downarrow i}(I, b) \leq \delta$. By summing the first inequality over $x \in \{0, 1\}^n$
 2290 with $x_i = 0$ (and the second over x with $x_i = 1$), we obtain

$$\begin{aligned}\mathbf{P}[x_i = 0, y_i = 0, I] &= 2^{-n} \sum_{x: x_i = 0} \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,0})] \\ &\leq 2^{-n} \sum_{x: x_i = 1} (e^\epsilon \mathbf{P}[\mathcal{A}(x) \in (I, \mathcal{Y}_{i,0})] + \delta_{x \uparrow i}(I, 0)) \\ &= e^\epsilon \mathbf{P}[x_i = 1, y_i = 0, I] + \delta_{i,1}^\uparrow(I, 0),\end{aligned}$$

2291 where $\delta_{i,a}^\uparrow(I, b)$ is defined to be $2^{-n} \sum_{x: x_i = a} \delta_{x \uparrow i}(I, b)$. Rearranging, we have

$$\mathbf{P}[x_i = 1, y_i = 0, I] \geq e^{-\epsilon} \mathbf{P}[x_i = 0, y_i = 0, I] - e^{-\epsilon} \delta_{i,1}^\uparrow(I, 0).$$

2292 Similarly, we can write

$$\mathbf{P}[x_i = 0, y_i = 1, I] \geq e^{-\epsilon} \mathbf{P}[x_i = 1, y_i = 1, I] - e^{-\epsilon} \delta_{i,0}^\downarrow(I, 1),$$

2293 where $\delta_{i,a}^\downarrow(I, b) = 2^{-n} \sum_{x: x_i=a} \delta_{x \downarrow i}(I, b)$. This means that

$$\begin{aligned}
& \mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I] \\
&= \frac{1}{1+e^\epsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{e^\epsilon}{1+e^\epsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\geq \frac{1}{1+e^\epsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{e^\epsilon}{1+e^\epsilon} (e^{-\epsilon} \mathbf{P}[x_i = 0, y_i = 0, I] - e^{-\epsilon} \delta_{i,1}^\uparrow(I, 0) \\
&\quad + e^{-\epsilon} \mathbf{P}[x_i = 1, y_i = 1, I] - e^{-\epsilon} \delta_{i,0}^\downarrow(I, 1)) \\
&= \frac{1}{1+e^\epsilon} (\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]) \\
&\quad + \frac{1}{1+e^\epsilon} (\mathbf{P}[x_i = 0, y_i = 0, I] - \delta_{i,1}^\uparrow(I, 0) + \mathbf{P}[x_i = 1, y_i = 1, I] - \delta_{i,0}^\downarrow(I, 1)) \\
&= \frac{1}{1+e^\epsilon} \mathbf{P}[I] - \frac{1}{1+e^\epsilon} (\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)),
\end{aligned}$$

2294 where the last step follows from the law of total probability. Thus,

$$\begin{aligned}
\mathbb{E}[|y_i - x_i| | I] &= \frac{\mathbf{P}[x_i = 1, y_i = 0, I] + \mathbf{P}[x_i = 0, y_i = 1, I]}{\mathbf{P}[I]} \\
&\geq \frac{1}{1+e^\epsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\epsilon)\mathbf{P}[I]}.
\end{aligned}$$

2295 Finally, we bound the entire expectation as

$$\begin{aligned}
\mathbb{E}\left[\sum_{i \in I} \mathbb{E}[|y_i - x_i| | I]\right] &\geq \mathbb{E}\left[\sum_{i \in I} \left(\frac{1}{1+e^\epsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\epsilon)\mathbf{P}[I]}\right)\right] \\
&= \sum_{I \subseteq [n]} \mathbf{P}[I] \sum_{i \in I} \left(\frac{1}{1+e^\epsilon} - \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{(1+e^\epsilon)\mathbf{P}[I]}\right) \\
&= \frac{1}{1+e^\epsilon} \sum_{I \subseteq [n]} \mathbf{P}[I] \cdot |I| - \frac{1}{1+e^\epsilon} \sum_{I \subseteq [n]} \mathbf{P}[I] \sum_{i \in I} \frac{\delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1)}{\mathbf{P}[I]} \\
&= \frac{1}{1+e^\epsilon} \mathbb{E}[|I|] - \frac{1}{1+e^\epsilon} \sum_{I \subseteq [n]} \sum_{i \in I} \delta_{i,1}^\uparrow(I, 0) + \delta_{i,0}^\downarrow(I, 1). \tag{6}
\end{aligned}$$

2296 The second term simplifies to

$$\begin{aligned}
& \sum_{I \subseteq [n]} \sum_{i \in I} \delta_{i,1}^{\uparrow}(I, 0) + \delta_{i,0}^{\downarrow}(I, 1) \\
&= \sum_{I \subseteq [n]} \sum_{i \in I} 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&= \sum_{i=1}^n \sum_{I \subseteq [n]} \mathbb{1}[i \in I] \cdot 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&\leq \sum_{i=1}^n \sum_{I \subseteq [n]} 2^{-n} \left(\sum_{x: x_i=1} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \delta_{x \downarrow i}(I, 1) \right) \\
&= \sum_{i=1}^n 2^{-n} \left(\sum_{x: x_i=1} \sum_{I \subseteq [n]} \delta_{x \uparrow i}(I, 0) + \sum_{x: x_i=0} \sum_{I \subseteq [n]} \delta_{x \downarrow i}(I, 1) \right) \\
&\leq \sum_{i=1}^n 2^{-n} \left(\sum_{x: x_i=1} \delta + \sum_{x: x_i=0} \delta \right) = \sum_{i=1}^n \delta = n\delta.
\end{aligned}$$

2297 Plugging back into (6), we obtain

$$\mathbb{E} \left[\sum_{i \in I} \mathbb{E}[|y_i - x_i| | I] \right] \geq \frac{\mathbb{E}|I|}{1 + e^\epsilon} - \frac{1}{1 + e^\epsilon} n\delta,$$

2298 giving the answer. $\square$

2299 5.7.2 Proof of Lemma 5.6

2300 Let $E_0 \subseteq E$ denote the embedded edges where $x_{\ell(e)} = 0$. By the embedding
 2301 procedure, we know E_0 is a Bernoulli subsample of E with probability $\frac{1}{2}$. It suffices
 2302 to show that all cuts in the subgraph induced by E_0 are non-empty, as this will
 2303 mean it is possible to construct a spanning tree entirely in E_0 . In other words, we
 2304 show that each cut in the graph has at least one zero-edge sampled with high
 2305 probability, which, by a standard argument, implies that there must exist a zero-
 2306 cost spanning tree. Now let $\mathcal{E}$ denote the *complement* event; i.e., that there exists a
 2307 cut A, B such that $E_0(A, B) = 0$. We will upper bound $\mathbf{P}[\mathcal{E}]$ by a union bound over
 2308 all cuts. Let λ denote the minimum cut size in (V, E) , and let N_i denote the number
 2309 of cuts A, B such that $E(A, B) = i$. The probability that a cut with value i has no
 2310 edges sampled is given by 2^{-i} . By the union bound, we have

$$\mathbf{P}[\mathcal{E}] \leq \sum_{i=\lambda}^{|E|} 2^{-i} N_i.$$

2311 To bound this, we use [Kar93, Theorem 6.2] which shows that $N_i \leq n^{\lceil 2i/\lambda \rceil} \leq$
 2312 $n \cdot n^{2i/\lambda}$. Plugging this in, we have by a geometric sum

$$\begin{aligned}
\sum_{i=\lambda}^{|E|} 2^{-i} N_i &\leq \sum_{i=\lambda}^{\infty} 2^{-i} n \cdot n^{2i/\lambda} \leq n \sum_{i=\lambda}^{\infty} \left(\frac{n^{2/\lambda}}{2} \right)^i \\
&= n \frac{\left(\frac{n^{2/\lambda}}{2} \right)^\lambda}{1 - \left(\frac{n^{2/\lambda}}{2} \right)} = \frac{n^3}{\left(1 - \frac{1}{2} (n^{2/\lambda}) \right) \cdot 2^\lambda}.
\end{aligned}$$

2313 Using $\lambda \geq 5 \log(n)$, it holds that $n^3 2^{-\lambda} < \frac{1}{n^2}$, and $\frac{1}{2} \cdot n^{2/\lambda} \leq \frac{2^{2/5}}{2} < \frac{2}{3}$, so the
 2314 denominator $(1 - \frac{1}{2} n^{2/\lambda}) \geq \frac{1}{3}$. Thus, the probability of failure is at most $\frac{3}{n^2}$. $\square$

5.7.3 Proof of Lemma 5.20

We are using some standard results in random graph theory to prove the following result. Additionally, we need the following definitions.

Definition 5.2 (Stochastic domination). *A random variable X is stochastically dominated by a random variable Y , if $\mathbf{P}[X \geq a] \leq \mathbf{P}[Y \geq a]$ for all a . A coupling of X and Y is a random vector $(\tilde{X}, \tilde{Y})$ such that the marginal distributions coincide with X and Y respectively.*

The following theorem due to Strassen [Str65] shows that stochastic dominance is equal to the existence of a monotone coupling.

Lemma 5.27 ([Str65]). *A random variable X stochastically dominates a random variable Y , if and only if there exists a (monotone) coupling $(\tilde{X}, \tilde{Y})$ of X and Y such that $\mathbf{P}[\tilde{X} \geq \tilde{Y}] = 1$.*

Now, we are ready to prove Lemma 5.20.

Lemma 5.20 (Largest components). *Suppose $F = (V, E)$ is a graph and let H be the subgraph obtained by keeping each edge independently with probability $p \leq \frac{1-\alpha}{d_{\max}(F)}$ for some $\alpha \in (0, 1)$. Then the maximal size of a connected component in H is $\frac{2-\alpha}{\alpha^2} 4 \log n$ with probability at least $1 - n^{-3}$.*

Proof. We prove this result assuming that the graph is d -regular, so $d_{\max}(F) = d$. The proof for a general graph is similar. Let $\alpha \in (0, 1)$, set $p := \frac{1-\alpha}{d}$ and let C_v denote the connected component of a fixed vertex v in H and WLOG assume that our edge subsampling is exploring C_v by BFS starting in v . If $0 \leq m_t \leq d$ is the number of neighbors of the currently explored vertex at step $t \geq 1$ that have not already been explored, then observe that the size of the frontier conditioned on the past $\mathcal{F}$, can be described as

$$X_0 = 1, \quad X_t = X_{t-1} - 1 + Z_t \quad \text{where } Z_t \mid \mathcal{F}_t \sim \text{Bin}(m_t, p),$$

because we keep each neighbor that is not already included with probability p independently.

Intuitively, cycles can only reduce the number of newly discovered vertices, so the BFS exploration is dominated by the process that simply ignores them. Therefore, define the process

$$Y_0 = 1, \quad Y_t = Y_{t-1} - 1 + \tilde{Z}_t \quad \text{where } \tilde{Z}_t \sim \text{Bin}(d, p), \quad (7)$$

We claim that for each $t \geq 1$, the random variable Y_t conditional on the past, stochastically dominates X_t . Therefore, we will explicitly give a coupling $(\tilde{X}, \tilde{Y})$ of X and Y and apply Lemma 5.27. Set $\tilde{X} = X$ and draw $U_t \mid \mathcal{F}_t \sim \text{Bin}(d - m_t, p)$ independently of Z_t . Now, let $\tilde{Y}_t = \tilde{Y}_{t-1} - 1 + Z_t + U_t$ and consider the coupling $(\tilde{X}, \tilde{Y})$. Clearly, the marginal distributions match because $Z_t + U_t \sim \text{Bin}(d, p)$ by the convolution of binomials. We show by induction that for each $t \geq 0$, $\tilde{X}_t \leq \tilde{Y}_t$. The base cases $X_0, Y_0 = 1$ are trivial. Now, observe that for $Z_t \sim \text{Bin}(m_t, p)$, we have

$$\begin{aligned} \tilde{X}_t &= \tilde{X}_{t-1} - 1 + Z_t \leq \tilde{Y}_{t-1} - 1 + Z_t \\ &\leq \tilde{Y}_{t-1} - 1 + Z_t + U_t = \tilde{Y}_t, \end{aligned}$$

where the last line follows from the non-negativity of the binomial distribution.

Hence, Y_t gives a valid upper bound on how large the component grows, i.e.

$$\mathbf{P}[|C_v| \geq t] = \mathbf{P}[X_t \geq 1] \leq \mathbf{P}[Y_t \geq 1].$$

Because Y_t is a sum of t binomials, by a standard convolution result, we get

$$Y_t = \sum_{k=1}^t \tilde{Z}_k - t,$$

which is distributed as $Y_t \sim \text{Bin}(td, p) - t$. Now, let $\mu = t(1 - \alpha)$. By a standard

Chernoff bound, $\mathbf{P}[\text{Bin}(td, p) \geq (1 + \gamma)\mu] \leq \exp(-\frac{\gamma^2 \mu}{2 + \gamma})$ for all $\gamma > 0$. Rearranging the term, yields

$$\mathbf{P}[\text{Bin}(td, p) \geq t] \leq \exp\left(-\frac{(t - \mu)^2}{t + \mu}\right) = \exp\left(-t \cdot \frac{\alpha^2}{2 - \alpha}\right). \quad (8)$$

To make this decrease with $\frac{1}{n^3}$, we choose $t = \frac{2 - \alpha}{\alpha^2} \cdot 4 \ln(n)$. Then we get by a union bound

$$\begin{aligned} \mathbf{P}[\exists v : |C_v| \geq t] &\leq n \exp(-4 \ln(n)) \\ &= n^{-3}. \end{aligned}$$

Hence, with probability at least $1 - n^{-3}$, no component has size larger than $\frac{2 - \alpha}{\alpha^2} \cdot 4 \log n$. $\square$

5.7.4 Proof of Theorem 5.25

Theorem 5.25 ([Aam26]). *Let $G = (V, E, \mathbf{w})$ be a graph and let $\mathcal{S}(G)$ be a family of solutions such that every $S \in \mathcal{S}(G)$ satisfies $S \subseteq E$ and $|S| = k$. Consider the problem of minimizing $w(S)$ among all $S \in \mathcal{S}(G)$. Then there exists an ε -DP algorithm $\mathcal{A}$ that outputs a feasible solution $S \in \mathcal{S}(G)$, such that with probability at least $1 - n^{-\Omega(1)}$,*

$$w(S) - w(S^*) \leq \mathcal{O}\left(\frac{k}{\varepsilon} \cdot \ln\left(\frac{m}{k}\right)\right),$$

where $S^* = \arg \min_{\tilde{S} \in \mathcal{S}(G)} w(\tilde{S})$ is any optimal solution.

The following proof is due to [Aam26]. Consider the algorithm that adds noise $Z_e \sim \text{Lap}(1/\varepsilon)$ to each edge independently. Denote these weights with $\mathbf{w}'$. This mechanism is ε -DP under the ℓ_1 neighboring relationship by a standard result [Dwo+o6b]. Now let $T \subseteq \mathcal{S}(G)$ be an optimal solution over the noisy weights $\mathbf{w}'$ and T^* be an optimal over the original weights $\mathbf{w}$. Let $Z_T = \sum_{t \in T} Z_t$ be the total noise of any T and let $M = \max_{T \in \mathcal{S}(G)} |Z_T|$. Then,

$$w(T') \leq w'(T') + M \leq w'(T^*) + M \leq w(T^*) + 2 \cdot M,$$

by the same inequalities as used in [Sea16]. Note that the sub-exponential norm of each of the i.i.d. Laplace random variables Z_e is $\|Z_e\|_{\psi_1} \leq \frac{2}{\varepsilon}$. By a standard Bernstein inequality [Ver18], we get for each T for any $t > 0$ and an absolute constant $c > 0$,

$$\mathbf{P}[|Z_T| \geq t] \leq 2 \exp\left(-c \min\left(\frac{\varepsilon^2 t^2}{n}, \varepsilon t\right)\right),$$

2370 Now setting $t = c \cdot (k/\varepsilon) \ln(m/k)$ for a sufficiently large constant c , the tail above
 2371 becomes dominated by the linear term $\exp(-\Omega(n \ln(m/k)))$. Clearly, $|\mathcal{S}(G)| \leq$
 2372 $\binom{n}{k} \leq \left(\frac{em}{k}\right)^k$. A union bound over $\mathcal{S}(G)$ yields,

$$\mathbf{P} \left[\max_{T \in \mathcal{S}(G)} |Z_T| \geq C \frac{k}{\varepsilon} \ln(m/k) \right] \leq n^{-\Omega(1)},$$

2373 Combined with the chain of inequalities above, this gives

$$w(T') - w(T^*) \leq \mathcal{O}((k/\varepsilon) \ln(m/k))$$

2374 with probability at least $1 - n^{-\Omega(1)}$.

We may optimally describe a message with a particular amount of distortion and later decide that the message needs to be specified more accurately. Then, when an addendum to the original message is sent we hope that this refinement is as efficient as if the more strict requirements had been known at the start.

Successive Refinement of Information — [EC91]

LOSSLESS PRIVATE MULTIPLE RELEASE

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Joint work with Joel Daniel Andersson, Rasmus Pagh and Boel Nelson

Koufogiannis et al. (2016) showed a *gradual release* result for Laplace noise-based differentially private mechanisms: given an ϵ -DP release, a new release with privacy parameter $\epsilon' > \epsilon$ can be computed such that the combined privacy loss of both releases is at most ϵ' and the distribution of the latter is the same as a single release with parameter ϵ' . They also showed gradual release techniques for Gaussian noise, later also explored by Whitehouse et al. (2022). In this chapter, we consider a more general *multiple release* setting in which analysts hold private releases with different privacy parameters corresponding to different access/trust levels. These releases are determined one by one, with privacy parameters in arbitrary order. A multiple release is *lossless* if having access to a subset S of the releases has the same privacy guarantee as the least private release in S , and each release has the same distribution as a single release with the same privacy parameter. Our main result is that lossless multiple release is possible for a large class of additive noise mechanisms. For the Gaussian mechanism we give a simple method for lossless multiple release with a short, self-contained analysis that does not require knowledge of the mathematics of Brownian motion. We also present lossless multiple release for the Laplace and Poisson mechanisms. Finally, we consider how to efficiently do gradual release of sparse histograms, and present a mechanism with running time independent of the number of dimensions.

In deployments of differential privacy, it may be hard to determine the appropriate privacy budget to grant an analyst since it depends on trust and accuracy assumptions that may change over time. A system may also have different security clearance levels—a data analyst might have a higher clearance level than a developer, but both might require access to *some* statistics. Similarly, a company that wants to release, for example, user statistics could make an accurate release for their own data analysts, a less accurate release for external consultants, and include an even less accurate release in a report for shareholders or other external actors. A different example setting is users that want to sell their data on data markets: users could let the accuracy of the release depend on how much they are paid, and use different budgets for different releases.

Another usage scenario, relevant in distributed or federated settings, is *local differential privacy* where the data owner could be sharing a differentially private function of their data with multiple servers, and the set of servers might change over time. Here the privacy budget might depend on the server: for example, a

patient may trust their local hospital more than their national hospitals, but might not trust the hospitals not to collude by sharing data among themselves.

MULTIPLE RELEASES These scenarios motivate creating *multiple releases* with different privacy budgets aimed at different analysts. However, these releases should be *coordinated* such that a group of analysts who combine their information do not gain more knowledge about the input than the most knowledgeable member of the group. This kind of collusion resilience was first studied by Xiao, Tao, and Chen [XTC09] with a non-DP privacy objective—we refer to their work for additional motivation.

A related aspect is that we may want to provide an analyst with a less private, more accurate release after the trust we place in them increases. In this case, we want the accuracy of the latest release to match the accuracy that can be obtained, given the combined privacy budget of both releases. That is, no additional cost should be incurred for making two releases rather than one. For example, an external consultant that later gets employed directly by the company should be able to get access to the more accurate release without constituting a privacy violation or requiring an increased privacy budget to reach the same accuracy.

Figuratively speaking, imagine turning a “knob” that controls the privacy level.

BASELINE It is possible to create multiple releases for *any* differentially private mechanism if we are willing to increase the privacy budget by a constant factor. In particular, we can create a sequence of independent releases with geometrically increasing privacy parameters, referred to as “ ϵ -doubling” by Ligett, Neel, Roth, Waggoner, and Wu [Lig+17], and provide each analyst with the most accurate release they are entitled to. Using composition results, the combined privacy budget for the information given to a set of analysts is a constant factor from the highest budget of a single analyst in the group. In this chapter, we study how to make multiple releases in a *lossless* way without accuracy or privacy penalty.

6.0.1 Related Work

Koufogiannis, Han, and Pappas [KHP16] introduced the concept of (lossless) *gradual release*, which makes it possible to increase the privacy budget with *no loss* in accuracy. They also considered privacy tightening for the Laplace mechanism, where successive releases are increasingly private. In the journal version of the paper, they also introduce gradual release for the Gaussian mechanism under approximate DP which is based on the machinery of Brownian motion.

This technique was later applied in a noise reduction framework by Ligett, Neel, Roth, Waggoner, and Wu [Lig+17] with the goal of producing mechanisms with ex-post privacy, i.e., where the privacy budget is set based on what is required to achieve a desired accuracy level. Follow-up work by Whitehouse, Ramdas, Wu, and Rogers [Whi+22] gave results for noise reduction under approximate DP using Brownian motion. These works were motivated by work on privacy filters and odometers [Rog+16], keeping track of privacy budgets over time, rather than multiple release settings. More recently, Pan [Pan24] demonstrated lossless gradual release for *Randomized Response*.

Girgis, Data, Chaudhuri, Fragouli, and Diggavi [Gir+20] studied privacy-utility-randomness trade-offs for a multi-analyst setting under local differential privacy. Here each user has a sample $X_i \sim D$ and makes a single public release of X_i . Each analyst then utilizes the public release (together with shared randomness among the users) to estimate D at a more accurate level than what the public release

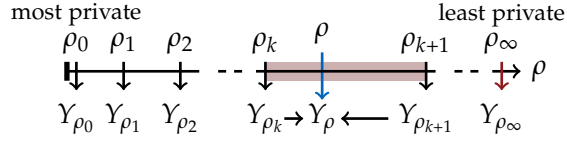


Figure 6.1: The idea behind lossless multiple release. For concreteness we consider additive noise mechanism and zero-concentrated differential privacy. Each Y_{ρ_i} denotes a noisy estimate, and to release a new estimate with $\rho > 0$, we can combine the adjacent estimates for ρ_k and ρ_{k+1} together with some fresh noise to obtain a new release Y_ρ that is exactly ρ -zero concentrated DP (ρ -zCDP). Note that these estimates do not need to be strictly increasing (or decreasing) but can be released in any order. Furthermore, releasing any subset of these estimates is exactly $\max(S)$ - ρ -zCDP, where S is the set of privacy parameters.

alone would allow. Later Zhang and He [ZH23] also used lossless gradual release techniques for the Gaussian mechanism in designing a database system supporting analysts with different privilege levels.

Similar research questions have been investigated outside of the DP literature. Xiao, Tao, and Chen [XTC09] studied releasing a sensitive dataset where each element is kept with probability p , and otherwise sampled uniformly from the universe. Li, Chen, Li, and Zhang [Li+12] considered privatizing data by additive Gaussian noise of scale σ . Both works dealt with arbitrary sequences of parameters (probabilities p or noise scalings σ), and demonstrated how to correlate releases to guarantee that (1) each release matches the single-release case and, (2) limiting the sensitive information derived from combining releases. Our method for adaptively producing Gaussian releases deviates from [Li+12], in that ours does not need to maintain any covariance matrix for past releases. This makes our approach more time- and space-efficient.

Finally, we mention the classic work of Equitz and Cover [EC91] who studied *successive refinement of information* where a source is encoded once and then refined in later stages without decreasing the rate compared to a one-shot optimal code. They showed that this is possible precisely when the sequence of reconstructions can be written as a Markov chain. While their work did not focus on privacy, successive refinement is conceptually related to lossless gradual release. Indeed, parts of the mathematical machinery was independently re-discovered by subsequent work on lossless gradual release.

BASIC TECHNIQUE We illustrate our framework in the setting of additive Gaussian noise and ρ -zero-concentrated differential privacy (ρ -zCDP). We want to release a private estimate of a real-valued query $f : \mathcal{X} \rightarrow \mathbb{R}$ with ℓ_2 -sensitivity $\Delta_2 = 1$ using the Gaussian mechanism. Now consider the simple case where one wants to privately release an estimate with two privacy levels $\rho < \rho'$. Then releasing $Y_\rho = f(\mathbf{x}) + \mathcal{N}(0, \frac{1}{2\rho})$ together with $Y_{\rho'-\rho} = f(\mathbf{x}) + \mathcal{N}(0, \frac{1}{2(\rho'-\rho)})$ is ρ' - ρ -zCDP by composition. That is, instead of using a privacy budget of $\rho + \rho'$ for independently releasing both estimates, the overall budget spent is just the maximum of both, which is ρ' .

To do multiple lossless releases for more general additive noise mechanisms, it turns out that one can always combine existing releases with fresh noise as illustrated in Figure 6.1. In fact, to make any new release with privacy parameter ρ , it suffices to have saved the two “adjacent” releases whose privacy parameters are closest to ρ . Our approach applies more generally to a class of (independent) additive noise mechanisms that support gradual release, but the releases can be

see definitions in
Section 5.1, and
recall that releasing
 $\mathcal{N}(0, 1/(2\rho))$
satisfies ρ -zCDP.

made in any order, and we do not require the set of releases to be known in advance.

OUR CONTRIBUTIONS We introduce a framework for *lossless multiple release*, generalizing past work on gradual release. In addition to allowing for making multiple private releases and having the privacy loss only scale with the least private release, we impose no specific ordering on the privacy parameters used. Furthermore, we formalize a general theorem (Section 6.3.2) showing that lossless multiple release is possible for a large class of mechanisms based on adding i.i.d. noise to each coordinate whose distribution satisfies a *convolution preorder*, as defined next.

Definition 6.1 (Convolution preorder). *A family of real-valued distributions $\mathcal{D}(\rho)$ parameterized by $\rho \in \mathbb{R}_+$ is said to satisfy a convolution preorder, if given $\mathcal{D}(\rho_1)$ and $\mathcal{D}(\rho_2)$ for $\rho_1 < \rho_2$, there exists a distribution $\mathcal{C}(\rho_2, \rho_1)$ such that $\mathcal{D}(\rho_2) * \mathcal{C}(\rho_2, \rho_1) = \mathcal{D}(\rho_1)$.*

This relation is a natural way of stating that the distributions become more noisy as the parameter ρ gets smaller. We are unaware of any existing term in the literature but the definition is closely related to *convolution order* [Sto]. Examples of noise distributions satisfying Definition 6.1 include the Laplace, Poisson, and the Gaussian mechanisms, parameterized by a decreasing function of their variance.

Theorem 6.1 (Meta theorem, informal). *Let $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ be a mechanism that adds independent, identically distributed noise to coordinates of a function $f : \mathcal{X} \rightarrow \mathbb{R}^d$. If the noise distribution of $\mathcal{A}_{f,\rho}$ satisfies convolution preorder, then there exists an algorithm enabling lossless multiple release. Also, any invertible post-processing of $\mathcal{A}_{f,\rho}$ preserves this property.*

For the Gaussian mechanism in particular, we show how to get lossless multiple release by only using basic properties of Gaussians, and we show similar results for the Laplace and Poisson mechanisms from first principles. We give concrete instantiations of *Gaussian sparse histograms* (Section 6.4.2) and *factorization mechanisms* (Section 6.4.1).

6.1 THREAT MODEL AND GOALS

Our setting is a multi-user system with a set of analysts/users S , all able to query the same dataset using a differentially private mechanism. User s has their own security clearance level with corresponding privacy budget ρ_s . This setting allows for multi-level security where each user belongs to a security clearance level, like in the classic Bell-LaPadula model [BLP76], where users with high clearance levels have higher values of ρ . A common example of such clearance levels is using increasing levels with labels such as *public*, *restricted*, *confidential*, and *top secret*. The user with the highest clearance in the system has a privacy budget of $\max_{s \in S}(\rho_s)$, which we denote $\rho_{\max}$.

We consider an adversary who gains partial knowledge, that is, an adversary that sees *some* of the releases. Our goal is to design a differentially private mechanism such that an adversary with access to releases from some users $S' \subseteq S$, can learn *at most* what they could have learned from a release with privacy parameter

2539 $\max_{s' \in S'}(\rho_{s'})$. This means that even by compromising or colluding with more users, the
 2540 adversary's knowledge may not increase. In case an adversary observes all releases
 2541 (e.g., by compromising all users), the privacy loss would be bounded by $\rho_{\max}$.

2542 However, our goal is to design a mechanism where the releases are *lossless*
 2543 in the sense that the noise distribution from multiple releases with a combined
 2544 budget $\rho_{\max}$ would be indistinguishable from one single release with the privacy
 2545 budget $\rho_{\max}$. In other words, the privacy loss should be determined by $\rho_{\max}$,
 2546 while independent releases would usually have privacy parameter $\sum_{s \in S} \rho_s$ due to
 2547 composition.

2548 6.2 GAUSSIAN LOSSLESS MULTIPLE RELEASE

2549 Extending the work in [KHP16; Li+12], we demonstrate next that in the case of
 2550 the Gaussian mechanism, providing lossless multiple release is clean and follows
 2551 immediately from simple properties of Gaussians. Throughout the paper, we use
 2552 the symbol Y to be a random variable that depends on the private dataset and Z
 2553 to be one that does not.

2554 We first consider the gradual release setting where an increasingly accurate
 2555 estimate is released, and the overall privacy loss is determined solely by the latest,
 2556 least private release. In Lemma 6.4, we drop this restriction, allowing releases
 2557 in any order while still guaranteeing that the overall privacy guarantee is the
 2558 maximum ρ value provided. For simplicity, we consider the one-dimensional case,
 2559 where we want to release some query $f : \mathcal{X} \rightarrow \mathbb{R}$. The d -dimensional setting is
 2560 handled by sampling each coordinate independently.

2561 **GETTING STARTED** Foreshadowing the usage of ρ -zCDP, we initially use $1/(2\rho)$
 2562 for denoting the variance of a Gaussian. Our inquiry starts with a basic observation
 2563 about inverse-variance weighting of Gaussians.

Lemma 6.2. Let $Y_\rho \sim \mathcal{N}(\beta, \frac{1}{2\rho})$ and $Y_{\rho'} \sim \mathcal{N}(\beta, \frac{1}{2\rho'})$ where $\beta \in \mathbb{R}$ and $\rho, \rho' > 0$.
 Then,

$$Y = \frac{\rho}{\rho + \rho'} Y_\rho + \frac{\rho'}{\rho + \rho'} Y_{\rho'} \sim \mathcal{N}\left(\beta, \frac{1}{2(\rho + \rho')}\right).$$

2564 *Proof.* The mean of Y is immediate by the fact that Y is a weighted-average of Y_ρ
 2565 and $Y_{\rho'}$, both of mean β . For the variance, direct computation yields:

$$\text{Var}[Y] = \frac{\rho^2/(2\rho) + \rho'^2/(2\rho')}{(\rho + \rho')^2} = \frac{1}{2(\rho + \rho')}.$$

2566 As the sum of two Gaussians is itself Gaussian, we are done. $\square$

2567 The essence of what is being claimed is that a Gaussian of variance $\frac{1}{2\rho}$ and another
 2568 Gaussian of variance $\frac{1}{2\rho'}$ with the same mean can be combined into a new Gaussian
 2569 with variance $\frac{1}{2(\rho + \rho')}$ and the same mean. Inspired by this, we can repeatedly invoke
 2570 the lemma for the following result.

Lemma 6.3. Given $0 < \rho_1 < \dots < \rho_m$ and $\beta \in \mathbb{R}$ define $Y_{\rho_1}, \dots, Y_{\rho_m} \in \mathbb{R}$ where $Y_{\rho_1} \sim \mathcal{N}(\beta, \frac{1}{\rho_1})$, and for $k > 1$:

$$Y_{\rho_{k+1}} = \frac{\rho_k}{\rho_{k+1}} Y_{\rho_k} + \frac{\rho_{k+1} - \rho_k}{\rho_{k+1}} \cdot \mathcal{N}\left(\beta, \frac{1}{2(\rho_{k+1} - \rho_k)}\right).$$

Then for any $i \in [m] : Y_{\rho_i} \sim \mathcal{N}(\beta, \frac{1}{2\rho_i})$ and for any $j \in [m] : \text{Cov}(Y_{\rho_i}, Y_{\rho_j}) = \frac{1}{2\max(\rho_i, \rho_j)}.$

Proof. We first show the distribution of Y_{ρ_k} by induction on k . Assume that $Y_{\rho_k} \sim \mathcal{N}(\beta, \frac{1}{\rho_k})$, which is true for the base case of $k = 1$. Note that the inductive step follows immediately from invoking Lemma 6.2. For the covariance, note that we can expand the expressions inside the covariance and “throw away” the independent noise added in each recurrence. Assuming $i \leq j$ we get $\text{Cov}(Y_{\rho_i}, Y_{\rho_j}) = \text{Cov}(Y_{\rho_i}, \frac{\rho_i}{\rho_j} Y_{\rho_i}) = \frac{\rho_i}{\rho_j} \text{Var}[Y_{\rho_i}] = \frac{1}{2\rho_j}$, implying the stated covariance. $\square$

RELEASES IN ARBITRARY ORDER The Gaussian sequence in Lemma 6.3 will be the basis for our lossless multiple release version of the Gaussian mechanism. What the lemma does not address is generating the sequence in arbitrary order: Statically, given the full sequence $(\rho_k)_{k \in [m]}$, we can generate the Gaussians, but what if we receive them one-by-one and in arbitrary order? We will address this next.

Lemma 6.4. For $\rho_\infty \in \mathbb{R}_+ \cup \{\infty\}$ and $\beta \in \mathbb{R}$, let $M = \{(0, \infty), (\rho_\infty, Y_{\rho_\infty})\}$ where $Y_{\rho_\infty} \in \mathcal{N}(\beta, \frac{1}{2\rho_\infty})$. Consider a finite subset $S \subset (0, \rho_\infty)$ and the following process that runs for $|S|$ iterations:

1. Pick an arbitrary $\rho \in S$ and delete it from S ;
2. Let $(\rho_l, Y_{\rho_l}), (\rho_r, Y_{\rho_r}) \in M$ where $\rho \in (\rho_l, \rho_r)$ and $\rho_r - \rho_l$ is minimal;
3. Sample $Z \sim \mathcal{N}\left(0, \frac{(1 - \rho_l/\rho)(1/\rho - 1/\rho_r)}{2(1 - \rho_l/\rho_r)}\right)$, let

$$Y_\rho = \frac{1 - \rho_l/\rho}{1 - \rho_l/\rho_r} Y_{\rho_r} + \frac{\rho_l/\rho - \rho_l/\rho_r}{1 - \rho_l/\rho_r} Y_{\rho_l} + Z,$$

and add (ρ, Y_ρ) to M .

Then the sequence of random variables generated by the process has the same distribution as described in Lemma 6.3.

Proof sketch. The argument is inductive. Under the hypothesis that all values generated up to the given point have the distribution described by Lemma 6.3, we argue that the newly generated value does too. The argument considers the four different cases for ρ_l, ρ_r , e.g., $\rho_l = 0, \rho_r \neq \rho_\infty$. As the proof involves tedious computation we refer the reader to Section 6.6.3 for the formal proof. $\square$

2589 **FORMALIZING LOSSLESS MULTIPLE RELEASE** Lemma 6.4 will constitute the
 2590 basis for our implementation of lossless multiple release, but we have yet to
 2591 formally define this notion. We do so next.

Definition 6.2 (Lossless multiple release). Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ be a family of mechanisms on a domain $\mathcal{X}$, indexed by a privacy parameter $\rho \in \mathbb{R}_+$. We say that $M : \mathcal{X} \times \mathbb{R}_+ \rightarrow \mathcal{Y}$ implements $\mathcal{M}_\rho$ with lossless multiple release if for every $x \in \mathcal{X}$ it satisfies:

1. $\forall \rho: M(x, \rho)$ and $\mathcal{M}_\rho(x)$ are identically distributed.
2. For every finite subset $S \subset \mathbb{R}_+$, processed in arbitrary order by M , and $y \in \mathcal{Y}$, conditioned on $M(x, \max(S)) = y$ the joint distribution of $(M(x, \rho))_{\rho \in S}$ is uniquely determined by y and S .

2592 Functionally, a mechanism meets the definition if its outputs can be correlated such
 2593 that for any set of outputs, their joint distribution can be viewed as (randomized)
 2594 post-processing of the least private release. An implementation necessarily has to
 2595 store information about releases that have been made, i.e., the sequence of inputs
 2596 to M and the corresponding outputs, to fulfill the requirement that releases for
 2597 different privacy parameters are correlated. When M only supports outputting
 2598 releases for a sequence of *increasing* privacy parameters, we call it *lossless gradual*
 2599 *release*.

2600 Having stated the definition for lossless multiple release, consider Algorithm 9.
 2601 It implements Lemma 6.4, and the idea is visualized in Figure 6.1.

Algorithm 9 GaussianMultipleRelease

Parameters: ℓ_2 sensitivity Δ_2

Require: Set of releases M , privacy parameter ρ

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that
 - 2: $\rho \in [\rho_k, \rho_{k+1})$ and $\forall (\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$
 - 3: Sample $Z_\rho \sim \mathcal{N}(0, \Delta_2^2 \cdot \frac{(1-\rho_k/\rho) \cdot (1/\rho - 1/\rho_{k+1})}{2(1-\rho_k/\rho_{k+1})})$
 - 4: Let $Y_\rho \doteq Z_\rho + \frac{(1-\rho_k/\rho)Y_{\rho_{k+1}} + (\rho_k/\rho - \rho_k/\rho_{k+1})Y_{\rho_k}}{1-\rho_k/\rho_{k+1}}$
 - 5: Add $M = M \cup \{(\rho, Y_\rho)\}$
 - 6: **Return** Y_ρ
-

2602 **Corollary 6.5.** With M initialized as $M = \{(0, \infty), (\infty, f(x))\}$, Algorithm 9 implements
 2603 the Gaussian mechanism with lossless multiple release.

2604 *Proof.* Let $\{Y_\rho\}_{\rho \in S}$ be the set of outputs produced by Algorithm 9 on receiving
 2605 the set S of privacy parameters in arbitrary order. Observe that the algorithm is
 2606 implementing the (adaptive) sampling in Lemma 6.4, and so it produces outputs
 2607 with the same distribution as Lemma 6.3. Property 1 of Definition 6.2 follows
 2608 immediately from observing that $Y_\rho \sim \mathcal{N}(f(x), \frac{1}{2\rho})$. For property 2, note that every
 2609 release in Lemma 6.3 can be viewed as randomized post-processing of the least
 2610 private release. One way to see this is to note that for any two releases Y_ρ and $Y_{\rho'}$
 2611 where $\rho < \rho'$, we have that $\text{Cov}(Y_\rho, Y_{\rho'}) = \text{Var}[Y_{\rho'}]$, implying that $Y_\rho = Y_{\rho'} + Z$ for
 2612 aptly scaled zero-mean Gaussian noise Z . $\square$

6.3 EXTENDING TO INDEPENDENT ADDITIVE NOISE

We will next show that lossless multiple release holds for a larger class of mechanisms. To proceed we recall the notion of an Independent Additive Noise Mechanism (IANM).

Definition 6.3. We define an IANM $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ as a mechanism of the form

$$\mathcal{A}_{f,\rho}(\mathbf{x}) = f(\mathbf{x}) + Z, \quad Z \sim \mathcal{D}(\rho),$$

where $\mathcal{D}(\rho)$ is a probability distribution parameterized in ρ , that draws a d -dimensional vector with i.i.d. samples.

It turns out that any IANM mechanism $\mathcal{A}_{f,\rho}$ satisfying Definition 6.1 supports lossless multiple release.

Lemma 6.6. Any IANM mechanism $\mathcal{A}_{f,\rho}$ with noise distribution $\mathcal{D}(\rho)$ satisfying convolution preorder can be implemented with lossless multiple release.

Proof. The proof is for the one-dimensional case, but the same argument can be invoked for the multidimensional case as each coordinate has independent noise. We begin by proving that given $S = \{\rho_k : k \in [m]\} \subset \mathbb{R}^+$, where $\rho_1 < \dots < \rho_m$, it is possible to construct a set of releases $\{Y_{\rho_k} : k \in [m]\}$ that satisfy Definition 6.2. By Definition 6.1, it is possible to express each Y_{ρ_k} as

$$Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_m} + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}, \quad (1)$$

where $Z_{\rho_m} \sim \mathcal{D}(\rho_m)$ and $W_{\rho_{j+1}, \rho_j} \sim \mathcal{C}(\rho_{j+1}, \rho_j)$ are sampled independently. To prove that $Y_{\rho_k} \stackrel{d}{=} \mathcal{A}_{f,\rho_k}(\mathbf{x})$, observe that Y_{ρ_k} is distributed as $f(\mathbf{x})$ plus a random variable drawn from $\mathcal{D}(\rho_m) * \mathcal{C}(\rho_m, \rho_{m-1}) * \dots * \mathcal{C}(\rho_{k+1}, \rho_k) = \mathcal{D}(\rho_k)$, as needed. For the second property in Definition 6.2, observe that conditioning on $Y_{\rho_m} = y$ we can express every other Y_{ρ_k} for $k \in [m-1]$ as

$$Y_{\rho_k} = y + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}.$$

As a result, the joint distribution on $(Y_{\rho_1}, \dots, Y_{\rho_m})_{|Y_{\rho_m}=y}$ is uniquely determined by y , proving the second property.

We will now argue (inductively) that we can produce these releases adaptively. Let $S_1 \subset S_2 \dots \subset S_m = S$ be a sequence of subsets of S where $|S_i| = i$. For the base case of S_1 we can make a single release using $\mathcal{A}_{f,\rho}$. For our inductive hypothesis, assume we have produced releases corresponding to all the privacy parameters in S_n . Now, for S_{n+1} , we re-label the privacy parameters such that $S_{n+1} = \{\rho_k\}_{k \in [n+1]}$ where $\rho_1 < \dots < \rho_{n+1}$. For the unique $\rho_i \in S_{n+1} \setminus S_n$, we can conditionally sample it from the joint distribution over $(Y_{\rho_k})_{k \in [n+1]}$, conditioned on the value of each release made in the previous round. By induction, it follows that we can *adaptively* release S . $\square$

2640 6.3.1 *Sampling in Concrete Settings*

2641 Lemma 6.6 proves the existence of a sampling procedure for lossless multiple
 2642 release. In Section 6.2 we showed a sampling procedure in the particular case of
 2643 Gaussian noise. In Section 6.6.1 we show that the structure of Algorithm 9 holds
 2644 for general IANM mechanisms. Namely, if the privacy parameters we support come
 2645 from a bounded range $(\rho_0, \rho_\infty) \subset \mathbb{R}_+$, then the corresponding algorithm has a
 2646 similar structure (see Algorithm 13).

2647 More precisely, consider two neighboring releases Y_{ρ_k} and $Y_{\rho_{k+1}}$ with noise
 2648 parameters ρ_k and ρ_{k+1} , respectively, and a new lossless release Y_ρ with parameter
 2649 $\rho \in [\rho_k, \rho_{k+1}]$. We can use (1) on this set of $m+1$ releases and condition on the
 2650 values of the m previous releases $Y_{\rho_1}, \dots, Y_{\rho_m}$. Next, write $W_{\rho_{k+1}, \rho_k} = W_1 + W_2$ where
 2651 $W_1 = Y_\rho - Y_{\rho_{k+1}}$ and $W_2 = Y_{\rho_k} - Y_\rho$. In Section 6.6.1, we show that sampling Y_ρ can
 2652 be reduced to the following task:

$$\text{Sample } W_1 \text{ conditioned on } W_1 + W_2 = Y_{\rho_k} - Y_{\rho_{k+1}}. \quad (2)$$

2653 **EXAMPLE: POISSON MECHANISM** Consider the IANM mechanism using the
 2654 Poisson distribution, $\text{Poi}(\lambda)$. This mechanism has the property that noise is always
 2655 a non-negative integer, making it a natural noise distribution for integer vectors
 2656 in settings where negative noise is undesirable. Appendix 6.6.5 states some basic
 2657 privacy properties of the Poisson mechanism.

2658 The family of Poisson distributions parameterized by $\rho = 1/\lambda$ satisfies convo-
 2659 lution preorder since for any $\lambda_1 > \lambda_2$, $\text{Poi}(\lambda_1) = \text{Poi}(\lambda_1 - \lambda_2) * \text{Poi}(\lambda_2)$. To adap-
 2660 tively perform private lossless multiple release of a value $f(\mathbf{x})$ using the Poisson
 2661 mechanism and parameters $\lambda_1 > \dots > \lambda_m$ we notice that the “bridging” distri-
 2662 bution in the k^{th} term of the sum in (1) has distribution $\text{Poi}(\lambda_k - \lambda_{k+1})$. Note that
 2663 without conditioning on $Y_{\rho_1}, \dots, Y_{\rho_m}$, $W_1 \sim \text{Poi}(\lambda - \lambda_{k+1})$ and $W_2 \sim \text{Poi}(\lambda_k - \lambda)$.
 2664 By Lemma 6.22 in the appendix we have that the sampling in (2) reduces to
 2665 $W_1 \sim \text{Bin}(N, p)$ for $N = Y_{\rho_k} - Y_{\rho_{k+1}}$ and $p = (\lambda - \lambda_{k+1})/(\lambda_k - \lambda_{k+1})$.

2666 **EXAMPLE: LAPLACE MECHANISM** Koufogiannis, Han, and Pappas [KHP16]
 2667 have already shown that the Laplace distribution parameterized by $\rho = 1/b$ satisfies
 2668 convolution preorder for any scale parameters $b_1 > b_2$. Let $\text{LapBridge}(b_2, b_1)$ be
 2669 the probability distribution that draws 0 with probability b_2^2/b_1^2 and from $\text{Lap}(0, b_1)$
 2670 with the remaining probability. Then $\text{Lap}(0, b_2) * \text{LapBridge}(b_2, b_1) = \text{Lap}(0, b_1)$
 2671 exactly. To implement lossless release via the sampling in (2), it turns out that the
 2672 conditional distribution is a mixture $W_1 \sim \text{LapBridge}(b, b_2)$ of three distributions:
 2673 With some probability W_1 is equal to either 0 or $Y_{\rho_k} - Y_{\rho_{k+1}}$, and otherwise it is
 2674 sampled from the convolution of two Laplace distributions. We also show that the
 2675 related exponential distribution satisfies convolution preorder, see Section 6.6.6 for
 2676 the full details.

2677 6.3.2 *Lossless Multiple Release as a Blackbox*

2678 Inspired by Corollary 15 in Koufogiannis, Han, and Pappas [KHP16], we provide a
 2679 meta theorem for lossless multiple release based on IANM mechanisms. The central
 2680 component is the following lemma, showing that the class of lossless multiple
 2681 release mechanisms is closed under invertible post-processing.

Lemma 6.7. *Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ satisfy lossless multiple release, and let $H : \mathcal{Y} \rightarrow \mathcal{Y}'$ be an invertible function. Then $\mathcal{M}'_\rho = H \circ \mathcal{M}_\rho$ also satisfies lossless multiple release.*

Proof. Let $M : \mathcal{X} \times \mathbb{R}_+ \rightarrow \mathcal{Y}$ be an implementation of $\mathcal{M}_\rho$ satisfying lossless multiple release, and let $M' = H \circ M$, which we will argue implements lossless multiple release. The only property that does not trivially hold for M' is the second property of Definition 6.2. For a set $S \subset \mathbb{R}_+$, note that to condition on $M'(\mathbf{x}, \max(S)) = y$ is equivalent to conditioning on $H^{-1}(M'(\mathbf{x}, \max(S))) = M(\mathbf{x}, \max(S)) = H^{-1}(y) \in \mathcal{Y}$. We thus have that conditioning $M'(\mathbf{x}, \max(S)) = y$ implies that the joint distribution over $(M(\mathbf{x}, \rho))_{\rho \in S}$ is fully determined, and consequently,

$$(H(M(\mathbf{x}, \rho)))_{\rho \in S} = (M'(\mathbf{x}, \rho))_{\rho \in S}.$$

This finishes the proof. $\square$

Theorem 6.8 (Meta theorem). *Let $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$ be a family of mechanisms on a domain $\mathcal{X}$, indexed by a privacy parameter $\rho \in \mathbb{R}_+$. Furthermore, assume $\mathcal{M}_\rho$ can be decomposed as $\mathcal{M}_\rho = H \circ \mathcal{A}_{f,\rho}$ where*

- $\mathcal{A}_{f,\rho} : \mathcal{X} \rightarrow \mathbb{R}^d$ is an independent add. noise mechanism for releasing $f : \mathcal{X} \rightarrow \mathbb{R}^d$ with privacy parameter ρ and with noise distribution satisfying a convolution preorder (Definition 6.1);
- $H : \mathbb{R}^d \rightarrow \mathcal{Y}$ is an invertible post-processing step;

Then $\mathcal{M}_\rho$ can be implemented with lossless multiple release.

Proof. The theorem follows from invoking Lemma 6.6 together with Lemma 6.7. $\square$

SUPPORTING NON-INVERTIBLE POST-PROCESSING To support non-invertible post-processing, we also introduce a weaker notion: *weakly lossless multiple release*.

Definition 6.4 (Weakly Lossless Multiple Release). *A mechanism $\mathcal{M}_\rho$ supports weakly lossless multiple release if it can be written as $\mathcal{M}_\rho = H \circ \mathcal{M}'_\rho$ where $\mathcal{M}'_\rho$ supports lossless multiple release and H is an arbitrary function (possibly chosen from some distribution).*

We define *weakly lossless gradual release* analogously. It follows that these classes of mechanisms are closed under all post-processing. While this notion is indeed weaker, any algorithm $M(\mathbf{x}, \rho)$ implementing weakly lossless multiple release for a ρ -private mechanism $\mathcal{M}_\rho(\mathbf{x})$, will have the property that the set of releases $(M(\mathbf{x}, \rho))_{\rho \in S}$ are $\max(S)$ -private, if ρ -privacy is closed under post-processing. Examples of such privacy notions are ϵ -DP and ρ -zCDP. We get the following immediate corollary to Theorem 6.8.

Corollary 6.9. *If the function H in Theorem 6.8 is not invertible, then $\mathcal{M}_\rho = H \circ \mathcal{A}_{f,\rho}$ supports weakly lossless multiple release.*

Weakly lossless multiple release will play a role in the next section, where we consider non-invertible post-processing such as truncation.

2705 6.4 APPLICATIONS

2706 In this section, we describe two applications supported by our framework for
2707 lossless multiple release:

- 2708 • Factorization mechanisms [Li+15], where we want to privately release a
2709 linear query $A\mathbf{x}$, and
- 2710 • Sparse Gaussian histograms, also known as stability histograms [Wil+24;
2711 GAT20].

2712 We will make use of both *lossless* multiple release (Definition 6.2), and its weaker
2713 variant (Definition 6.4). This will be necessary since, e.g., the truncation used
2714 for sparse Gaussian histograms is not an invertible function, and so not covered
2715 by Theorem 6.8.

2716 Because the noise generation for histograms on large domains is expensive, we
2717 give a dimension-independent algorithm that works in the gradual release setting.
2718 Throughout, we state our results as post-processings of the Gaussian mechanism,
2719 but analogous results hold for any other IANM mechanism meeting Definition 6.1.

2720 6.4.1 Lossless Multiple Release Factorization Mechanism

2721 Let A be a query matrix and fix a public factorization $A = LR$. Denote the *factor-*
2722 *ization mechanism* [Li+15] as $\mathcal{F}_\rho(\mathbf{x}) = A\mathbf{x} + L\mathbf{z} = L(R\mathbf{x} + \mathbf{Z})$ on some private dataset
2723 $\mathbf{x}$ where $\mathbf{Z}$ is a vector drawn from a distribution $\mathcal{D}(\rho)$ satisfying Definition 6.1
2724 parameterized by ρ , typically Gaussian or Laplace noise.

Lemma 6.10. $\mathcal{F}_\rho$ can be implemented with weakly lossless multiple release, and
lossless multiple release if L has a left-inverse.

2725 *Proof.* Observe that $\mathcal{F}_\rho(\mathbf{x}) = L \circ \mathcal{A}_{R,\rho}$ for an IAN mechanism $\mathcal{A}_{R,\rho}(\mathbf{x}) = R\mathbf{x} + \mathbf{Z}$ where
2726 $\mathbf{Z} \sim \mathcal{D}(\rho)$ for a $\mathcal{D}(\rho)$ satisfying Definition 6.1. The result follows from Theorem 6.8
2727 and Corollary 6.9, and noting that the linear map L is invertible exactly when L
2728 has a left-inverse. $\square$

2729 Besides proving existence in Lemma 6.10, we also give an explicit instantiation
2730 in Algorithm 10 using the Gaussian mechanism.

Algorithm 10 FactorizationMultipleRelease

Parameters: Factorization $A = LR$, ℓ_2 sensitivity Δ_2

Require: Set of releases M , privacy parameter ρ

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that
 - 2: $\rho \in [\rho_k, \rho_{k+1})$ and $\forall(\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$.
 - 3: $Z_\rho \sim \mathcal{N}\left(0, \Delta_2^2 \cdot \frac{(1-\rho_k/\rho) \cdot (1/\rho - 1/\rho_{k+1})}{2(1-\rho_k/\rho_{k+1})}\right)^d$.
 - 4: Let $Y_\rho \doteq L \cdot Z_\rho + \frac{(1-\rho_k/\rho)Y_{\rho_{k+1}} + (\rho_k/\rho - \rho_k/\rho_{k+1})Y_{\rho_k}}{1-\rho_k/\rho_{k+1}}$.
 - 5: Set $M = M \cup \{(\rho, Y_\rho)\}$.
 - 6: **Return** Z .
-

Lemma 6.11. *With initialization $M = \{(0, \infty), (\infty, A\mathbf{x})\}$, Algorithm 10 implements the Gaussian noise factorization mechanism $\mathcal{F}_\rho$ with (weakly) lossless multiple release.*

Proof sketch. Note that the algorithm is practically a copy of Algorithm 9, but for the specific sensitive function $R\mathbf{x}$. The only structural difference is that the linearity of the post-processing L allows for storing correlated Gaussian noise directly in M . $\square$

6.4.2 Weakly Lossless Multiple Release of Histograms

We will now describe an example where the post-processing is neither linear nor invertible: releasing sparse (Gaussian) histograms [Kor+09; GAT20; BW18]. Let $\mathbf{x} = (X_i)^n$ be a dataset of n users, and we want to privately release the histogram $H(\mathbf{x}) = \sum_{i=1}^n X_i$ where $X_i \in \{0, 1\}^d$ and a user can contribute to l distinct counts and the Gaussian mechanism which scales proportional to $\sqrt{l}$ is preferable. In many natural settings, the domain size d can be very large, and therefore, the resulting histogram is usually very sparse, $k = \|H(\mathbf{x})\|_0 \ll d$. Releasing a noisy histogram where noise has been added to *each* coordinate would destroy the sparsity. One can cope by introducing a threshold $\tau > 0$ where only noisy counts that exceed τ are released. τ is usually set high enough such that the noise to a zero count will (after thresholding with high probability) still be zero. We denote the *support* of the histogram as $U(H(\mathbf{x})) = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ as the index of the non-zero coordinates. Define the thresholding function $T_\tau : \mathbb{R}^d \rightarrow \mathbb{R}^d$ as:

$$T_\tau(\mathbf{x}) = (y_i)_{i=1}^d, \quad \text{where } y_i = \begin{cases} x_i, & \text{if } x_i \geq \tau \\ 0, & \text{otherwise} \end{cases}.$$

Denote the sparse histogram mechanism as $\mathcal{H}_\rho(\mathbf{x}) = T_\tau(H(\mathbf{x}) + Z)$ where $Z \sim \mathcal{D}(\rho)$ is a distribution satisfying a convolution preorder (Definition 6.1).

Lemma 6.12. *$\mathcal{H}_\rho$ can be implemented with weakly lossless multiple release.*

Proof. Observe that $\mathcal{H}_\rho$ can be expressed as a composite function $T_\tau \circ \mathcal{A}_{\rho,f}(\mathbf{x})$ for an IANM mechanism $\mathcal{A}_{\rho,f}(\mathbf{x})$ meeting Definition 6.1. The statement thus follows from Corollary 6.9. $\square$

It is straightforward to implement weakly lossless gradual release for stability histograms using our framework with time and space complexity linear in dimension d ; see Algorithm 11. The following lemma is proved in Section 6.6.4.

Lemma 6.13. *With initialization $M = \{(0, 0)\}$, Algorithm 11 implements $\mathcal{H}_\rho$ with weakly lossless gradual release.*

Algorithm 11 HistogramGradualRelease

Parameters: ℓ_2 sensitivity Δ_2

Require: histogram $H(\mathbf{x})$, set of releases M ,
privacy parameter ρ , threshold τ

- 1: Extract the single element $(\rho', Z') \in M$.
 - 2: Sample $\tilde{Z} \sim \mathcal{N}\left(0, \frac{1}{2}\Delta_2^2(\rho - \rho')\right)^d$.
 - 3: Let $Z = \frac{\rho'}{\rho}Z' + \frac{1}{\rho}\tilde{Z}$.
 - 4: **for** each $i \in [d]$ **do**
 - 5: **if** $H(\mathbf{x})_i + Z_i > \tau$ **then** $Y_i = H(\mathbf{x})_i + Z_i$.
 - 6: **else** $Y_i = 0$.
 - 7: Set $M = \{(\rho, Z)\}$
 - 8: **Return** Y
-

2757 IMPROVING EFFICIENCY FOR WEAKLY LOSSLESS GRADUAL RELEASE Ko-
 2758 rolova, Kenthapadi, Mishra, and Ntoulas [Kor+09] showed that under approximate
 2759 differential privacy, one can skip the noise generation for zero counts if thresh-
 2760 olding is enabled, adding only a small probability for infinite privacy loss if this
 2761 coordinate is non-zero in a neighboring dataset. Unfortunately, this approach
 2762 does not fit our meta theorem, so instead, we apply a technique due to Cormode,
 2763 Procopiuc, Srivastava, and Tran [Cor+12] that efficiently simulates the noise dis-
 2764 tribution of those zero counts that would exceed the threshold and thus gives an
 2765 identical distribution to the truncated noisy histogram. The procedure is described
 2766 in the following lemma.

Lemma 6.14. *For a fixed noise distribution $\mathcal{D}$, the output distribution of releasing $\mathcal{T}_\tau(H(\mathbf{x}) + Z)$ with $Z \sim \mathcal{D}$ are independent noise samples is equal to the following process: (a) add noise to every non-zero coordinate in $H(\mathbf{x})_i$ for $i \in U(H(\mathbf{x}))$ and then (b) draw $q \sim \text{Binomial}(d - k, p)$ where $p = \mathbf{P}_{Z \sim \mathcal{D}}[Z > \tau]$. (c) Sample a subset $Q \subseteq [d] \setminus U(H(\mathbf{x}))$ of size $|Q| = q$ uniformly at random and (d) set the noise for $H(\mathbf{x})$ to $Z \sim \mathcal{D}$ conditioned on being above the threshold τ .*

2767 *Proof sketch.* A formal argument can be found in Cormode, Procopiuc, Srivastava,
 2768 and Tran [Cor+12], here we just provide a sketch. It is clear that every entry in
 2769 the support gets the correct output distribution. Furthermore, the number of zero
 2770 counts q whose noise exceeds τ follows a binomial distribution, and we can add
 2771 conditional noise to a uniformly drawn subset of size q . $\square$

2772 Algorithm 14 in Section 6.6.4 implements the efficient routine described by
 2773 Lemma 6.14 under weakly lossless gradual release. The key challenge is that the
 2774 probability for a histogram count to exceed the threshold in a given round now
 2775 depends on whether it exceeded the threshold in any prior round. Intuitively,
 2776 all zero counts that have never exceeded the threshold have an equal probability
 2777 to exceed the threshold in any given round, and so the simulation technique

in Lemma 6.14 can be used for these counts (with different probabilities and conditional distributions). However, once any zero count exceeds the threshold, it will have to be treated the same as non-zero counts in all future rounds.

The utility and privacy of Algorithm 14 is proved by arguing that its output distribution matches that of Algorithm 11. A proof sketch for the following lemma is given in Section 6.6.4.

Lemma 6.15. *Let $H(\mathbf{x})$ be a histogram over $[d]$, $\rho_1, \dots, \rho_m$ be a sequence increasing of privacy budgets, $\tau_1, \dots, \tau_m$ be a sequence of thresholds and $\Delta_2 > 0$ the ℓ_2 -sensitivity of the histogram. Also let the sequences of outputs $(Y^{(1)}, \dots, Y^{(m)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(m)})$ be derived from running Algorithm 11 and Algorithm 14 respectively with the preceding parameters as input. Then the sequences $(Y^{(1)}, \dots, Y^{(m)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(m)})$ are identically distributed.*

The benefit of the efficient sampling technique is that the number of sampled Gaussians will *never* exceed that of the naïve approach and can potentially be in the order of the sparsity depending on the parameter regime.

Lemma 6.16. *Let c be the number of zero-counts in the histogram output by Algorithm 14 least once over its execution. Then Algorithm 14 will sample $(k + c)m$ (truncated or otherwise) Gaussians.*

We again refer the reader to Section 6.6.4 for a proof.

6.5 CONCLUSION AND OPEN QUESTIONS

We have initiated a systematic study of differential privacy with multiple releases, motivated by settings in which many levels of privacy or trust may co-exist. The main message is that it is possible to generalize a large class of lossless gradual release techniques to this setting, even when releases are determined online and in no particular order. For mechanisms based on Gaussian, Laplace, or Poisson additive noise we give simple and efficient sampling procedures for creating new lossless releases. In particular, we are able to do lossless multiple release for any factorization mechanism using invertible matrices. Finally, we consider algorithmic challenges related to lossless gradual release and show that private sparse histograms may be computed much more efficiently than what a direct application of our general results would imply.

There are still many open questions concerning mechanisms that do not inherit their privacy guarantee from an IANMM mechanism. In particular, it would be interesting to determine under which conditions the exponential mechanism [MT07] supports lossless multiple release. Koufogiannis, Han, and Pappas [KHP16] conjecture that this is always possible. Other central private algorithms, such as report noisy max [DR14] also do not have known lossless multiple release mechanisms, even in the gradual release setting. A final challenge we would like to mention is implementing our mechanisms on a finite computer, e.g., creating a multiple release version of the discrete Gaussian mechanism [CKS20]. An appealing approach would be to base this on Poisson noise, which meets the technical conditions for our framework and approaches the Gaussian distribution in the limit.

Algorithm 12 GenericMultipleRelease

Parameters: Noise distribution family $\mathcal{D}(\rho)$ satisfying a convolution preorder, bridging noise distribution $\mathcal{C}(\rho, \rho')$ where $\mathcal{D}(\rho') * \mathcal{C}(\rho', \rho) = \mathcal{D}(\rho)$ for $0 < \rho < \rho'$.

Require: Sensitive value $f(\mathbf{x})$, new privacy parameter ρ_k , set of past releases $M = \{(\rho_i, Y_{\rho_i}) : i \in [m] \setminus \{k\}\}$

- 1: **if** $M = \emptyset$ **then**
- 2: Sample Z_{ρ_k} from $\mathcal{D}(\rho_k)$.
- 3: Set $Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_k}$.
- 4: **else if** $k = 1$ **then**
- 5: Sample W_{ρ_{k+1}, ρ_k} from $\mathcal{C}(\rho_{k+1}, \rho_k)$.
- 6: Set $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$.
- 7: **else if** $k \in (1, m)$ **then**
- 8: Draw $W_{\rho_{k+1}, \rho_k} \sim \mathcal{C}(\rho_{k+1}, \rho_k)$ conditioned on $W_{\rho_{k+1}, \rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - Y_{\rho_{k+1}}$.
- 9: Set $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$.
- 10: **else if** $k = m$ **then**
- 11: Sample Z_{ρ_k} from $\mathcal{D}(\rho_k)$ conditioned on $Z_{\rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - f(\mathbf{x})$.
- 12: Set $Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_k}$.
- 13: Set $M = M \cup \{(\rho_k, Y_{\rho_k})\}$.
- 14: **Return** Y_{ρ_k} .

2811 6.6 APPENDIX

2812 6.6.1 On Lossless Multiple Release Sampling

2813 Lemma 6.6 makes no claim about how easy it is to sample noise from the condi-
 2814 tional noise distribution, but guarantees its existence. We will use this section for
 2815 discussing this sampling in greater detail. Consider the joint distribution of the
 2816 releases defined by Equation (1) from the proof of Lemma 6.6, re-stated below for
 2817 convenience:

$$Y_{\rho_k} = f(\mathbf{x}) + Z_{\rho_m} + \sum_{j=k}^{m-1} W_{\rho_{j+1}, \rho_j}, \quad k = 1, \dots, m, \quad (1)$$

2818 where Y_{ρ_k} is the ρ_k -private release, $Z_{\rho_m} \sim \mathcal{D}(\rho_m)$ and $W_{\rho_{j+1}, \rho_j} \sim \mathcal{C}(\rho_{j+1}, \rho_j)$. Recall
 2819 that $\mathcal{C}(\rho', \rho)$ is the unique distribution where for any $0 < \rho < \rho'$ we have $\mathcal{D}(\rho) =$
 2820 $\mathcal{D}(\rho') * \mathcal{C}(\rho', \rho)$. As argued for in the proof of Lemma 6.6, to make a new ρ -private
 2821 release, we consider the joint distribution, and sample the new releases conditioned
 2822 on all past releases.

2823 Formally, for a set of privacy parameters $\rho_1 < \dots < \rho_m$, we are interested
 2824 in releasing ρ_k for a single $k \in [m]$, conditioned on the set of past releases
 2825 $M = \{(\rho_i, Y_{\rho_i}) : i \in [m] \setminus \{k\}\}$. We give pseudocode for this in Algorithm 12, and a
 2826 lemma for its correctness.

Lemma 6.17. *Initializing $M = \emptyset$ and then running Algorithm 12 for a set of privacy parameters $\{\rho_i : i \in [m]\}$ (processed in arbitrary order), will produce a set of outputs $\{Y_{\rho_i} : i \in [m]\}$ with distribution given by Equation (1).*

2827 *Proof.* For $M = \emptyset$, we have that $Y_{\rho_k} = f(\mathbf{x}) + \mathcal{D}(\rho_k) = \mathcal{A}_{f, \rho_k}(\mathbf{x})$ on lines 2-3, as
 2828 expected, where $\mathcal{A}$ is our IANM noise mechanism releasing f with ρ -privacy. For

Algorithm 13 SimplifiedGenericMultipleRelease

Parameters: Noise distribution family $\mathcal{D}(\rho)$ satisfying a convolution preorder, bridging noise distribution $\mathcal{C}(\rho, \rho')$ where $\mathcal{D}(\rho') * \mathcal{C}(\rho', \rho) = \mathcal{D}(\rho)$ for $0 < \rho < \rho'$.

Require: Privacy parameter ρ , set of releases M .

- 1: Find $(\rho_k, Y_{\rho_k}), (\rho_{k+1}, Y_{\rho_{k+1}}) \in M$ such that $\rho \in [\rho_k, \rho_{k+1})$ and $\forall(\rho', \cdot) \in M : \rho' \notin (\rho_k, \rho_{k+1})$.
- 2: Sample $W_{\rho_{k+1}, \rho}$ from $\mathcal{C}(\rho_{k+1}, \rho)$ conditioned on $W_{\rho_{k+1}, \rho} + W_{\rho, \rho_k} = Y_{\rho_k} - Y_{\rho_{k+1}}$.
- 3: Set $Y_\rho = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho}$.
- 4: Set $M = M \cup \{(\rho, Y_\rho)\}$.
- 5: **Return** Y_ρ

the remaining cases, assume M contains $m - 1$ releases with the correct joint distribution, and we are generating a new release at privacy level ρ_k . We will argue that Algorithm 12 produces a Y_{ρ_k} whose joint distribution with the releases in M matches (1). For $k = 1$, observe that $Y_{\rho_1} = Y_{\rho_2} + W_{\rho_2, \rho_1}$, and so lines 5-6 are correct. For $1 < k < m$, note that $Y_{\rho_k} = Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$ and $Y_{\rho_{k-1}} = Y_{\rho_k} + W_{\rho_k, \rho_{k-1}}$, which combined allows us to identify the correct conditional distribution. Y_{ρ_k} is given by $Y_{\rho_{k+1}} + W_{\rho_{k+1}, \rho_k}$, conditioned on $W_{\rho_{k+1}, \rho_k} + W_{\rho_k, \rho_{k-1}} = Y_{\rho_{k-1}} - Y_{\rho_{k+1}}$, matching lines 8-9. For the final case of $k = m$, then $Y_{\rho_m} = f(\mathbf{x}) + Z_{\rho_m}$ and we have that $Y_{\rho_{m-1}} = Y_{\rho_m} + W_{\rho_m, \rho_{m-1}}$. It follows that the correct noise distribution is $Y_{\rho_m} = f(\mathbf{x}) + Z_{\rho_m}$, conditioned on $Z_{\rho_m} + W_{\rho_m, \rho_{m-1}} = Y_{\rho_{m-1}} - f(\mathbf{x})$, matching lines 11-12. An inductive argument identical to that given in the proof of Lemma 6.6 completes the proof. $\square$

6.6.2 Simplification and Removing Dependency on the Dataset

Note that $f(\mathbf{x})$ is only used in Algorithm 12 when a more accurate release is generated (lines 1 and 10). In the remaining cases we are only adding noise to past releases. This already allows us to simplify the algorithm, and argue for not having to store $f(\mathbf{x})$ in memory indefinitely. The algorithm reduces to the case (see line 7 of Algorithm 12) where only the two closest releases are combined into a new one. Such an algorithm is given in Algorithm 13, together with Lemma 6.18.

Lemma 6.18. Let $M = \{(\rho_0, Y_{\rho_0}), (\rho_\infty, Y_{\rho_\infty})\}$ for $Y_{\rho_\infty} = \mathcal{A}_{\rho_\infty, f}(\mathbf{x})$, and $Y_{\rho_0} = Y_{\rho_\infty} + W_{\rho_\infty, \rho_0}$. Then running Algorithm 13 with input M and a set of privacy parameters $\{\rho_i : i \in [m]\} \subset (\rho_0, \rho_\infty)$ (processed in arbitrary order), will produce a set of outputs $\{Y_{\rho_i} : i \in [m]\}$ with distribution given by Equation (1). Moreover, the set M will at all times satisfy ρ_∞ -privacy.

Proof. The statement follows from carefully comparing to Algorithm 12. Let $S = (\rho_1, \dots, \rho_m)$ be a sequence of noise values from the lemma statement, and let $O = (Y_{\rho_1}, \dots, Y_{\rho_m})$ be the outputs. Consider a second sequence $S' = (\rho_\infty, \rho_0, \rho_1, \dots, \rho_m)$, and consider the corresponding sequence of outputs $O' = (Y'_{\rho_\infty}, Y'_{\rho_0}, Y'_{\rho_1}, \dots, Y'_{\rho_m})$ from inputting S' and $M' = \emptyset$ to Algorithm 12. We can directly check that $Y_{\rho_0} \stackrel{d}{=} Y'_{\rho_0}$ and $Y_{\rho_\infty} \stackrel{d}{=} Y'_{\rho_\infty}$. For the remaining outputs, note that just before each algorithm is called, their internal states M and M' are also identically distributed. Now, the fact that each $\rho_i \in (\rho_0, \rho_\infty)$ implies that the remaining outputs $Y'_{\rho_1}, \dots, Y'_{\rho_m}$ are generated from lines 10-12 in Algorithm 12. Checking carefully, lines 1-3 in Algorithm 13 are implementing the same routine, and since $M \stackrel{d}{=} M'$, it follows

that $(Y_{\rho_1}, \dots, Y_{\rho_m}) \stackrel{d}{=} (Y'_{\rho_1}, \dots, Y'_{\rho_m})$, and so the statement follows from Lemma 6.17. The last statement on the ρ_∞ -privacy of M follows from Algorithm 13 implements Equation (1), and so each release in M can at any time during execution be viewed as randomized post-processing of Y_∞ . $\square$

Essentially, if we commit to supporting a bounded range of privacy parameters then we get a simpler algorithm. Lemma 6.18 also says something more: If we commit to supporting a lowest level of privacy ρ_∞ , then Algorithm 13 can be implemented in such a way that its internal state is ρ_∞ -private. After initializing M using the sensitive function $f(\mathbf{x})$, we can erase $f(\mathbf{x})$ from memory and Y_∞ will contain enough private information to generate all future releases. This could prove useful in settings where the time between releases is large, and we want to limit the private information leaked if the state of the algorithm were to be compromised. This can be compared with the gradual release setting, where natural implementations would require consistent access to $f(\mathbf{x})$.

6.6.3 Proof for Gaussian Lossless Multiple Release

Proof of Lemma 6.4. Throughout the proof we will ignore the factor 2 in the denominator of the variance. To argue that the values generated by the process match the distribution Lemma 6.3, we will argue for increasing subsets of releases. Let $S_n = \{\rho_k : k \in [n-1]\}$ where $0 < \rho_1 < \dots < \rho_n < \rho_\infty$ be the set of values in S for which we have generated Gaussians at the start of the n^{th} round of the process. Note that we re-label the ρ 's between the rounds such that $\rho_k \in S_n$ always is the k^{th} smallest value in S_n . Our argument will proceed by induction: assume that all the Gaussians $\{Y_{\rho_k} : k \in [n]\}$ generated after n rounds have the distribution given by Lemma 6.3. Then we will show that $\{Y_{\rho_k} : k \in [n-1]\} \cup \{Y_\rho\}$ has the same distribution as predicted by $S_n \cup \{\rho\}$ from invoking Lemma 6.3.

We begin with our base case. For $n = 1$, we have that $\rho_l = 0$ and $\rho_r = \rho_\infty$, and so $Y_\rho = Y_{\rho_\infty} + \mathcal{N}(0, 1/\rho - 1/\rho_\infty)$. Since $Y_{\rho_\infty} \sim \mathcal{N}(\beta, 1/\rho_\infty)$, we have that $Y_\rho \sim \mathcal{N}(\beta, 1/\rho)$, as expected, and so the base case passes.

For $n \geq 2$, we first consider the following cases.

Case 1: $\rho_l = 0, \rho_r = \rho_1 \in S_n$. In this case, $Y_\rho = Y_{\rho_1} + \mathcal{N}(1/\rho - 1/\rho_1)$, and so $Y_\rho \sim \mathcal{N}(\beta, 1/\rho)$. Since $\rho < \rho_1$, we have that $\forall i \in [n-1] : \text{Cov}(Y_\rho, Y_{\rho_i}) = \text{Cov}(Y_{\rho_1}, Y_{\rho_i}) = 1/\rho_i$, as expected for the release with the smallest value in $\{\rho\} \cup S_n$, and so the case is complete.

Case 2: $\rho_l = \rho_{n-1}, \rho_r = \rho_\infty = \infty$. We have to deal with the case where we have set $\rho_\infty = \infty$ separately, as it is a bit of a trick. Note that we get $Y_\rho = \frac{(1-\rho_{n-1})\beta + \rho_{n-1}Y_{\rho_{n-1}}}{\rho} + \mathcal{N}(0, \frac{1-\rho_{n-1}/\rho}{\rho})$, since $Y_{\rho_\infty} = \beta$ in this case. The end result is once more a sum of Gaussians, with mean β , and for the variance we can explicitly compute that $\text{Var}[Y_\rho] = \frac{\rho_{n-1}^2}{\rho_{n-1}^2 \rho^2} - \frac{\rho - \rho_{n-1}}{\rho^2} = 1/\rho$, as expected. For the covariance, for $i \in [n-1] : \text{Cov}(Y_\rho, Y_{\rho_i}) = \frac{\rho_{n-1}}{\rho} \text{Cov}(Y_{\rho_{n-1}}, Y_{\rho_i}) = 1/\rho$, as expected for the largest value in $\{\rho\} \cup S_n$, and so we are done. Now we consider the most general case.

Case 3: $\rho_l \in S_{n-1}$ and $\rho_r \neq \infty$. We begin with computing the variance of Y using the hypothesis $\text{Var}[Y_{\rho_r}] = 1/\rho_r$ and $\text{Var}[Y_{\rho_l}] = 1/\rho_l$

$$\begin{aligned} \text{Var}[Y_\rho] &= \frac{(\rho_l^{-1} - \rho^{-1})(\rho^{-1} - \rho_r^{-1})}{\rho_l^{-1} - \rho_r^{-1}} + \frac{(\rho_l^{-1} - \rho^{-1})^2 \text{Var}[Y_{\rho_r}] + (\rho^{-1} - \rho_r^{-1})^2 \text{Var}[Y_{\rho_l}]}{(\rho_l^{-1} - \rho_r^{-1})^2} \\ &= \frac{1}{(\rho_l^{-1} - \rho_r^{-1})^2} \left[(\rho_l^{-1} - \rho^{-1})(\rho^{-1} - \rho_r^{-1})(\rho_l^{-1} - \rho_r^{-1}) + \rho_r^{-1}(\rho_l^{-1} - \rho^{-1})^2 \right. \\ &\quad \left. + \rho_l^{-1}(\rho^{-1} - \rho_r^{-1})^2 \right] \\ &= \frac{1}{(\rho_l^{-1} - \rho_r^{-1})^2} \cdot \rho^{-1}(\rho_l^{-1} - \rho_r^{-1})^2 = \frac{1}{\rho}, \end{aligned}$$

where the third equality follows from applying the identity

$$(a - b)(b - c)(a - c) + c(a - b)^2 + a(b - c)^2 = b(a - c)^2,$$

to the expression in the brackets for $a = \rho_l^{-1}$, $b = \rho^{-1}$ and $c = \rho_r^{-1}$, proving the correctness of the variance. Furthermore, Y_ρ is a sum of Gaussians and a convex combination of two Gaussians with mean β , so $Y_\rho \sim \mathcal{N}(\beta, 1/\rho)$. What remains to show is that the covariances match up, which we do next.

We start with the case where $\rho_r \neq \rho_\infty$, and so there exists $\rho_k, \rho_{k+1} \in S_n$ such that $\rho \in (\rho_k, \rho_{k+1})$. For $j \in [n - 1]$, we therefore have that

$$\begin{aligned} \text{Cov}(Y_\rho, Y_{\rho_j}) &= \text{Cov}\left(\frac{(\rho_l^{-1} - \rho^{-1})Y_{\rho_r} + (\rho^{-1} - \rho_r^{-1})Y_{\rho_l}}{\rho_l^{-1} - \rho_r^{-1}}, Y_{\rho_j}\right) \\ &= \frac{1}{\rho_l^{-1} - \rho_r^{-1}} \left[(\rho_l^{-1} - \rho^{-1})\text{Cov}(Y_{\rho_r}, Y_{\rho_j}) + (\rho^{-1} - \rho_r^{-1})\text{Cov}(Y_{\rho_l}, Y_{\rho_j}) \right] \\ &= \frac{1}{\rho_l^{-1} - \rho_r^{-1}} \left[(\rho_l^{-1} - \rho^{-1}) \min(\rho_r^{-1}, \rho_j^{-1}) + (\rho^{-1} - \rho_r^{-1}) \min(\rho_l^{-1}, \rho_j^{-1}) \right]. \end{aligned}$$

We consider the cases of $j \leq k$ and $j > k$ separately. For $j \leq k$, we have that

$$\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{1}{\rho_l^{-1} - \rho_r^{-1}} \left[(\rho_l^{-1} - \rho^{-1})\rho_r^{-1} + (\rho^{-1} - \rho_r^{-1})\rho_l^{-1} \right] = \frac{1}{\rho},$$

and similarly for $j \geq k + 1$ we get

$$\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{1}{\rho_k^{-1} - \rho_{k+1}^{-1}} \left[(\rho_k^{-1} - \rho^{-1})\rho_j^{-1} + (\rho^{-1} - \rho_{k+1}^{-1})\rho_k^{-1} \right] = \frac{1}{\rho_j}.$$

It follows that $\text{Cov}(Y_\rho, Y_{\rho_j}) = \frac{\Delta_2^2}{2\max(\rho, \rho_j)}$ for $j \in [m]$, and so we are done with this part of the covariances.

For the last step, we consider what happens when $\rho_r = \rho_\infty < \infty$. In this case, $\rho_l = \rho_{n-1}$ and so for $j \in [n - 1]$:

$$\begin{aligned} \text{Cov}(Y_\rho, Y_{\rho_j}) &= \text{Cov}\left(\frac{(1 - \rho_{n-1}/\rho)Y_{\rho_\infty} + (\rho_{n-1}/\rho - \rho_{n-1}/\rho_\infty)Y_{\rho_{n-1}}}{1 - \rho_{n-1}/\rho_\infty}, Y_{\rho_j}\right) \\ &= \frac{1 - \rho_{n-1}/\rho}{1 - \rho_{n-1}/\rho_\infty} \text{Cov}(Y_{\rho_\infty}, Y_{\rho_j}) + \frac{\rho_{n-1}/\rho - \rho_{n-1}/\rho_\infty}{1 - \rho_{n-1}/\rho_\infty} \text{Cov}(Y_{\rho_{n-1}}, Y_{\rho_j}) \\ &= \frac{1 - \rho_{n-1}/\rho}{\rho_\infty - \rho_{n-1}} + \frac{1/\rho - 1/\rho_\infty}{1 - \rho_{n-1}/\rho_\infty} = \frac{1 - \rho_{n-1}/\rho + \rho_\infty/\rho - 1}{\rho_\infty - \rho_{n-1}} = 1/\rho, \end{aligned}$$

and now we are done. $\square$

2913 6.6.4 *Details on (Efficient) Weakly Lossless Gradual Release of Sparse Gaussian His-*
 2914 *tograms*

2915 Algorithm 11 implements *weakly lossless gradual release* of private histograms. We
 2916 prove this next.

2917 *Proof of Lemma 6.13.* Observe that for any increasing sequence $(\rho_k)_{k \in m}$, the corre-
 2918 sponding sequence of variables Z on line 3 produced by the algorithm, have the
 2919 same distribution as in Lemma 6.2 for $\beta = 0$. The Y that is ultimately returned, how-
 2920 ever, is a post-processing of $H(\mathbf{x}) + Z$, which has the same distribution as Lemma 6.2
 2921 for $\beta = H(\mathbf{x})$. Therefore Algorithm 11 is implementing lossless gradual release
 2922 for the Gaussian mechanism applied to $H(\mathbf{x})$, combined with a non-invertible
 2923 post-processing. The lemma statement follows from Corollary 6.9. $\square$

2924 Note that one *only* has to store the noisy terms from the preceding round to
 2925 implement Algorithm 11. Nevertheless, it might be infeasible, say, when the domain
 2926 is really large, to sample the noise for the zero coordinates. Algorithm 14 uses the
 2927 computational trick in Lemma 6.14 to speed up this computation. The algorithm is
 2928 static, where the privacy budgets are fixed upfront but can easily be converted to

an online algorithm. Recall that $U(H(\mathbf{x})) = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ is the support of the histogram.

Algorithm 14 EfficientHistogramGradualRelease

Parameters: ℓ_2 -sensitivity Δ_2

Require: Private histogram $H(\mathbf{x})$, privacy budgets $\rho_1 < \dots < \rho_m$, thresholds $\tau_1, \dots, \tau_m$

```

1: Let  $S^{(0)} = U(H(\mathbf{x}))$  // Tracking counts that have already been released
2: Let  $Z^{(0)} = \{0\}^d$  and  $\rho_0 = 0$ .
3:  $\forall r \in [m]$ : let distribution  $\mathcal{D}^{(r)} = \mathcal{N}\left(0, \frac{1}{2}\Delta_2^2(\rho_r - \rho_{r-1})\right)$ .
4: for each round  $r \in [m]$  do
5:   Initialize  $Y^{(r)} = \{0\}^d$ .
6:   for each tracked count in preceding round  $j \in S^{(r-1)}$  do
7:     Draw fresh noise  $\tilde{Z}_j^{(r)} \sim \mathcal{D}^{(r)}$ .
8:     Update aggregate noise  $Z_j^{(r)} = \frac{\rho_{r-1}}{\rho_r} Z_j^{(r-1)} + \frac{1}{\rho_r} \tilde{Z}_j^{(r)}$ .
9:     if  $H(\mathbf{x})_j + Z_j^{(r)} > \tau_r$  then  $Y_j^{(r)} = H(\mathbf{x})_j + Z_j^{(r)}$ .
10:  Let  $\tilde{Z}^{(k)} \sim \mathcal{D}^{(k)}$  for  $k \in [r]$  be random variables.
11:  Compute  $p^{(r)} = \mathbf{P}\left[\sum_{k=1}^r \tilde{Z}^{(k)} > \tau_r \rho_r \mid \{\forall \ell \in [r-1] : \sum_{k=1}^\ell \tilde{Z}^{(k)} \leq \rho_\ell \tau_\ell\}\right]$ .
12:  Draw  $q \sim \text{Binomial}\left(d - |S^{(r-1)}|, p^{(r)}\right)$ .
13:  Select a subset  $Q \subseteq [d] \setminus S^{(r-1)}$  uniformly at random of size  $q$ .
14:  for each index  $j \in Q$  do
15:    Initialize  $Z_j^{(r)} = 0$ .
16:    for  $k \in [r-1]$  do
17:      Draw fresh noise  $\tilde{Z}_j^{(k)} \sim \mathcal{D}^{(k)}$  conditioned on  $\tilde{Z}_j^{(k)} \leq \rho_k \tau_k - Z_j^{(r)}$ .
18:      Update aggregate noise  $Z_j^{(r)} = Z_j^{(r)} + \tilde{Z}_j^{(k)}$ .
19:      Draw fresh noise  $\tilde{Z}_j^{(k)} \sim \mathcal{D}^{(k)}$  conditioned on  $\tilde{Z}_j^{(k)} > \rho_k \tau_k - Z_j^{(r)}$ .
20:      Update aggregate noise  $Z_j^{(r)} = Z_j^{(r)} + \tilde{Z}_j^{(k)}$ .
21:     $Y_j^{(r)} = H(\mathbf{x})_j + Z_j^{(r)}$  // Guaranteed to be above the threshold.
22:  Set  $S^{(r)} = S^{(r-1)} \cup Q$ .
23:  output private histogram  $Y^{(r)}$ .
```

2931 We proceed to give proofs for Lemma 6.15 and 6.16.

2932 *Proof sketch for Lemma 6.15.* The claim directly follows an inductive argument. For
 2933 the base case, note that the first iteration of Algorithm 14 is running the same
 2934 routine as described by Lemma 6.14, and so $\hat{Y}^{(1)}$ must be identically distributed
 2935 to $Y^{(1)}$. For our inductive hypothesis, assume that the subsequences $(Y^{(1)}, \dots, Y^{(k)})$
 2936 and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(k)})$ are identically distributed. Note that for the $(k+1)^{\text{st}}$ release,
 2937 Algorithm 14 handles all the true nonzero counts and every zero count that has
 2938 ever exceeded the threshold in a past round in the same way. These counts in $\hat{Y}^{(k+1)}$
 2939 and $Y^{(k+1)}$ clearly have the same distribution.

2940 Now, note that each of the remaining zero counts have up to and including
 2941 the k^{th} round been reported as zero in each round, and so their probability of
 2942 exceeding the threshold, $p^{(k+1)}$, should be equal across all of them.

2943 The sampling performed by Algorithm 14 is structured in the same manner as
 2944 Lemma 6.14, but with sampling probability $p^{(k+1)}$, and the noise terms added for
 2945 any zero count exceeding the threshold, are different. For the distributions to match,
 2946 the probability of exceeding the threshold and the noise distribution for an element
 2947 chosen to exceed the threshold are more complex. The probability of exceeding the
 2948 threshold is conditioned on prefixes of Gaussians, which correctly simulates the
 2949 probability of not exceeding the threshold at any prior round until first doing so in
 2950 the $(k+1)^{\text{st}}$ one. The noise added once selected to exceed the threshold is a sum of
 2951 truncated Gaussians, which simulate the same event. As the principle is built on
 2952 the same logic as Lemma 6.14, we have that $(Y^{(1)}, \dots, Y^{(k+1)})$ and $(\hat{Y}^{(1)}, \dots, \hat{Y}^{(k+1)})$
 2953 are identically distributed, and so the claim is true by induction. $\square$

2954 *Proof of Lemma 6.16.* Note that each of the k non-zero true counts will, in each
 2955 round, have fresh noise added on line 7. If a non-zero count is selected by the
 2956 binomial sampling in the k^{th} round, the for-loop starting on line 17 will result
 2957 in sampling k noise terms, after which for future rounds it will get treated as a
 2958 non-zero count. It follows that non-zero entries contribute km samples, and zero
 2959 counts exceeding the threshold contribute cm samples. $\square$

2960 6.6.5 The Poisson Mechanism

2961 We are not aware of any explicit statements in the literature on the privacy guar-
 2962 antees obtained by adding Poisson distributed noise to a d -dimensional vector of
 2963 integers. However, since the Poisson distribution is the limiting distribution of bi-
 2964 nomial distributions with the same mean $\lambda = Np$, where N is the number of trials,
 2965 such bounds can be derived from existing bounds on the binomial mechanism.
 2966 For the sake of completeness, we include such statements based on the following
 2967 theorem from [Aga+18]:

Theorem 6.19 (Agarwal, Suresh, Yu, Kumar, and McMahan [Aga+18]). For any δ , parameters N, p and sensitivity bounds $\Delta_1, \Delta_2, \Delta_\infty$ such that

$$Np(1-p) \geq \max \left(23 \log \left(\frac{10d}{\delta} \right), 2\Delta_\infty \right),$$

the d -dimensional Binomial mechanism is (ϵ, δ) -differentially private for

$$\epsilon = \frac{\Delta_2 \sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{Np(1-p)}} + \frac{\Delta_2 c_p \sqrt{\log \frac{10}{\delta}} + \Delta_1 b_p}{Np(1-p)(1-\delta/10)} + \frac{\frac{2}{3} \Delta_\infty \log \frac{1.25}{\delta} + \Delta_\infty d_p \log \frac{20d}{\delta} \log \frac{10}{\delta}}{Np(1-p)},$$

where

$$d_p \triangleq \frac{4}{3} \cdot (p^2 + (1-p)^2), \quad b_p \triangleq \frac{2(p^2 + (1-p)^2)}{3} + (1-2p), \\ c_p \triangleq \sqrt{2} \left(3p^3 + 3(1-p)^3 + 2p^2 + 2(1-p)^2 \right).$$

2968

Setting $p = \lambda/N$ and considering the limiting bound when $N \rightarrow \infty$, we get:

Lemma 6.20 (Privacy guarantees of the Poisson Mechanism). The d -dimensional Poisson mechanism with parameter $\lambda > \max(23 \log(10d/\delta), 2\Delta_\infty)$, is (ϵ, δ) -DP with

$$\epsilon = \frac{\Delta_2 \sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{\lambda}} + \frac{5\sqrt{2} \Delta_2 \sqrt{\log \frac{10}{\delta}} + \frac{5}{3} \Delta_1}{\lambda(1-\delta/10)} \\ + \frac{\frac{2}{3} \Delta_\infty \log \frac{1.25}{\delta} + \frac{4}{3} \Delta_\infty \log \frac{20d}{\delta} \log \frac{10}{\delta}}{\lambda}.$$

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A simpler expression can be derived for unit sensitivities by relaxing the constants and assuming that δ is not too large:

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2971

Corollary 6.21 (Simplified upper bound with unit sensitivities). Assume that $\Delta_1 = \Delta_2 = \Delta_\infty = 1$. Then for $\delta < 1/100$, the d -dimensional Poisson mechanism with parameter $\lambda > 23 \log(10d/\delta)$ is (ϵ, δ) -differentially private for

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2973

$$\epsilon = \frac{\sqrt{2 \log \frac{1.25}{\delta}}}{\sqrt{\lambda}} + \frac{2 \log \frac{20d}{\delta} \log \frac{10}{\delta}}{\lambda}.$$

2974

We will need the following lemma that determines the sampling step of private lossless multiple release for the Poisson mechanism:

2975

Lemma 6.22. *Let $X_1 \sim \text{Poi}(\lambda_1)$ and $X_2 \sim \text{Poi}(\lambda_2)$ be independent Poisson random variables. Then, for any nonnegative integer k , the conditional distribution of X_1 given $X_1 + X_2 = k$ is*

$$X_1 \mid (X_1 + X_2 = k) \sim \text{Binomial} \left(k, \frac{\lambda_1}{\lambda_1 + \lambda_2} \right).$$

Proof. Since X_1 and X_2 are independent, their joint probability mass function is for all $x = 0, 1, \dots, k$

$$PX_1 = x, X_2 = k - x = e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^x}{x!} \frac{\lambda_2^{k-x}}{(k-x)!}.$$

Moreover, the sum $X_1 + X_2$ is Poisson with parameter $\lambda_1 + \lambda_2$, so that

$$\mathbf{P}[X_1 + X_2 = k] = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}.$$

Thus, by the definition of conditional probability,

$$\begin{aligned} \mathbf{P}[X_1 = x \mid X_1 + X_2 = k] &= \frac{\mathbf{P}[X_1 = x, X_2 = k - x]}{\mathbf{P}[X_1 + X_2 = k]} \\ &= \frac{e^{-(\lambda_1 + \lambda_2)} \frac{\lambda_1^x}{x!} \frac{\lambda_2^{k-x}}{(k-x)!}}{e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}} \\ &= \binom{k}{x} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^x \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{k-x}. \end{aligned}$$

This is precisely the probability mass function of a Binomial random variable with parameters k and $\frac{\lambda_1}{\lambda_1 + \lambda_2}$. $\square$

6.6.6 The Laplace Mechanism

Koufogiannis, Han, and Pappas [KHP16] showed how to do lossless *gradual* releases for the Laplace mechanism and supports either tightening the privacy guarantees or loosening them. We will now strengthen this result by exactly showing how to do them in arbitrary order, similar to what was done in Section 6.3 for the Gaussian mechanism. We will first show that the Laplace distribution satisfies Definition 6.1. This already implies the existence of an algorithm supporting gradual lossless release via Lemma 6.6. After, we will derive how to sample a new release with scale parameter b given two distinct releases with scaling $b_2 < b$ and $b < b_1$.

Lemma 6.23 ([KKP01] Characteristic function of Laplace). *Let X be Laplace random variable with probability density function as in Definition A.2, then for all $t \in \mathbb{R}$, the characteristic function of X is*

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} \frac{1}{2b} e^{-|x|/b} dx = \frac{1}{1 + b^2 t^2}.$$

2991 *Proof.* By a simple integration:

$$\begin{aligned}\mathbb{E}[e^{itX}] &= \frac{1}{2b} \int_{-\infty}^{\infty} e^{itx - |x|/b} dx = \frac{1}{2b} \left(\int_{-\infty}^0 e^{(1/b+ti)x} dx + \int_0^{\infty} e^{(-1/b+ti)x} dx \right) \\ &= \frac{1}{2b} \left(\frac{1}{1/b+ti} + \frac{1}{1/b-ti} \right) = \frac{1}{1+b^2t^2}.\end{aligned}\quad \square$$

2992 We will now show that the zero-centered Laplace distribution with scale parameter
2993 b satisfies convolution preorder (Definition 6.1). Note that a larger value of b
2994 corresponds to a more private release by definition. Building on this, we will
2995 condition on the two closest releases to create a new one in the middle, as done in
2996 Lemma 6.4 for the Gaussian. The following fact was already shown in Koufogiannis,
2997 Han, and Pappas [KHP16], but we give a proof here for completeness using a
2998 technique based on characteristic functions as done by Equitz and Cover [EC91].

Claim 6.24 (Convolution preorder: Laplace). *Fix $b_2 < b_1 \in \mathbb{R}^+$ let $X \sim \text{Lap}(0, b_2)$ and draw*

$$W = \begin{cases} 0 & \text{with probability } b_2^2/b_1^2, \\ \text{Lap}(0, b_1) & \text{otherwise} \end{cases}, \quad \text{then } X + W \sim \text{Lap}(0, b_1).$$

2999 *Proof.* We know that the characteristic function of X is $\varphi_X(t) = \frac{1}{1+b_2^2t^2}$ and for the
3000 convolution $\varphi_{X+W} = \frac{1}{1+b_1^2t^2}$. Because of independence, the convolution is defined
3001 for all t as:

$$\begin{aligned}\varphi_X(t)\varphi_W(t) &= \varphi_{X+W}(t) \\ \Leftrightarrow \varphi_W(t) &= \frac{1+b_2^2t^2}{1+b_1^2t^2} = \frac{b_1^2+b_1^2b_2^2t^2}{b_1^2(1+b_1^2t^2)} = \frac{b_2^2(1+b_1^2t^2)+b_1^2-b_2^2}{b_1^2(1+b_1^2t^2)} \\ &= \frac{b_2^2}{b_1^2} + \left(1 - \frac{b_2^2}{b_1^2}\right) \cdot \frac{1}{1+b_1^2t^2}.\end{aligned}$$

3002 The last expression encodes the convex combination of the claimed mixture
3003 distribution because $\frac{1}{1+b_1^2t^2}$ is again the characteristic function of a zero-centered
3004 Laplace distribution with scaling parameter b_1 . $\square$

3005 We next prove a simple result about the convolution of two Laplace distributions
3006 (compare also eq., 2.3.23 of Kotz, Kozubowski, and Podgórski [KKP01]).

Lemma 6.25 (Convolution). *For two fixed scaling parameters $b_1 \neq b_2 \in \mathbb{R}^+$, let $X_1 \sim \text{Lap}(0, b_1)$ and $X_2 \sim \text{Lap}(0, b_2)$. Then the density of $X_1 + X_2$ is given by*

$$(f_{b_1} * f_{b_2})(t) = \frac{1}{2(b_1^2 - b_2^2)} \left(b_1 e^{-|t|/b_1} - b_2 e^{-|t|/b_2} \right).$$

Proof. Assume $t \geq 0$ and note that the other case follows by symmetry. We compute the density of the convolution by a straightforward integration:

$$\begin{aligned} (f_{b_1} * f_{b_2})(t) &= \int_{-\infty}^{\infty} f_{b_1}(x) f_{b_2}(t-x) dx = \frac{1}{4b_1 b_2} \int_{-\infty}^{\infty} \exp\left(-\frac{|x|}{b_1} - \frac{|t-x|}{b_2}\right) dx \\ &= \frac{1}{4b_1 b_2} \cdot \left(\int_{-\infty}^0 e^{\frac{x}{b_1} - \frac{t-x}{b_2}} dx + \int_0^t e^{-\frac{x}{b_1} - \frac{t-x}{b_2}} dx + \int_t^{\infty} e^{-\frac{x}{b_1} - \frac{x-t}{b_2}} dx \right) \\ &= \frac{1}{4b_1 b_2} \cdot \left(\left[\frac{\exp(x/b_1 - (t-x)/b_2)}{b_1^{-1} + b_2^{-1}} \right]_{-\infty}^0 \right. \\ &\quad \left. + \left[\frac{\exp(-x/b_1 - (t-x)/b_2)}{b_2^{-1} - b_1^{-1}} \right]_0^t + \left[\frac{\exp(-x/b_1 - (x-t)/b_2)}{-b_2^{-1} - b_1^{-1}} \right]_t^{\infty} \right) \\ &= \frac{1}{4} \cdot \left(\frac{\exp(-t/b_2)}{b_1 + b_2} + \frac{\exp(-t/b_1) - \exp(-t/b_2)}{b_1 - b_2} + \frac{\exp(-t/b_1)}{b_1 + b_2} \right) \\ &= \frac{1}{2(b_1^2 - b_2^2)} \left(b_1 e^{-t/b_1} - b_2 e^{-t/b_2} \right). \quad \square \end{aligned}$$

Now, we are ready to show that the Laplace mechanism can be implemented with multiple releases.

Lemma 6.26 (Multiple release Laplace). *For fixed $0 < b_2 < b < b_1$, let $\mu_1 =$*

b^2/b_1^2 and $\mu_2 = b_2^2/b^2$ and $D_1 \sim \text{Ber}(\mu_1)$ and $D_2 \sim \text{Ber}(\mu_2)$. Furthermore, let

$$X_1 = \begin{cases} 0 & \text{if } D_1 = 1 \\ \text{Lap}(0, b) & \text{otherwise} \end{cases} \quad \text{and} \quad X_2 = \begin{cases} 0 & \text{if } D_2 = 1 \\ \text{Lap}(0, b_2) & \text{otherwise} \end{cases}.$$

Then we have that $\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0] = 1$, and for every real number $k \neq 0$ the distribution of X_1 conditioned on $X_1 + X_2 = k$ is:

$$X_1 \mid (X_1 + X_2 = k) \sim \begin{cases} 0 & w.p. \mu_1 \cdot (1 - \mu_2) \cdot \frac{\exp(-|k|/b_2)}{2b_2 \cdot f_{X_1+X_2}(k)} \\ k & w.p. (1 - \mu_1) \cdot \mu_2 \cdot \frac{\exp(-|k|/b)}{2b \cdot f_{X_1+X_2}(k)} \\ H(b, b_2, k) & w.p. (1 - \mu_1) \cdot (1 - \mu_2) \cdot \frac{(f_b * f_{b_2})(k)}{f_{X_1+X_2}(k)} \end{cases} \quad (3)$$

where 0 and k denote the constant distributions, $H(b, b_2, k)$ is the probability distribution with probability density function $h(x) = \frac{f_b(x)f_{b_2}(k-x)}{(f_b * f_{b_2})(k)}$,

$$\begin{aligned} f_{X_1+X_2}(k) &= \frac{(1 - \mu_1)\mu_2}{2b} e^{-|k|/b} \frac{(1 - \mu_1)(1 - \mu_2)}{2(b^2 - b_2^2)} \left(b e^{-|k|/b} - b_2 e^{-|k|/b_2} \right) \\ &\quad + \mu_1 \mu_2 \delta_0(x) + \frac{\mu_1(1 - \mu_2)}{2b_2} e^{-|k|/b_2}, \text{ and} \\ (f_b * f_{b_2})(k) &= \frac{1}{2(b_1^2 - b_2^2)} \left(b_1 e^{-t/b_1} - b_2 e^{-t/b_2} \right), \end{aligned}$$

where δ_0 is the Dirac delta function.

Proof. First we consider $\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0]$, arguing that

$$\mathbf{P}[X_1 = 0 \mid X_1 + X_2 = 0] = \frac{\mathbf{P}[X_1 = 0] \mathbf{P}[X_2 = 0]}{\mathbf{P}[X_1 + X_2 = 0]} = \frac{\mu_1 \mu_2}{\mu_1 \mu_2 + 0} = 1.$$

To see why the first equality holds, split $\mathbf{P}[X_1 + X_2 = 0]$ into each of the four combinations of discrete/continuous for X_1, X_2 . The only non-zero contribution to the probability mass comes from the discrete/discrete case, as the remaining cases contribute mass proportional to the probability that a continuous distribution assumes an exact value, which is zero.

Next we turn to the distribution of X_1 conditioned on $X_1 + X_2 = k$ where $k \neq 0$. Denote by f_b the probability density function of $\text{Lap}(0, b)$. Note that the mixture densities become

$$\begin{aligned} f_{X_1}(x) &= \mu_1 \delta_0(x) + (1 - \mu_1) f_b(x), \\ f_{X_2}(x) &= \mu_2 \delta_0(x) + (1 - \mu_2) f_{b_2}(x). \end{aligned}$$

Before analyzing the different cases separately, we compute the pdf of the convolution $X_1 + X_2$.

Convolution $f_{X_1+X_2}$.

Note that we have to take care of a subtle technicality: the case that both X_1 and X_2 are zero from the discrete part can only happen when we condition on

3027 $X_1 + X_2 = 0$. We can now give the density of the convolution $X_1 + X_2$ at any real
3028 point x :

$$\begin{aligned}
 f_{X_1+X_2}(x) &= \sum_{(d_1, d_2) \in \{0,1\}^2} f_{X_1+X_2|D_1=d_1, D_2=d_2}(x) \cdot \mathbf{P}[D_1 = d_1 \wedge D_2 = d_2] \\
 &= \mu_1(1 - \mu_2) \cdot f_{b_2}(x) + (1 - \mu_1)\mu_2 \cdot f_b(x) + (1 - \mu_1)(1 - \mu_2) \cdot (f_b * f_{b_2})(x) \\
 &\quad + \mu_1\mu_2\delta_0(x) \\
 &= \frac{\mu_1(1 - \mu_2)}{2b_2} e^{-|x|/b_2} + \frac{(1 - \mu_1)\mu_2}{2b} e^{-|x|/b} \\
 &\quad + (1 - \mu_1)(1 - \mu_2) \frac{1}{2(b^2 - b_2^2)} \left(b e^{-|x|/b} - b_2 e^{-|x|/b_2} \right) + \mu_1\mu_2\delta_0(x),
 \end{aligned}$$

3029 where the third term follows from Lemma 6.25. Note that the last term only
3030 contributes when $x = 0$.

3031 We can now show how the sampling procedure in the claim is justified. We first
3032 assume $k \neq 0$ and analyze three possible cases how $X_1 + X_2$ is built up: Either both
3033 of them are drawn from the (continuous) Laplace distribution or exactly one. (The
3034 case where both are from their respective discrete parts can only happen if $k = 0$,
3035 analyzed above.)

3036 **Case 1:** $X_1 = 0$ and $X_2 = k$.

3037 $X_2 = k$ is necessarily from its continuous part. By the definition of conditional
3038 probability, we have for $k \neq 0$:

$$\begin{aligned}
 \mathbf{P}[X_1 = 0 | X_1 + X_2 = k, k \neq 0] &= \frac{\mathbf{P}[X_1 = 0 \wedge X_1 + X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} = \frac{\mathbf{P}[X_1 = 0] \mathbf{P}[X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} \\
 &= \mu_1 \frac{f_{X_2}(k)}{f_{X_1+X_2}(k)} = \frac{\mu_1(1 - \mu_2)}{2b_2 f_{X_1+X_2}(k)} \cdot e^{-|k|/b_2} := p_1.
 \end{aligned}$$

3039 For the third equality, we simply used definition of a probability density function
3040 via its limit:

$$\lim_{\Delta \rightarrow 0^+} \frac{\mathbf{P}[X_2 \in [k - \Delta, k + \Delta]]}{\mathbf{P}[X_1 + X_2 \in [k - \Delta, k + \Delta]]} = \frac{f_{X_2}(k)}{f_{X_1+X_2}(k)}.$$

3041 **Case 2:** $X_1 = k$ and $X_2 = 0$.

3042 Now assume the flipped case. By a similar argument:

$$\begin{aligned}
 p_2 := \mathbf{P}[X_1 = k | X_1 + X_2 = k] &= \frac{\mathbf{P}[X_1 = k, X_1 + X_2 = k]}{\mathbf{P}[X_1 + X_2 = k]} \\
 &= \frac{\mathbf{P}[X_1 = k] \mathbf{P}[X_2 = 0]}{\mathbf{P}[X_1 + X_2 = k]} = \frac{(1 - \mu_1)\mu_2}{2b f_{X_1+X_2}(k)} \cdot e^{-|k|/b}.
 \end{aligned}$$

3043 **Case 3:** $X_1 \neq 0, X_2 \neq 0, X_1 + X_2 = k$.

3044 In the remaining case, both X_1 and X_2 are independently sampled from continuous
3045 Laplace distributions with probability density functions $f_b(x)$ and $f_{b_2}(x)$. Therefore,
3046 with the remaining probability $1 - (p_1 + p_2)$, we know that X_1 is sampled according
3047 to the following conditional probability density function:

$$f_{X_1}(x) |_{X_1+X_2=k} = \frac{f_{X_1, X_1+X_2}(x, k)}{(f_b * f_{b_2})(k)} = \frac{f_b(x) f_{b_2}(k - x)}{(f_b * f_{b_2})(k)},$$

where the last line follows from the trivial identity $X_1 + X_2 = k \Leftrightarrow X_2 = k - X_1$. This is a valid probability density function because f is trivially non-negative due to its parts being non-negative and furthermore

$$\begin{aligned} \int_{-\infty}^{\infty} \frac{f_b(x)f_{b_2}(x-k)}{(f_b * f_{b_2})(k)} dx &= \frac{1}{(f_b * f_{b_2})(k)} \int_{-\infty}^{\infty} f_b(x)f_{b_2}(k-x) dx \\ &= \frac{(f_b * f_{b_2})(k)}{(f_b * f_{b_2})(k)} = 1. \end{aligned}$$

What is left is to verify that the probabilities in Equation (3) indeed add up to one for $k \neq 0$:

$$\begin{aligned} &\frac{(1-\mu_1)(1-\mu_2)(f_b * f_{b_2})(k)}{f_{X_1+X_2}(k)} + \frac{(1-\mu_1)\mu_2 \exp(-|k|/b)}{2bf_{X_1+X_2}(k)} + \frac{\mu_1(1-\mu_2) \exp(-|k|/b_2)}{2b_2f_{X_1+X_2}(k)} \\ &= \frac{1}{f_{X_1+X_2}(k)} \cdot \\ &\quad \underbrace{\left((1-\mu_1)(1-\mu_2)(f_b * f_{b_2})(k) + \frac{(1-\mu_1)\mu_2 \exp(-|k|/b)}{2b} + \frac{\mu_1(1-\mu_2) \exp(-|k|/b_2)}{2b_2} \right)}_{f_{X_1+X_2}(k) \text{ for } k \neq 0} \\ &= 1 \end{aligned} \quad \square$$

6.6.7 Showing Convolution Preorder for Exponential Noise

The exponential distribution is closely related to the Laplace distribution but gives poor differential privacy guarantees. Nevertheless, it still serves as a building block for some private mechanisms, e.g., Report-Noisy-Max [Din+21]. We show next that it also satisfies a convolution preorder.

Definition 6.5 (Exponential distribution characteristic function). *The characteristic function of the Exponential Distribution is given by $\varphi_X(t) = \mathbb{E}[e^{itX}] = \frac{\lambda}{\lambda - it}$*

Claim 6.27 (Convolution preorder: Exponential distribution). *Fix $\lambda_1 < \lambda_2 \in \mathbb{R}_+$, let $X \sim \text{Exp}(\lambda_2)$ and draw*

$$W = \begin{cases} 0 & \text{with probability } \lambda_1/\lambda_2, \\ \text{Exp}(\lambda_1) & \text{otherwise} \end{cases}, \quad \text{then } X + W \sim \text{Exp}(\lambda_1).$$

Proof. Using the same trick as in the proof of Lemma 6.24, we have that

$$\begin{aligned} \varphi_W(t) &= \frac{\varphi_{X+W}(t)}{\varphi_X(t)} = \frac{\lambda_1}{\lambda_2} \cdot \frac{\lambda_2 - it}{\lambda_1 - it} \\ &= \frac{\lambda_1}{\lambda_2} \left(1 + \frac{\lambda_2 - \lambda_1}{\lambda_1 - it} \right) = \frac{\lambda_1}{\lambda_2} + \left(1 - \frac{\lambda_1}{\lambda_2} \right) \varphi_{X+W}(t), \end{aligned}$$

where the final expression is the characteristic function of the claimed mixture distribution. $\square$

3061 6.6.8 Convolution Preorder for Discrete Laplace

3062 Alongside the Discrete Gaussian, the Discrete Laplace is fundamental with huge
 3063 practical impact because it enables implementation on a finite machine. The exis-
 3064 tence of a convolution preorder for the Laplace distribution already suggests that
 3065 something similar might be possible for the Discrete Laplace as well. Indeed, the
 3066 following proof uses the same base idea of studying the characteristic function.

3067 The following result is not included in the published version of the paper. The
 3068 discrete Laplace distribution $\text{DLap}(\alpha)$ is a two-sided geometric distribution with
 3069 probability mass function $p(x) = \frac{1-\alpha}{1+\alpha} \cdot \alpha^{|x|}$ for $\alpha \in (0, 1)$ and $x \in \mathbb{Z}$. It is well known
 3070 that the discrete counterpart to the continuous Laplace mechanism satisfies pure
 3071 DP if one releases $f(\mathbf{x}) + \text{DLap}(\alpha)$ for $\alpha = \exp(-\varepsilon/\Delta)$ where Δ is the ℓ_1 sensitivity
 3072 of the query f [GRSo9]. In the following, we will need the characteristic function
 3073 of the Discrete Laplace which we quickly reprove for convenience.

Claim 6.28 (Characteristic function of the Discrete Laplace). *For $\alpha \in (0, 1)$, the characteristic function of $X \sim \text{DLap}(\alpha)$ is for all t :*

$$\phi_X(t) = \frac{(1 - \alpha)^2}{1 - 2\alpha \cos(t) + \alpha^2}.$$

Proof.

$$\begin{aligned} \phi_X(t) &= \mathbb{E}[\exp(itX)] = \frac{1 - \alpha}{1 + \alpha} \cdot \left(1 + \sum_{k \geq 1} e^{itk} \alpha^k + \sum_{k \geq 1} e^{-itk} \alpha^k \right) \\ &= \frac{1 - \alpha}{1 + \alpha} \cdot \left(1 + \sum_{k \geq 1} (e^{it} \alpha)^k + \sum_{k \geq 1} (e^{-it} \alpha)^k \right) \\ &= \frac{1 - \alpha}{1 + \alpha} \cdot \left(1 + \frac{e^{it} \alpha}{1 - e^{it} \alpha} + \frac{e^{-it} \alpha}{1 - e^{-it} \alpha} \right) \\ &= \frac{1 - \alpha}{1 + \alpha} \cdot \left(\frac{(1 - e^{it} \alpha)(1 - e^{-it} \alpha) + (e^{it} \alpha)(1 - e^{-it} \alpha) + e^{-it} \alpha(1 - e^{it} \alpha)}{(1 - e^{it} \alpha)(1 - e^{-it} \alpha)} \right) \\ &= \frac{(1 - \alpha)^2}{1 - 2\alpha \cos(t) + \alpha^2}, \end{aligned}$$

3074 where we used the identity $e^{it} + e^{-it} = 2 \cos(t)$ by Euler's formula, and the
 3075 convergence of the geometric sum is justified because the convergence radius
 3076 is $|\alpha e^{it}| = \alpha \sqrt{\cos(t)^2 + \sin(t)^2} = \alpha < 1$ by assumption on α (and similarly for
 3077 $|\alpha e^{-it}|$). $\square$

Claim 6.29 (Deconvolution Laplace). *Fix two scaling parameters $0 < \alpha_x \leq \alpha_z < 1$. Let $X \sim \text{DLap}(\alpha_x)$ and $Z \sim \text{DLap}(\alpha_z)$. Then $\phi_X(t)\phi_Y(t) = \phi_Z(t)$ if we set,*

$$\phi_Y(t) = \psi(t)\psi(-t) \quad \text{with } \psi(t) = \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)}. \quad (4)$$

3078 *Proof.* Plugging in Lemma 6.28 gives,

$$\phi_Y(t) = \frac{\phi_Z(t)}{\phi_X(t)} = \frac{(\alpha_z - 1)^2}{(\alpha_x - 1)^2} \cdot \frac{1 - 2\alpha_x \cos(t) + \alpha_x^2}{1 - 2\alpha_z \cos(t) + \alpha_z^2}.$$

3079 Observe that the characteristic function of this deconvolution inherently is a
 3080 convolution of some other distribution. Therefore, using the identity $2 \cos(t) =$
 3081 $\exp(it) + \exp(-it)$ gives the factorization in the claim:

$$\begin{aligned}\phi_Y(t) &= \left(\frac{1 - \alpha_z}{1 - \alpha_x} \right)^2 \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \cdot \frac{1 - \alpha_x \exp(-it)}{1 - \alpha_z \exp(-it)} \\ &= \left(\frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \right) \left(\frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(-it)}{1 - \alpha_z \exp(-it)} \right) \\ &= \psi(t)\psi(-t) \quad \text{for } \psi(t) = \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)}.\end{aligned}$$

3082

□

Theorem 6.30 (Sampling from Deconvolution). *Fix $0 < \alpha_x \leq \alpha_z < 1$ scaling parameters as before. Let $X \sim \text{DLap}(\alpha_x)$ and Y_1, Y_2 be iid, independent of X as*

$$\mathbf{P}[Y_i = 0] = \frac{1 - \alpha_z}{1 - \alpha_x}, \quad \mathbf{P}[Y_i = k] = \left(1 - \frac{1 - \alpha_z}{1 - \alpha_x}\right) (1 - \alpha_z) \alpha_z^{k-1}, \quad k \geq 1.$$

Equivalently,

$$Y_1, Y_2 \sim \begin{cases} 0 & \text{with probability } \frac{1 - \alpha_z}{1 - \alpha_x}, \\ \text{Geom}(1 - \alpha_z) & \text{otherwise} \end{cases}.$$

Then $X + (Y_1 - Y_2) \sim \text{DLap}(\alpha_z)$.

3083 *Proof.* We will now analyze the distribution with the characteristic function $\psi(t)$
 3084 more carefully.

$$\begin{aligned}\psi(t) &= \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \frac{1 - \alpha_x \exp(it)}{1 - \alpha_z \exp(it)} \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} \cdot \left(1 + \frac{(\alpha_z - \alpha_x) \exp(it)}{1 - \alpha_z \exp(it)} \right) \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} \exp(0 \cdot it) \cdot \left(1 + (\alpha_z - \alpha_x) \sum_{k=1}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right) \right) \quad (\text{Geom series}) \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=1}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right) \\ &= \underbrace{\frac{1 - \alpha_z}{1 - \alpha_x}}_{a_1} + \underbrace{\frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \exp(it)}_{a_2} \\ &\quad + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=2}^{\infty} \left(\alpha_z^{k-1} \exp(kit) \right).\end{aligned}\tag{5}$$

Observe that these coefficients a_i (without the complex exponential part) are by definition non-negative and sum up to

$$\begin{aligned} \sum_{i=1}^{\infty} a_i &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \sum_{k=1}^{\infty} \alpha_z^{k-1} \\ &= \frac{1 - \alpha_z}{1 - \alpha_x} + \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \frac{1}{1 - \alpha_z} \\ &= \frac{1 - \alpha_z + \alpha_z - \alpha_x}{1 - \alpha_x} = 1. \end{aligned}$$

Recall that $\exp(it\alpha)$ is the characteristic function of the Dirac delta δ_a , the distribution that outputs a with probability one. Therefore, we can interpret $\psi(t)$ in Equation (5) as the distribution that draws 0 with probability $\frac{1 - \alpha_z}{1 - \alpha_x}$ and otherwise some $k \in \mathbb{N}_{>0}$ with probability

$$\begin{aligned} \tilde{p} &= \alpha_z^{k-1} \left(\frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \right) \cdot \left(1 - \frac{1 - \alpha_z}{1 - \alpha_x} \right)^{-1} \\ &= \alpha_z^{k-1} \frac{(1 - \alpha_z)(\alpha_z - \alpha_x)}{1 - \alpha_x} \cdot \frac{1 - \alpha_x}{\alpha_z - \alpha_x} \\ &= \alpha_z^{k-1} (1 - \alpha_z). \end{aligned}$$

Observe that the second case (conditioned on not zero) is actually a draw from a geometric distribution $\text{Geom}(1 - \alpha_z)$! Observe by independence,

$$\phi_{Y_1 - Y_2}(t) = \phi_{Y_1}(t) \phi_{Y_2}(-t) = \psi(t) \psi(-t).$$

Hence, it follows that $X + (Y_1 - Y_2) \sim \text{DLap}(\alpha_z)$. □

6.7 EXTENSION: NON-ADDITIVE NOISE MECHANISMS

The following results are ongoing work that had not yet been published when this thesis was submitted. Nevertheless, we believe that they complement the previous results as they describe an alternative way of viewing Private Lossless Multiple Release (PLMR) in terms of a Markov chain. The main intuition is that the *memoryless property* of a Markov chain exactly corresponds to the post-processing characterization we had before.

6.7.1 Markov Chain Formulation

Recall that lossless multiple release requires that, for any subset of the privacy levels and in any order, the entire collection of releases can be reproduced by post-processing the least private one alone. This is precisely the memoryless property of a Markov chain, which lets us replace reasoning about the joint distribution of all releases with a single transition kernel between adjacent privacy levels.

Lemma 6.31 (Lossless Multiple Release $\Leftrightarrow$ Markov process). *Consider a private mechanism $\mathcal{M}_\rho : \mathcal{X} \rightarrow \mathcal{Y}$. Then $\mathcal{M}_\rho$ can be implemented with lossless multiple release, if and only if, for every $x \in \mathcal{X}$ and every finite $S \subset \mathbb{R}_+$, there exists a Markov process $(Y_\rho)_{\rho \in S}$ where $Y_\rho \stackrel{d}{=} \mathcal{M}_\rho(x)$.*

Proof. $PLMR \Rightarrow \text{Markov process}$. Let M denote an implementation of PLMR for $\mathcal{M}_\rho$. Consider the sequence $(Y_\rho)_{\rho \in S} \equiv (M(x, \rho))_{\rho \in S}$. By Property 1, $M(x, \rho) \stackrel{d}{=} \mathcal{M}_\rho(x)$. To check the Markov property, consider any $\rho_1 < \rho_2 < \rho_3$ in S . Using Property 2 for every $S' \subseteq \{\rho_1, \rho_2, \rho_3\} \subseteq S$ the joint distribution over $(Y_\rho)_{\rho \in S'}$ is determined by Y_{ρ_3} , hence. Thus, considering every input order, the following objects are determined:

$$Y_{\rho_1} \mid Y_{\rho_2}, \quad Y_{\rho_1} \mid Y_{\rho_3}, \quad Y_{\rho_2} \mid Y_{\rho_3},$$

and so, from first principles, $Y_{\rho_1} \mid Y_{\rho_2}$ is independent of Y_{ρ_3} , proving that the Markov property holds. Hence $(Y_\rho)_{\rho \in S}$ is a Markov process.

$\text{Markov process} \Rightarrow PLMR$. Define M as the mechanism that, for a fixed $x \in \mathcal{X}$ and on receiving a privacy parameter ρ outputs $M(x, \rho) \equiv Y_\rho$. Property 1 of Definition 6.2 immediately holds as $Y_\rho \stackrel{d}{=} \mathcal{M}_\rho(x)$. Property 2 is implied by the Markov property of $(Y_\rho)_{\rho \in S}$ together with the chain rule. $\square$

We can use this result to give a very elegant characterization of k -ary *Randomized Response* under arbitrary releases. This immediately extends the (arguably more complicated) analysis for gradual release [Pan24]. Recall the following definition of randomized response.

Definition 6.6 ([War65] k -ary Randomized Response). *Let $x \in [k]$ be a single private value and fix some privacy parameter $\epsilon > 0$. We define k -ary randomized response $k\text{RandResp}(\epsilon, x)$ to be the mechanism that releases*

$$Y_\epsilon = \begin{cases} x & \text{with probability } \frac{\exp(\epsilon)}{\exp(\epsilon) + k - 1} \\ z \in [k] \setminus \{x\} \text{ uniformly at random} & \text{otherwise.} \end{cases}$$

We will now show how an output $Y_1 \sim \text{RandResp}(\epsilon_2, X)$ can be (data-independently) post-processed, into a sample of $\text{RandResp}(\epsilon_1, X)$ using a stochastic kernel. In the following, we denote by $\tanh$ the hyperbolic tangent.

Lemma 6.32 (Stochastic Kernel k -ary RR). *Let $x \in [k]$ and fix two privacy parameters $0 \leq \epsilon' \leq \epsilon$. Now draw $Y_\epsilon \sim k\text{RandResp}(\epsilon, x)$ and sample conditionally*

$$Y_{\epsilon'} \mid (Y_\epsilon = y_\epsilon) \sim \begin{cases} y_\epsilon & \text{with probability } 1 - p_{\text{flip}}(\epsilon, \epsilon') \\ z \in [k] \setminus \{y_\epsilon\} \text{ uni. at random} & \text{with total probability } p_{\text{flip}}(\epsilon, \epsilon') \end{cases},$$

where

$$p_{\text{flip}}(\epsilon, \epsilon') = \frac{q_\epsilon - r'_\epsilon}{q_\epsilon - r_\epsilon} = \frac{k-1}{k} \cdot \left(1 - \frac{(\exp(\epsilon') - 1) \cdot (\exp(\epsilon) + k - 1)}{(\exp(\epsilon) - 1) \cdot (\exp(\epsilon') + k - 1)} \right).$$

Then we have $Y_{\epsilon'} \sim \text{RandResp}(\epsilon', x)$.

Proof. Fix some private dataset $x \in [k]$. Rewrite the mechanism in terms of a stochastic kernel with some unknown flip probabilities $p_{\text{flip}}(\epsilon, \epsilon')$ that only depend on the two privacy parameters ϵ and ϵ' :

$$K^{\varepsilon \rightarrow \varepsilon'} := \begin{pmatrix} 1 - p_{\text{flip}}(\varepsilon, \varepsilon') & \frac{p_{\text{flip}}(\varepsilon, \varepsilon')}{k-1} & \frac{p_{\text{flip}}(\varepsilon, \varepsilon')}{k-1} \\ \frac{p_{\text{flip}}(\varepsilon, \varepsilon')}{k-1} & 1 - p_{\text{flip}}(\varepsilon, \varepsilon') & \cdots \\ \vdots & \vdots & \vdots \\ \frac{p_{\text{flip}}(\varepsilon, \varepsilon')}{k-1} & \cdots & 1 - p_{\text{flip}}(\varepsilon, \varepsilon') \end{pmatrix}.$$

3128 Observe that the law of the mechanism is exactly

$$M_\varepsilon(x) = \begin{pmatrix} (\exp(\varepsilon) + k - 1)^{-1} \\ \vdots \\ \frac{\exp(\varepsilon)}{\exp(\varepsilon) + k - 1} \\ \vdots \\ (\exp(\varepsilon) + k - 1)^{-1} \end{pmatrix} = \begin{pmatrix} q_\varepsilon \\ \vdots \\ r_\varepsilon \\ \vdots \\ q_\varepsilon \end{pmatrix}.$$

3129 ,

3130 Now fix any arbitrary coordinate $l \in [k]$ and observe

$$(K^{\varepsilon \rightarrow \varepsilon'} \cdot M_\varepsilon(x))_l = (1 - p)r_\varepsilon + pq_\varepsilon = r_\varepsilon - p(q_\varepsilon - r_\varepsilon).$$

3131 Solving this for $r_{\varepsilon'}$ and substituting back yields

$$p_{\text{flip}}(\varepsilon, \varepsilon') = \frac{q_\varepsilon - r'_\varepsilon}{q_\varepsilon - r_\varepsilon} = \frac{k-1}{k} \cdot \left(1 - \frac{(\exp(\varepsilon') - 1) \cdot (\exp(\varepsilon) + k - 1)}{(\exp(\varepsilon) - 1) \cdot (\exp(\varepsilon') + k - 1)} \right).$$

3132 By symmetry, all other coordinates are equal. $\square$

3133 It is easy to verify that this kernel forms a *consistent* Markov family, i.e. $K^{\varepsilon \rightarrow \varepsilon} =$
 3134 $K^{\varepsilon \rightarrow \tilde{\varepsilon}} \cdot K^{\tilde{\varepsilon} \rightarrow \varepsilon'}$ for all $0 \leq \varepsilon' \leq \tilde{\varepsilon} \leq \varepsilon$. This immediately leads to a simple closed form
 3135 for the binary case.

Lemma 6.33 (Stochastic Kernel Binary RR). *Let $x \in \{0, 1\}$ and fix two privacy parameters $0 \leq \varepsilon' \leq \varepsilon$. Now draw $Y_\varepsilon \sim \text{RandResp}(\varepsilon, x)$ and sample conditionally*

$$Y_{\varepsilon'} = \begin{cases} 1 - Y_\varepsilon & \text{with probability } p_{\text{flip}}(\varepsilon, \varepsilon') \\ Y_\varepsilon & \text{otherwise} \end{cases}, \text{ where } p_{\text{flip}}(\varepsilon, \varepsilon') := \frac{1}{2} \left(1 - \frac{\tanh(\varepsilon'/2)}{\tanh(\varepsilon/2)} \right).$$

Then we have $Y_{\varepsilon'} \sim \text{RandResp}(\varepsilon', x)$.

3136 *Proof.* Note that, by definition of the hyperbolic tangent $\tanh(a/2) := \frac{\exp(a) - 1}{\exp(a) + 1}$.

3137 Hence for $k = 2$, we immediately get by basic algebra

$$p_{\text{flip}}(\varepsilon, \varepsilon') = \frac{1}{2} \left(1 - \frac{\tanh(\varepsilon'/2)}{\tanh(\varepsilon/2)} \right).$$

3138 $\square$

RELEASE IN ARBITRARY ORDER We will now show that we can also produce them in any order. Consider the case where two releases $Y_1 \rightarrow Y_2$ with corresponding privacy levels $0 < \varepsilon_1 < \varepsilon_2$ have been published by the Markov kernel, $K^{\varepsilon_2 \rightarrow \varepsilon_1}$ and both of them are distributed as single releases from $Y_1 \sim \text{kRandResp}(\varepsilon_1, x)$ and $Y_2 \sim \text{kRandResp}(\varepsilon_2, x)$ respectively. We will now show how to create a new release whose privacy level $\varepsilon \in (\varepsilon_1, \varepsilon_2)$ lies in between but preserves the Markov chain $Y_2 \rightarrow Y \rightarrow Y_1$ while also $Y \sim \text{kRandResp}(\varepsilon, x)$ as desired.

Lemma 6.34 (Markov Bridge k -ary RR). *Let $x \in [k]$ and fix privacy parameters $0 < \varepsilon_1 \leq \varepsilon \leq \varepsilon_2$. As already defined in Lemma 6.32,*

$$K^{\eta \rightarrow \eta'}(i, j) = \begin{cases} 1 - p_{\text{flip}}(\eta, \eta') & i = j \\ p_{\text{flip}}(\eta, \eta') / (k - 1) & \text{otherwise} \end{cases}, \text{ with}$$

$$p_{\text{flip}}(\varepsilon, \varepsilon') = \frac{q_\varepsilon - r'_\varepsilon}{q_\varepsilon - r_\varepsilon} = \frac{k - 1}{k} \cdot \left(1 - \frac{(\exp(\varepsilon') - 1) \cdot (\exp(\varepsilon) + k - 1)}{(\exp(\varepsilon) - 1) \cdot (\exp(\varepsilon') + k - 1)} \right).$$

Assume a Markov process $Y_2 \rightarrow Y_1$ has produced

$$Y_2 \sim \text{kRandResp}(\varepsilon_2, x), \text{ and}$$

$$Y_1 \mid Y_2 = y_2 \sim K^{\varepsilon_2 \rightarrow \varepsilon_1}(y_2, \cdot)$$

If we conditionally on sample

$$\mathbf{P}[Y = y \mid Y_1 = a, Y_2 = b] = \frac{K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a)}{K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a)},$$

then

1. $Y_2 \rightarrow Y \rightarrow Y_1$ forms a Markov chain, and furthermore,
2. the marginal distribution $Y \sim \text{kRandResp}(\varepsilon, x)$ is correct.

Proof. Fix the private dataset $x \in [k]$ and let Y_1, Y, Y_2 sampled as described. Furthermore, denote with $M_\varepsilon(x)$ the probability vector of the mechanism given x . Note that by Lemma 6.33, we know that Y_1 and Y_2 are distributed as independent releases of $\text{kRandResp}(\varepsilon_1, x)$ and $\text{kRandResp}(\varepsilon_2, x)$. Furthermore, by construction, $Y_1 \rightarrow Y_2$ forms a Markov chain, and hence the joint distribution of Y_1, Y_2 factorizes, i.e. $\mathbf{P}[Y_1 = a, Y_2 = b] = \mathbf{P}[Y_1 = a \mid Y_2 = b] \mathbf{P}[Y_2 = b]$. We first show that $Y_1 \rightarrow Y \rightarrow Y_2$ forms a Markov chain with corresponding kernels $K^{\varepsilon_2 \rightarrow \varepsilon}, K^{\varepsilon \rightarrow \varepsilon_1}$.

To do so, we show that the joint distribution satisfies the Markov property, i.e., it factorizes out. Therefore, fix $y, a, b \in [k]$ and observe by Bayes's Theorem,

$$\mathbf{P}[Y_1 = a, Y = y, Y_2 = b] = \mathbf{P}[Y = y \mid Y_1 = a, Y_2 = b] \mathbf{P}[Y_1 = a, Y_2 = b] \quad (6)$$

$$= \frac{K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a)}{K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a)} \cdot K^{\varepsilon_2 \rightarrow \varepsilon_1}(b, a) \cdot \mathbf{P}[Y_2 = b] \quad (7)$$

$$= K^{\varepsilon_2 \rightarrow \varepsilon}(b, y) K^{\varepsilon \rightarrow \varepsilon_1}(y, a) \mathbf{P}[Y_2 = b], \quad (8)$$

where the last line shows the desired factorization.

It is left to show that $Y \sim \text{kRandResp}(\epsilon, x)$ also has the correct marginal distribution. By marginalization, we have for all y ,

$$\begin{aligned} \mathbf{P}[Y = y] &= \sum_{ab} \frac{K^{\epsilon_2 \rightarrow \epsilon}(b, y) K^{\epsilon \rightarrow \epsilon_1}(y, a)}{K^{\epsilon_2 \rightarrow \epsilon_1}(b, a)} \cdot K^{\epsilon_2 \rightarrow \epsilon_1}(b, a) \cdot \mathbf{P}[Y_2 = b] \\ &= \sum_{ab} K^{\epsilon_2 \rightarrow \epsilon}(b, y) K^{\epsilon \rightarrow \epsilon_1}(y, a) = \sum_a K^{\epsilon \rightarrow \epsilon_1}(y, a) \cdot \mathbf{P}[Y = y] \\ &= \mathbf{P}[Y = y]. \end{aligned}$$

□

Find a short empirical evaluation on Figure 6.2. In this example, we adaptively call the k -ary *Randomized Response* with three different privacy parameters, where the privacy level of the last one lies in between the first two. We then compare the empirical histogram with the direct releases.

6.7.2 Impossibility Results

One open question from the published version was whether there are mechanisms where PMLR is impossible. Here, we answer this question in the affirmative while again using the Markov chain formulation.

Lemma 6.35. *There exists a mechanism for which Private Multiple Lossless Releases is not possible.*

Proof. Consider an output space of three elements and let $M : \mathcal{X} \rightarrow \mathcal{Y}$ be the exponential mechanism that on dataset D_i , releases $r \in [3]$ with probability $\propto \exp\left(\frac{\epsilon}{2} u_{D_i}(r)\right)$. Furthermore, let D_1, D_2, D_3 be three neighboring datasets together with utilities, $u_1 = (0, 0, 0)$ and $u_2 = (0, 0, 1)$ and $u_3 = (1, 1, 0)$. We will show that multiple releases are not possible by trying to construct a Markov chain that supports PMLR. Assume $\epsilon_1 = \ln 8$ and $\epsilon_2 = \ln 9$. Then the marginal probabilities are

$$p_{D_1}^{\epsilon_1} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \quad p_{D_2}^{\epsilon_1} = \left(\frac{1}{5}, \frac{1}{5}, \frac{3}{5}\right) \quad p_{D_3}^{\epsilon_1} = \left(\frac{3}{7}, \frac{3}{7}, \frac{1}{7}\right) \quad (9)$$

$$p_{D_1}^{\epsilon_2} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \quad p_{D_2}^{\epsilon_2} = \left(\frac{1}{4}, \frac{1}{4}, \frac{2}{4}\right) \quad p_{D_3}^{\epsilon_2} = \left(\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\right) \quad (10)$$

Now consider the following subset of the constraints of the probability of releasing element 3 under ϵ_2 . Assume for contradiction there exists a (data-independent) stochastic kernel $K = (k_{ij})_{ij} \in \mathbb{R}^{3 \times 3}$ such that

$$p_{D_i}^{\epsilon_1} = K \cdot p_{D_i}^{\epsilon_2} \quad \forall i \in [3].$$

One can easily verify that this system of linear equations is infeasible. □

We leave a further exploration of this direction to future work. It may also be worth investigating possible connections to the lower bounds in [EC91] in the context of *Successive Refinement of Information*.

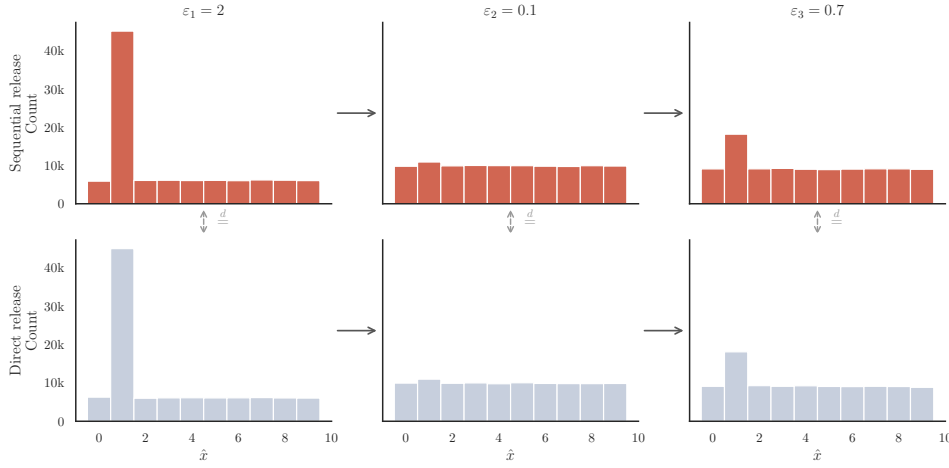


Figure 6.2: Empirical evaluation of sampling from the explicit kernel for 10-ary Randomized Response on 10^6 independent runs on the dataset $\mathbf{x} = 1$. **(top)** The correlated sampling procedure from the kernel. The overall privacy budget for releasing all samples is purely determined by the least private release, i.e. ϵ_3 . **(bottom)** independent releases using overall privacy budget of $\epsilon_1 + \epsilon_2 + \epsilon_3 = 2.8$.

FUTURE WORK Furthermore, one could relax PLMR by dropping the strict requirement that every subset of releases must be as private as the least private release within that subset. This requirement was primarily motivated by the on-line setting, where privacy levels arrive one at a time and the set of releases is dynamically changing. One could also imagine a setting, where all of those levels are already specified (offline setting), and one only requires the whole output sequence to be as private as the least private release, without imposing any stricter constraints on the subsets of this sequence. this relaxed problem can be formulated as a linear program over the full joint distribution of (a finite output space of) releases. The constraints enforce (a) the prescribed marginal distribution of each individual release and (b) the desired privacy guarantee for the joint output. One can then minimize the smallest privacy parameter ϵ' for which the constraints are feasible. The resulting linear program has a size exponential in the number of requested privacy levels, but it may still be useful for small sets of releases.

“Yeah, yeah,” I say. “Invisible.
When you call me, I immediately know that it is you
because you’re the only one I know who hides his number.”

— The Kangaroo Chronicles, Marc-Uwe Kling

THE CORRELATED SPARSE GAUSSIAN HISTOGRAM MECHANISM

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Joint work with Christian Lebeda

We consider the problem of releasing a sparse histogram under (ϵ, δ) -differential privacy. The stability histogram independently adds noise from a Laplace or Gaussian distribution to the non-zero entries and removes those noisy counts below a threshold. Thereby, the introduction of new non-zero values between neighboring histograms is only revealed with probability at most δ , and typically, the value of the threshold dominates the error of the mechanism. We consider the variant of the stability histogram with Gaussian noise.

Recent works ([Joseph and Yu, COLT ’24] and [Lebeda, SOSA ’25]) reduced the error for private histograms using correlated Gaussian noise. However, these techniques can not be directly applied in the very sparse setting. Instead, we adopt Lebeda’s technique and show that adding correlated noise to the non-zero counts only allows us to reduce the magnitude of noise when we have a sparsity bound. This, in turn, allows us to use a lower threshold by up to a factor of $1/2$ compared to the non-correlated noise mechanism. We then extend our mechanism to a setting without a known bound on sparsity. Additionally, we show that correlated noise can give a similar improvement for the more practical discrete Gaussian mechanism.

Releasing approximate counts is a common task in differential private data analysis. For example, the task of releasing a search engine log, where one wants to differentially private release the number of users who have searched each possible search term. The full domain of search terms is too large to work with directly, but the vector of all search counts is extremely sparse as most phrases are never searched. The standard approach to exploit sparsity while ensuring differential privacy is the *stability histogram* [BV19; Kor+09; BNS16; LAP22; GAT20; Göt+12; Wil+20]. The idea is to preserve sparsity by filtering out zero counts and then adding noise from a suitable probability distribution to the remaining ones. Unfortunately, some filtered-out counts could be non-zero in a neighboring dataset, and thus, revealing any of them would violate privacy. To limit this privacy violation, the stability histogram introduces a threshold τ and releases only those noisy counts that exceed $1 + \tau$. This way, we might still reveal the true dataset among a pair of neighboring datasets if one of these counters exceeds the threshold, but with an appropriate choice of τ , this is extremely unlikely to happen.

In our setting, one single user may contribute to many counts; hence, using the *Gaussian Sparse Histogram Mechanism* (GSHM) [GAT20; Wil+24] might be preferable. The GSHM is similar to the stability histogram, but it replaces noise from the Laplace distribution with Gaussian noise. To analyze the (ϵ, δ) guarantees of the

GSHM, one has to find an upper bound for δ , which is influenced by the standard Gaussian mechanism and the small probability of infinite privacy loss that can happen when the noise exceeds τ . We denote these two quantities as δ_{gauss} and δ_{inf} . Recently, Wilkins, Kifer, Zhang, and Karrer [Wil+24] gave exact privacy guarantees of the GSHM, which improves over the analysis by GAT [GAT20] where δ_{gauss} and δ_{inf} were simply summed. Their main contribution is a more intricate case distinction of a single user's impact on the values of δ_{gauss} and δ_{inf} . This allows them to use a lower threshold with the same privacy parameters. Their improvement lies in the constants, but it can be significant in practical applications because the tighter analysis essentially gives better utility at no privacy cost.

Our goal is to reduce the error further. Because the previous analysis is exact, we must exploit some additional structure of the problem. Consider a d -dimensional histogram $H(\mathbf{x}) = \sum_{i=1}^n X_i$ for $\mathbf{x} = (X_1, \dots, X_n)$ for users with data $X_i \in \{0, 1\}^d$. Notice that *adding* a single user to $\mathbf{x}$ can only increase counts in $H(\mathbf{x})$, whereas *removing* one can only decrease counts. Lebeda Lebeda [Leb25] recently showed how to exploit this monotonicity by adding a small amount of correlated noise. This reduces the total magnitude of noise almost by a factor of 2 compared to the standard Gaussian Mechanism. So, it is natural to ask if we can adapt this technique to the setting when the histogram is sparse. This motivates our main research question:

Question: When can we take advantage of monotonicity and use correlated noise to improve the *Gaussian Sparse Histogram Mechanism*?

OUR CONTRIBUTIONS We answer this question by introducing the *Correlated Stability Histogram* (CSH). Building on the work by Lebeda [Leb25], we extend their framework to the setting of releasing a histogram under a sparsity constraint, that is $\|H(\mathbf{x})\|_0 \leq k$ for some known k . This is a natural setting that occurs e.g. for Misra-Gries sketches, where k is the size of the sketch [MG82; LT23]. Furthermore, enforcing this sparsity constraint is necessary to achieve the structure required to benefit from correlated noise.

Correlated GSHM. We introduce the *Correlated Stability Histogram* (CSH), a variant of the GSHM using correlated noise. Our algorithm achieves a better utility-privacy trade-off than GSHM for k -sparse histograms. Similar to the result of Lebeda Lebeda [Leb25], we show that correlated noise can reduce the error by almost a factor of 2 at no additional privacy cost. In particular, our main result lies in the following (informally stated) theorem:

Theorem 7.1 (The Correlated Stability Histogram Mechanism (Informal)). *Let $H(\mathbf{x}) = \sum_i^n X_i$ denote a histogram with bounded sparsity, where $\mathbf{x} = (X_1, \dots, X_n)$ and $X_i \in \{0, 1\}^d$. If the GSHM privately releases $H(\mathbf{x})$ under (ϵ, δ) -DP with noise magnitude σ and removes noisy counters below a threshold τ , then the Correlated Stability Histogram Mechanism (CSH) releases $H(\mathbf{x})$ also under (ϵ, δ) -DP with noise of magnitude $\sigma/2 + o(1)$ and removes noisy counters below a threshold $\tau/2 + o(1)$.*

As a baseline, we first analyze our mechanism using the *add-the-deltas* [GAT20] approach and show that even this approach outperforms the exact analysis in Wilson, Zhang, Lam, Desfontaines, Simmons-Marengo, and Gipson [Wil+20] in many settings. We then turn our attention to a tighter analysis, which uses an intricate

case distinction to upper bound the parameter δ . Furthermore, we complement our approach with the following results:

Generalization & Extensions. We extend our mechanism to multiple other settings, including extensions considered in Wilkins, Kifer, Zhang, and Karrer [Wil+24]. First, we generalize our approach to allow an additional threshold that filters out infrequent data in a pre-processing step. Second, we discuss how other aggregate database queries can be included in our mechanism. Last, we generalize to top- k counting queries when we have no bound for $\|H(\mathbf{x})\|_0$.

Discrete Gaussian Noise. Our mechanism achieves the same improvement over GSHM when noise is sampled from the discrete Gaussian rather than the continuous distribution. We present a simple modification to make our mechanism compatible with discrete noise. This has practical relevance even for the dense setting considered by Lebeda Lebeda [Leb25].

Organization. The rest of the section is organized as follows. Section 7.1 introduces the problem formally and reviews the required background. Section 7.2 introduces the *Correlated Stability Histogram*, our main algorithmic contribution for the sparse case. Furthermore, in Section 7.3 we generalize our approach to the non-sparse setting. The extensions of our technique to match the setting of Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] are discussed in Section 7.5. In Section 7.6, we show how to adapt our mechanism for discrete Gaussian noise. Finally, we conclude the paper in Section 6.5 and discuss open problems for future work.

Additionally, numerical evaluations in Section 7.4 confirm our claim that we improve the error over previous techniques.

7.1 PRELIMINARIES AND BACKGROUND

Given a dataset $\mathbf{x} \in \mathcal{U}^*$, where $\mathcal{U}^* := \bigcup_{m=0}^{\infty} \mathcal{U}^m$ is the set of datasets with any size, we want to perform an aggregate query $\mathcal{A}$ under differential privacy. We focus on settings where each user has a set of elements, and we want to estimate the count of all elements over the dataset. Therefore, we consider a dataset $\mathbf{x} = (X_1, \dots, X_n)$ of n data points where each $X_i \in \{0, 1\}^d$. Our goal is to output a private estimate of the histogram $H(\mathbf{x}) \in \mathbb{N}^d$, where $H(\mathbf{x}) = \sum_{i=1}^n X_i^1$. For any vector $\mathbf{h} \in \mathbb{R}^d$, we define the *support* as $U(\mathbf{h}) = \{i \in [d] : H_i \neq 0\}$ and denote the ℓ_0 norm as $\|\mathbf{h}\|_0 := |U(\mathbf{h})|$, being the number of non-zeroes in $\mathbf{h}$. In this work, we focus on settings where the dimension d is very large or infinite.

Proposed by Dwork, McSherry, Nissim, and Smith [Dwo+06b], differential privacy (DP) is a property of a randomized mechanism. The intuition behind differential privacy is that privacy is preserved by ensuring that the output distribution does not depend much on any individual's data. Note that by this definition $|\mathbf{x}| = n$ and $|\mathbf{x}'| = n \pm 1$:

Definition 7.1 (Neighboring datasets). *A pair of datasets $\mathbf{x} \sim \mathbf{x}'$ is neighboring if there exists an i such that either*

$$\mathbf{x} = (X'_1, \dots, X'_{i-1}, X'_{i+1}, \dots, X'_{n+1}) \quad \text{or} \quad \mathbf{x}' = (X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n).$$

¹ Note that in the differential privacy literature the term *histogram* is often used for the special case where users each have a single element. We use it for consistency with related work (e.g. [Wil+24]). The setting we consider is closely related to the problem of releasing one-way marginals.

In this chapter, we consider both the ℓ_2 sensitivity and, more generally, the exact sensitivity space of the queries.

Definition 7.2 (Sensitivity space and ℓ_2 sensitivity). *The sensitivity space of a deterministic function $f : \mathcal{U}^* \rightarrow \mathbb{R}^d$ is the set $\Delta f = \{f(\mathbf{x}) - f(\mathbf{x}') \in \mathbb{R}^d \mid \mathbf{x} \sim \mathbf{x}'\}$ and denote $\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^d x_i^2}$ as the ℓ_2 norm of any $\mathbf{x} \in \mathbb{R}^d$. Then the ℓ_2 sensitivity of f is defined as*

$$\Delta_2 f = \max_{\mathbf{x} \sim \mathbf{x}'} \|f(\mathbf{x}) - f(\mathbf{x}')\|_2 = \max_{\mathbf{x} \in \Delta f} \|\mathbf{x}\|_2.$$

The standard Gaussian mechanism adds continuous noise from a Gaussian distribution to $f(\mathbf{x})$ with magnitude scaled according to Lemma 7.2.

Lemma 7.2 ([CKS20, Theorem 8] The Analytical Gaussian Mechanism). *Let $f : \mathcal{U}^* \rightarrow \mathbb{R}^d$ denote a function with ℓ_2 sensitivity at most Δ_2 . Then the mechanism that outputs $f(X) + (Z_1, \dots, Z_d)$, where each $Z_i \sim \mathcal{N}(0, \sigma^2)$ are independent and identically distributed, satisfies (ϵ, δ) -differential privacy if the following inequality holds*

$$\delta \geq \Phi\left(\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right) - e^\epsilon \Phi\left(-\frac{\Delta_2}{2\sigma} - \frac{\epsilon\sigma}{\Delta_2}\right),$$

where $\Phi(X)$ denotes the cdf of the Gaussian distribution.

7.1.1 The Gaussian Sparse Histogram Mechanism

We consider the problem of privately releasing the histogram $H(\mathbf{x})$ of a dataset $\mathbf{x} = (X_1, \dots, X_n)$ where each $X_i \in \{0, 1\}^d$ under differential privacy. The standard techniques for releasing a private histogram are the well-known Laplace mechanism [Dwo+06b] and Gaussian mechanism [Dwo+06a; CKS20], which achieve ϵ - and (ϵ, δ) -DP, respectively. Although this works well for dense data, it is unsuited for very sparse data where $\|H(\mathbf{x})\|_0 \ll d$ because adding noise to *all entries* increases both the maximum error as well as the space and time requirements. Furthermore, the mechanisms are undefined for infinite domains ($d = \infty$).



Figure 7.1: Examples of different kinds of neighboring datasets for the Gaussian Sparse Histogram Mechanism where a single user can contribute to at most four counters, thus $\|X_i\|_0 \leq 4$. These counters are depicted in green. **a)** For the example on the left, the mechanism behaves exactly as running the Gaussian mechanism on a restricted domain. **b)** In the case in the middle, we only have to bound the probability that one of the green elements together with the additive noise term exceeds the threshold $1 + \tau$. **c)** The case on the right is the most difficult case for the privacy analysis because the overall δ value depends on both kinds of changes.

Algorithm 15 The Gaussian Sparse Histogram Mechanism (GSHM)**Require:** Parameters σ and τ , histogram $H(\mathbf{x}) \in \mathbb{N}^d$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^d$.
- 2: **for** each $i \in [d]$ where $H(\mathbf{x})_i \neq 0$ **do**
- 3: Sample $Z_i \sim \mathcal{N}(0, \sigma^2)$.
- 4: **if** $H(\mathbf{x})_i + Z_i > 1 + \tau$ **then**
- 5: Set $\tilde{H}_i = H(\mathbf{x})_i + Z_i$.
- 6: **Release** $\tilde{\mathbf{h}}$.

In the classic sparse histogram setting where $\|X_i\|_0 = 1$, the preferred technique under (ϵ, δ) -DP is the *stability histogram* which combines Laplace noise with a thresholding technique (see [BV19; Kor+09; BNS16; LAP22; GAT20; Göt+12; Wil+20]).

In this chapter, we consider the *Gaussian Sparse Histogram Mechanism* (see [GAT20; Wil+24]), which replaces Laplace noise in the stability histogram with Gaussian noise. This is often preferred when users can contribute multiple items as the magnitude of noise is scaled by the ℓ_2 sensitivity instead of the ℓ_1 sensitivity. The GSHM adds Gaussian noise to each non-zero counter of $H(\mathbf{x})$ and removes all counters below a threshold $1 + \tau$. Find the pseudocode in Algorithm 15. We discuss the impact of the parameter τ below.

To get (ϵ, δ) differential privacy guarantees, we observe that there are two sources of privacy loss that have to be accounted for by the value of δ : δ_{gauss} from the Gaussian noise itself and the probability of infinite privacy loss δ_{inf} , when a zero count is deterministically ignored in one dataset, but possibly released in a neighboring one. These events have infinite privacy loss because they can only occur for one of the datasets. Therefore, δ_{inf} bounds the possibility of outputting a counter that is not present in the neighboring dataset. Figure 7.1 gives some intuition about the role of δ_{gauss} and δ_{inf} . Throughout this section, we assume that $\|X_i\|_0 \leq k$ for some known integer $k \ll d$.

7.1.2 The add-the-deltas Approach

The following approach appeared in a technical report in the *Google differential privacy library* [GAT20]. Similar techniques have been used elsewhere in the literature (e.g. [Wil+20]). We refer to this technique as *add-the-deltas*. We have to account for the privacy loss due to the magnitude of the Gaussian noise. Therefore, the value of δ_{gauss} is typically found by considering the worst-case effect of a user that only changes non-zero counters in both histograms. The value of δ_{gauss} follows from applying Lemma 7.2 (compare also Figure 7.1a).

The event of infinite privacy loss is captured by δ_{inf} . This is bounded by considering the worst-case scenario of changing k zero-counters to 1 and the fact that infinite privacy loss occurs exactly when any of these k counters exceed the threshold in the non-zero dataset (compare Figure 7.1b).

The observation for the *add-the-deltas* approach is that $\delta_{\text{gauss}} + \delta_{\text{inf}}$ is a valid upper bound on the overall δ value, and hence the condition of Lemma 7.3 is sufficient to achieve differential privacy.

Lemma 7.3 ([GAT20] *add-the-deltas*). *If pairs of neighboring histograms differs in*

at most k counters then the Gaussian Sparse Histogram Mechanism with parameters σ and τ satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -differential privacy where

$$\delta_{\text{gauss}} = \Phi\left(\frac{\sqrt{k}}{2\sigma} - \frac{\epsilon\sigma}{\sqrt{k}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{k}}{2\sigma} - \frac{\epsilon\sigma}{\sqrt{k}}\right), \quad \text{and} \quad \delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma}\right)^k.$$

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7.1.3 Exact Analysis by Taking the Max over the Sensitivity Space

While the *add-the-deltas* approach is sufficient to bound δ it does not give the tightest possible parameters. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] were able to derive an exact value for δ . Compared to the *add-the-deltas* approach above, the key insight is that the two worst-case scenarios cannot occur simultaneously for any pair of neighboring histograms. This means that each counter either contributes to δ_{gauss} or δ_{inf} , but never to both. In the extreme cases, either all k counters flip from zero to one, from one to zero, or none of them do between neighboring datasets. Thus, we only need to consider a single source of privacy loss. In the other (mixed) cases, we have to consider both sources of privacy loss, but the change is smaller than for the worst-case pair of datasets. A small example is depicted in Figure 7.1. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] used this fact to reduce the threshold required to satisfy the given privacy parameters using a tighter analysis. We now restate their main result: the exact privacy analysis of the GSHM.

Lemma 7.4 ([Wil+24] Exact Privacy Analysis of the GSHM). *If pairs of neighboring histograms differs in at most k counters then the Gaussian Sparse Histogram Mechanism with parameters σ and τ satisfies (ϵ, δ) -differential privacy where $\gamma(j) = (k - j) \log \Phi\left(\frac{\tau}{\sigma}\right)$ and*

$$\begin{aligned} \delta \geq & \max \left[1 - \Phi\left(\frac{\tau}{\sigma}\right)^k, \right. \\ & \max_{j \in [k]} 1 - \Phi\left(\frac{\tau}{\sigma}\right)^{k-j} + \Phi\left(\frac{\tau}{\sigma}\right)^{k-j} \\ & \quad \cdot \left[\Phi\left(\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon - \gamma(j))\sigma}{\sqrt{j}}\right) - e^{\epsilon - \gamma(j)} \Phi\left(-\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon - \gamma(j))\sigma}{\sqrt{j}}\right) \right], \\ & \left. \max_{j \in [k]} \Phi\left(\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon + \gamma(j))\sigma}{\sqrt{j}}\right) - e^{\epsilon + \gamma(j)} \Phi\left(-\frac{\sqrt{j}}{2\sigma} - \frac{(\epsilon + \gamma(j))\sigma}{\sqrt{j}}\right) \right]. \end{aligned}$$

Note that we use a slightly different convention from Wilkins, Kifer, Zhang, and Karrer [Wil+24]. We use a threshold of $1 + \tau$ rather than τ . Furthermore, they also consider a more general mechanism that allows for optional aggregation queries and an additional threshold parameter. Although our work can easily be adapted to this setting as well, these extensions are not the focus of our main work to simplify presentation. A discussion of the extensions can be found in Section 7.5.

7.1.4 The Correlated Gaussian Mechanism

Our work builds on a recent result by Lebeda Lebeda [Leb25] about using correlated noise to improve the utility of the Gaussian mechanism under the add-remove

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neighboring relationship. They show that when answering d counting queries, adding a small amount of correlated noise to all queries can reduce the magnitude of Gaussian noise by almost half. We restate their main result for (ϵ, δ) -differential privacy with a short proof as they use another privacy definition.

Lemma 7.5 ([Leb25] The Correlated Gaussian Mechanism). *Let $H(\mathbf{x}) := \sum_{i=1}^n X_i$ where $X_i \in \{0, 1\}^d$. Then the mechanism that outputs $H(\mathbf{x}) + Z_{\text{cor}} \mathbf{1}^d + (Z_1, \dots, Z_d)$ satisfies (ϵ, δ) -DP. Here $\mathbf{1}^d$ is the d -dimensional vector of all ones, $Z_{\text{cor}} \sim \mathcal{N}(0, \sigma^2/\gamma)$, and each $Z_i \sim \mathcal{N}(0, \sigma^2)$ where*

$$\delta \geq \Phi\left(\frac{\sqrt{d+\gamma}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{d+\gamma}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{d+\gamma}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{d+\gamma}}\right).$$

Proof. This follows from combining the inequality for the standard Gaussian mechanism (see Lemma 7.2) with [Leb25, Lemma 3.5]. Furthermore, if we set $\gamma = \sqrt{d}$ as in [Leb25, Theorem 3.1] we minimize the total magnitude of noise. Notice that the value of σ scales with $\sqrt{d}$ for the standard Gaussian mechanism and it scales with $\frac{1}{2}\sqrt{d+\gamma} = \frac{1}{2}\sqrt{d+\sqrt{d}}$ here. We add two noise samples to each query, and the total error for each data point scales with

$$\frac{1}{2}\sqrt{d+\gamma+d/\gamma+1} = (\sqrt{d}+1)/2.$$

□

Concurrently with the result discussed above, Joseph and Yu [JY24] considered the setting where user contributions are sparse such that $\|X_i\|_0 \leq k$ for some $k \leq d/2$. They give an algorithm that adds the optimal amount of correlated Gaussian noise for any d and $k \leq d/2$. It is natural to ask whether this algorithm can improve error for our setting since they focus on a sparse setting. However, their improvement factor over the standard Gaussian mechanism depends on the sparsity. Naively applying their technique for the setting of the GSHM where $k \ll d$ yields no practical improvements. We instead focus on adapting the technique from [Leb25]. We discuss a more restrictive setting where the technique of Joseph and Yu [JY24] applies in Section 6.5.

7.2 ALGORITHMIC FRAMEWORK

We are now ready to introduce our main contribution, a variant of the Gaussian Sparse Histogram Mechanism using correlated noise. We first introduce our definition of k -sparse monotonic histograms. Throughout this section, we assume that all histograms are both k -sparse and monotonic. Intuitively, we define monotonicity on a histogram in a way that captures the setting where the counts are either all increasing or all decreasing. Observe that due to the monotonicity constraint, we have that the supports U and U' for two neighboring histograms satisfy either $U \subseteq U'$ or $U' \subseteq U$. We use this property in the privacy proofs later. The histograms are also monotonic in [Wil+24], but they do not require k -sparsity. We provide a mechanism for a setting where the histograms are not k -sparse in Section 7.3. There we enforce the sparsity constraint using a simple pre-processing step. Both sparsity and monotonicity are required in order to achieve the structure between neighboring histograms required to benefit from correlated noise.

Algorithm 16 Correlated Stability Histogram (CSH)

Require: Parameters k, σ and τ , histogram $H(\mathbf{x}) \in \mathbb{N}^d$ where $\|H(\mathbf{x})\|_0 \leq k$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^d$.
- 2: Sample $Z_{\text{corr}} \sim \mathcal{N}(0, \sigma^2 / \sqrt{k})$.
- 3: **for** each $i \in [d]$ where $H(\mathbf{x})_i \neq 0$ **do**
- 4: Sample $Z_i \sim \mathcal{N}(0, \sigma^2)$.
- 5: **if** $H(\mathbf{x})_i + Z_i + Z_{\text{corr}} > 1 + \tau$ **then**
- 6: Set $\tilde{H}_i = H(\mathbf{x})_i + Z_i + Z_{\text{corr}}$.
- 7: **Release** $\tilde{H}$.

Definition 7.3 (k -sparse monotonic histogram). *We assume that the input histogram is k -sparse. That is, for any dataset $\mathbf{x}$ we have $\|H(\mathbf{x})\|_0 \leq k$. Furthermore, the sensitivity space of H is $\{0, 1\}^d \cup \{0, -1\}^d$. That is, the difference between counters of neighboring histograms are either non-decreasing or non-increasing.*

One example of k -sparse monotonic histograms is Misra-Gries sketches. Merging Misra-Gries sketches is common in practical applications. The sensitivity space of merged Misra-Gries sketches of size k exactly matches Definition 7.3 (See Lebeda and Tetek [LT23]). Our algorithm is more general than Misra-Gries sketches and satisfies differential privacy as long as the structure between neighboring histograms holds for all pairs of neighboring datasets.

Notice that Definition 7.3 implies that neighboring histograms differ in at most k counters. As such, we can release the histogram using the standard Gaussian Sparse Histogram Mechanism. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] already uses the fact that counters are either non-decreasing or non-increasing in their analysis. We intend to further take advantage of the monotonicity by adding a small amount of correlated noise to all non-zero counters. This allows us to reduce the total magnitude of noise similar to [Leb25]. The reduced magnitude of noise in turn allows us to reduce the threshold required for privacy. The pseudocode for our mechanism is in Algorithm 16.

We first give privacy guarantees for Algorithm 16 in a (relatively) simple closed form similar to the *add-the-deltas* approach. Later, we give tighter bounds using a more complicated analysis similar to Lemma 7.4. Unfortunately, the proofs from previous work in either case rely on the fact that all noisy counters are independent. That is clearly not the case for our mechanism because the value of Z_{corr} is added to all entries. Instead, we use different techniques to give similar results, starting with the *add-the-deltas* approach. The following lemma gives a general bound for the event that one of j correlated noisy terms exceeds a threshold τ . We use this in the proof for both approaches later.

Lemma 7.6 (Upper bound for Correlated Noise). *Let $Z_{\text{corr}} \sim \mathcal{N}(0, \sigma^2 / \sqrt{k})$ be a single sample for a real $k > 0$ together with j additional samples $Z_1, \dots, Z_j \sim \mathcal{N}(0, \sigma^2)$. Then*

for any $\tau > 0$, we have $\mathbf{P}[\exists i \in [j] : Z_{\text{corr}} + Z_i > \tau] \leq 1 - \Phi\left(\frac{\tau}{\sigma(k^{-1/4} + 1)}\right)^{j+1}$.

Proof. We first give a bound for Z_{corr} and the Z_i 's independently, which is sufficient for us to bound the probability of this event. First, observe that the j uncorrelated

terms $Z_i \sim \mathcal{N}(0, \sigma^2)$ are independent, and we are interested in bounding the maximum of them and hence:

$$\begin{aligned} \mathbf{P} \left[\max_{Z_1, \dots, Z_j} Z_i \leq \frac{\tau}{1 + k^{-1/4}} \right] &= \mathbf{P} \left[\mathcal{N}(0, \sigma^2) \leq \frac{\tau}{1 + k^{-1/4}} \right]^j \\ &= \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right)^j. \end{aligned}$$

Similarly, we have for $Z_{\text{corr}} \sim N(0, \sigma^2/\sqrt{k})$ that

$$\mathbf{P} \left[Z_{\text{corr}} \leq \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \right] = \Phi \left(\frac{\tau k^{-1/4}}{\sigma (1 + k^{-1/4}) \cdot k^{-1/4}} \right) = \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right).$$

Notice that if both $Z_i \leq \tau/(1 + k^{-1/4})$ and $Z_{\text{corr}} \leq \tau k^{-1/4}/(1 + k^{-1/4})$ hold then $Z_{\text{corr}} + Z_i \leq \tau$. As such, we can prove that the lemma holds since,

$$\mathbf{P} \left[Z_{\text{corr}} + \max_{Z_1, \dots, Z_j} Z_i > \tau \right] \leq \mathbf{P} \left[Z_{\text{corr}} > \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \vee \max_{Z_1, \dots, Z_j} Z_i > \frac{\tau}{1 + k^{-1/4}} \right] \quad (1)$$

$$= 1 - \mathbf{P} \left[Z_{\text{corr}} \leq \frac{\tau k^{-1/4}}{1 + k^{-1/4}} \right] \cdot \mathbf{P} \left[\max_{Z_1, \dots, Z_j} Z_i \leq \frac{\tau}{1 + k^{-1/4}} \right] \quad (2)$$

$$= 1 - \Phi \left(\frac{\tau}{\sigma (1 + k^{-1/4})} \right)^{j+1},$$

where step (1) holds by a union bound and step (2) holds because the random variables Z_{corr} and $\tilde{Z} = \max_{Z_1, \dots, Z_j} Z_i$ are independent. $\square$

7.2.1 The add-the-deltas Analysis

The following Theorem 7.7 proves privacy guarantees of our mechanism using a similar technique than the one proposed by the GAT [GAT20] and Wilson, Zhang, Lam, Desfontaines, Simmons-Marengo, and Gipson [Wil+20] which is known as *add-the-deltas*. The total value for δ is split between δ_{gauss} and δ_{inf} which accounts for the two types of privacy loss that are relevant to the mechanism. Like with the GSHM, these values are found by considering worst-case pairs of neighboring histograms for each case. However, a pair of neighboring histograms cannot be worst-case for both values as seen in Figure 7.1 which motivates the tighter analysis later in the section.

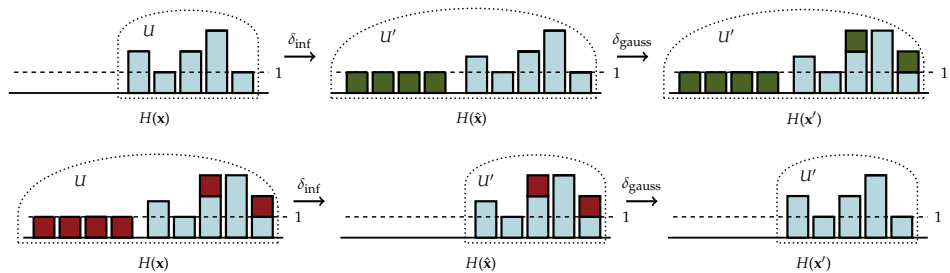


Figure 7.2: The separation technique of the δ_{gauss} and δ_{inf} used in Theorems 7.7 and 7.9. The idea is to construct an intermediate histogram $H(\hat{\mathbf{x}})$ with the same support U' as $H(\mathbf{x}')$ but only reflect the changes that can cause infinite privacy loss between $H(\mathbf{x})$ and $H(\mathbf{x}')$.

Theorem 7.7 (*add-the-deltas technique*). *Alg. 16 satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -dp for k -sparse monotonic histograms where*

$$\delta_{\text{gauss}} = \Phi\left(\frac{\sqrt{k+\sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k+\sqrt{k}}}\right) - e^\epsilon \Phi\left(-\frac{\sqrt{k+\sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k+\sqrt{k}}}\right)$$

$$\delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma(1+k^{-1/4})}\right)^{k+1}.$$

Proof. By Definition 2.4, the lemma holds if for any pair of neighboring k -sparse monotonic histograms $H(\mathbf{x})$ and $H(\mathbf{x}')$ and all sets of outputs Y we have

$$\mathbf{P}[CSH(H(\mathbf{x})) \in Y] \leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}.$$

We prove that the inequality above holds by introducing a third histogram. This new histogram is constructed such that it is between $H(\mathbf{x})$ and $H(\mathbf{x}')$. We first state the desired properties of this histogram and then show that such a histogram must exist for all neighboring histograms. Assume for now that there exists a histogram $H(\hat{\mathbf{x}}) \in \mathbb{N}^d$ where the following two inequalities hold for any set of outputs Y :

$$\mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] \leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}, \quad (3)$$

$$\mathbf{P}[CSH(H(\mathbf{x})) \in Y] \leq \mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}}. \quad (4)$$

Then the inequality we need for the lemma follows immediately since

$$\begin{aligned} \mathbf{P}[CSH(H(\mathbf{x})) \in Y] &\leq \mathbf{P}[CSH(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}} \\ &\leq e^\epsilon \mathbf{P}[CSH(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}. \end{aligned}$$

Next, we show how to construct $H(\hat{\mathbf{x}})$ from $H(\mathbf{x})$ and $H(\mathbf{x}')$, and finally we show that each of two inequalities holds using this construction. Let $U = \{i \in [d] : H(\mathbf{x})_i \neq 0\}$ denote the support of $H(\mathbf{x})$ and define similarly U' and $\hat{U}$. We construct $H(\hat{\mathbf{x}})$ such that it has the same support as $H(\mathbf{x}')$, that is $U' = \hat{U}$. For all $i \in (U \cap U')$ we set $H(\hat{\mathbf{x}})_i = H(\mathbf{x})_i$ and for all $i \notin (U \cap U')$ we set $H(\hat{\mathbf{x}})_i = H(\mathbf{x}')_i$. In other words, we construct $H(\hat{\mathbf{x}})$ such that (1) $H(\mathbf{x}')$ and $H(\hat{\mathbf{x}})$ only differ in entries that are in both U and U' (2) $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ only differ in entries that are in only one of U and U' . This allows us to analyze each case separately to derive our values for δ_{gauss} and δ_{inf} . An example of this construction is shown in Figure 7.2.

We start with the case of δ_{gauss} using $H(\mathbf{x}')$ and $H(\hat{\mathbf{x}})$. Let CSH' denote a new mechanism equivalent to Algorithm 16 except that the condition on line 5 is removed. Notice that with $H(\mathbf{x}')$ or $H(\hat{\mathbf{x}})$ as inputs CSH' is equivalent to the Correlated Gaussian Mechanism restricted to U' . Since $|U'| \leq k$ we have by Lemma 7.5 that

$$\mathbf{P}[CSH'(H(\hat{\mathbf{x}})) \in Y] \leq e^\epsilon \mathbf{P}[CSH'(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}}.$$

Notice that if we post-process the output of CSH' by removing entries below $1 + \tau$, then the output distribution is equivalent to CSH . Equation (3) therefore holds because post-processing does not affect differential privacy guarantees (see Definition 2.4).

The histograms $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ only differ in entries that all have a count of 1 in one of the histograms while they all have a count of 0 in the other. δ_{inf} accounts for the event where any such counter exceeds the threshold because the

distributions are identical for the shared support. The probability of this event is increasing in the number of different elements between $H(\mathbf{x})$ and $H(\hat{\mathbf{x}})$ and therefore, the worst case happens for neighboring datasets such that $H(\mathbf{x}) = \mathbf{1}^k$ and $H(\hat{\mathbf{x}}) = \mathbf{0}^k$. Note that this bound also hold when zero counters in $H(\mathbf{x})$ are non-zero in $H(\hat{\mathbf{x}})$. We focus on one direction because the proof is almost identical for the symmetric case. We bound the probability of outputting any entries from $H(\mathbf{x})$, that is $\tilde{H}_i \doteq 1 + Z_{\text{corr}} + Z_i > 1 + \tau$ for at least one $i \in (U \setminus U')$. The bound for δ_{inf} follows from setting $j = k$ in Lemma 7.6. $\square$

In the Lemma 7.8 below, we show how to directly compute the required threshold based on the result in Theorem 7.7.

Lemma 7.8 (Computing the threshold τ). *For a fixed privacy budget ϵ, δ and parameters k and σ , the add-the-deltas technique requires*

$$\tau \geq \Phi^{-1} \left(\sqrt[k+1]{1 - \delta - \delta_{\text{gauss}}} \right) \cdot \left(1 + k^{-1/4} \right) \sigma,$$

where Φ^{-1} is the inverse CDF of the normal distribution and δ_{gauss} is defined as in Theorem 7.7.

Proof. Observe $\delta - \delta_{\text{gauss}} \geq \delta_{\text{inf}} = 1 - \Phi\left(\frac{\tau}{\sigma(1+k^{-1/4})}\right)^{k+1}$. Taking Φ^{-1} on both sides yields $\frac{\tau}{\sigma(1+k^{-1/4})} \geq \Phi^{-1} \left(\sqrt[k+1]{1 - \delta - \delta_{\text{gauss}}} \right)$. Multiplying by $\sigma \left(1 + k^{-1/4} \right)$ proves the claim. $\square$

7.2.2 Tighter Analysis

Next, we carry out a more careful analysis that considers all elements from the sensitivity space similar to Wilkins, Kifer, Zhang, and Karrer [Wil+24]. As discussed above, we cannot directly translate the analysis by Wilkins et al. because they rely on independence between each entry. We refer the full proof to the full version of the paper [LR24], and we only give a short intuition behind the theorem here.

Theorem 7.9 (Tighter Analysis). *First define $\gamma(j) = \min \left(\sqrt{j}, \frac{1}{2} \sqrt{j + \sqrt{k}} \right)$, $\psi(m) = \Phi \left(\frac{\tau}{(1+k^{-1/4})\sigma} \right)^{m+1}$, and $\hat{\epsilon}(j) = \epsilon + \ln(\psi(k-j))$. Then Algorithm 16 with parameters k, σ , and τ satisfies (ϵ, δ) -differential privacy for k -sparse monotonic histograms, where*

$$\begin{aligned} \max \left[1 - \psi(k), \Phi \left(\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k + \sqrt{k}}} \right) - e^\epsilon \Phi \left(-\frac{\sqrt{k + \sqrt{k}}}{4\sigma} - \frac{2\epsilon\sigma}{\sqrt{k + \sqrt{k}}} \right), \right. \\ \max_{j \in [k-1]} 1 - \psi(k-j) + \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\epsilon\sigma}{\gamma(j)} \right) - e^\epsilon \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\epsilon\sigma}{\gamma(j)} \right), \\ \left. \max_{j \in [k-1]} \Phi \left(\frac{\gamma(j)}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\gamma(j)} \right) - e^{\hat{\epsilon}(j)} \Phi \left(-\frac{\gamma(j)}{2\sigma} - \frac{\hat{\epsilon}(j)\sigma}{\gamma(j)} \right) \right] \leq \delta. \end{aligned}$$

The result relies on a case by case analysis where each term in the maximum corresponds to a specific difference between neighboring histograms. The first two terms covers cases when we only have to consider either the infinite privacy loss event or the Gaussian noise, respectively. The remaining terms cover the differences when we have to account for both Gaussian noise and the threshold similar to

Algorithm 17 Top-k Mechanism using Correlated Gaussian Noise

Require: Parameters k , σ , and τ , dataset $\mathbf{x} = (X_1, \dots, X_n)$ where $X_i \in \{0, 1\}^d$.

- 1: Let $H(\mathbf{x}) = \sum_{i=1}^n X_i$.
- 2: Let $\tilde{H}(\mathbf{x}) = \{0\}^d$.
- 3: **for** each $i \in [d]$ **do**
- 4: Set $\tilde{H}(\mathbf{x})_i = \max(0, H(\mathbf{x})_i - H(\mathbf{x})^{(k+1)})$.
- 5: **Release** $\text{CSH}(k, \sigma, \tau, \tilde{H}(\mathbf{x}))$.

case c) in Figure 7.1. The first of the internal maximum covers the cases where $U \supset U'$ and $|U'| \leq j$, while the second covers the cases where $U \subset U'$ and $|U| \leq j$. For each case in our analysis, we split up the impact from Gaussian noise and the threshold. Together, these cases cover all elements in the sensitivity space for k -sparse monotonic histograms.

7.3 TOP-K COUNTING QUERIES

The privacy guarantees of Algorithm 16 are conditioned on the histogram being k -sparse for all datasets. Here we present a technique when we do not have this guarantee. Specifically, we consider the setting when the input is a dataset $\mathbf{x} = (X_1, \dots, X_n)$ where $X_i \in \{0, 1\}^d$. We want to release a private estimate of $H(\mathbf{x}) = \sum_{i=1}^n X_i$, and X_i can have any number of non-zero entries. We use superscript to denote the elements of the histogram in descending order with ties broken arbitrarily such that $H(\mathbf{x})^{(1)} \geq H(\mathbf{x})^{(2)} \geq \dots \geq H(\mathbf{x})^{(d-1)} \geq H(\mathbf{x})^{(d)}$. This setting is studied in a line of work for *Private top-k selection*: [DR19; Ros97; QSZ21; MT07; BU17; WZ24].

Our algorithm in this setting relies on a simple pre-processing step. We first find the value of the $(k+1)$ 'th largest entry in the histogram. We then subtract that value from all entries in the histogram and remove negative counts. This gives us a new histogram which we use as input for Algorithm 16. We show that this histogram is both k -sparse and monotonic which implies that the mechanism has the same privacy guarantees as Algorithm 16.

Lemma 7.10. *The function $\tilde{H} : \{0, 1\}^{n \times d} \rightarrow \mathbb{N}^d$ in Algorithm 17 produces k -sparse monotonic histograms.*

Proof. By Definition 7.3 we have to show two properties of $\tilde{H}$. It must hold for any $\mathbf{x}$ that $\|\tilde{H}(\mathbf{x})\|_0 \leq k$ and for any neighboring pair $\mathbf{x} \sim \mathbf{x}'$ we have $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, 1\}^d$ or $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, -1\}^d$.

The sparsity claim is easy to see. Any counter that is not strictly larger than $H(\mathbf{x})^{k+1}$ is removed in line 4. By definition of $H(\mathbf{x}')^{k+1}$, there are at most k such entries.

To prove the monotonicity property, we must show that counters in $\tilde{H}$ either all increase or all decrease by at most one. We only give the proof for the case where $\mathbf{x}$ is constructed by adding one data point to $\mathbf{x}'$. The proof is symmetric for the case where $\mathbf{x}$ is created by removing one data point from $\mathbf{x}'$.

We first partition $H(\mathbf{x}')$ into three sets: Let $U = \{i \in [d] \mid H(\mathbf{x}')_i > H(\mathbf{x}')^{(k+1)}\}$, $M = \{i \in [d] \mid H(\mathbf{x}')_i = H(\mathbf{x}')^{(k+1)}\}$ and $L = [d] \setminus \{U, M\}$. Because we have $H(\mathbf{x})_i - H(\mathbf{x}')_i \in \{0, 1\}$ for all $i \in [d]$, also $H(\mathbf{x})^{(k+1)} - H(\mathbf{x}')^{(k+1)} \in \{0, 1\}$. Note that the $(k+1)$ 'th largest entry might not represent the same element, but we only care about the value.

Consider the case where $H(\mathbf{x})^{(k+1)} = H(\mathbf{x}')^{(k+1)}$. We see that for all $l \in L$: $\tilde{H}(\mathbf{x})_l = \tilde{H}(\mathbf{x}')_l = 0$, and for all $u \in U$ we have $\tilde{H}(\mathbf{x}')_u - \tilde{H}(\mathbf{x})_u = H(\mathbf{x}')_u - H(\mathbf{x})^{(k+1)} - H(\mathbf{x})_u + H(\mathbf{x}')^{(k+1)} \in \{0, 1\}$ because we subtract the same value and increment $H(\mathbf{x}')_u$ by at most one. Some elements $m \in M$ might now be exactly one larger than $H(\mathbf{x}')^{(k+1)}$ in $H(\mathbf{x})$, but still $\tilde{H}(\mathbf{x})_m \in \{0, 1\}$ by definition. Thus we have that $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, 1\}^d$.

Now, consider the case where $H(\mathbf{x})^{(k+1)} = H(\mathbf{x}')^{(k+1)} + 1$. No element $l \in L$ can become larger or equal to the new $(k+1)$ 'th largest element, so still $\tilde{H}(\mathbf{x}')_l = \tilde{H}(\mathbf{x})_l = 0$. Elements in $u \in U$ get reduced by at most one for $H(\mathbf{x})$, because they either are increased together with the $H(\mathbf{x}')^{(k+1)}$ and then $\tilde{H}(\mathbf{x})_u - \tilde{H}(\mathbf{x}')_u = 0$, or they stay the same in which case $\tilde{H}(\mathbf{x})_u = \tilde{H}(\mathbf{x}')_u - 1$. One can also see that for $m \in M$, $H(\mathbf{x})_m - H(\mathbf{x})^{(k+1)} \in \{0, -1\}$, depending on whether or not they increase together with $H(\mathbf{x})^{(k+1)}$. As such we have $\tilde{H}(\mathbf{x})_m = 0$ for all $m \in M$. Therefore, $\tilde{H}(\mathbf{x}) - \tilde{H}(\mathbf{x}') \in \{0, -1\}^d$. $\square$

Since Algorithm 17 simply returns the output of running Algorithm 16 with a k -sparse monotonic histogram it has the same privacy guarantees.

Corollary 7.11. *The privacy guarantees specified by Theorems 7.7 and 7.9 hold for Algorithm 17.*

Algorithm 17 introduces bias when subtracting $H(\mathbf{x})^{(k+1)}$ from each counter as a pre-processing step. If we have access to a private estimate of $H(\mathbf{x})^{(k+1)}$ we can add it to all counters of the output as post-processing. Since this estimate would be used for multiple counters, similar to Z_{corr} , we might want to use additional privacy budget to get a more accurate estimate. This can be included directly in the privacy analysis. If we e.g. release $\tilde{H}(\mathbf{x})^{(k+1)} = H(\mathbf{x})^{(k+1)} + \mathcal{N}(0, \sigma^2 / \sqrt{k})$ the privacy guarantees from Theorem 7.7 hold if we change $\sqrt{k + \sqrt{k}}$ to $\sqrt{k + 5\sqrt{k}}$ in δ_{gauss} .

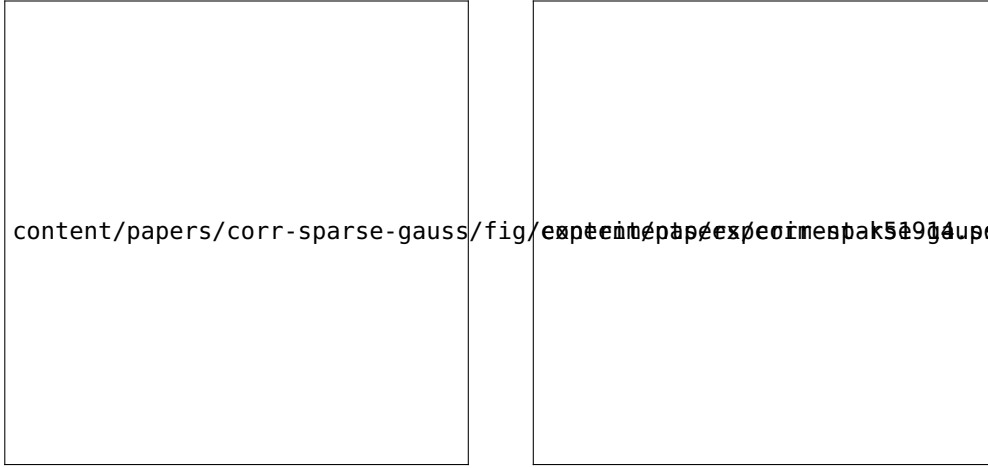
7.4 NUMERICAL EVALUATIONS

To back up our theoretical claims, we compare the error of the *Correlated Stability Histogram* against both the *add-the-deltas* approach [GAT20] and the *exact* analysis [Wil+24] of the uncorrelated GSHM. We consider the error in both privacy analysis approaches for our mechanisms as well (Theorems 7.7 and 7.9).

We compare the parameters of the mechanisms when releasing k -sparse monotonic histograms. We ran the experiments shown in Figure 7.3 with the same privacy parameters as in Wilkins, Kifer, Zhang, and Karrer [Wil+24]², which is $\epsilon = 0.35$ and $\delta = 10^{-5}$. Following their approach, we plot the minimum τ such that each mechanism satisfies (ϵ, δ) -DP for a given magnitude of noise. Note that our setting of $\|H(\mathbf{x})\|_0 \leq k$ differs from [Wil+24], so the experiments are not directly comparable.

RESULTS. The *first plot* shown in Figure 7.3a) resembles the same $k = 51914$ used in [Wil+24, Figure 1 (A)]. In this setting, one can lower the threshold by approximately 43%. Because our technique favors large k (we have to scale the noise with $\frac{1}{2}\sqrt{k + \sqrt{k}}$ instead of $\sqrt{k}$), we also include the case where $k = 10$ is small in Figure 7.3b). The plots show that we make some small improvements even in that case. Even in that case, our looser *add-the-deltas* analysis for the CSH beats Wilkins, Kifer, Zhang, and Karrer [Wil+24]. Note that the values of

² in turn based on the privacy guarantees of the *Facebook URL dataset* [Mes+20].



(a) Same ϵ, δ, k as in Wilkins, Kifer, Zhang, and Karrer [Wil+24]

(b) Some improvement also for smaller k .

Figure 7.3: The results of our experiments. Using the same parameters as in [Wil+24], the graphs show the minimum τ required to get $(0.35, 10^{-5})$ -DP guarantees for a noise level σ . The green line denotes the tight analysis of Wilkins, Kifer, Zhang, and Karrer [Wil+24], the red shows the add-the-deltas [GAT20] approach and the blue and orange lines are our results. The marked points denote the minimum τ for each technique. **a)** Uses the same parameters as in Wilkins, Kifer, Zhang, and Karrer [Wil+24]. As high values of k are preferable for our mechanism, we bring down the threshold from ≈ 13950 to ≈ 7860 , lowering it by $\approx 43\%$. **b)** We get some small improvement even for small k values. Note that since our mechanism adds two noise samples the plot shows the total magnitude of noise.

σ in Algorithm 15 and Algorithm 16 are not directly comparable, because we add noise twice in Algorithm 16. For a fair comparison, we instead use the total magnitude of noise in the plots. For the **Correlated Sparse Histogram Mechanism!** (Correlated Sparse Histogram Mechanism!) we plot the value of $(\sigma^2 + \sigma^2/\sqrt{k})^{1/2} = (1 + 1/\sqrt{k})^{1/2}\sigma$. The dotted lines indicated the minimum magnitude of total noise for which GSHM and Correlated Sparse Histogram Mechanism! satisfy (ϵ, δ) -DP.

7.5 EXTENSIONS

The setting in Wilkins, Kifer, Zhang, and Karrer [Wil+24] is slightly different from ours. Here we discuss how to adapt our technique to their setting and consider two extensions of the GSHM we did not include in the pseudocode of Algorithm 15.

ADDITIONAL SPARSITY THRESHOLD. The mechanism of Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24] employs a second threshold $\tau' < \tau$ that allows to filter out infrequent data in a pre-processing step: All counters below τ' are removed before adding noise. The high threshold τ is then used to remove noisy counters similar to both Algorithms 15 and 16. This generalized setup allows them to account for pre-processing steps that the privacy expert has no control over. Our setting corresponds to the case where $\tau' = 1$, and it is straightforward to incorporate this constraint into our mechanism: For a given τ' , we simply have to replace τ with the difference between the two thresholds in all theorems. In fact, this situation is very similar to our result in Section 7.3 where we remove all values below a lower threshold $\tau' = 1 + H(\mathbf{x})^{(k+1)}$ before adding noise.

Note that the lower threshold of Wilkins, Kifer, Zhang, and Karrer [Wil+24] is assumed to be data independent. If it is data-dependent, one must take care not to violate privacy. The privacy guarantees hold if we can give guarantees about the structure of pre-processed neighboring datasets similar to Lemma 7.10.

AGGREGATOR FUNCTIONS. Wilkins et al. Wilkins, Kifer, Zhang, and Karrer [Wil+24], consider a setting where we are given an aggregator function $\mathcal{A}$ which returns a vector when applied to any dataset $\mathbf{x}$.

If any $j \in [d]$ in the privatized released histogram $\tilde{H}(\mathbf{x})_j$, exceeds the threshold, the aggregator function is applied to all data points where $\mathbf{x}_{i,j} = 1$. The aggregated values are then privatized by adding Gaussian noise which shape and magnitude can be set independently of the magnitude of noise added to the count. The privatized aggregate is then released along with $\tilde{H}(\mathbf{x})_j$. This setup is motivated by *group-by* database operations where we first want to privately estimate the count of each group together with some aggregates modeled by $\mathcal{A}$.

We did not consider aggregate functions in our analysis and instead focused on the classical setting, where we only wanted to privately release a sparse histogram while ignoring zero counters. Wilkins, Kifer, Zhang, and Karrer account for the impact of the aggregator functions in the equivalent of δ_{gauss} in [Wil+24, Theorem 5.4, Corollary 5.4.1].

In short, the effect of aggregators can be accounted for as an increase in the ℓ_2 sensitivity using a transformation of the aggregate function. The same technique could be applied to our mechanism. This would give us a similar improvement as Lemma 7.8 over the standard GSHM. It's not as clear how the mechanisms compare under the tighter analyses.

The core idea behind the correlated noise can also be used to improve some aggregate queries. This can be used even in settings where we do not want to add correlated noise to the counts. The mechanism of Lebeda uses an estimate of the dataset size $\tilde{n}$ to reduce the independent noise added to each query. They spent part of the privacy budget to estimate $\tilde{n}$. However, in our setting, we already have a private estimate for the dataset size for the aggregate query in $\tilde{H}(\mathbf{x})_j$.

If each query is a *sum* query where users have a value between 0 and some value $C \in \mathbb{R}$, we can reduce the magnitude of noise by a factor of 2 by adding $(\tilde{n} - n)C/2$ to each sum (It follows from [Leb25, Lemma 3.6]). We can use a similar trick if $\mathcal{A}$ computes a sum of values in some range $[L, U]$. The standard approach would be to add noise with magnitude scaled to $\max(|L|, |U|)$ (see e.g. [Wil+20, Table 1]). Instead, we can recenter values around zero. As an example, $\mathcal{A}$ contains a sum query with values in $[100, 200]$. We subtract $(L + U)/2$ from each point so that we instead sum values in $[-50, 50]$ and reduce the sensitivity from 200 to 50. As post-processing, we add $150 \cdot \tilde{H}(\mathbf{x})_j$ to the estimate of the new sum. The error then depends on $\tilde{H}(\mathbf{x})_j - H(\mathbf{x})_j$. As with the Correlated Gaussian Mechanism, we gain an advantage over estimating the sum directly if $\mathcal{A}$ contains multiple such sums because we can reuse the estimate $\tilde{H}(\mathbf{x})_j$ for each sum.

7.6 FROM THEORY TOWARDS PRACTICE: DISCRETE GAUSSIAN NOISE

All mechanisms discussed in this chapter achieve privacy guarantees using continuous Gaussian noise which is a standard tool in the differential privacy literature. However, real numbers cannot be represented exactly on computers which makes the implementation challenging. We can instead use the Discrete Gaussian Mechanism [CKS20]. In this section, we modify the technique by Lebeda Lebeda [Leb25]

Algorithm 18 The Discrete Correlated Gaussian Mechanism**Require:** Parameters k and σ , monotonic histogram $H(\mathbf{x}) \in \mathbb{N}^k$.

- 1: Let $\tilde{\mathbf{h}} = \{0\}^k$.
- 2: Sample $Z_{\text{corr}} \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k})$.
- 3: **for** each $i \in [k]$ **do**
- 4: Sample $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2)$.
- 5: Set $\tilde{H}_i = H(\mathbf{x})_i + (Z_i + Z_{\text{corr}})/2$.
- 6: **Release** $\tilde{H}$.

to make the mechanism compatible with discrete Gaussian noise $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ and then provide privacy guarantees for the discrete analogue to Algorithm 16 using ρ -zero concentrated DP (ρ -zCDP). We first list required privacy guarantees of the *Multivariate Discrete Gaussian* as defined in Definition A.6.

Lemma 7.12 ([CKS20, Theorem 2.13] Multivariate Discrete Gaussian Mechanism). *Let $\sigma_1, \dots, \sigma_d > 0$ and $\rho > 0$ and let $q : \mathcal{U}^* \rightarrow \mathbb{Z}^d$ satisfy $\sum_{i \in [d]} (q(\mathbf{x})_i - q(\mathbf{x}')_i)^2 / \sigma_i^2 \leq 2\rho$ for all neighboring datasets $\mathbf{x} \sim \mathbf{x}'$. Define a randomized algorithm $M : \mathcal{U}^* \rightarrow \mathbb{Z}^d$ by $M(\mathbf{x}) = q(\mathbf{x}) + Z$ where $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_i^2)$ independently for each $i \in [d]$. Then M satisfies ρ -zCDP.*

Next, we present a discrete variant of the Correlated Gaussian Mechanism and prove that it satisfies ρ -zCDP. We can then convert the privacy guarantees to approximate differential privacy. We combine this with the *add-the-deltas* technique for a discrete variant of Algorithm 16. This approach does not give the tightest privacy parameters, but it is significantly simpler than a direct analysis of approximate differential privacy for multivariate discrete Gaussian noise (see [CKS20, Theorem 2.14]). The primary contribution in this section is a simple change that makes the correlated Gaussian mechanism compatible with the discrete Gaussian. We leave providing tighter analysis of the privacy guarantees open for future work.

Lemma 7.13. *Algorithm 18 satisfies ρ -zCDP where $\rho = (k + \sqrt{k})/(8\sigma^2)$.*

Proof. Fix any pair of neighboring histograms $H(\mathbf{x})$ and $H(\mathbf{x}')$. We prove the lemma for the case where $H(\mathbf{x}) - H(\mathbf{x}') \in \{0, 1\}^k$. The proof is symmetric when $H(\mathbf{x}) - H(\mathbf{x}') \in \{0, -1\}^k$.

Construct a new pair of histograms $\hat{H}(\mathbf{x}), \hat{H}(\mathbf{x}') \in \mathbb{N}^{k+1}$ such that $\hat{H}(\mathbf{x})_i = 2 \cdot H(\mathbf{x})_i$ for all $i \in [k]$ and $\hat{H}(\mathbf{x})_{k+1} = 0$. We set $\hat{H}(\mathbf{x}')_i = 2 \cdot \hat{H}(\mathbf{x}')_i - 1$ for all $i \in [k]$ and finally $\hat{H}(\mathbf{x}')_{k+1} = 1$.

We clearly have that $\hat{H}(\mathbf{x}) \in \mathbb{Z}^{k+1}$ and $\hat{H}(\mathbf{x}') \in \mathbb{Z}^{k+1}$ for all possible $H(\mathbf{x})$ as required by Lemma 7.12. Now, we set $\sigma_i^2 = 4\sigma^2$ for $i \in [k]$ and $\sigma_{k+1}^2 = 4\sigma^2/\sqrt{k}$. We constructed $\hat{H}(\mathbf{x})$ and $\hat{H}(\mathbf{x}')$ such that they differ by at most 1 in all entries for any pair of neighboring histograms. We therefore have that

$$\sum_{i \in [k+1]} (\hat{H}(\mathbf{x})_i - \hat{H}(\mathbf{x}')_i)^2 / \sigma_i^2 \leq \sum_{i \in [k+1]} 1 / \sigma_i^2 = k / (4\sigma^2) + \sqrt{k} / (4\sigma^2) = 2\rho.$$

By Lemma 7.12 we have that $D_\alpha(M(\mathbf{x}) || M(\mathbf{x}')) \leq \rho\alpha$ for all $\alpha > 1$, where $M(\mathbf{x}) = \hat{H}(\mathbf{x}) + Z$ and $Z_i \sim \mathcal{N}(0, \sigma_i^2)$. The output of $M(\mathbf{x})$ can be post-processed such that $\tilde{H}_i = (M(\mathbf{x})_i + M(\mathbf{x})_{k+1})/2$. Notice that such a post-processing gives us the

same output distribution as Algorithm 18 for both $H(\mathbf{x})$ and $H(\mathbf{x}')$. The algorithm therefore satisfies ρ -zCDP because the α -Rényi divergence cannot increase from post-processing by Definition 2.9. $\square$

Note that the scaling step is crucial for the privacy proof. It would not be sufficient to simply replace $Z_i \sim \mathcal{N}(0, \sigma^2)$ with $Z_i \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$. We need to sample noise in discrete steps of length $1/2$ instead of length 1. Otherwise, the trick of centering the differences between the histograms using the $k+1$ entry as an offset does not work. If we prefer a mechanism that always outputs integers, we can simply post-process the output. Next, we give the privacy guarantees of the Correlated Stability Histogram when using discrete noise.

Lemma 7.14. *Let $Z_{\text{corr}} \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k})$ and $Z_1, \dots, Z_k \sim \mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2)$. Then for any $\tau > 0$ we have that:*

$$\mathbf{P}[\exists i : Z_{\text{corr}} + Z_i > 2\tau] \leq 1 - \mathbf{P}\left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k}) \leq \frac{2\tau k^{-1/4}}{k^{-1/4} + 1}\right] \cdot \mathbf{P}\left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2) \leq \frac{2\tau}{k^{-1/4} + 1}\right]^k.$$

Proof. Similar to the proof of Lemma 7.6: if $Z_{\text{corr}} \leq 2\tau k^{-1/4}/(k^{-1/4} + 1)$ and $\max_{Z_1, \dots, Z_j} Z_i = \tilde{Z} \leq 2\tau/(k^{-1/4} + 1)$ then the sums cannot be above 2τ . The only difference is that we have to split up the probabilities. In the discrete case the probabilities slightly change when we rescale. $\square$

Finally, we can replace the Gaussian noise in Algorithm 16 with discrete Gaussian noise. Using the *add-the-deltas* approach and the lemmas above we get that.

Theorem 7.15. *For parameters k, σ , and τ , consider the mechanism $\mathcal{M}: \mathbb{N}^d \rightarrow \mathbb{Q}^d$ that given a histogram $H(\mathbf{x})$: (1) runs Algorithm 18 with parameters k and σ restricted to the support of $H(\mathbf{x})$ (2) removes all noisy counts less than or equal to $1 + \tau$. Then $\mathcal{M}$ satisfies $(\epsilon, \delta_{\text{gauss}} + \delta_{\text{inf}})$ -DP for k -sparse monotonic histograms, where δ_{gauss} is such that $(k + \sqrt{k})/(8\sigma^2)$ - ρ -zCDP implies $(\epsilon, \delta_{\text{gauss}})$ -DP and*

$$\delta_{\text{inf}} = 1 - \mathbf{P}\left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2/\sqrt{k}) \leq \frac{2\tau k^{-1/4}}{k^{-1/4} + 1}\right] \cdot \mathbf{P}\left[\mathcal{N}_{\mathbb{Z}}(0, 4\sigma^2) \leq \frac{2\tau}{k^{-1/4} + 1}\right]^k.$$

Proof. The proof relies on the construction of an intermediate histogram from the proof of Theorem 7.7. We have that

$$\begin{aligned} \mathbf{P}[\mathcal{M}(H(\mathbf{x})) \in Y] &\leq \mathbf{P}[\mathcal{M}(H(\hat{\mathbf{x}})) \in Y] + \delta_{\text{inf}} \\ &\leq e^\epsilon \mathbf{P}[\mathcal{M}(H(\mathbf{x}')) \in Y] + \delta_{\text{gauss}} + \delta_{\text{inf}}, \end{aligned}$$

where the first inequality follows from Lemma 7.14 and the second inequality follows from Lemma 7.13. Both inequalities rely on the fact that the histograms are k -sparse. $\square$

7.7 CONCLUSION AND OPEN PROBLEMS

We introduced the *Correlated Stability Histogram* for the setting of k -sparse monotonic histograms and provided privacy guarantees using the *add-the-deltas* approach and a more fine-grained case-by-case analysis. We show that our mechanism outperforms the state-of-the-art – the *Gaussian Sparse Histogram Mechanism* – and improves the utility by up to a factor of 2. In addition to various extensions, we enriched our theoretical contributions with a step towards practice by including a version that works with discrete Gaussian noise.

Unlike the previous work [Wil+24], our bound in Theorem 7.9 is not tight. It would be interesting to derive exact bounds for our mechanism as well. Finally, we point out that the uncorrelated GSHM is still preferred in some settings. The *Correlated Sparse Histogram Mechanism* requires an upper bound on $\|H(\mathbf{x})\|_0$, whereas GSHM requires a bound on $\|X_i\|_0$. If we have access to a bound $\|X_i\|_0 \leq k$ but no sparsity bound, we can apply our technique from Section 7.3. This approach works well when the dataset is close to k -sparse, but if the dataset is large with many high counters the pre-processing step can introduce high error. Furthermore, if histograms are k -sparse, but we additionally know that $\|X_i\|_0 \leq m$, for some $m < k$, our improvement factor changes. If $m \leq k/4$, the GSHM has a lower error than the CSH. However, in that setting, it might still be possible to reduce the error using the technique of Joseph and Yu [JY24]. We leave exploring this regime for future work.

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Part III

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Throughout the thesis, we use multiple probability distributions, which we will quickly define in this section.

Definition A.1 (Gaussian Distribution). *The zero-centered Gaussian (normal) distribution $\mathcal{N}(0, \sigma^2)$ with variance σ^2 has pdf $\phi(x, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{x^2}{2\sigma^2})$. We denote the cdf of the Gaussian distribution as $\Phi(x)$. We refer to Φ^{-1} as the inverse CDF.*

Definition A.2 (Laplace distribution). *The zero-centered Laplace distribution $\text{Lap}(0, b)$ with scale parameter $b > 0$ has probability density function $f_b(x) = \frac{1}{2b} \exp(-|x|/b)$ for all $x \in \mathbb{R}$.*

Definition A.3 ([Ros18] Beta Distribution). *The beta distribution $\text{Beta}(\alpha, \beta)$ is a distribution defined on $[0, 1]$ whose density is given by $p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{\text{B}(\alpha, \beta)}$, $\forall x \in [0, 1]$, where $\alpha, \beta > 0$ are shape parameters, $\text{B}(\alpha, \beta) \doteq \int_0^1 x^{\alpha-1}(1-x)^{\beta-1} dx$ is a normalization constant.*

Definition A.4 (Exponential Distribution). *Given $\lambda > 0$, a random variable X follows exponential distribution $\text{Exp}(\lambda)$, denoted by $X \sim \text{Exp}(\lambda)$, if it has density $p_X(x) \doteq \lambda e^{-\lambda x}$, $\forall x \geq 0$, and $p_X(x) \doteq 0$, $\forall x < 0$.*

Definition A.5 (Gumbel Distribution). *Given parameter $b \in \mathbb{R}$, the Gumbel distribution, $\text{Gumbel}(b)$, has pdf $p(z) = \frac{1}{b} \cdot \exp(-(\frac{z}{b} + \exp(-\frac{z}{b})))$, $\forall z \in \mathbb{R}$, and cumulative distribution function $F(z; b) = \exp(-e^{-z/b})$, $\forall z \in \mathbb{R}$.*

Definition A.6 (Discrete Gaussian Distribution). *Let $\sigma \in \mathbb{R}$ with $\sigma > 0$. The discrete Gaussian distribution with mean 0 and scale σ is denoted $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$. It is a probability distribution supported on the integers and defined by*

$$\forall x \in \mathbb{Z} : \mathbf{P}_{X \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma^2)}[X = x] = \frac{e^{-x^2/(2\sigma^2)}}{\sum_{y \in \mathbb{Z}} e^{-y^2/(2\sigma^2)}}.$$

Definition A.7 (Geometric Distribution). We say X has distribution $\text{Geom}(p)$ with parameter $0 < p \leq 1$, if $\mathbf{P}[X = k] = (1 - p)^{k-1}p$ for all $k \in \mathbb{N}_{>0}$.

Definition A.8 (Discrete Laplace). The noise distribution for the discrete Laplace (two-sided geometric) $\text{DLap}(\alpha)$ is a two-sided geometric distribution with pmf $p(x) = \frac{1-\alpha}{1+\alpha} \cdot \alpha^{|x|}$ for $\alpha \in (0, 1)$ and $x \in \mathbb{Z}$.

ABBREVIATIONS

ANM Additive Noise Mechanism

DP Differential Privacy

IANM Independent Additive Noise Mechanism

HC Minimum-Cost Hierarchical Clustering

MST Minimum-Weight Spanning Tree

MWPM Minimum-Weight (Perfect) Matching

PLMR Private Lossless Multiple Release

PPSACR Probability Proportional to Sizes with Adaptive Candidate Removal

ρ -zCDP ρ -zero concentrated DP

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